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To our students

Foreword

This work reflects the lectures of the authors at the Civil Engineering School of Barcelona, Technical University of Catalonia-BarcelonaTech.

This book corresponds to the courses of Strength of Materials that are included in various Engineering and Architecture degrees.

The book is structured in two parts. The first one, Chapters 1 to 4, presents the fundamental concepts common to the Theory of Elasticity and Strength of Materials; the second, Chapters 5 to 9, studies the mechanical effects of the different internal resultants acting on a cross-section.

The understanding of the different topics is favored for the inclusion of numerous exercises and worked examples.

The authors thank Prof. Antonia Larese for her collaboration in the compilation, edition and revision of the Spanish edition of the book. We are indebted to Miss Uxue Chasco, Dr. Christian Narvaez and Dr. Alejandro Cornejo for their indispensable teamwork in the English edition. Likewise, we thank Mr. Raúl Giménez for the drawing of the figures and schemes that are included. We thank the Civil Engineering School of Barcelona and to the International Center for Numerical Methods (CIMNE) for the support received from through their respective incentive programs for the production of lecturing material.

Finally, we thank our families for their support and understanding, without which this work would not have been possible.

Miguel Cervera and Elena Blanco
Barcelona, July 2023

Contents

1	Stress and Strain	1
1.1	Introduction	1
1.2	The concept of stress	2
1.3	Stress components	5
1.4	Plane stress state	6
1.4.1	Cauchy internal equilibrium equations	6
1.4.2	Stress in an arbitrary direction	7
1.4.3	Transformation of the reference system	8
1.4.4	Principal stresses	10
1.5	Three-dimensional stress state	14
1.6	Concept of strain	15
1.6.1	Displacements	15
1.6.2	Strain	19
1.7	Plane state of strain	23
1.7.1	Displacements and strains	23
1.7.2	Strain tensor	26
1.7.3	Transformation of the reference system	27
1.7.4	Principal strains	29
1.8	Three-dimensional state of strain	32
2	Elasticity and material behavior	35
2.1	Introduction	35
2.2	Elasticity and linearity. Hooke's law	35
2.3	Principle of Superposition	36
2.4	Generalized Hooke's Law	37
2.4.1	Plane stress state	48
2.4.2	Plane strain state	50
2.5	Stress-strain relationship. Experimental study	53
2.6	Limit stress, allowable stress and safety factor	60
3	Concepts and Basic Principles	63
3.1	Introduction	63
3.2	Objectives of the Strength of Materials	63

3.3	Concept of prismatic element	65
3.4	Concept of Bar Structure	68
3.5	Structural elements	70
3.6	Supports and links in bar structures	71
3.6.1	Supports	71
3.6.2	Supports in mid-plane truss structures	72
3.6.3	Links	73
3.7	Principles of Strength of Materials	77
3.7.1	The Principle of Stiffness	77
3.7.2	The Principle Superposition	77
3.7.3	The Saint-Venant Principle	79
3.7.4	Geometrical constraints	80
3.8	Internal resultants	81
3.9	Relationship between internal resultants and stresses	84
3.10	Internal resultants in mid-plane elements	86
3.11	Equilibrium equations in straight elements	87
3.12	Statically indeterminacy	91
3.12.1	Static equilibrium and solution of structures	91
3.12.2	Truss structures	92
3.12.3	Frames	93
4	Internal resultants	97
4.1	Introduction	97
4.2	Internal forces and moments	97
4.3	Internal forces in truss structures	98
4.4	Internal resultants in midplane structures	104
5	Axial Force	123
5.1	Introduction	123
5.2	Axial force in a straight element	124
5.3	Cables	136
5.4	Axial force and temperature	140
6	Plane Bending	143
6.1	Introduction	143
6.2	Pure plane bending	143
6.2.1	Pure bending in midplane beams	144
6.2.2	Pure bending about a principal plane of inertia	150
6.2.3	Maximum allowable bending moments	151
6.2.4	Section Modulus	152
6.2.5	Section Geometrical Efficiency	153
6.3	Non-uniform bending	161
6.4	Composed plane bending	168

7	Skew bending	181
7.1	Introduction	181
7.2	Pure skew bending	181
7.2.1	Stresses in pure skew bending	182
7.2.2	Deformations in pure skew bending	184
7.3	Simple skew bending	190
7.4	Skew composed bending	196
7.5	Central core of a cross-section	202
7.5.1	Circular cross-section	203
7.5.2	Circular crown cross-section	204
7.5.3	Rectangular cross-section	204
7.5.4	Wide flange W-profile	205
8	Shear Force	207
8.1	Introduction	207
8.2	Elementary shear theory	207
8.3	Collignon shear theory	210
8.4	Rectangular cross section	216
8.5	Thin-walled cross-sections	218
8.5.1	Open thin-walled cross section	218
8.5.2	Closed thin-walled cross sections	224
8.6	Shear center	226
8.6.1	Position of the shear centre	230
9	Torsion	235
9.1	Introduction	235
9.2	Coulomb theory	237
9.2.1	Circular sections	237
9.2.2	Hollow circular cross-section	241
9.2.3	Closed thin-walled cross-sections	241
9.3	Saint-Venant torsion	245
9.3.1	General formulation	245
9.3.2	Membrane analogy	247
9.3.3	Rectangular cross-section	249
9.3.4	Narrow rectangular cross-sections	250
9.3.5	Open rolled profiles	252
A	Geometric characteristics of the section	259
A.1	Axial first moment of area of a cross section	259
A.2	Moments of inertia of a cross-section	262
A.2.1	Axial moment of inertia and rotation radius	262
A.2.2	Second polar moment of area	263
A.2.3	Product of inertia	264
A.3	Translation of reference axes	267

A.3.1	Axial moment of inertia	267
A.3.2	Second polar moment of area	268
A.3.3	Product of inertia	268
A.4	Reference axis rotation	270
A.5	Principal axes of inertia	272
A.6	Thin cross sections	275
B	Structural Steel Profiles	279
C	Tables	285
C.1	Magnitudes and Units of Mechanics	285
C.2	Mechanical properties of materials	286
C.3	Multiples and Submultiples	288
C.4	Greek alphabet	288

Chapter 1

Stress and Strain

1.1 Introduction

The main objective of Strength of Materials is to provide the engineer with the means to analyze and design structures or components capable of withstanding the loads and actions to which they are or may be subjected during their service life. Both analysis and design of any structural component involve the determination of *stresses* and *strains*. Therefore, the concepts of stress and strains are basic concepts in the exposition of this subject.

Strength of Materials and the Theory of Elasticity, as parts of the Mechanics of Deformable Solids, are two disciplines with common objectives: both deal with the study of strength (stress state) and stiffness (strain state) of deformable solid bodies subjected to the action of force systems in static equilibrium. Likewise, both start from the principle of linearity between action and response; this, implies that the behavior of the materials is elastic and that the displacements that occur are small.

Strength of Materials limits its field of application to certain types of structural elements (beams, columns, etc.) supported in certain predetermined ways (simple supports, joints, clappings, etc.) and subjected to certain types of actions (point and distributed forces, generally, and other actions defined appropriately). These restrictions regarding geometries, conditions of support, and considered actions, allow the formulation of certain hypotheses and a simplified approach for the analytical solution of several engineering problems.

Theory of elasticity addresses the elastic mechanical problem in its most general form in terms of geometries, boundary conditions and types of actions considered. This leads to a rigorous mathematical formulation that prevents obtaining analytical solutions, except for a limited number of cases, requiring the use of approximate numerical methods (finite differences, finite elements, etc.) to the solution of most hands-on problems.

Given their common objectives (and principles), Strength of Materials and the Theory of Elasticity follow parallel paths from their common begin-

nings and, from the systematization of both, it is not easy to clearly delimit the respective fields. From the current perspective, it is usual to consider the Strength of Materials as a sub-field of the, more general, Theory of Elasticity. In any case, both disciplines handle a multitude of common concepts, such as force, displacement, stress, strain, equilibrium, compatibility, linearity, etc..

In this Chapter, the concepts of stress and strain are presented and developed, which will be used throughout Chapters. The conception of stress as applied force per surface unit and its relation to structural strength is undoubtedly due to Galileo. The modern definition of strain as the unitary change of length is attributed to Cauchy, although it certainly was used previously. This Chapter also establishes the existing equilibrium relationships between the applied forces and the its subsequent stress state, and the compatibility relationships between displacements and strains.

1.2 The concept of stress

Let us consider a solid body subjected to the action of a system of external forces (applied loads and reactions) in equilibrium, and imagine it cut by a section S that divides it into two parts, (A) and (B) , located on both sides of the section S (Figure 1.1). For the resulting parts $((A)$ and $(B))$ to be in equilibrium, there must be some interaction forces across the surface S that we denote as $\mathbf{F}$. These interaction forces are equal but in opposite directions, as required by the principle of action and reaction.

Let us consider a point on the surface S and its surroundings area ΔS . Let us call $\Delta \mathbf{F}$ the force that part (B) exerts on part (A) through the area ΔS . The force per unit of area is:

$$\mathbf{t}_m = \frac{\Delta \mathbf{F}}{\Delta S} \quad (1.1)$$

This force per unit area $\mathbf{t}_m$ is called the average stress on the surface S at the point considered. If the area ΔS tends to a differential element of area dS , one obtains what is defined as stress at a point according to a given surface S :

$$\mathbf{t} = \lim_{\Delta S \rightarrow 0} \frac{\Delta \mathbf{F}}{\Delta S} = \frac{d\mathbf{F}}{dS} \quad (1.2)$$

Regarding the concept of stress, the following observations can be made:

- The stress dimensions are $[FL^{-2}]$, that is, force over surface unit. In the international system, stress is measured in Pascals (Pa), i.e. in N/m^2 .

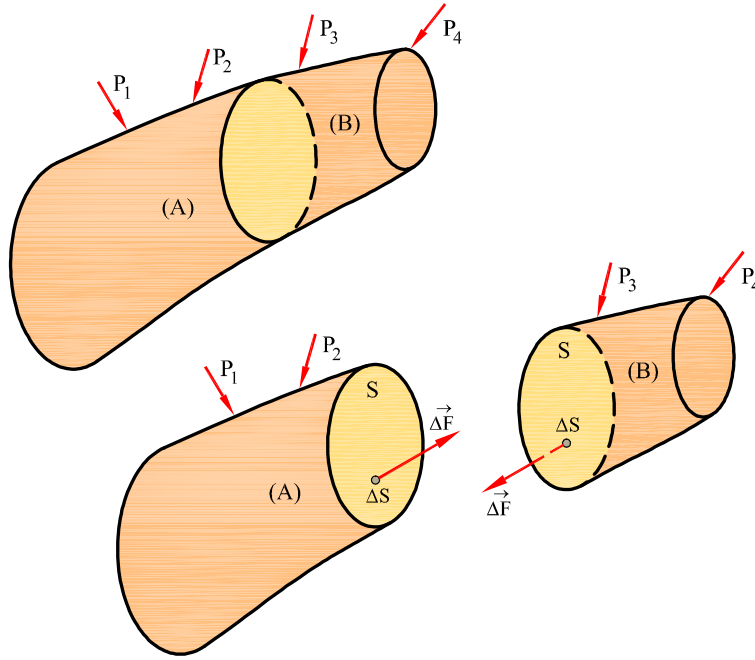


Figure 1.1: Concept of average stress

- In general, the stress is not normal to the cutting plane; instead, it can be decomposed into two components: the normal and shear stress to the plane of the section; σ and τ , respectively, as shown in Figure 1.2. The modulus of the stress $\mathbf{t}$ is equal to:

$$|\mathbf{t}| = \sqrt{\sigma^2 + \tau^2} \quad (1.3)$$

- The stress depends on the point and the orientation of the chosen section. Thus, at a given point there will be different stresses according to the considered orientation (Figure 1.3), and for a given section, there will be different stresses for different points.

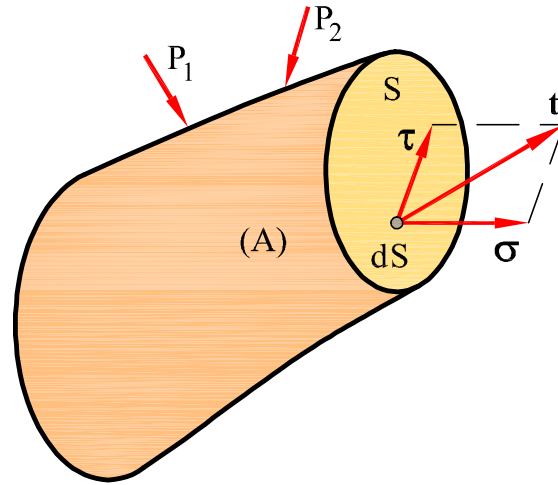


Figure 1.2: Normal and tangential components of stress

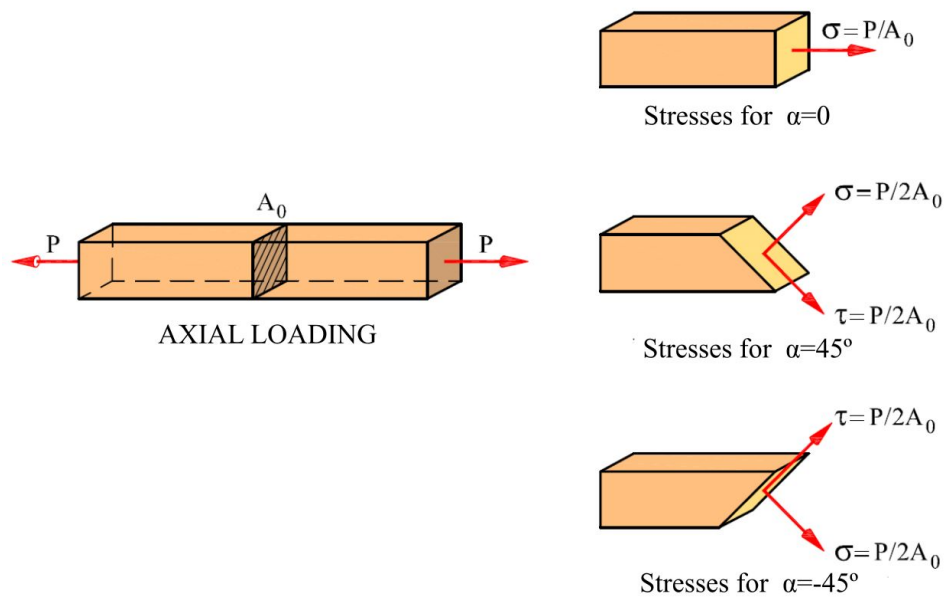


Figure 1.3: Components of the stress according to the orientation of the cutting plane

1.3 Stress components

Let us consider a body subjected to the action of a system of forces in static equilibrium. Let O be a point in its interior where the stress state is determined. Let us analyze a differential volume element, of hexahedral shape and dimensions (dx, dy, dz) , with vertex at O and with faces parallel to a certain Cartesian reference system (x, y, z) with origin at O (Figure 1.4). On each face of the differential element there acts different stresses $\mathbf{t}$ that can be decomposed according to the axes of reference. We will call σ to the components of the stress vector normal to each face and τ to the tangential components. The tangential components can be decomposed in turn into the directions of the two axes parallel to the face.

Thus, on faces normal to the axis $\begin{Bmatrix} x \\ y \\ z \end{Bmatrix}$ the components $\begin{Bmatrix} \sigma_x, \tau_{xy}, \tau_{xz} \\ \sigma_y, \tau_{yx}, \tau_{yz} \\ \sigma_z, \tau_{zx}, \tau_{zy} \end{Bmatrix}$ act.

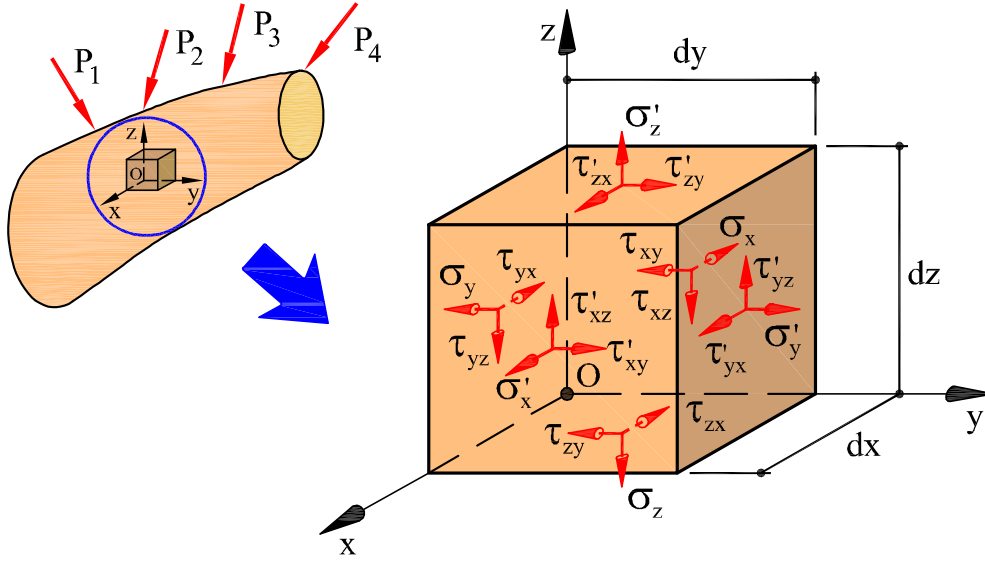


Figure 1.4: Reference system and cartesian components of the stress

The subscript of the normal stresses represents the coordinate axis that is normal to the surface on which they act. On the other hand, the two subscripts of the tangential stresses indicate the axis that is normal and parallel to the surface on which they act, respectively. In the faces that do not form the coordinate axes, positive components are considered to be those that have the directions of the axes. By contrast, on the faces that form the coordinate axes. Thereby, traction stresses are positive normal

stresses and those negative are compression stresses.

1.4 Plane stress state

If the stress components on normal planes to the z axis, σ_z , τ_{zx} , and τ_{zy} , are zero, the stresses $\mathbf{t}$ are contained in the plane xy . In this situation, the plane-stress state is achieved (Figure 1.5). Here, the stress state is completely defined by the components σ_x , σ_y , τ_{xy} . In this case knowing these components of the stress tensor in any point O , it is possible to determine the stress on an arbitrary surface, provided its direction cosines are known.

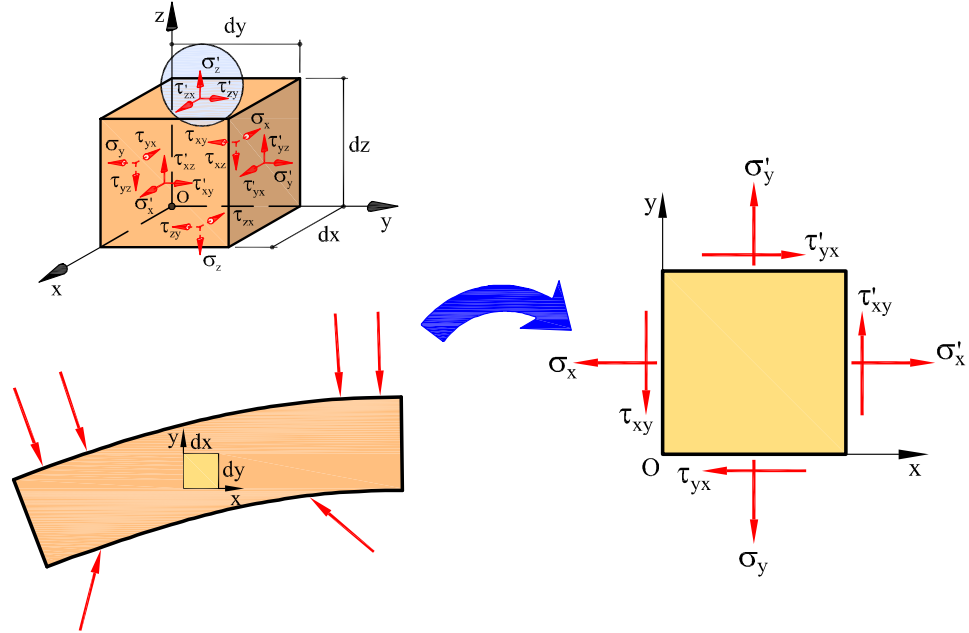


Figure 1.5: Plane stress state

1.4.1 Cauchy internal equilibrium equations

Distinguishing with a “prime” the stresses that act on the sides that do not form the coordinate axes (Figure 1.5), the relationships that exist between the stresses corresponding to parallel sides, for example, the sides of the differential element perpendicular to the x axis are, by continuity of the stress field:

$$\begin{aligned}
\sigma'_x &= \sigma_x + d\sigma_x = \sigma_x + \frac{\partial \sigma_x}{\partial x} dx \\
\tau'_{xy} &= \tau_{xy} + d\tau_{xy} = \tau_{xy} + \frac{\partial \tau_{xy}}{\partial x} dx
\end{aligned} \tag{1.4}$$

Similarly, the relations for the components of stress acting on the sides perpendicular to the y axis.

Calling $\mathbf{b}$ to the forces per unit area whose Cartesian components are $\mathbf{b}$ (b_x, b_y) and posing the static equilibrium conditions to the isolated differential element, by balancing of forces according to the two Cartesian directions, the internal equilibrium or Cauchy equations are obtained:

$$\begin{aligned}
\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + b_x &= 0 & (\sum F_x = 0) \\
\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + b_y &= 0 & (\sum F_y = 0)
\end{aligned} \tag{1.5}$$

In addition, from the equilibrium of moments about the point O (1.5) the principle of reciprocity of the tangential stress is obtained:

$$(\tau_{xy} dy) dx - (\tau_{yx} dx) dy = 0 \quad (\sum M_O = 0) \tag{1.6}$$

that is,

$$\tau_{xy} = \tau_{yx} \tag{1.7a}$$

Therefore, the principle of reciprocity of the tangential stress is deduced, which establish that "in two perpendicular planes, the normal tangential stresses to the common edge at a point are equal in modulus, and both concur or separate simultaneously from the edge" (Figure 1.6).

1.4.2 Stress in an arbitrary direction

Let AB be an arbitrary orientation and $\mathbf{n}(l, m)$ the normal vector that defines it, with $l = \cos \alpha$ and $m = \sin \alpha$. From the equilibrium of forces in the differential element, it is possible to calculate the stress components:

$$t_x = \sigma_x \cos \alpha + \tau_{yx} \sin \alpha \tag{1.8a}$$

$$t_y = \tau_{xy} \cos \alpha + \sigma_y \sin \alpha \tag{1.8b}$$

where t_x and t_y are the Cartesian components of the stress $\mathbf{t}$ on O along the plane AB . In matrix form it can be written:

$$\begin{bmatrix} t_x \\ t_y \end{bmatrix} = \begin{bmatrix} \sigma_x & \tau_{yx} \\ \tau_{xy} & \sigma_y \end{bmatrix} \begin{bmatrix} l \\ m \end{bmatrix} \tag{1.9}$$

$$\mathbf{t} = \mathbf{T} \mathbf{n} \quad (1.10)$$

where $\mathbf{T}$ is the stress tensor corresponding to the plane state.

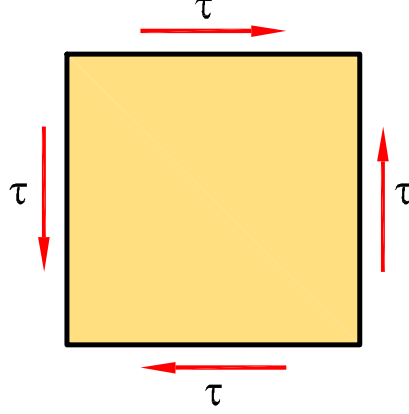


Figure 1.6: Plane state: reciprocity of stresses

From the principle of reciprocity of tangential stresses, the tensor of stress is symmetric.

From the above it can be deduced that, once the stress tensor is known at a point and the direction cosines of the normal to any orientation AB passing through the point, we can calculate the stress $\mathbf{t}$ for that orientation.

Knowing the stress $\mathbf{t}$ in a point according to an arbitrary orientation AB (Figure 1.7a), the normal and tangential components of such stress are obtained by projecting the components t_x and t_y on the normal and tangential directions to the plane AB (Figure 1.7b), respectively:

$$\begin{aligned} \sigma &= t_x \cos \alpha + t_y \sin \alpha \\ &= \sigma_x \cos^2 \alpha + \sigma_y \sin^2 \alpha + 2\tau_{xy} \sin \alpha \cos \alpha \end{aligned} \quad (1.11a)$$

$$\begin{aligned} \tau &= t_y \cos \alpha - t_x \sin \alpha \\ &= \tau_{xy} (\cos^2 \alpha - \sin^2 \alpha) + (\sigma_y - \sigma_x) \sin \alpha \cos \alpha \end{aligned} \quad (1.11b)$$

1.4.3 Transformation of the reference system

The components of the stress tensor vary according to the system of reference in which they are expressed. If the stress state is plane (Figure 1.8), the stresses referred to the rotated system x', y' will vary with respect to those corresponding to the system x, y . Let $\mathbf{N}$ be the matrix of direction cosines of the axes x', y' with respect to the axes x, y :

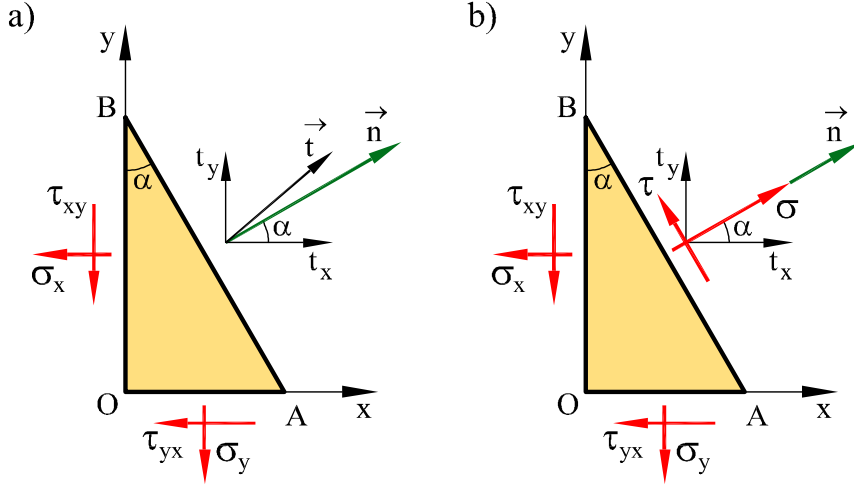


Figure 1.7: Stress according to an arbitrary direction

$$\mathbf{N} = \begin{bmatrix} \cos(x'x) & \cos(y'x) \\ \cos(x'y) & \cos(y'y) \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} = \begin{bmatrix} l & -m \\ m & l \end{bmatrix} \quad (1.12)$$

The stress $\mathbf{t}'_x(t'_{x'x}, t'_{x'y})$ acting on faces normal to the x' axis expressed as a function of the stress tensor of the system x, y are:

$$\mathbf{t}'_x = \begin{bmatrix} t'_{x'x} \\ t'_{x'y} \end{bmatrix} = \begin{bmatrix} \sigma_x & \tau_{yx} \\ \tau_{xy} & \sigma_y \end{bmatrix} \begin{bmatrix} l \\ m \end{bmatrix} \quad (1.13)$$

Analogously for, the stresses $\mathbf{t}'_y(t'_{y'x}, t'_{y'y})$ acting on the faces normal to y' , so it can be written:

$$\begin{bmatrix} t'_{y'x} & t'_{y'y} \end{bmatrix} = \begin{bmatrix} \sigma_x & \tau_{yx} \\ \tau_{xy} & \sigma_y \end{bmatrix} \begin{bmatrix} l & -m \\ m & l \end{bmatrix} \quad (1.14)$$

$$\mathbb{T}' = \mathbf{T} \mathbf{N} \quad (1.15)$$

where $\mathbf{T}$ is the stress tensor in the system x, y of reference and $\mathbf{N}$ is the matrix of direction cosines corresponding to a rotation α of such reference system. The elements of the matrix $\mathbb{T}'$ are the stresses acting on the faces normal to x', y' referred to system x, y .

The components of the tensor in the new reference system are obtained projecting the tensor $\mathbb{T}'$ in the directions x', y' :

$$\begin{bmatrix} \sigma_{x'} & \tau_{y'x'} \\ \tau_{x'y'} & \sigma_{y'} \end{bmatrix} = \begin{bmatrix} l & m \\ -m & l \end{bmatrix} \begin{bmatrix} \sigma_x & \tau_{yx} \\ \tau_{xy} & \sigma_y \end{bmatrix} \begin{bmatrix} l & -m \\ m & l \end{bmatrix} \quad (1.16)$$

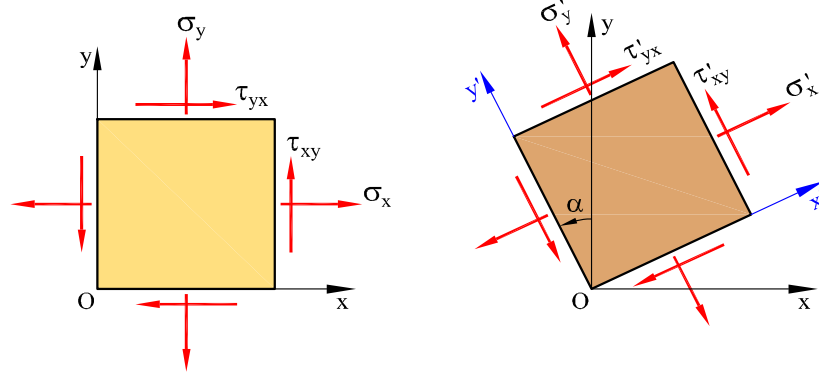


Figure 1.8: Transformation of the reference system

$$\mathbf{T}' = \mathbf{N}^T \mathbf{T} \mathbf{N} \quad (1.17)$$

Matrix $\mathbf{N}$ is an orthogonal matrix and it holds $\mathbf{N}^T \mathbf{N} = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix; therefore, it can be written:

$$\mathbf{T} = \mathbf{N} \mathbf{T}' \mathbf{N}^T \quad (1.18)$$

1.4.4 Principal stresses

Knowing the expression that provides the value of the Cartesian components of the stress $\mathbf{t}$, for a given direction $\mathbf{n}$ is known, those directions where the stress vector is parallel to $\mathbf{n}$ (Figure 1.9), can be found;

$$\mathbf{t} = \sigma \mathbf{n} \quad (1.19)$$

where σ is a scalar value corresponding to the stress modulus in the direction defined by $\mathbf{n}$. Combining the Eqs. (1.10) and (1.19), it must be fulfilled that:

$$(\mathbf{T} - \sigma \mathbf{I}) \mathbf{n} = \mathbf{0} \quad (1.20)$$

For this system of linear and homogeneous equations to have a solution, it is necessary that the determinant of the system must be null:

$$\begin{vmatrix} \sigma_x - \sigma & \tau_{yx} \\ \tau_{xy} & \sigma_y - \sigma \end{vmatrix} = 0 \quad (1.21)$$

expanding:

$$(\sigma_x - \sigma)(\sigma_y - \sigma) - \tau_{xy}^2 = 0 \quad (1.22)$$

Solving, we get:

$$\sigma = \sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad (1.23)$$

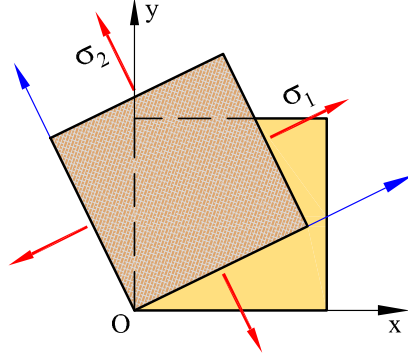


Figure 1.9: Plane state: principal stresses

where σ_1 and σ_2 are the maximum and minimum principal stresses, respectively.

The same result can be reached starting from the normal and tangential components according to an arbitrary $\mathbf{n}$ direction. From Eq. (1.11b) it is deduced that there is a certain angle α for which the tangential stress τ vanishes, that is:

$$\tau_{xy} (\cos^2 \alpha - \sin^2 \alpha) + (\sigma_y - \sigma_x) \sin \alpha \cos \alpha = 0 \quad (1.24)$$

where

$$\frac{\tau_{xy}}{\sigma_x - \sigma_y} = \frac{\sin \alpha \cos \alpha}{\cos^2 \alpha - \sin^2 \alpha} = \frac{1}{2} \tan 2\alpha \quad (1.25)$$

From this equation it is possible to find two directions perpendicular to each other for which the tangential stress is cancelled; these directions are the principal directions of the stress and the corresponding normal stresses are the principal stresses.

Similarly, it can be calculated that the maximum stress in the point occurs in the directions that form 45° with the principal directions:

$$\tau_{max} = \frac{1}{2}(\sigma_1 - \sigma_2) = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad (1.26)$$

Example 1.4.1

In a plane-stress state the normal and tangential stresses which act in the planes OA and OB are known (Figure 1.7). Calculate the values of these stresses for the plane AB that forms an angle $\alpha = 30^\circ$ with the y axis. Data: $\sigma_x = 28.48$ MPa, $\sigma_y = 15$ MPa, $\tau_{xy} = 3$ MPa.

Applying the previous equations and substituting the values of the example, we obtain for the plane AB :

$$\begin{aligned}\sigma &= 28.48 \cdot 0.866^2 + 15 \cdot 0.5^2 + 3 \cdot 0.866 = 27.71 \text{ MPa} \\ \tau &= 3(0.866^2 - 0.5^2) + (15 - 28.48)0.866 \cdot 0.5 = -4.34 \text{ MPa}\end{aligned}$$

where σ and τ are the normal and tangential components, respectively.

Example 1.4.2

For the plane stress state of Example 1.4.1, calculate the value and orientation of the principal stresses.

Data: $\sigma_x = 28.48$ MPa, $\sigma_y = 15$ MPa, $\tau_{xy} = 3$ MPa.

Principal stresses occur for directions α and $\alpha + 90^\circ$, in which the tangential stress τ is zero. Applying Eq. (1.25) we obtain the values of α and $\alpha + 90^\circ$ corresponding to the principal directions:

$$\tan 2\alpha = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{2 \cdot 3}{28.48 - 15} = 0.4451 \quad \Rightarrow \quad \alpha_1 = 12^\circ \quad \alpha_2 = 102^\circ$$

The value of the normal stresses according to these directions is obtained by Eq. (1.23):

$$\sigma = \sigma_{1,2} = \frac{28.48 + 15}{2} \pm \sqrt{\left(\frac{28.48 - 15}{2}\right)^2 + 3^2}$$

Solving, the following values $\sigma_1 = 29.11$ MPa, $\sigma_2 = 14.36$ MPa are obtained, corresponding to the maximum and minimum stress, respectively.

Example 1.4.3

A steel rivet is subjected to a tangential stress τ_{xy} and a normal stress σ_x (Figure 1.10). Calculate the value and direction of the principal stresses. Data: $\sigma_x = -30$ MPa, $\tau_{xy} = 90$ MPa.

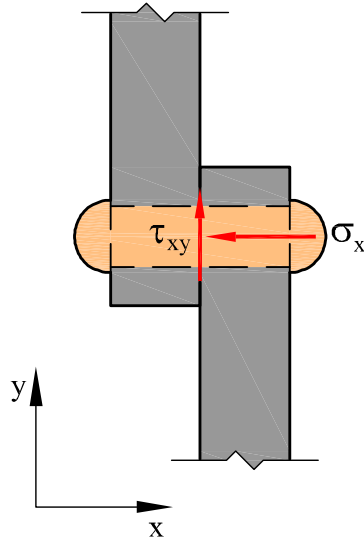


Figure 1.10: Example 1.4.3: joining of two pieces by a steel rivet

The directions corresponding to the principal stresses can be obtained from the expression:

$$\tan 2\alpha = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

Replacing the values in the example, the principal directions are:

$$\tan 2\alpha = \frac{2 \cdot 90}{-30} = -6 \quad \Rightarrow \quad \alpha_1 = -40.27^\circ \quad ; \quad \alpha_2 = \alpha_1 + 90^\circ = 49.73^\circ$$

The principal stresses can be calculated using the following equation and taking into account that $\sigma_y = 0$:

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \frac{-30}{2} \pm \sqrt{\left(\frac{-30}{2}\right)^2 + 90^2}$$

Solving, the following values are obtained: $\sigma_1 = -106$ MPa , $\sigma_2 = 76$ MPa.

1.5 Three-dimensional stress state

In a three-dimensional stress state (Section 1.3), the stress $\mathbf{t}(t_x, t_y, t_z)$ acting on an arbitrary orientation defined by the face ABC (Figure 1.11), determined by its exterior vector $\mathbf{n}(l, m, n)$ is calculated analogously to the plane stress state (Section 1.4.2). It can be written:

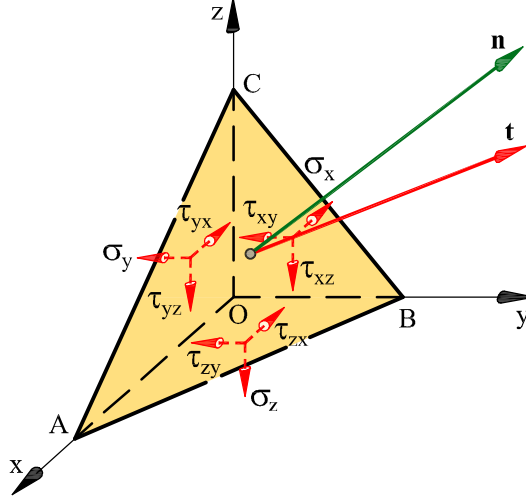


Figure 1.11: Cauchy's tetrahedron

$$\begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} = \begin{bmatrix} \sigma_x & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & \sigma_y & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} \begin{bmatrix} l \\ m \\ n \end{bmatrix} \quad (1.27)$$

or in the compact form:

$$\mathbf{t} = \mathbf{T} \mathbf{n} \quad (1.28)$$

where $\mathbf{T}$ is the stress tensor which is equal to:

$$\mathbf{T} = \begin{bmatrix} \sigma_x & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & \sigma_y & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} \quad (1.29)$$

For the principle of reciprocity of tangential stresses, the stress tensor is symmetric.

As the stress tensor is symmetric, there are always three directions orthogonal to each other in which tangential stresses do not occur. These directions are the *principal stress directions*. Expressed in axes along these directions (Section 1.4.3), the tensor $\mathbf{T}$ is diagonal.

1.6 Concept of strain

1.6.1 Displacements

A displacement is the change of position of a body resulting from rigid solid movements and deformations. Some examples are presented:

Example 1.6.1.1

Let us consider a two-dimensional body that moves without rotating and deforming to a new position. This displacement of the body is a *rigid solid translation* (Figure 1.12).

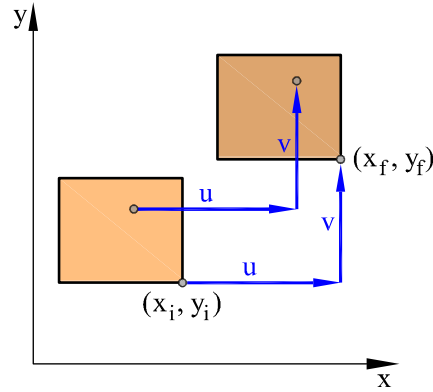


Figure 1.12: Translation of a rigid solid

The components of the displacement of a point P , u y v , in the directions x, y , respectively, can be expressed by the corresponding differences between the final position (x_f, y_f) and initial position (x_i, y_i) :

$$u = x_f - x_i$$

$$v = y_f - y_i$$

The components u and v are the same for any part of the body, there is no change in the size or shape of it, that is, there is no deformation.

Example 1.6.1.2

Let us consider the same two-dimensional body but now the displacements are the result of a rotation θ around an axis. This movement of the body is a *rigid solid rotation* (Figure 1.13)

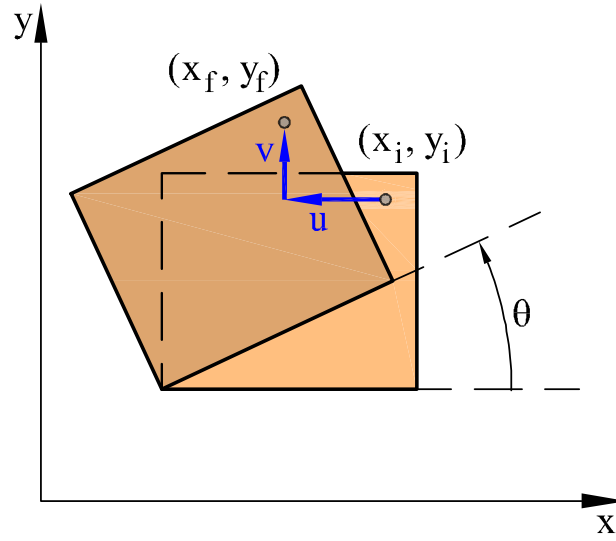


Figure 1.13: Rotation of a rigid solid

The components u and v of displacement in directions x, y are the same as in the previous example:

$$\begin{aligned} u &= x_f - x_i \\ v &= y_f - y_i \end{aligned}$$

In this case, the components u and v are not the same for any point in the body. The orientation of the body changes, but there is no change in the size or shape of the body, that is, *there is no deformation*.

From the above examples, the following conclusions are presented:

- The movement of a rigid solid only displace the body to a new position.
- Any movement of a rigid solid can be analyzed as a combination of a translation and a rotation of a rigid body.
- In the movements of a rigid solid *there is no deformation* because there is no variation of the distance between any two points of the body.

Example 1.6.1.3

Let us analyze the rectangular body $ABCD$ that is deformed by undergoing a modification of the length in direction x (Figure 1.14). This type of deformation is called *longitudinal deformation*.

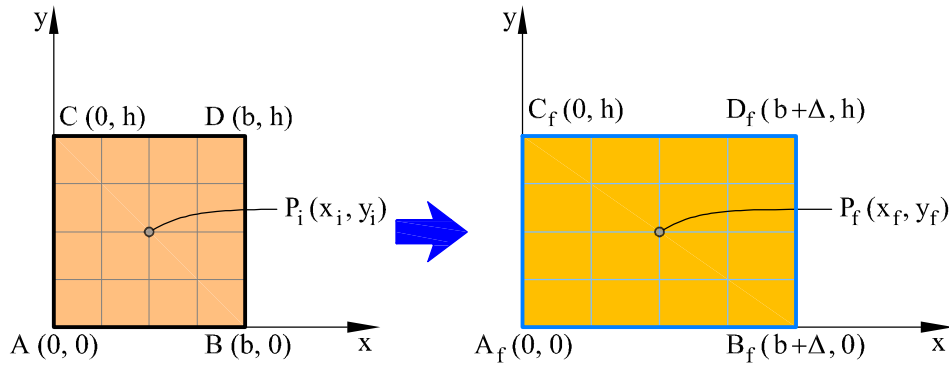


Figure 1.14: Longitudinal deformation

An arbitrary point $P_i (x_i, y_i)$ of the non-deformed body moves to the final position $P_f (x_f, y_f)$. If there is no deformation in direction y , in the longitudinal deformation the components u and v of the displacement x, y are:

$$\begin{aligned} u &= x_f - x_i = u(x) = \Delta \frac{x}{b} \\ v &= y_f - y_i = 0 \end{aligned}$$

Note that the displacement u only depend on the original coordinate x ($u_{(x=0)} = 0, u_{(x=b)} = \Delta$). In addition, the points located on lines parallel to the axis y suffer the same displacement in direction x and do not move in direction y , (null transverse deformation). Finally, the longitudinal deformation produces a change in the size of the body, due to the variation of the longitudinal distance between two points (Figure 1.14).

Example 1.6.1.4

The rectangular body $ABCD$ is deformed by undergoing a modification of the angles between its edges (Figure 1.15). This type of deformation is called *shear deformation*.

An arbitrary point $P_i (x_i, y_i)$ of the non-deformed body moves to the final position $P_f (x_f, y_f)$. In shear deformation, the components u and v of the displacement in directions x, y are:

$$\begin{aligned} u &= x_f - x_i = u(y) = \Delta \frac{y}{h} \\ v &= y_f - y_i = 0 \end{aligned}$$

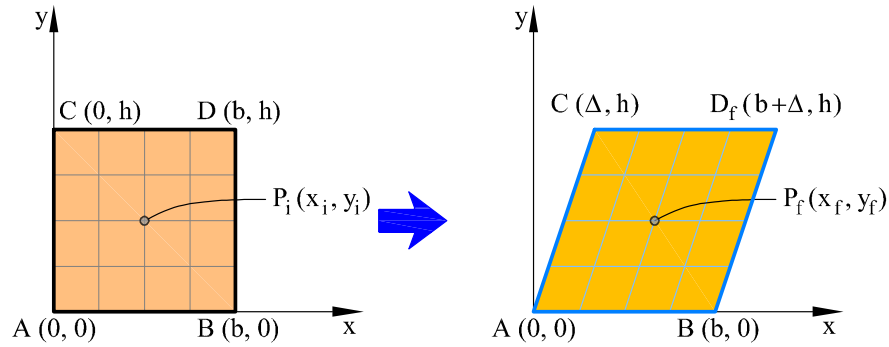


Figure 1.15: Shear deformation

In this case, the displacement u only depend on the original coordinate y . The points located on lines parallel to the x axis at a distance y suffer the same displacement in direction x , and move in direction y . Shear deformation produces a change in the shape of the body, due to the variation of the angle that originally formed the edges. (Figure 1.15).

Taking into account the previous examples, it is possible to obtain the following conclusions (Figure 1.16):

- In a deformable body that undergoes a *longitudinal deformation* (Figure 1.16a) there is a **change of volume** in which the distance between two points can change.
- In a deformable body that undergoes a *shear deformation* (Figure 1.16b) a **shape change** occurs in which the angle between two lines can change.
- These changes in volume and shape are what distinguish a deformable body from a rigid body.

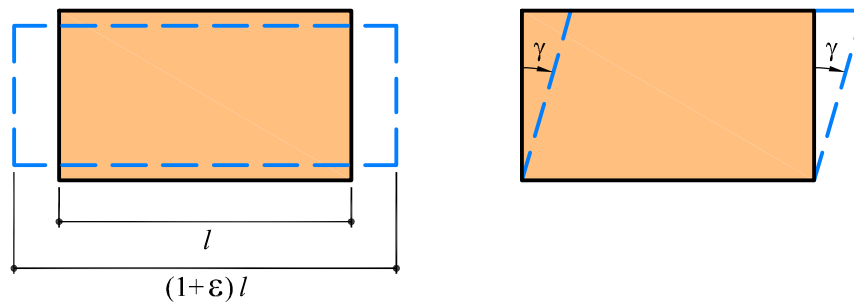


Figure 1.16: (a) Longitudinal deformation (b) Shear deformation

1.6.2 Strain

Let us consider a body subjected to the action of applied forces, and with the necessary links to prevent rigid solid movements. Since there is no perfectly rigid material, the action of the forces results in the body being deformed, that is, it changes size and shape. We will assume in what follows that the body is sufficiently rigid so that the displacements that occur in the deformation process are small compared to the dimensions of the body (rigidity principle).

(a) Longitudinal deformation

Let us consider a straight bar of initial length l that undergoes an elongation Δl to a final length $l_f = l + \Delta l$ (Figure 1.17):

$$\Delta l = l_f - l \quad (1.30)$$

The *average longitudinal deformation* or average strain in the bar is calculated by dividing the elongation by the initial length:

$$\varepsilon_m = \frac{\Delta l}{l} \quad (1.31)$$

Let us analyze the elongation suffered by a bar element of length Δx . As a consequence of the stretching, the points N and M are moved to their new positions N_f and M_f , respectively. The displacements suffered by these points are u_N and u_M , respectively. Being Δu the difference $\Delta u = u_M - u_N$, the average longitudinal deformation suffered by the element is equal to the relation:

$$\varepsilon_m(NM) = \frac{u_M - u_N}{x_M - x_N} = \frac{\Delta u}{\Delta x} \quad (1.32)$$

If the element of length Δx tends to a differential element of length dx , the *strain* or *longitudinal deformation at a point N* is the:

$$\varepsilon_x(N) = \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} = \frac{du}{dx} = u' \quad (1.33)$$

The longitudinal deformation is considered positive when it is of elongation, while the shortening is negative. The longitudinal deformation or strain is *dimensionless* since it is the ratio of two length quantities. The relationship between the longitudinal deformation and displacement $\varepsilon = u'$ is based on the hypothesis that the displacements u are small $u \ll l$ and that this deformation ε is small $\varepsilon \ll 1$.

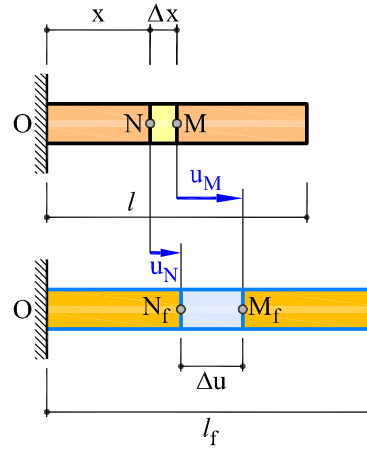


Figure 1.17: Bar subjected to longitudinal Strain.

Example 1.6.2.1

Consider a straight bar that undergoes an elongation u_B in the longitudinal direction due to a uniform increase in temperature (Figures 1.18).

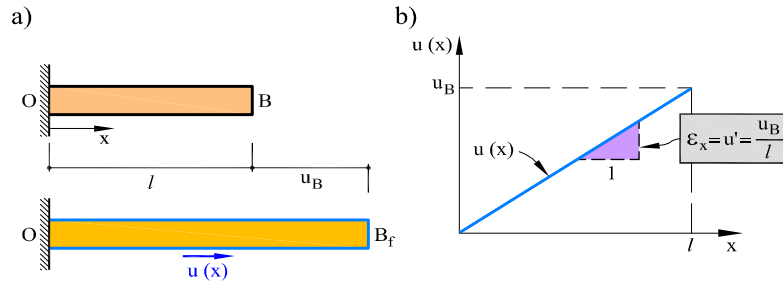


Figure 1.18: Deformation of a bar due to a uniform increase in temperature

The displacement of a point in direction x is a linear function of the displacement u_B suffered by the end B of the bar (Figure 1.18b):

$$u(x) = u_B \frac{x}{l}$$

The longitudinal deformation is:

$$\varepsilon_x = u' = \frac{u_B}{l}$$

and the average longitudinal deformation is:

$$\varepsilon_m = \frac{\Delta u}{l} = \frac{u_B - u_O}{l} = \frac{u_B}{l}$$

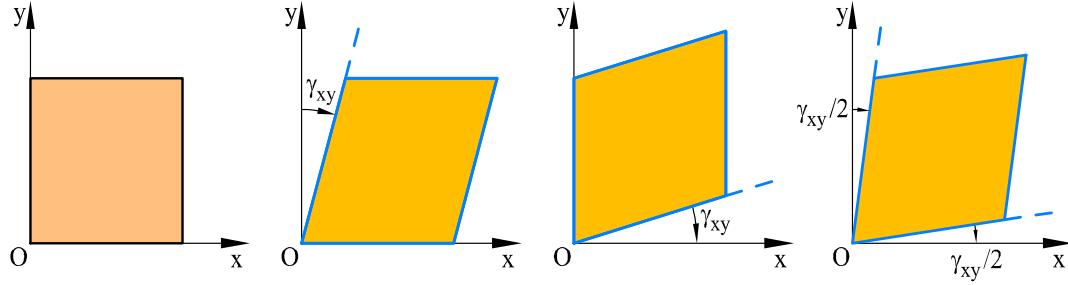


Figure 1.19: Angular deformation

Note that in this case the strain ε_x and the average longitudinal deformation ε_m are same. This kind of deformation is called *uniform deformation*

(b) Angular deformation

The *shape changes* of a body are associated with the variation of the angles that form the edges of the body that originally are straight.

Figure 1.19 shows the shear deformation suffered by a square body deformed by shear into a rhombus. The difference between the three deformed blocks (Figures 1.19b-1.19d) is a rigid solid rotation. However, in each deformed body, the variation of the angle between the two edges (which was initially straight) is the angle γ_{xy} . This is an example of an angular deformation.

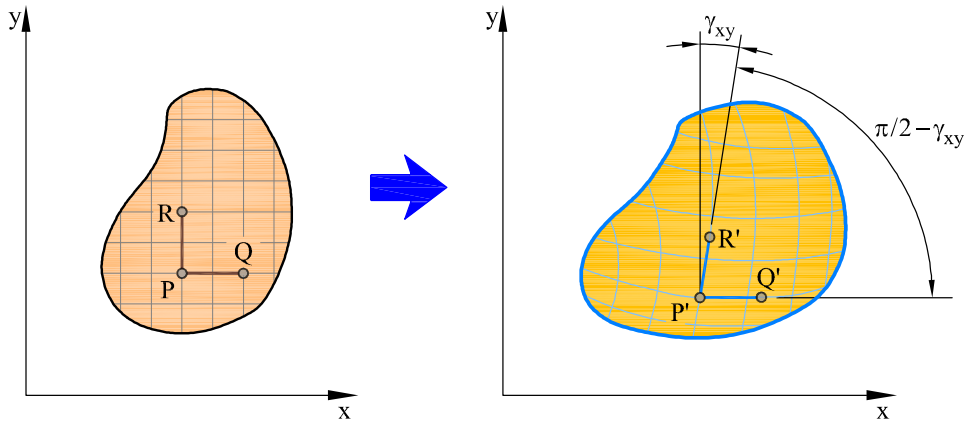


Figure 1.20: Angular deformation at a point

The angular deformation or distortion at a point is defined as the angular variation between two initially perpendicular segments, as consequence of a

change in shape (Figure 1.20).

For example, considering the perpendicular segments $\overline{PQ}$ and $\overline{PR}$, the angular deformation at point P can be expressed by:

$$\gamma_{xy}(P) = \lim_{\substack{Q \rightarrow P \\ R \rightarrow P}} (\widehat{QPR} - \widehat{Q'P'R'}) = \lim_{\substack{Q \rightarrow P \\ R \rightarrow P}} \left(\frac{\pi}{2} - \widehat{Q'P'R'} \right) \quad (1.34)$$

The angular deformation is considered *positive* when the angle that form two perpendicular segments *decreases*. The deformation angular or distortion is *dimensionless*, since the variation angular is measured in radians. Finally, the angular deformation measures the *change of shape* of a body in the vicinity of a point P . All of the above is based on the hypothesis that the angular deformation γ is small $\gamma \ll 1$ rad.

Example 1.6.2.2

The support of a beam formed by two rigid plates and a block of neoprene of height h is deformed as shown in Figure 1.21. Calculate the angular deformation in the neoprene.

Data: $u = 1.035 \cdot 10^{-3}$ m, $h = 0.15$ m.

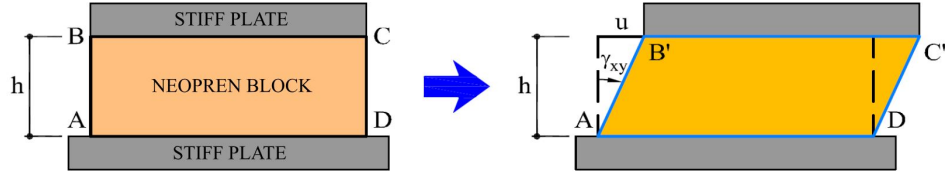


Figure 1.21: Support of the Example 1.6.2.2

The angular distortion is:

$$\begin{aligned} \gamma_{xy} &= \widehat{BAD} - \widehat{B'AD} = \frac{\pi}{2} - \left(\frac{\pi}{2} - \widehat{BAB'} \right) \\ &= \arctan \frac{u}{h} = \arctan \frac{1.035 \cdot 10^{-3}}{0.15} = 6.9 \cdot 10^{-3} \text{ rad} \end{aligned}$$

1.7 Plane state of strain

1.7.1 Displacements and strains

A plane state of strains is defined as one in which the displacements along the axes x, y, z are:

$$u = u(x, y) \quad v = v(x, y) \quad w = 0$$

Let us consider the deformation of a differential segment dx (Figure 1.22) where u is the displacement in direction x suffered by the origin point of the segment. For the end point of such a segment:

$$u + du = u + \frac{\partial u}{\partial x} dx = u + \varepsilon_x dx \quad (1.35)$$

which defines the strain along the x axis, ε_x :

$$\varepsilon_x = \frac{\partial u}{\partial x} \quad (1.36)$$

Similarly, the strain can be defined according to the y axis as:

$$\varepsilon_y = \frac{\partial v}{\partial y} \quad (1.37)$$

where v is the displacement along the y axis. Let us now consider the deformation of a differential orthogonal element $dx dy$ (Figure 1.23). It can be written:

$$\alpha + \beta = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} = \gamma_{xy} \quad (1.38)$$

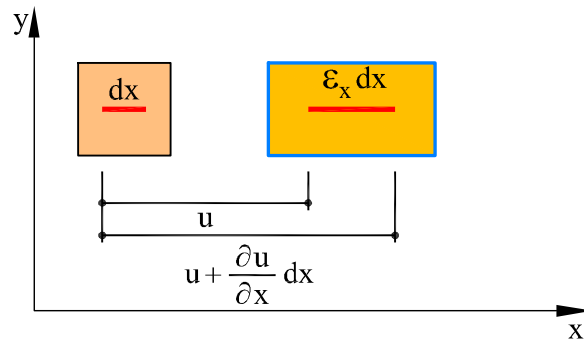


Figure 1.22: Relationship between displacement and longitudinal strain

where γ_{xy} is the angular deformation or distortion of the originally rectangular element.

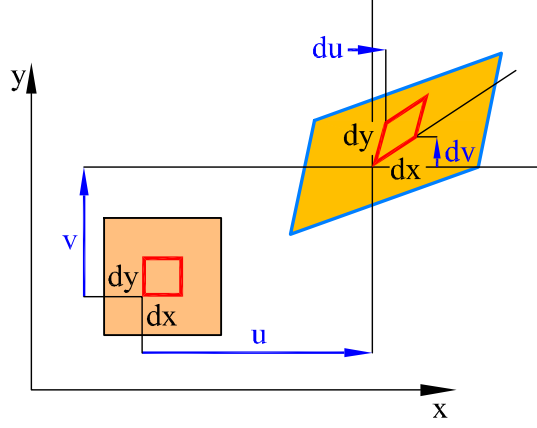


Figure 1.23: Relationship between displacement and angular deformation

Therefore, in a plane problem with displacements $u(x, y)$ and $v(x, y)$ in directions x, y , respectively, there are three independent strains, two longitudinal and one angular, which are expressed by:

$$\varepsilon_x = \frac{\partial u}{\partial x} \quad , \quad \varepsilon_y = \frac{\partial v}{\partial y} \quad , \quad \gamma_{xy} = \gamma_{yx} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \quad (1.39)$$

The relationships between strains and displacements are based on the hypothesis that *displacements* and *deformations are small* (rigidity principle).

For solving the inverse problem, that is, in order to find the displacements u, v given $\varepsilon_x, \varepsilon_y, \gamma_{xy}$, these must satisfy the compatibility equation of Saint-Venant:

$$\frac{\partial^2 \varepsilon_x}{\partial y^2} + \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y} \quad (1.40)$$

Example 1.7.1.1

Calculate a plane displacement field that is composed of one rigid translation $\mathbf{a}_0 = (u_0, v_0)$, one rigid rotation of modulus ω_0 around the point (x_0, y_0) , constant longitudinal deformations $\varepsilon_{x_0}, \varepsilon_{y_0}$, and one angular constant deformation γ_{xy_0} ((Figure 1.24).

The field is:

$$\begin{aligned} u &= u_0 - (y - y_0) \omega_0 + \varepsilon_{x_0} x + \frac{1}{2} \gamma_{xy_0} y \\ v &= v_0 + (x - x_0) \omega_0 + \varepsilon_{y_0} y + \frac{1}{2} \gamma_{xy_0} x \end{aligned}$$

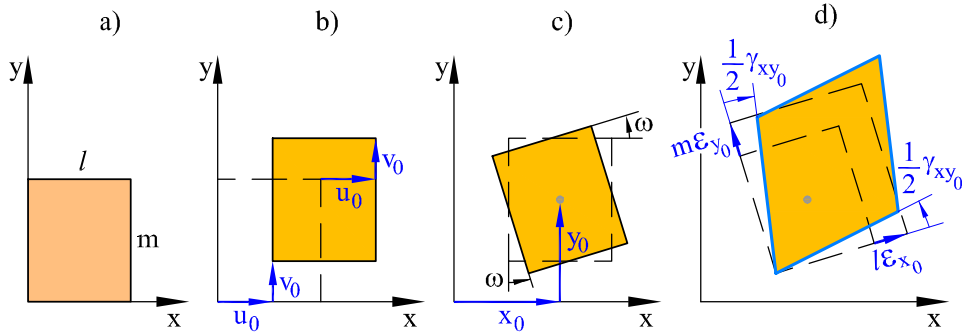


Figure 1.24: Displacements and strains of Example 1.7.1.1

It can be verified that:

$$\begin{aligned}\varepsilon_x &= \frac{\partial u}{\partial x} = \varepsilon_{x0} \\ \varepsilon_y &= \frac{\partial v}{\partial y} = \varepsilon_{y0} \\ \gamma_{xy} &= \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = \gamma_{xy0}\end{aligned}$$

and that the compatibility condition is fulfilled, Eq. (1.40), as the deformations are constant.

Example 1.7.1.2

A plane strain field is given by: $\varepsilon_x = 8x$; $\varepsilon_y = x$; $\gamma_{xy} = -y$. Verify that the equation of compatibility is fulfilled, Eq. (1.40), and find the displacement field.

The compatibility equation is fulfilled identically as the strain field is linear deformation field, therefore, its second derivatives are zero.

The displacement field is found by integrating the equations that relate such a field to the deformation field:

$$\frac{\partial u}{\partial x} = 8x \qquad \frac{\partial v}{\partial y} = x \qquad \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = -y$$

$$\text{From 1}^a: \quad u = 4x^2 + f(y)$$

$$\text{From 2}^a: \quad v = xy + g(x)$$

$$\text{From 3}^a: \quad \frac{df}{dy} + y + \frac{dg}{dx} = -y \quad \Rightarrow \quad f(y) = -y^2 \quad ; \quad g(x) = 0$$

and then:

$$\begin{aligned} u &= 4x^2 - y^2 \\ v &= xy \end{aligned}$$

1.7.2 Strain tensor

Let us consider a unitary vector $\overrightarrow{OD} = \mathbf{n}$, with director cosines (l, m) . The deformation δ from element $\overrightarrow{OD}$ is defined as the displacement $\overrightarrow{DD'}$ of the point D , caused by the deformation of the differential rectangular element of area with dimensions l and m , which has a $\overrightarrow{OD}$ per diagonal, disregarding rotations and translations of rigid solid (Figure 1.25). To calculate the

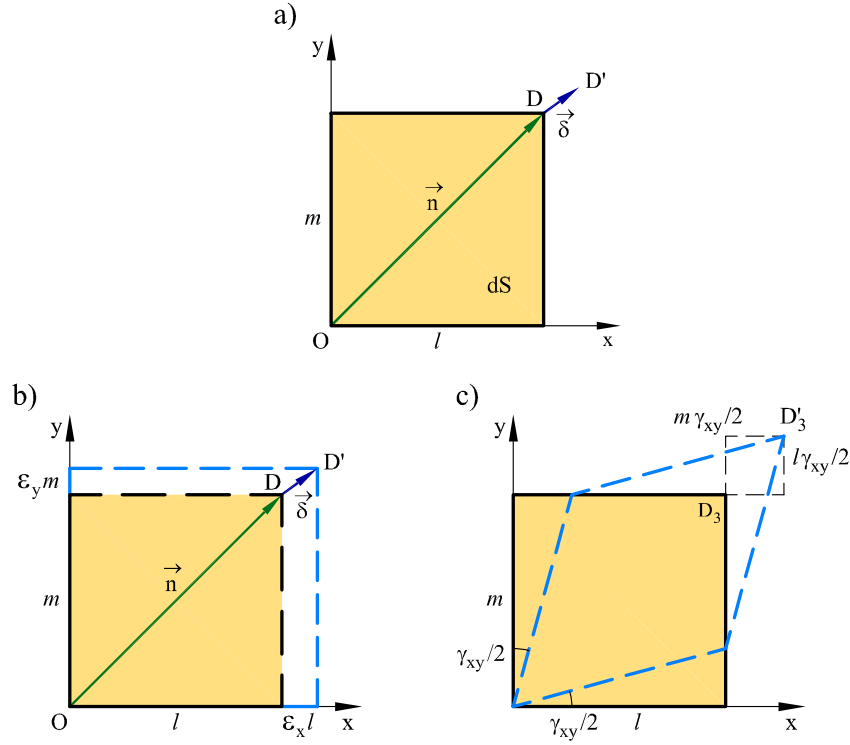


Figure 1.25: Plane strain state: (a) Deformation vector (b) Displacements due to elongations (c) Displacements due to distortions

Cartesian components of the vector δ , that is (δ_x, δ_y) , we add the components of the displacement of the point D due to the elongations unit ϵ_x, ϵ_y and to the distortion γ_{xy} (Figures 1.25b y c):

$$\begin{aligned} \delta_x &= \epsilon_x l + \frac{\gamma_{xy}}{2} m \\ \delta_y &= \frac{\gamma_{xy}}{2} l + \epsilon_y m \end{aligned} \tag{1.41}$$

or in matrix form:

$$\delta = \mathbf{D} \mathbf{n} \quad (1.42)$$

where $\mathbf{D}$ is the strain tensor which is equal to:

$$\mathbf{D} = \begin{bmatrix} \varepsilon_x & \frac{1}{2}\gamma_{xy} \\ \frac{1}{2}\gamma_{xy} & \varepsilon_y \end{bmatrix} \quad (1.43)$$

The strain tensor is symmetric, having dispensed in its construction of rigid solid rotations.

From the expression (1.42) it is deduced that, knowing the components of the deformation state of a point (strain tensor) and the direction cosines of any linear element, the deformation of such element can be known (Figure 1.25):

$$\varepsilon = \varepsilon_x \cos^2 \phi + \varepsilon_y \sin^2 \phi + \gamma_{xy} \sin \phi \cos \phi \quad (1.44a)$$

$$\gamma = 2(\varepsilon_y - \varepsilon_x) \sin \phi \cos \phi + \gamma_{xy} (\cos^2 \phi - \sin^2 \phi) \quad (1.44b)$$

1.7.3 Transformation of the reference system

Analogously to the formulation of the stress tensor (Section 1.4.3), in a plane state the strain tensor transforms when rotating the system of reference x, y to become x', y' .

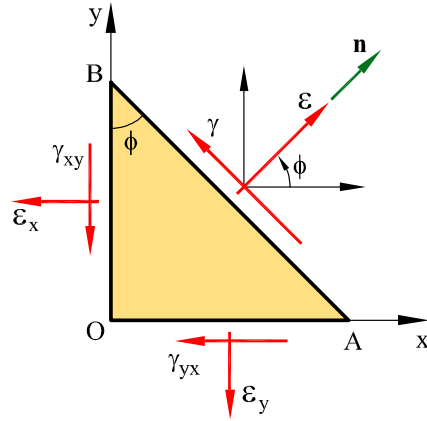


Figure 1.26: Deformations along an arbitrary $\vec{n}$ direction

Let $\mathbf{N}$ be the matrix of direction cosines:

$$\mathbf{N} = \begin{bmatrix} \cos(x'x) & \cos(y'x) \\ \cos(x'y) & \cos(y'y) \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} = \begin{bmatrix} l & -m \\ m & l \end{bmatrix} \quad (1.45)$$

and $\mathbf{D}$ is the strain tensor referred to the system x, y . The expression that

allows to obtain the deformation tensor $\mathbf{D}'$, referred to the reference system x', y' , is:

$$\begin{bmatrix} \varepsilon_{x'} & \frac{1}{2}\gamma_{x'y'} \\ \frac{1}{2}\gamma_{x'y'} & \varepsilon_{y'} \end{bmatrix} = \begin{bmatrix} l & m \\ -m & l \end{bmatrix} \begin{bmatrix} \varepsilon_x & \frac{1}{2}\gamma_{xy} \\ \frac{1}{2}\gamma_{xy} & \varepsilon_y \end{bmatrix} \begin{bmatrix} l & -m \\ m & l \end{bmatrix} \quad (1.46)$$

or, in matrix form

$$\mathbf{D}' = \mathbf{N}^T \mathbf{D} \mathbf{N} \quad (1.47)$$

Example 1.7.3.1

At point O of a two-dimensional solid, the strains along the x, y axes (Figure 1.27a) are known. It is requested to calculate the values of these strains for the directions x', y' that form an angle $\alpha = 40^\circ$ with the axes x, y . Data: $\varepsilon_x = 3.6 \cdot 10^{-3}$, $\varepsilon_y = 2.4 \cdot 10^{-3}$, $\gamma_{xy} = 5.0 \cdot 10^{-3}$.

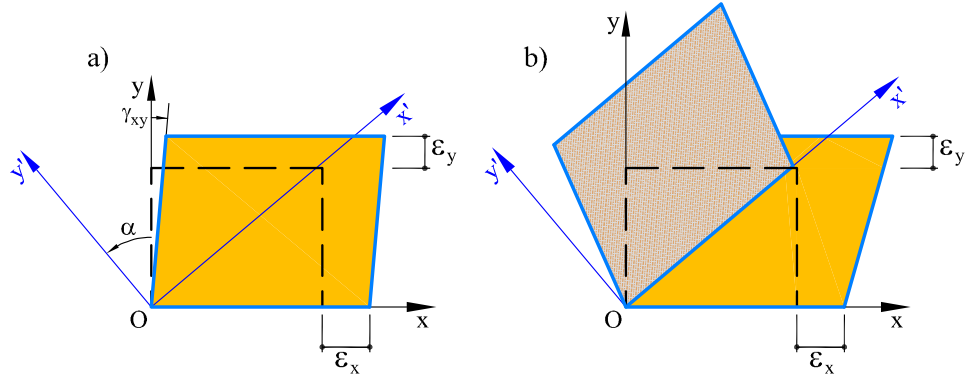


Figure 1.27: (a) Plane strain of Example 1.7.3.1 (b) Strains referred to the system x', y'

The components of the strain tensor $\mathbf{D}$ are known, then applying Eq. (1.46) we obtain the strain tensor $\mathbf{D}'$ referred to the axes x', y' .

$$\begin{aligned} \begin{bmatrix} \varepsilon_{x'} & \frac{1}{2}\gamma_{x'y'} \\ \frac{1}{2}\gamma_{x'y'} & \varepsilon_{y'} \end{bmatrix} &= \begin{bmatrix} 0.766 & 0.643 \\ -0.643 & 0.766 \end{bmatrix} \begin{bmatrix} 3.6 & 2.5 \\ 2.5 & 2.4 \end{bmatrix} 10^{-3} \begin{bmatrix} 0.766 & -0.643 \\ 0.643 & 0.766 \end{bmatrix} \\ &= \begin{bmatrix} 5.57 & -0.16 \\ -0.16 & 0.43 \end{bmatrix} 10^{-3} \end{aligned}$$

The values of the unit strains according to the directions x', y' (Figure 1.27b) are:

$$\begin{aligned} \varepsilon_{x'} &= 5.6 \cdot 10^{-3} \\ \varepsilon_{y'} &= 4.3 \cdot 10^{-4} \\ \gamma_{x'y'} &= -3.2 \cdot 10^{-4} \end{aligned}$$

1.7.4 Principal strains

Analogous to what was proposed when studying the stresses (Section 1.4), there are some directions for which it is true that the deformation is exclusively longitudinal:

$$\delta = \varepsilon \mathbf{n} \quad (1.48)$$

where ε is a scalar value corresponding to the modulus of the deformation in the direction defined by $\mathbf{n}$. Combining the Eqs. (1.48) and (1.42) we have:

$$(\mathbf{D} - \varepsilon \mathbf{I}) \mathbf{n} = 0 \quad (1.49)$$

where $\mathbf{I}$ is a unitary matrix 2×2 . For this system of linear and homogeneous equations to have a solution, it is necessary that the determinant of the system is zero, that is:

$$\begin{vmatrix} \varepsilon_x - \varepsilon & \frac{1}{2}\gamma_{xy} \\ \frac{1}{2}\gamma_{xy} & \varepsilon_y - \varepsilon \end{vmatrix} = 0 \quad (1.50)$$

developing the determinant and solving:

$$\varepsilon = \varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \frac{1}{2} \sqrt{(\varepsilon_x - \varepsilon_y)^2 + \gamma_{xy}^2} \quad (1.51)$$

where ε_1 and ε_2 are the *principal deformations*.

Alternatively, the directions of the principal deformations can be determined from the expressions (1.35) that give the unit elongation and distortion along one arbitrary direction $\mathbf{n}$. If Eq. (1.44b) is set equal to zero

$$2(\varepsilon_y - \varepsilon_x) \sin \phi \cos \phi + \gamma_{xy} (\cos^2 \phi - \sin^2 \phi) = 0 \quad (1.52)$$

is obtained:

$$\frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \frac{2 \sin \phi \cos \phi}{\cos^2 \phi - \sin^2 \phi} = \tan 2\phi \quad (1.53)$$

which gives the directions of the principal deformations.

The maximum shear strain at the point is in the directions that form 45° with the principal directions and it is worth:

$$\gamma_{\max} = \varepsilon_1 - \varepsilon_2 = \sqrt{(\varepsilon_x - \varepsilon_y)^2 + \gamma_{xy}^2} \quad (1.54)$$

Example 1.7.4.1

The plate in Figure 1.28 deforms according to the strains $\varepsilon_x = 5.6 \cdot 10^{-3}$, $\varepsilon_y = 2.4 \cdot 10^{-3}$, $\gamma_{xy} = 8.4 \cdot 10^{-3}$. Calculate: a) the corresponding strain tensor, and b) the principal strains and their directions.

With the strains referred to the system x, y , the strain tensor $\mathbf{D}$ is :

$$\mathbf{D} = \begin{bmatrix} \varepsilon_x & \frac{1}{2}\gamma_{xy} \\ \frac{1}{2}\gamma_{xy} & \varepsilon_y \end{bmatrix} = \begin{bmatrix} 5.6 & 4.2 \\ 4.2 & 2.4 \end{bmatrix} 10^{-3}$$

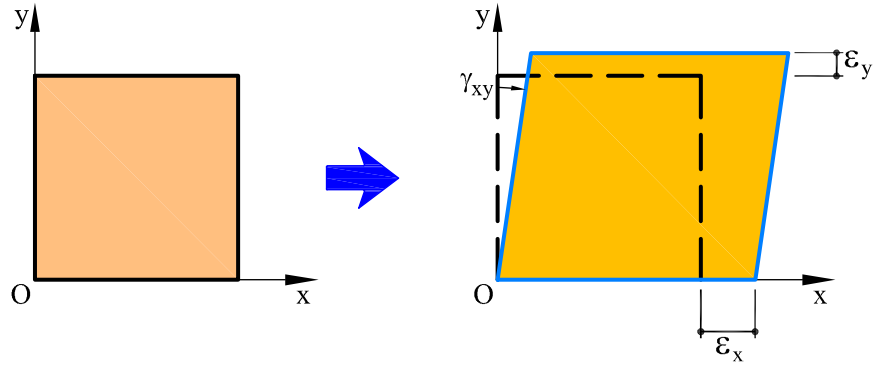


Figure 1.28: Strains in the plate of Example 1.7.4.1

The principal strains can be calculated by finding the roots of the equation:

$$\begin{vmatrix} \varepsilon_x - \varepsilon & \frac{1}{2}\gamma_{xy} \\ \frac{1}{2}\gamma_{xy} & \varepsilon_y - \varepsilon \end{vmatrix} = \begin{vmatrix} 5.6 - \varepsilon & 4.2 \\ 4.2 & 2.4 - \varepsilon \end{vmatrix} 10^{-3} = 0$$

Solving the determinant, the equation is:

$$\varepsilon^2 - 8\varepsilon - 4.2 = 0$$

The solution of this equation gives the values of ε_1 and ε_2 which are the principal strains:

$$\begin{aligned} \varepsilon_1 &= 8.5 \cdot 10^{-3} \\ \varepsilon_2 &= -0.5 \cdot 10^{-3} \end{aligned}$$

The directions in which the principal strains occur are calculated using the expression:

$$\tan 2\phi = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \frac{8.4}{5.6 - 2.4} = 2.625$$

so that $\phi_1 = 34.57^\circ$ and $\phi_2 = 124.57^\circ$ are the principal directions of the strains.

Example 1.7.4.2

At point O of a two-dimensional solid we have the strains $\varepsilon_x = 4.2 \cdot 10^{-3}$, $\varepsilon_y = -3.0 \cdot 10^{-3}$, and $\gamma_{xy} = 6.0 \cdot 10^{-3}$ (Figure 1.29). Calculate the principal strains and the maximum distortion.

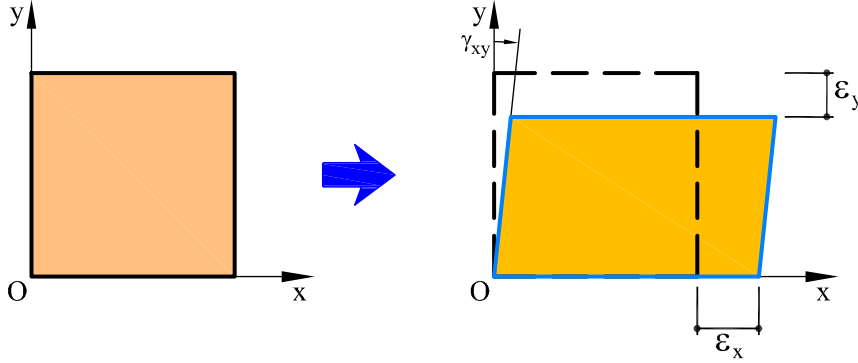


Figure 1.29: Strains of the two-dimensional solid of Example 1.7.4.2

The principal strain directions are obtained using the expression:

$$\tan 2\phi = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \frac{6.0}{4.2 + 3.0} = 0.833$$

Therefore, the principal directions are: $\phi_1 = 20^\circ$ y $\phi_2 = 110^\circ$. The principal strains, ε_1 y ε_2 , can be obtained from the expression:

$$\begin{aligned} \varepsilon &= \varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \frac{1}{2} \sqrt{(\varepsilon_x - \varepsilon_y)^2 + \gamma_{xy}^2} \\ &= \left[\frac{4.2 + 3.0}{2} \pm \frac{1}{2} \sqrt{(7.2)^2 + (6.0)^2} \right] 10^{-3} \end{aligned}$$

where, $\varepsilon_1 = 5.2 \cdot 10^{-3}$, $\varepsilon_2 = -4.0 \cdot 10^{-3}$.

The orientation of the maximum shear strain can be determined by the expression:

$$\tan 2\beta = -\frac{\varepsilon_x - \varepsilon_y}{\gamma_{xy}} = -\frac{4.2 + 3.0}{6.0} = -1.2$$

Thus, $\beta_1 = -25^\circ$ y $\beta_2 = \beta_1 + 90^\circ = 65^\circ$.

Note that the principal directions and the directions of shear deformation form an angle of 45° . The maximum distortion value can be calculated as:

$$\gamma_{\max} = \sqrt{(\varepsilon_x - \varepsilon_y)^2 + \gamma_{xy}^2} = 9.4 \cdot 10^{-3}$$

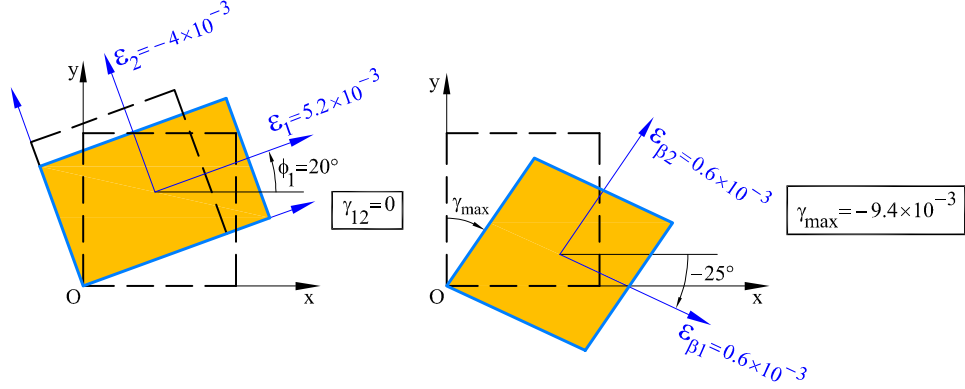


Figure 1.30: (a) Strains (b) Maximum distortion

The unit elongation that occurs in the direction β_1 can be calculated by changing the reference system:

$$\begin{aligned}
 \varepsilon_{\beta_1} &= \varepsilon_x \cos^2 \beta_1 + \varepsilon_y \sin^2 \beta_1 + \gamma_{xy} \sin \beta_1 \cos \beta_1 \\
 &= [4.2 \cdot 0.82139 - 3.0 \cdot 0.1786 - 6.0 \cdot 0.4226 \cdot 0.9063] \cdot 10^{-3} \\
 &= 0.6 \cdot 10^{-3}
 \end{aligned}$$

It can be verified that the elongations in the directions β_1 and β_2 are the same.

1.8 Three-dimensional state of strain

In a three-dimensional problem in which the displacement components are $u(x, y, z)$, $v(x, y, z)$ and $w(x, y, z)$ in the directions x, y, z , respectively, there are six independent strains, three longitudinal and three angular, expressed by

$$\begin{aligned}
 \varepsilon_x &= \frac{\partial u}{\partial x} & \gamma_{xy} &= \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \\
 \varepsilon_y &= \frac{\partial v}{\partial y} & \gamma_{yz} &= \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \\
 \varepsilon_z &= \frac{\partial w}{\partial z} & \gamma_{zx} &= \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}
 \end{aligned} \tag{1.55}$$

The three-dimensional relationships between unit strains and displacements are based on the hypothesis that *the displacements* and *the deformations* are small (principle of stiffness).

Under these conditions, the deformation $\delta(\delta_x, \delta_y, \delta_z)$ of an element $\overrightarrow{OD} = \mathbf{n}(l, m, n)$ (Figure 1.31 a) is calculated analogously to the plane case (Section 1.7.2).

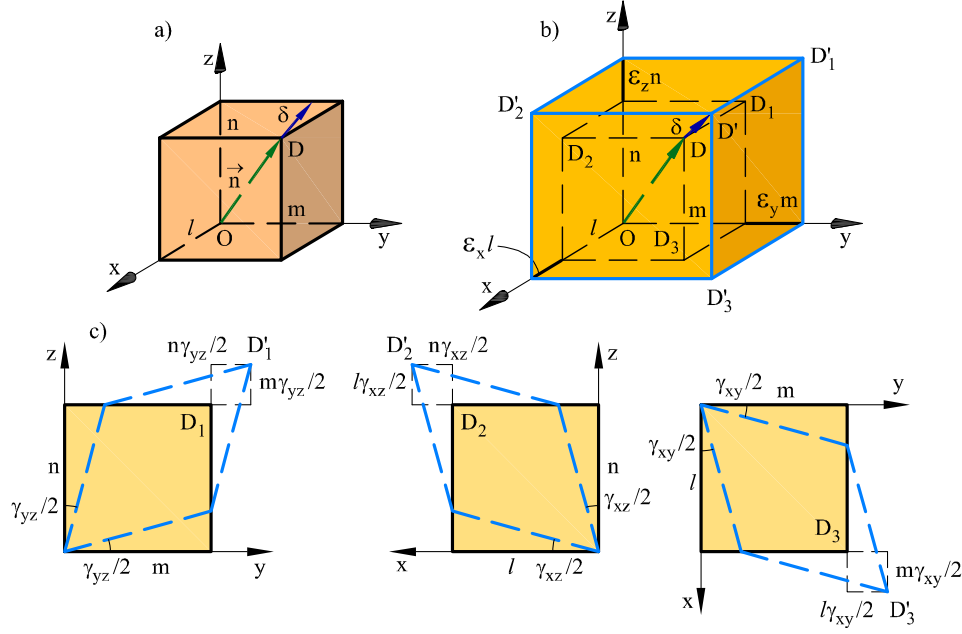


Figure 1.31: (a) Concept of deformation vector (b) Displacements due to elongations (c) Displacements due to distortions

In matrix form

$$\delta = \mathbf{D} \mathbf{n} \quad (1.56)$$

where $\mathbf{D}$ is the strain tensor, which is equal to:

$$\mathbf{D} = \begin{bmatrix} \epsilon_x & \frac{1}{2}\gamma_{xy} & \frac{1}{2}\gamma_{xz} \\ \frac{1}{2}\gamma_{xy} & \epsilon_y & \frac{1}{2}\gamma_{yz} \\ \frac{1}{2}\gamma_{xz} & \frac{1}{2}\gamma_{yz} & \epsilon_z \end{bmatrix} \quad (1.57)$$

The strain tensor is symmetric, having dispensed in its construction of the rigid solid rotations.

Since the strain tensor is symmetric, there are always three orthogonal directions to each other in which the deformation is exclusively longitudinal. They are called *principal directions of strain*. Expressed in these directions the tensor $\mathbf{D}$ is diagonal.

Chapter 2

Elasticity and material behavior

2.1 Introduction

The previous Chapter introduces and develops the concepts of stress and strain, as well as their respective relationships to the applied load and displacements. The necessary compatibility and equilibrium equations (between displacements and strains) have been established.

In this Chapter, the fundamentals of Mechanics of Deformable Solids are completed by relating stresses and strains. First, the fundamental principles of Elasticity are introduced: Hooke's law and the principle of superposition, which are the basis for the generalized Hooke's law. Then, the Chapter discusses how to link the experimental behavior of materials with the hypotheses of elasticity. Finally, in this context, the concepts of limit stress, allowable stress, and safety factor are introduced.

2.2 Elasticity and linearity. Hooke's law

Solid bodies deform under the action of external forces, and when these forces are removed, the bodies try to return to their original shape. This property, which all solids have to a greater or lesser degree, is known as elasticity.

In reality, solids are neither perfectly elastic nor perfectly inelastic. The deformations that they undergo, have an elastic component that disappears when the applied forces cease and a permanent component that persists afterwards. In a large number of solids, if the forces do not exceed a certain threshold, the permanent deformations are very small, and these bodies are therefore considered elastic.

Consider a simple experiment in which a simply-supported beam similar to the one depicted in (Figure 2.1) is studied; the vertical displacement δ

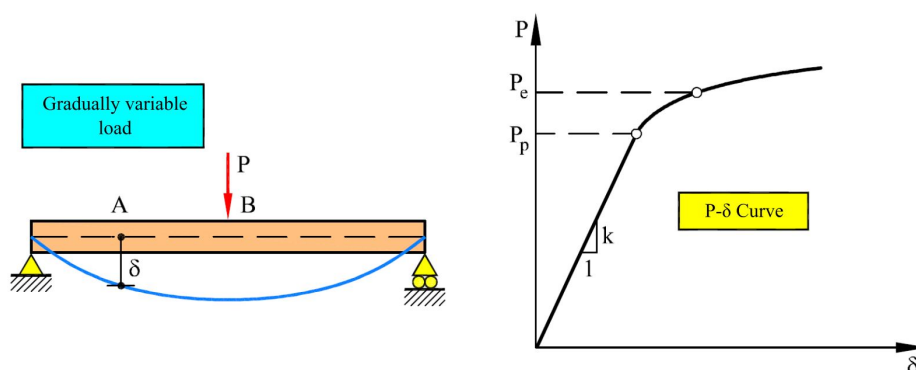


Figure 2.1: Experiment and load-displacement curve

of a certain point A of the beam produced by a force P applied to another point B of that beam is measured. The force P is increased gradually to a given value and the P - δ curve is drawn.

If the beam is gradually unloaded, the loading and unloading branches coincide (P - δ), so there is no permanent deformation, and the behavior is elastic for maximum values of P less than a specified limit value P_p , also known as the *elastic limit*. If the load P exceeds the elastic limit, the unloading branch will differ from the loading branch, permanent deformations will occur, and the behavior will not be perfectly elastic.

In the same experiment, another limit value P_e can be observed, which is smaller than P_p (and usually close to it), known as the *proportionality limit*, such that for loads less than this, the displacements are proportional to the forces that produce them. This statement defines linear elastic behavior and is known as the Hooke's law, after Robert Hooke established it in his work "De Potentia Restitutive (of Springs)" in 1678.

Mathematically, Hooke's law is expressed in the form:

$$P = k \delta \quad (2.1)$$

where k is the proportionality constant between the applied force P and the displacement δ produced. Obviously, this constant depends on by the geometry of the problems as well as the mechanical properties of the beam material.

2.3 Principle of Superposition

Taking into account Hooke's law and assuming that the displacements produced by the applied forces are very small compared to the dimensions of the body, then the Principle of Superposition or Principle of Linearity can be

applied. According to the Principle of Superposition, the effects produced by a system of forces applied in a body are equal to the sum of the effects produced by those same forces applied separately.

Considering a system defined by two forces P_1 and P_2 applied on a body (Figure 2.1), the Principle of Superposition states that the vertical displacement of a point C will be:

$$\delta = \delta_1 + \delta_2 \quad (2.2)$$

where δ_1 and δ_2 are the values of the corresponding vertical displacement when P_1 and P_2 are applied alone.

If Hooke's law is applicable, it will be:

$$P_1 = k_1 \delta_1 \quad P_2 = k_2 \delta_2 \quad (2.3)$$

and it can be written as:

$$\delta = \frac{1}{k_1} P_1 + \frac{1}{k_2} P_2 \quad (2.4)$$

The statement of the Principle of Superposition is equivalent to establishing that “the effects that a system of applied forces originates in a body are independent of the order application of forces” due to the commutative and associative properties of addition.

Mathematically, according to the Principle of Superposition, the action-response relationship is linear and therefore, the properties of a linear function apply.

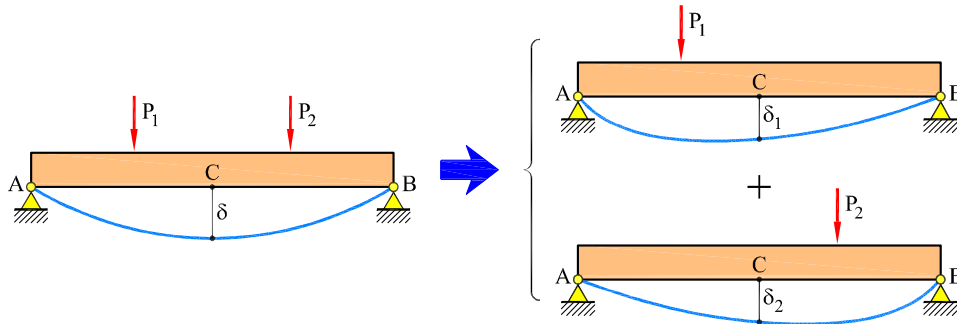


Figure 2.2: Superposition principle

2.4 Generalized Hooke's Law

In this section, Hooke's law will be applied to the relationship between stresses and strains acting at a point. In this form, it is known as the

Generalized Hooke's Law. The coefficients of proportionality are material constants that do not depend on the geometry of the body, since the state stress and strain are characteristic of a point.

Also, let us assume that in the following that the bodies are mechanically isotropic, that is, that their mechanical properties are the same in all directions. This hypothesis of mechanical isotropy is not exactly fulfilled in some construction materials, such as wood, which is fibrous, neither in stratified rocks, nor in materials made by a rolling process, etc. Nevertheless, the results obtained with this hypothesis are satisfactory in most of the cases.

A differential volume element on whose faces the corresponding stress components act is considered (Figure 2.1.a). To study the element's deformation of the element, the Principle of Superposition is applied by combining the strains caused by the different stress components acting independently. Based on the mechanical isotropy hypothesis, it is assumed that these strains are independent of the orientation of the chosen reference axes.

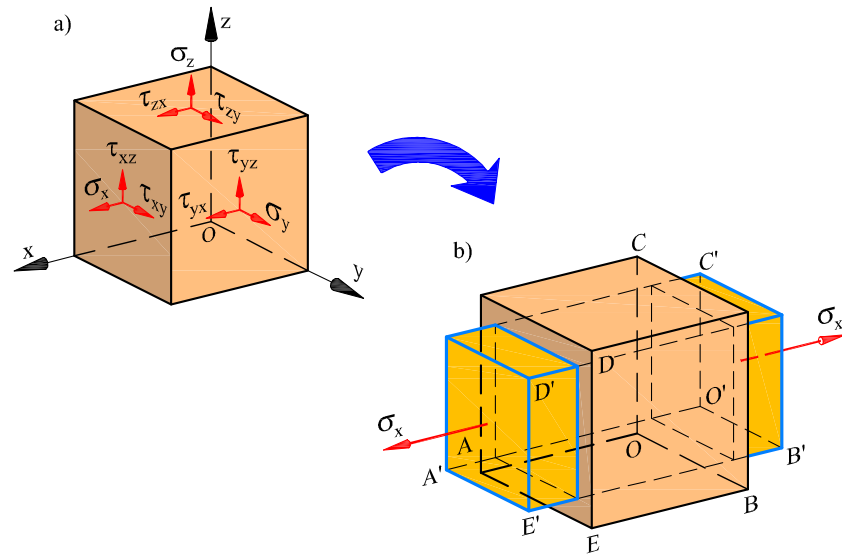


Figure 2.3: Differential volume element and strains under normal stress

Let us suppose first that only the normal stress σ_x component acts on two opposite faces of the differential element. Experimentally, normal stresses in isotropic materials do not produce angular deformation and only cause linear deformations along the edges of the parallelepiped (Figure 2.1.b). According to the generalized Hooke's law, these linear strains will be proportional to the stresses σ_x that produce them, that is:

$$\begin{aligned}
\varepsilon_{x_1} &= \frac{\overline{O'A'} - \overline{OA}}{\overline{OA}} = \frac{\sigma_x}{E} \\
\varepsilon_{y_1} &= \frac{\overline{O'B'} - \overline{OB}}{\overline{OB}} = -\nu\varepsilon_{x_1} = -\nu\frac{\sigma_x}{E} \\
\varepsilon_{z_1} &= \frac{\overline{O'C'} - \overline{OC}}{\overline{OC}} = -\nu\varepsilon_{x_1} = -\nu\frac{\sigma_x}{E}
\end{aligned} \tag{2.5}$$

where E is the longitudinal modulus of elasticity or Young's modulus, a property of the material that is determined experimentally. Although R. Young introduced the modulus of elasticity in 1803, L. Navier gave its modern definition (1826).

The dimensions of the modulus of elasticity are force per unit area $[FL^{-2}]$, as shown in expression (2.5). The mean values of the modulus of elasticity of some materials used in civil and mechanical engineering are shown in Table 2.4.

Material	E [GPa]	Material	E [GPa]
Steel	210	Concrete	20
Copper	120	Zinc	100
Brass	100	Duralumin	70
Bronze	110	Aluminum	76
Granite	50	Melting	170
Wood	10	Tin	40

Table 2.1: Young's modulus for different materials

In the preceding expressions, it can be seen that the longitudinal strain ε_{x_1} is accompanied by transversal strain $\varepsilon_{y_1}, \varepsilon_{z_1}$ proportional to ε_{x_1} , and defined by a physical constant of the material known as Poisson's ratio (S. D. Poisson, 1829). Poisson's ratio in steel ν is equal to 0.3 and in concrete it ranges from 0.15 to 0.20.

Similarly, by isotropy, the action of the normal stress component produces longitudinal and transversal strain of values:

$$\varepsilon_{y_2} = \frac{\sigma_y}{E} \quad \varepsilon_{z_2} = \varepsilon_{x_2} = -\nu\frac{\sigma_y}{E} \tag{2.6}$$

and the action of the stress normal component produces some strains equal to:

$$\varepsilon_{z_3} = \frac{\sigma_z}{E} \quad \varepsilon_{y_3} = \varepsilon_{x_3} = -\nu\frac{\sigma_z}{E} \tag{2.7}$$

Using the Principle of Superposition, the linear strain along the x axis caused by the joint action of the normal stresses $\sigma_x, \sigma_y, \sigma_z$ is $\varepsilon_x = \varepsilon_{x_1} + \varepsilon_{x_2} + \varepsilon_{x_3}$.

Similarly, it will occur for the linear strains ε_y and ε_z . Consequently, the following expressions can be derived:

$$\begin{aligned}\varepsilon_x &= \frac{\sigma_x}{E} - \frac{\nu}{E}(\sigma_y + \sigma_z) \\ \varepsilon_y &= \frac{\sigma_y}{E} - \frac{\nu}{E}(\sigma_z + \sigma_x) \\ \varepsilon_z &= \frac{\sigma_z}{E} - \frac{\nu}{E}(\sigma_x + \sigma_y)\end{aligned}\tag{2.8}$$

Consider now the strains that the tangential stress components cause in the elementary parallelepiped. Experimentally, the action of the reciprocal pairs of tangential stresses (that is, τ_{xy} and τ_{yx} , τ_{yz} and τ_{zy} , τ_{zx} and τ_{xz}) on a mechanically isotropic material produces an angular deformation (γ_{xy} , γ_{yz} , γ_{zx} , respectively) that transforms a straight parallelepiped into an oblique one (Figure 2.4).

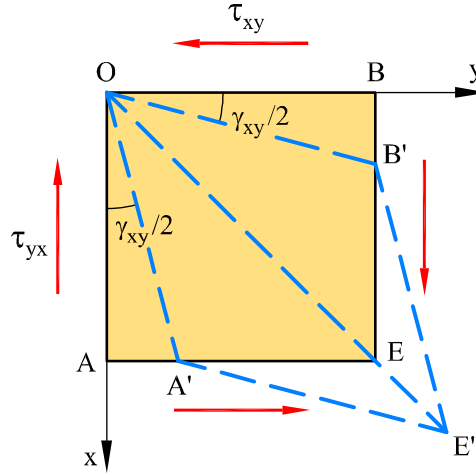


Figure 2.4: Strain under tangential stress

Hooke's generalized law states that the distortions produced are proportional to the acting tangential stress:

$$\begin{aligned}\gamma_{xy} &= \frac{\tau_{xy}}{G} \\ \gamma_{yz} &= \frac{\tau_{yz}}{G} \\ \gamma_{zx} &= \frac{\tau_{zx}}{G}\end{aligned}\tag{2.9}$$

where G is a physical constant of the material known as the transverse longitudinal modulus or shear modulus and has the dimensions $[FL^{-2}]$.

The analytical formulation of Generalized Hooke's law for an isotropic material is given by equations (2.8) and (2.9). The components of the stress state based on the strain state can be obtained by inverting these equations:

$$\begin{aligned}\sigma_x &= \lambda \varepsilon_{vol} + 2\mu \varepsilon_x & \tau_{xy} &= G \gamma_{xy} \\ \sigma_y &= \lambda \varepsilon_{vol} + 2\mu \varepsilon_y & \tau_{yz} &= G \gamma_{yz} \\ \sigma_z &= \lambda \varepsilon_{vol} + 2\mu \varepsilon_z & \tau_{zx} &= G \gamma_{zx}\end{aligned}\quad (2.10)$$

where $\varepsilon_{vol} = \varepsilon_x + \varepsilon_y + \varepsilon_z$ is the volumetric strain and λ and μ are the Lamé coefficients, defined by the following expressions:

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \quad \mu = \frac{E}{2(1+\nu)} \quad (2.11)$$

Shear and bulk modulus

It can be shown that:

$$G = \frac{E}{2(1+\nu)} = \mu \quad (2.12)$$

Therefore, an isotropic elastic material can be described by any of the following pairs of physical constants (E, ν) , (E, G) or (λ, μ) .

The shear modulus must be positive due to physical intuition and thermodynamic considerations, therefore it must be $\mu \geq -1.0$. However, no negative Poisson's ratios ν have been measured on any material. The minimum value of μ measured experimentally is that of beryllium ($\nu = 0.01 \div 0.05$).

To obtain the volumetric strain as a function of the acting stresses, the generalized Hooke's law equalities (2.8) are added, that is:

$$\varepsilon_{vol} = \varepsilon_x + \varepsilon_y + \varepsilon_z = \frac{1-2\nu}{E} (\sigma_x + \sigma_y + \sigma_z) \quad (2.13)$$

If the stress state is an hydrostatic pressure state, in which $\sigma_x = \sigma_y = \sigma_z = p$, then the volumetric strain value is equal to:

$$\varepsilon_{vol} = 3 \frac{(1-2\nu)}{E} p = \frac{p}{K} \quad (2.14)$$

where

$$K = \frac{E}{3(1-2\nu)} \quad (2.15)$$

is a material physical constant known as the *volumetric modulus of elasticity* or *bulk modulus*, and it has dimensions $[FL^{-2}]$.

For physical and thermodynamic reasons, the volumetric modulus of elasticity must be positive, which means that $\nu \leq 1/2$; that is, in any isotropic material, the Poisson's ratio must be less than 0.5, which corresponds to the limit of incompressible behavior ($K = \infty$).

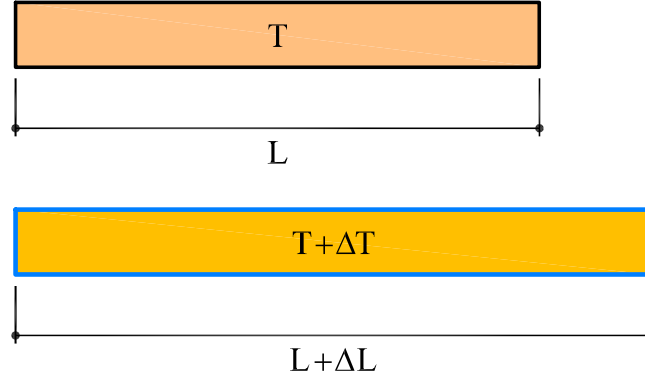


Figure 2.5: Elongation of a piece due to temperature variation

Thermal Effects

When solids are exposed to temperature changes, they vary in size and shape. When the temperature of an unrestricted bar changes, it is observed that the unitary thermal strain ε_T is approximately a linear function of the temperature variation (Figure 2.5).

$$\varepsilon_T = \frac{\Delta L}{L} = \alpha \Delta T \quad (2.16)$$

where α is the *thermal expansion coefficient* of the material.

The elongation of a bar caused by an increase in temperature is equal to:

$$\Delta L = \varepsilon_T L = \alpha \Delta T L \quad (2.17)$$

The strains caused by thermal effects are considered independent from those caused by other mechanical stresses during or after the temperature change. Being independent, the Principle of Superposition can be applied; thus, by adding the temperature-induced strains to Hooke's law expressions (2.8) and (2.9), it is obtained;

$$\begin{aligned} \varepsilon_x &= \frac{\sigma_x}{E} - \frac{\nu}{E}(\sigma_y + \sigma_z) + \alpha \Delta T & \gamma_{xy} &= \frac{\tau_{xy}}{G} \\ \varepsilon_y &= \frac{\sigma_y}{E} - \frac{\nu}{E}(\sigma_z + \sigma_x) + \alpha \Delta T & \gamma_{yz} &= \frac{\tau_{yz}}{G} \\ \varepsilon_z &= \frac{\sigma_z}{E} - \frac{\nu}{E}(\sigma_x + \sigma_y) + \alpha \Delta T & \gamma_{zx} &= \frac{\tau_{zx}}{G} \end{aligned} \quad (2.18)$$

Thermal strains cause variations in length (and volume, in fact), but not angular distortions, in isotropic solids without kinematic constraints.

Example 2.4.1

A compressive stress of $\sigma_z = 20$ MPa is applied to a cubic concrete block with the dimensions $100 \times 100 \times 100$ mm³. The body is constrained to move in the x, y directions (Figure 2.6). Calculate the following by using the generalized Hooke's law: (a) the value of the stresses in the x, y directions; (b) the elongation of the block in the z direction. Data: $E = 20$ GPa y $\nu = 0.20$.

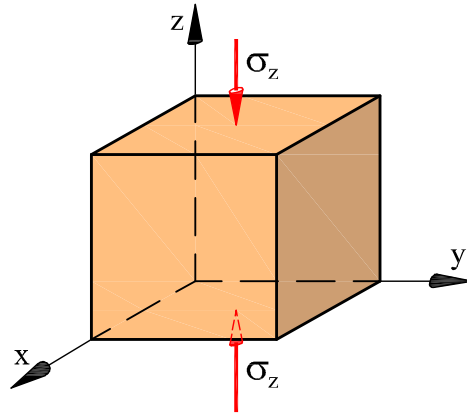


Figure 2.6: Concrete block of Example 2.4.1

The strains in the directions x, y are zero if the movements in those directions are restricted ($\varepsilon_x = \varepsilon_y = 0$). Applying Hooke's generalized law the following expressions can be derived:

$$\begin{aligned}\varepsilon_x &= \frac{\sigma_x - 0.2(\sigma_y - 20)}{E} = 0 \\ \varepsilon_y &= \frac{\sigma_y - 0.2(-20 + \sigma_x)}{E} = 0 \\ \varepsilon_z &= \frac{-20 - 0.2(\sigma_x + \sigma_y)}{E}\end{aligned}$$

From the first and second equations:

$$\begin{aligned}\sigma_x - 0.2\sigma_y &= -4 \text{ MPa} \\ \sigma_y - 0.2\sigma_x &= -4 \text{ MPa}\end{aligned}$$

Consequently, the values of the stresses σ_x and σ_y are identical and equal to:

$$\sigma_x = \sigma_y = -\frac{4}{0.8} = -5 \text{ MPa}$$

The unit shortage in the z direction is deduced from the third equation:

$$\varepsilon_z = \frac{[-20 - 0.2(-10)] 10^6}{20 \cdot 10^9} = -0.9 \cdot 10^{-3}$$

And the shortage of the concrete block in z direction is equal to:

$$\Delta L_z = L_z \varepsilon_z = 100 (-0.9 \cdot 10^{-3}) = -9 \cdot 10^{-2} \text{ mm}$$

Example 2.4.2

The rectangular plate in Figure 2.7 is subjected to $\sigma_x = 120$ MPa and $\sigma_y = 60$ MPa tensile stresses. Assuming a plane stress state, determine: (a) the elongation that it experiences in the x, y directions, (b) the variation in thickness $e = 20$ mm, (c) the variation in volume ΔV (d) check that $\frac{\Delta V}{V} = \varepsilon_x + \varepsilon_y + \varepsilon_z$.

Data: $E = 72$ GPa y $\nu = 0.30$.

Applying Eq. (2.8) with $\sigma_z = 0$, is derived:

$$\begin{aligned} \varepsilon_x &= \frac{(120 - 0.3 \cdot 60) 10^6}{72 \cdot 10^9} = 1.417 \cdot 10^{-3} \\ \varepsilon_y &= \frac{(60 - 0.3 \cdot 120) 10^6}{72 \cdot 10^9} = 0.333 \cdot 10^{-3} \end{aligned}$$

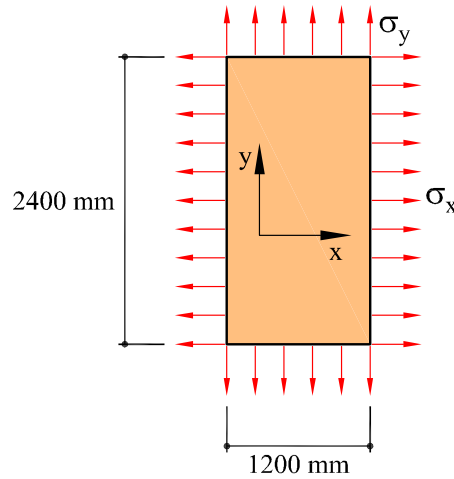


Figure 2.7: Rectangular plate of Example 2.4.2 under a plane stress state

(a) The elongation of the plate in the x, y directions is

$$\begin{aligned} \Delta L_x &= 1200 \cdot 1.417 \cdot 10^{-3} = 1.70 \text{ mm} \\ \Delta L_y &= 2400 \cdot 0.333 \cdot 10^{-3} = 0.80 \text{ mm} \end{aligned}$$

(b) The strains in the z -direction of the plate is:

$$\varepsilon_z = -\frac{\nu(\sigma_x + \sigma_y)}{E} = -\frac{0.3(120 + 60)10^6}{72 \cdot 10^9} = -0.75 \cdot 10^{-3}$$

Therefore, the thickness of the plate reduces by:

$$\Delta e = 20(-0.75 \cdot 10^{-3}) = -1.5 \cdot 10^{-2} \text{ mm}$$

(c) The initial volume of the plate is:

$$V = 1200 \cdot 2400 \cdot 20 = 57.6 \cdot 10^6 \text{ mm}^3$$

and taking into account the elongations it undergoes, the final volume is:

$$V_f = 1201.7 \cdot 2400.8 \cdot 19.985 = 57.6576 \cdot 10^6 \text{ mm}^3$$

Consequently, the change in volume of the plate is:

$$\Delta V = V_f - V = 0.0576 \cdot 10^6 \text{ mm}^3$$

(d) The unit volume change is equal to:

$$\frac{\Delta V}{V} = \frac{0.0576 \cdot 10^6}{57.6 \cdot 10^6} = 1 \cdot 10^{-3}$$

The sum of the unitary elongations is equal to:

$$\varepsilon_x + \varepsilon_y + \varepsilon_z = (1.417 + 0.333 - 0.75) 10^{-3} = 1 \cdot 10^{-3}$$

thus proving that the unit increase in volume is equal to the *volumetric deformation*:

$$\frac{\Delta V}{V} = \varepsilon_x + \varepsilon_y + \varepsilon_z = \varepsilon_{vol}$$

Example 2.4.3

An axial force P acts on the steel bar in Figure 2.8. The transversal dimensions are reduced to 49.982 mm under that load. Assuming that the stresses σ_y and σ_z are negligible or zero, calculate (a) the longitudinal elongation of the bar ΔL_x , and (b) the load P , assuming that the stresses σ_y and σ_z are negligible or zero.

Data: $\sigma_y = \sigma_z = 0$, $E = 200 \text{ GPa}$ y $\nu = 0,30$.

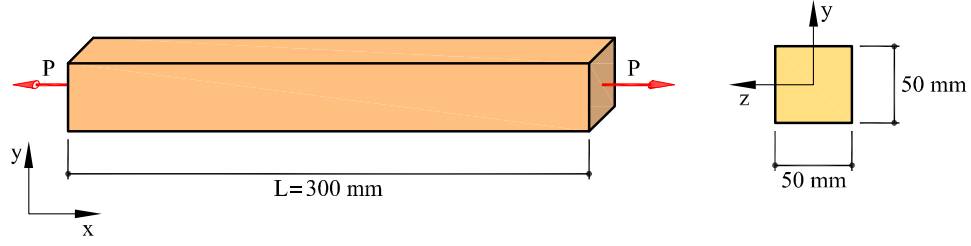


Figure 2.8: Steel bar of the Example 2.4.3

(a) The variation of the transversal dimensions of the bar is:

$$\Delta L_y = \Delta L_z = 50 - 49.982 = -1,8 \cdot 10^{-2} \text{ mm}$$

and the strain in directions x, y is:

$$\varepsilon_y = \varepsilon_z = \frac{-1,8 \cdot 10^{-2}}{50} = -3,6 \cdot 10^{-4}$$

Applying the generalised Hooke's law and being $\sigma_y = \sigma_z = 0$,

$$\varepsilon_y = \varepsilon_z = -\frac{\nu \sigma_x}{E} = -3,6 \cdot 10^{-4}$$

therefore:

$$\sigma_x = \frac{3,6 \cdot 10^{-4} \cdot 200 \cdot 10^9}{0,3} = 240 \text{ MPa}$$

and the strain in the x -direction results:

$$\varepsilon_x = \frac{\sigma_x}{E} = \frac{240 \cdot 10^6}{200 \cdot 10^9} = 1,2 \cdot 10^{-3}$$

Consequently, the bar's elongation in the x direction is equal to:

$$\Delta L_x = \varepsilon_x L_x = 1,2 \cdot 10^{-3} \cdot 300 = 0,36 \text{ mm}$$

(b) Given that the transversal cross section area is equal to $A = 5 \cdot 5 = 25 \text{ cm}^2$, the axial load required can be determined from the expression:

$$P = \sigma_x A = 240 \cdot 10^6 \cdot 25 \cdot 10^{-4} = 600 \text{ kN}$$

Example 2.4.4

Solve Example 2.4.3 considering equal decrease of the transversal dimensions and assuming a temperature decrease during the test with a value of $\Delta T = -15^\circ\text{C}$. The coefficient of expansion of the steel is $\alpha = 1 \cdot 10^{-5} \text{ }^\circ\text{C}^{-1}$.

Since the shrinkage ΔL_y and ΔL_z are the same as in the previous example, the corresponding strains are also the same: $\varepsilon_y = \varepsilon_z = -3.6 \cdot 10^{-4}$. Applying the generalised Hooke's law, and taking into account the temperature variation, it is:

$$\varepsilon_y = \varepsilon_z = -\frac{\nu \sigma_x}{E} + \alpha \Delta T = -3.6 \cdot 10^{-4}$$

Taking into account that $\alpha \Delta T = -1.5 \cdot 10^{-4}$, it follows:

$$-\frac{\nu \sigma_x}{E} = (-3.6 + 1.5) \cdot 10^{-4} = -2.1 \cdot 10^{-4}$$

where the stress σ_x is equal to:

$$\sigma_x = \frac{2.1 \cdot 10^{-4} \cdot 200 \cdot 10^9}{0.3} = 140 \text{ MPa}$$

The strain in the x -direction is :

$$\varepsilon_x = \frac{\sigma_x}{E} + \alpha \Delta T = \frac{140 \cdot 10^6}{200 \cdot 10^9} + (-1.5 \cdot 10^{-4}) = 5.5 \cdot 10^{-4}$$

Therefore, the piece's elongation in the x direction is equal to:

$$\Delta L_x = \varepsilon_x L_x = 5.5 \cdot 10^{-4} \cdot 300 = 0.165 \text{ mm}$$

The cross section area of the bar is $A = 25 \text{ cm}^2$, and the load in this case is:

$$P = \sigma_x A = 140 \cdot 10^6 \cdot 25 \cdot 10^{-4} = 350 \text{ kN}$$

Example 2.4.5

A neoprene block is subjected to a test to determine its shear modulus G . As shown in Figure 2.9, the block is fixed to a test table and a rigid plate. A load $P = 100 \text{ kN}$ is applied, and the horizontal displacement experienced is $\delta = 1 \text{ mm}$. With these data, (a) calculate the modulus of shear stiffness G and (b) establish a general relationship between the magnitudes G, h, a, b, P and δ .

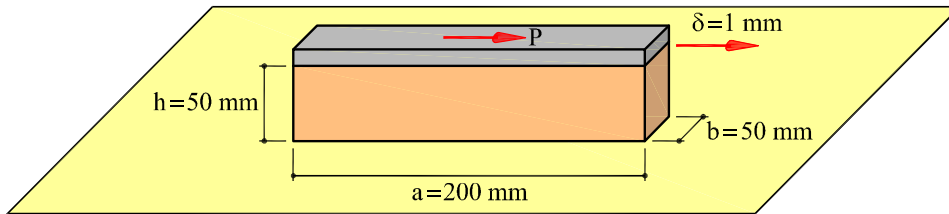


Figure 2.9: Neoprene block test Example 2.4.5

(a) The angular distortion or deformation of the neoprene block is equal to:

$$\gamma = \frac{\delta}{h} = \frac{0.1}{5} = 2 \cdot 10^{-2} \text{ rad}$$

The tangential stress can be calculated using the following expression:

$$\tau = \frac{P}{A} = \frac{100 \cdot 10^3}{(20 \cdot 5) 10^{-4}} = 10 \text{ MPa}$$

According to generalized Hooke's law, the shear modulus is equal to:

$$G = \frac{\tau}{\gamma} = \frac{10}{2 \cdot 10^{-2}} = 500 \text{ MPa}$$

(b) The relation between the magnitudes G, h, a, b, P y δ , is:

$$\frac{\delta}{h} = \frac{P}{Gab}$$

2.4.1 Plane stress state

A structure is in a plane stress state (Section 1.6) when one of its dimensions (z) is significantly smaller than the two others which define the plane of analysis (x, y). Moreover, the loads that the structure sustains, are contained within this plane (x, y) (Figure 2.10).

Two examples of structures under plane stress state are shown in Figure 2.11: (a) a wall loaded in its mid-plane and (b) a deep beam loaded as indicated.

The equations relating stresses with strain (generalized Hooke's law) obtained for the three-dimensional general case are simplified when applied to the two-dimensional problem. In a plane stress state, with:

$$\sigma_z = \tau_{xz} = \tau_{yz} = 0$$

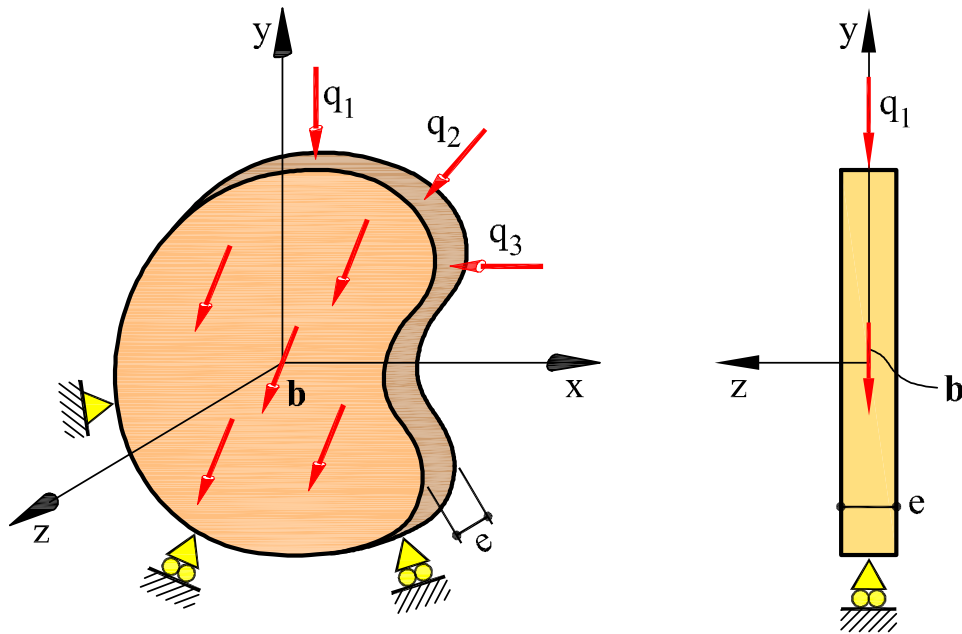


Figure 2.10: Plane stress state

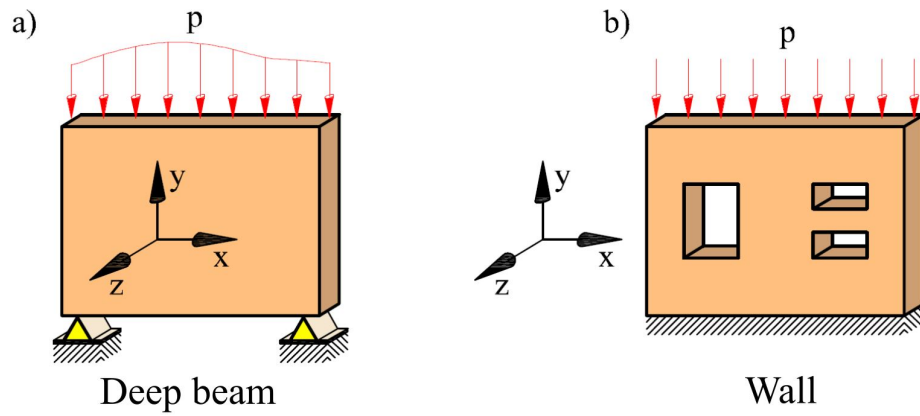


Figure 2.11: Structures under plane stress: (a) deep beam and (b) wall

The generalized Hooke's law (Eqs. (2.8) and (2.9)) can be written, for an

isotropic material:

$$\begin{aligned}\varepsilon_x &= \frac{1}{E}(\sigma_x - \nu\sigma_y) & \gamma_{xy} &= \frac{1}{G}\tau_{xy} \\ \varepsilon_y &= \frac{1}{E}(\sigma_y - \nu\sigma_x) & \gamma_{xz} &= 0 \\ \varepsilon_z &= -\frac{\nu}{E}(\sigma_x + \sigma_y) & \gamma_{yz} &= 0\end{aligned}\quad (2.19)$$

For a two-dimensional problem, these equations can be inverted to obtain the components of stress state as a function of strain state, as in the three-dimensional problem (Eqs. (2.10) and (2.11)):

$$\begin{aligned}\sigma_x &= \frac{E}{1-\nu^2}(\varepsilon_x - \nu\varepsilon_y) & \tau_{xy} &= G\gamma_{xy} \\ \sigma_y &= \frac{E}{1-\nu^2}(\varepsilon_y - \nu\varepsilon_x) & \tau_{xz} &= 0 \\ \sigma_z &= 0 & \tau_{yz} &= 0\end{aligned}\quad (2.20)$$

2.4.2 Plane strain state

A structure is in a plane strain state (Section 1.9) when one of its dimensions (z) is significantly larger than the others (x, y) and is subjected to loads perpendicular to the axis of the largest dimension (Figure 2.12). These conditions result in the following:

$$\varepsilon_z = \gamma_{xz} = \gamma_{yz} = 0$$

Some examples of plane strain state are shown in Figure 2.13: (a) loaded retaining wall and (b) piping under internal or external pressure.

In this case, the generalized Hooke's law (Eqs. (2.8) and (2.9)), reduces to the following expressions:

$$\begin{aligned}\varepsilon_x &= \frac{1}{E}[\sigma_x - \nu(\sigma_y + \sigma_z)] & \gamma_{xy} &= \frac{1}{G}\tau_{xy} \\ \varepsilon_y &= \frac{1}{E}[\sigma_y - \nu(\sigma_x + \sigma_z)] & \gamma_{xz} &= 0 \\ \varepsilon_z &= \frac{1}{E}[\sigma_z - \nu(\sigma_x + \sigma_y)] = 0 & \gamma_{yz} &= 0\end{aligned}\quad (2.21)$$

where:

$$\sigma_z = \nu(\sigma_x + \sigma_y)$$

Substituting the expressions (2.21), the following is obtained:

$$\begin{aligned}\varepsilon_x &= \frac{1-\nu^2}{E}\left[\sigma_x - \left(\frac{\nu}{1-\nu}\right)\sigma_y\right] \\ \varepsilon_y &= \frac{1-\nu^2}{E}\left[\sigma_y - \left(\frac{\nu}{1-\nu}\right)\sigma_x\right]\end{aligned}\quad (2.22)$$

If $\tilde{E} = E / (1 - \nu^2)$ y $\tilde{\nu} = \nu / (1 - \nu)$, the following expressions are obtained:

$$\begin{aligned}\varepsilon_x &= \frac{1}{\tilde{E}} [\sigma_x - \tilde{\nu}\sigma_y] \\ \varepsilon_y &= \frac{1}{\tilde{E}} [\sigma_y - \tilde{\nu}\sigma_x]\end{aligned}\quad (2.23)$$

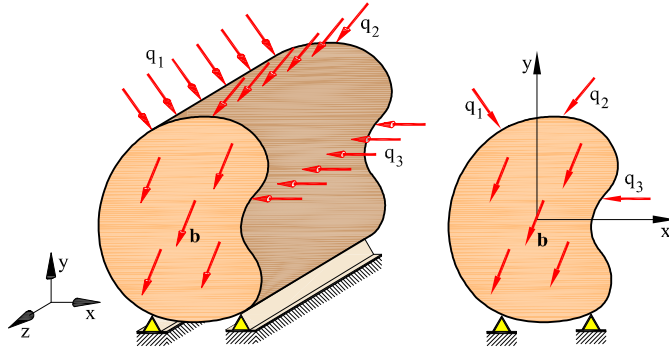


Figure 2.12: Plane strain state

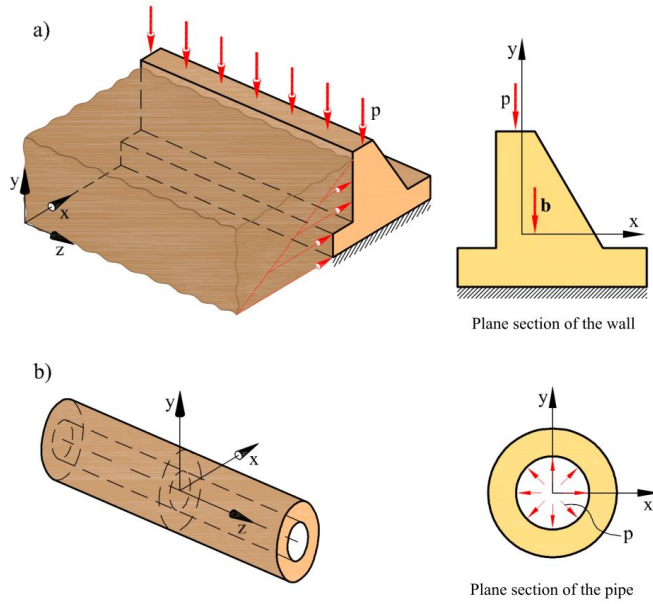


Figure 2.13: Plane strain state examples (a) retaining wall y (b) piping under internal pressure or external

By inverting the expressions (2.23), the stress state can be calculated as a function of the strain in a plane strain state:

$$\begin{aligned}
 \sigma_x &= \frac{\tilde{E}}{1 - \tilde{\nu}^2} (\varepsilon_x - \tilde{\nu} \varepsilon_y) & \tau_{xy} &= G \gamma_{xy} \\
 \sigma_y &= \frac{\tilde{E}}{1 - \tilde{\nu}^2} (\varepsilon_y - \tilde{\nu} \varepsilon_x) & \tau_{xz} &= 0 \\
 \sigma_z &= \nu (\varepsilon_x + \varepsilon_y) & \tau_{yz} &= 0
 \end{aligned} \tag{2.24}$$

Note that Eqs. (2.23) and (2.24) are respectively analogous to Eqs (2.21) and (2.23), substituting E and ν by $\tilde{E}$ and $\tilde{\nu}$.

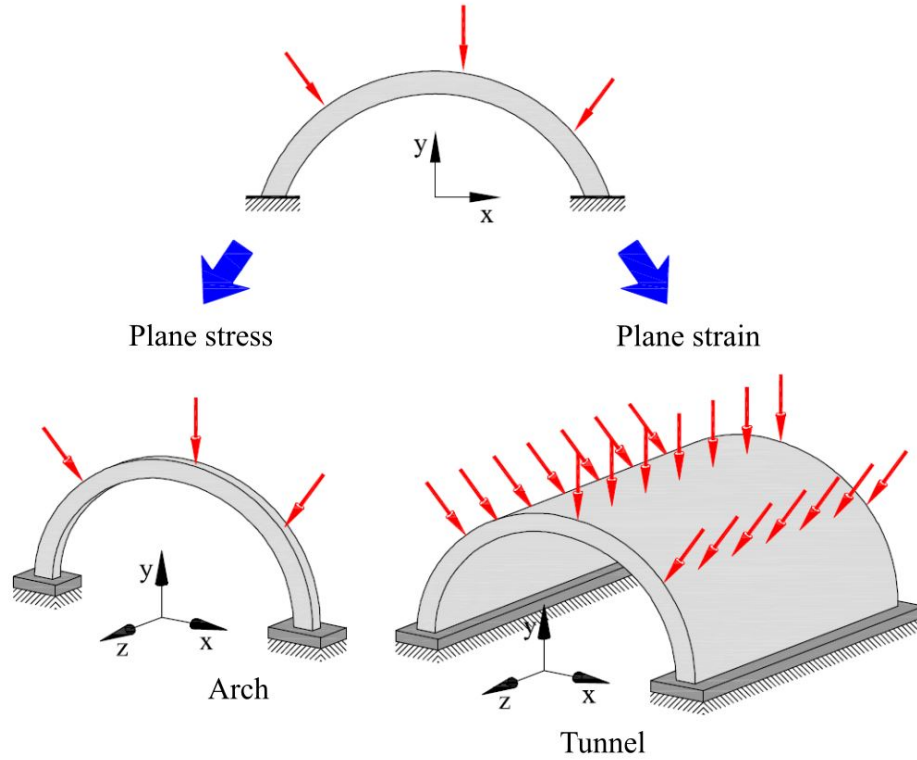


Figure 2.14: Difference between plane stress and plane strain states.

It can be seen from a comparison of the expressions obtained for the plane stress and plane strain states that, both are very similar when the analysis plane is considered (the xy plane in the description above). It should be noted, however, that outside this plane (along the z axis), reality is very different in both cases, as shown in 2.14.

2.5 Stress-strain relationship. Experimental study

The actual relationship between stress and strain of a material is determined experimentally, through laboratory tests. The simplest tests to perform are those of pure tension or compression on normalized cylindrical or prismatic specimens.

These tests are performed by applying a force P to its ends in the direction of the specimen's axis (Figure 2.15). If A is the initial specimen cross-sectional area, the nominal (mean) stress is defined as follows:

$$\sigma = \frac{P}{A} \quad (2.25)$$

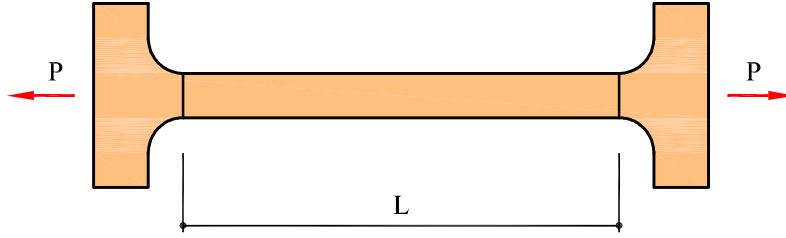


Figure 2.15: Specimen for uniaxial tensile test

On the other hand, the nominal axial strain is defined as the ratio of L , the calibrated length of the specimen, to ΔL , the elongation produced by the axial force:

$$\varepsilon = \frac{\Delta L}{L} \quad (2.26)$$

If the test also measures the variation Δd of one of the transversal dimensions of the specimen with initial length d , the nominal transversal strain can be calculated as follows:

$$\varepsilon_t = \frac{\Delta d}{d} \quad (2.27)$$

The test is performed by gradually increasing the applied force P , measuring the axial elongations ΔL and the transversal shrinkage Δd that correspond to each load level, and recording the results. The corresponding sets of nominal stress and strain values (σ , ε , ε_t) are calculated using the obtained set of values (P , ΔL , Δd). The stress-strain curve of the material is a curve on which the pairs (σ, ε) are plotted (Figure 2.16).

Although each material responds differently, the following limit zones and limit stresses can generally be distinguished in the stress-strain curve:

- *Proportional zone (OA)*. In this first zone, Hooke's law is satisfied; that is, for stresses less than the *proportionality limit* σ_p , the relationship between stress and strain is linear. The Young's modulus (E) or elastic modulus of the material is the slope of the curve. Poisson's ratio can be calculated as the ratio between longitudinal and transversal strain $\nu = -\varepsilon_t/\varepsilon$.
- *Elastic Zone (OB)*. In this zone, the unloading happens elastically, which means that there are no permanent deformations when the load is reduced until it is cancelled out. The maximum stress that can be applied without causing permanent deformations (also known as plastic, inelastic deformation) is known as the *elastic limit* (σ_e). Depending on the material, the zone AB of non-linear elastic behavior may or may not be observable.
- *Plastic zone (BD)*. Permanent deformation is observed in the specimen when the test is unloaded (OL in the diagram). The stress at which the material deforms with almost no additional stress is known as the yield limit (σ_f).
- *Yielding zone (CD)*. In this zone, the deformation increases without an appreciable increase in stress. Until reaching point D , the longitudinal strain and the transverse strain are substantially uniform along the length of the specimen.

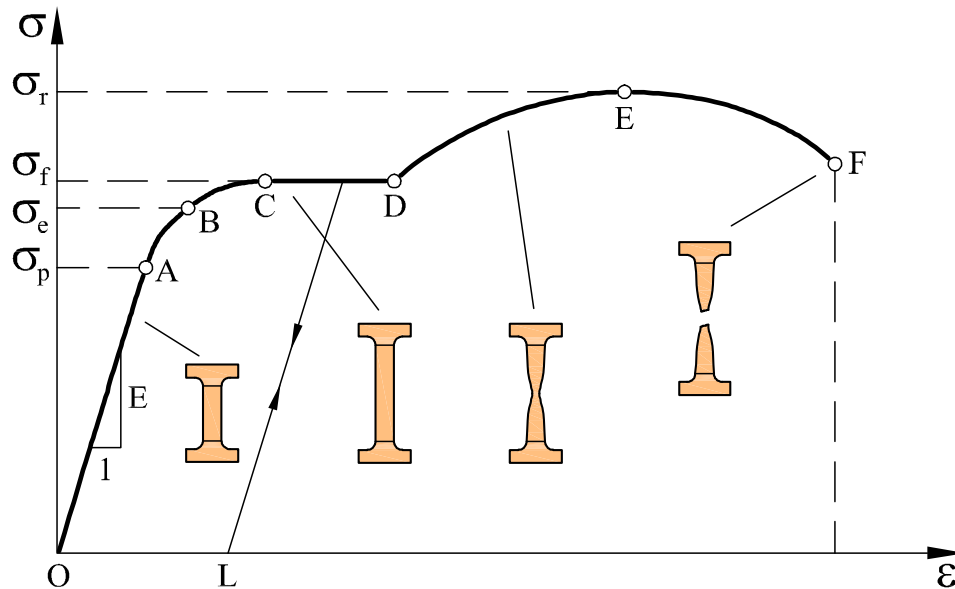


Figure 2.16: Uniaxial stress-strain curve

- *Strain hardening zone (DE)*. Starting from a certain strain value, the stress required to further increase the plastic strain increases. This phenomenon is known as strain hardening and is typically accompanied by the loss of the uniform strain state and the onset of strain localization in the center of the specimen.
- *Necking zone (EF)*. In this zone, a portion of the specimen's cross section starts to shrink considerably. Necking is the term for the phenomenon of localized longitudinal and transverse strain with loss of the prior uniform state (Figure 2.17). The nominal definition of the stress, which is related to the initial area of the specimen rather than the real area at each instant, and which is smaller due to the constriction, is the cause of the apparent stress loss in this zone. The *ultimate stress* (σ_r) is the maximum stress measured during the test.

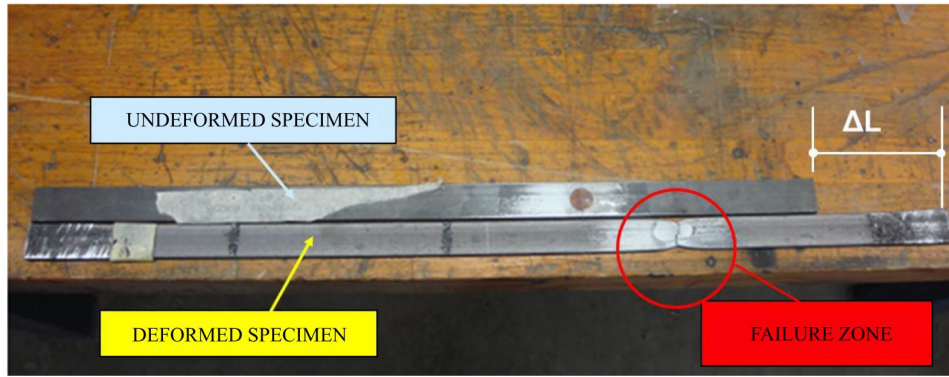


Figure 2.17: Undeformed and deformed steel bar after tensile test

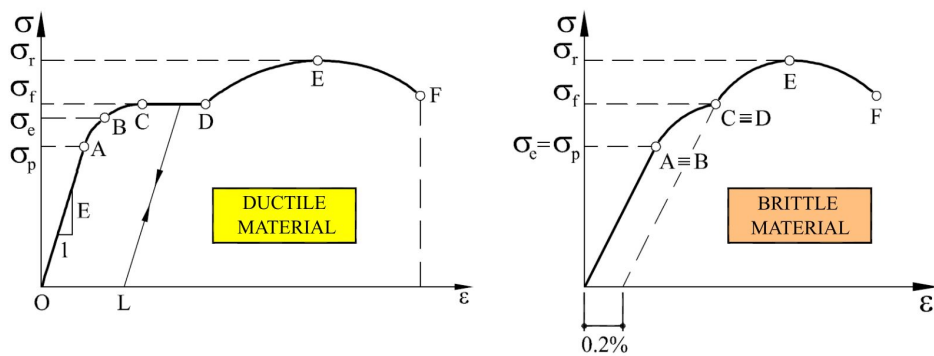


Figure 2.18: Stress-strain curves for materials: a) ductile and b) brittle

The described stress-strain curve corresponds to the so-called *ductile materials*, such as steel or copper, which have a pronounced plastic behavior and fail with a high level of deformation. In this materials, $\varepsilon_D \simeq 20 \varepsilon_A$ $\varepsilon_F \simeq 200 \varepsilon_A$, and the zones described are clearly distinguishable.

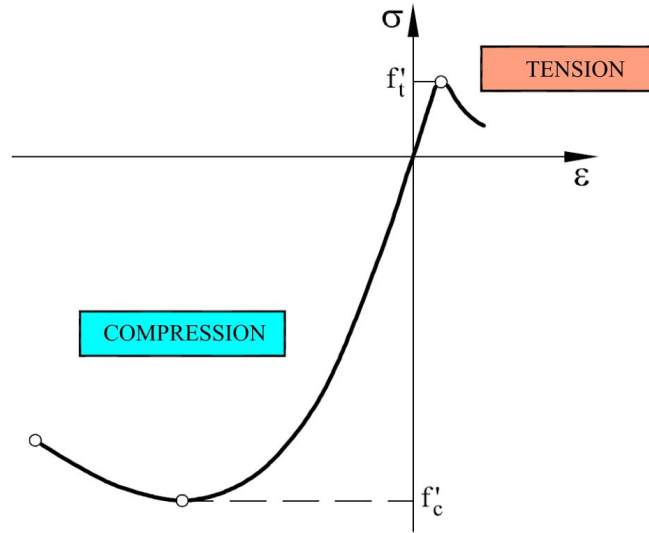


Figure 2.19: Strain-Stress curve for concrete

In other materials such as cast iron, concrete, or glass, known as brittle materials, there is no clearly defined plastic or yield zone.

In these materials, the yield stress, σ_f , is commonly assumed to be that corresponding to 0.2 % permanent deformation (Figure 2.18). These materials fail with little strain ($\varepsilon_F \simeq 5 \varepsilon_A$), abruptly, a potentially dangerous situation for the safety of structures.

Despite what has been said, classifying a particular material as ductile or brittle is not totally accurate since the mechanical behavior is significantly dependent on the loading conditions. Thus, a ductile steel under “normal” conditions, can experience brittle behavior at low temperatures or high loading rates. In contrast, glass and cast iron show ductile behavior at high temperatures. Many brittle materials have a very different strengths depending on whether they are subjected to tensile or compression stress. Concrete and most natural rocks, for example, have a very brittle behavior when subjected to tensile stress with a low strength. In contrast, its behavior under compression is more ductile, despite the lack of a distinguishable yield zone, and its compressive strength is higher, on the order of 10 times higher as shown in Figure 2.19.

Structural steels, on the other hand, behave similarly in tension and compression (for not very slender pieces), with a distinguished yielding zone

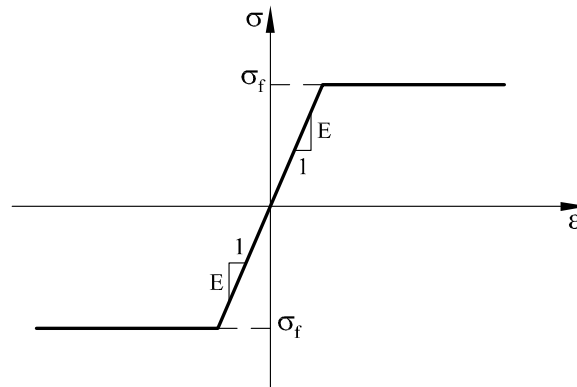


Figure 2.20: Idealized stress-strain curve for structural steel

in both cases. The stress-strain curve for these materials is typically represented by trilinear diagrams (Figure 2.20), in which only the proportional and yield zone are considered. On the other hand, other steel alloys, hardly show horizontal yielding zone but have a clearly different zone of strain hardening, which is usually linear.

Example 2.5.1

The bar in Figure 2.21, with a diameter of 12 mm and a length of 60 mm, is subjected to a tensile test until it breaks. Based on the results of the experiment, calculate: a) the proportionality limit σ_p , b) Young's modulus and c) the ultimate stress.

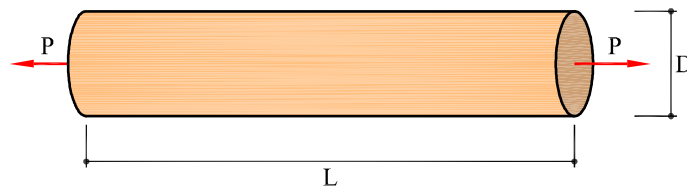


Figure 2.21: A bar for Example 2.5.1

From the test measurements, the stresses and strains

<i>Loads and elongation from the tensile test</i>										
P [kN]	0	6	12	18	24	30	36	42	48	57 (aprox)
$\Delta L \cdot 10^{-3}$ [mm]	0	12	24	36	48	60	98	144	240	Fracture

<i>Stress and strain from the tensile test</i>									
σ [MPa]	0	53	106	159	212	265	318	371	424
$\varepsilon \cdot 10^{-6}$	0	200	400	600	800	1000	1630	2400	4000

and taking into account that:

$$\begin{aligned} \text{Initial bar area} \quad A &= \frac{\pi D^2}{4} = \frac{\pi 12^2}{4} = 113 \text{ mm}^2 \\ \text{Nominal stress} \quad \sigma &= \frac{P}{A} \\ \text{Axial nominal strain} \quad \varepsilon &= \frac{\Delta L}{L} = \frac{\Delta L \text{ mm}}{60 \text{ mm}} \end{aligned}$$

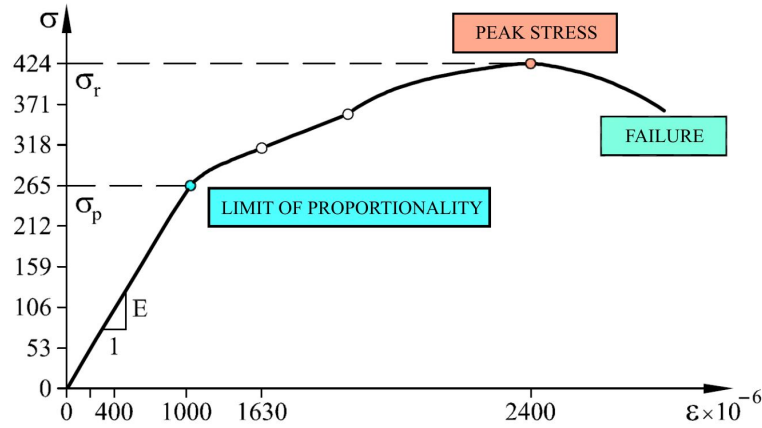


Figure 2.22: Stress-strain curve for the tested bar

The proportionality limit corresponds to the stress $\sigma_p = 265$ MPa, as shown in the corresponding stress-strain curve (Figure 2.22). When this limit of proportionality is reached, the unitary strain equals $\varepsilon = 10^{-3}$.

The modulus of elasticity, or Young's modulus, is the slope of the line that represents the proportional elastic behavior:

$$E = \frac{\Delta \sigma}{\Delta \varepsilon} = \frac{265}{10^{-3}} = 265 \text{ GPa}$$

The ultimate stress σ_r equals $\sigma_r = 424$ MPa, is the greatest stress measured during the test. According to the type of stress-strain diagram obtained, the tested material has ductile behavior.

Example 2.5.2

At the proportionality limit, the bar in Figure 2.23 increases in length by 0.34 mm and its transversal dimension $h = 18$ mm decreases by $7.1 \cdot 10^{-3}$ mm. If the corresponding load is 32 kN, calculate: a) the limit of proportionality σ_p , b) the elasticity modulus and c) Poisson's ratio.

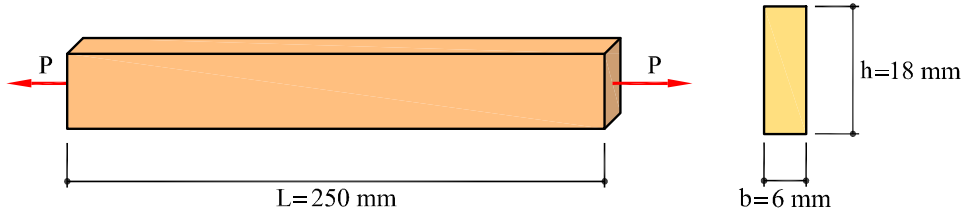


Figure 2.23: Bar for Example 2.5.2

The load at the limit of proportionality is $P_p = 32$ kN, and the bar cross-section area is: $A = 6 \cdot 18 = 108$ mm²; then the limit of proportionality is:

$$\sigma_p = \frac{P_p}{A} = \frac{32 \cdot 10^3}{108} = 296 \text{ MPa}$$

The nominal axial strain is :

$$\varepsilon = \frac{\Delta L}{L} = \frac{0.34}{250} = 1.36 \cdot 10^{-3}$$

Therefore, the elasticity modulus or Young's modulus is:

$$E = \frac{\sigma_p}{\varepsilon} = \frac{296 \cdot 10^6}{1.36 \cdot 10^{-3}} = 217 \text{ GPa}$$

The transverse dimension variation h is $\Delta h = -7.1 \cdot 10^{-3}$; hence, the nominal transverse strain is:

$$\varepsilon_t = \frac{\Delta h}{h} = \frac{-7.1 \cdot 10^{-3}}{18} = -0.394 \cdot 10^{-3}$$

By definition, Poisson's ratio is equal to:

$$\nu = -\frac{\varepsilon_t}{\varepsilon} = -\frac{-0.394 \cdot 10^{-3}}{1.36 \cdot 10^{-3}} = 0.289$$

2.6 Limit stress, allowable stress and safety factor

Throughout their service life, structures are subjected to permanent, variable, or exceptional actions and loads that can result in a total or partial structural failure.

Safety requirements for built and designed structures require the definition of the highest value of stress that the material can reach without compromising structural integrity. Therefore, the limit stress σ_{lim} is defined as the stress value at which a material reaches its strength limit, and below which neither excessive strain nor, of course, its failure take place. In ductile materials, such as steel, the yield stress ($\sigma_{\text{lim}} = \sigma_f$) is usually considered as the limiting stress, whereas in brittle materials, such as concrete, the ultimate stress ($\sigma_{\text{lim}} = \sigma_r$) is considered as the limiting stress.

In practice, however, structures are designed to work at stress levels significantly below the limit stress of the corresponding structural materials. The existence of uncertainties which the designer must take into account during the design phase is one of many reasons for this practice. These are:

- *Uncertainties associated with the characterization of the mechanical properties of materials* due to the quality and homogeneity of the materials, the dispersion in the experimental tests, the adequacy of these to the actual service conditions, creep and relaxation phenomena, degradation due to physical or chemical causes, cyclic or dynamic effects, fatigue, etc.
- *Uncertainties associated with the definition and characterization of actions*, either due to insufficient data to estimate them (climatic and environmental effects, seismic effects), due to their insufficient definition in the project phase (service or exceptional loads), due to the temporary variability of its nature, due to the lack of knowledge of its evolution during the useful life of the structure, etc.
- *Uncertainties associated with the calculation methods*, either as a result of the simplifying assumptions adopted in the calculation bases, the very nature of the analysis methods, or the difficulty associated with them if sophisticated methods are employed.
- *Uncertainties associated with the construction methods*, which may be associated with the planned construction process, the execution and quality control procedures employed, etc.
- *Uncertainties associated with conservation and maintenance techniques*, which can appear as phenomena of progressive deterioration, corrosion and environmental aggression, etc.

Consequently, the *allowable stress*, σ_{all} , is defined as:

2.6. LIMIT STRESS, ALLOWABLE STRESS AND SAFETY FACTOR 61

$$\sigma_{\text{all}} = \frac{\sigma_{\text{lim}}}{n} \leq \sigma_{\text{lim}} \quad (2.28)$$

where σ_{lim} is the limiting stress of the material and n is the safety coefficient, which is greater (or exceptionally equal) than unity. Structures are therefore designed with the criterion that the stress at any point is less than the allowable stress value; that is, if σ_{max} is the maximum value of stress in the structure, it must be fulfilled that

$$\sigma_{\text{max}} \leq \sigma_{\text{all}} \leq \sigma_{\text{lim}} \quad (2.29)$$

The estimation of the safety factor is dependent on a large number of variables, but it must be greater the greater the degree of design phase uncertainty or the greater the socioeconomic impact of a structural failure. The safety coefficient can be decreased for structures that are simple or have a short service life. In any case, this coefficient is determined based on the accumulated experience with comparable structures and the effective technological level. In building and construction engineering, safety coefficients ranging from 2 to 5 are employed, with higher values adopted for *on-site* construction than for prefabrication. In aeronautical engineering, safety coefficients between 1.5 and 2 are required, but are accompanied by significantly more complex methods of analysis, construction, conservation, and control.

Chapter 3

Concepts and Basic Principles

3.1 Introduction

Strength of Materials is the part of Mechanics of Deformable Solids that results from applying the Theory of Elasticity to a restricted type of problems that arise in the day to day of Structural Engineering to enable its analytical solution. This category of problems is mainly defined by:

- *A restriction in the definition of the static problem*, such that Strength of Materials is applied to prismatic elements (beams, columns, bars, shafts, etc.) that are supported, pin-jointed or clamped and exposed to the action of point or distributed forces, descents of supports, thermal loads, etc.
- *The adoption of certain fundamental hypotheses* that allow, on the one hand, to define a linear problem and, on the other hand, address an initially three-dimensional problem (the pieces have length, height and width) as a “assembly” of basically one-dimensional problems (elements are represented as lines).

This Chapter briefly presents the fundamentals of Strength of Materials. Basic concepts such as elements (bar, beam), structure, support, links, etc., are introduced. Afterward others of fundamental importance as those of internal forces on a cross-section, statically determinate and indeterminate structures, etc.

3.2 Objectives of the Strength of Materials

Strength of Materials, as an integral part of Mechanics of Structures, has as fundamental objective to determine the response of structures when they

are subjected to the different actions that they must withstand during their construction and service life.

By “structural response” is meant the determination of the stress and strain states to which the structure is subjected by the effect of the different states of load considered. On the other hand, the determination of the stress states is necessary to check the fulfillment of the *strength* criteria established by the corresponding regulations and codes of good practice, in order to guarantee the *security* of the structures. On the other hand, the determination of the strain states is necessary to satisfy the *stiffness* criteria, which are often linked to the requirements of the structures *functionality*.

Therefore, Strength of Materials aims to establish the conditions of *strength* and *stiffness* of the analyzed structures. Both aspects are addressed in the two types of fundamental problems that are raised in engineering practice: the *design* and the *verification* of structures.

The process of building a structure begins with its *conception*. In this part its typology is determined and the loads that will act on it are identified. Strength of Materials comes into play to evaluate the structural response of the corresponding “structural pre-design”. Based on the obtained results, to the *design* of the structural elements is done, that is, values are given to the dimensions of these elements. Later, the *definition* part of the project is addressed, the anticipated requirements are checked. In this phase, decisions made in the previous part are modified or ratified. In any case, the structural typology is defined and specified and the actions to consider (self weight, live loads, etc.) are specified. The *verification* of the resulting structural model follows. If this *a posteriori* test of the structural design is satisfactory, the *final project* is prepared; if it is not, it is necessary to return to the definition part. The procedure proceeds iteratively until all requirements are adequately satisfied.

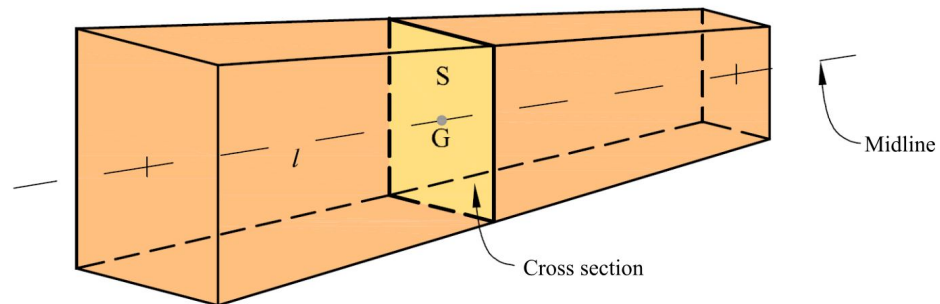


Figure 3.1: Concept of prismatic element

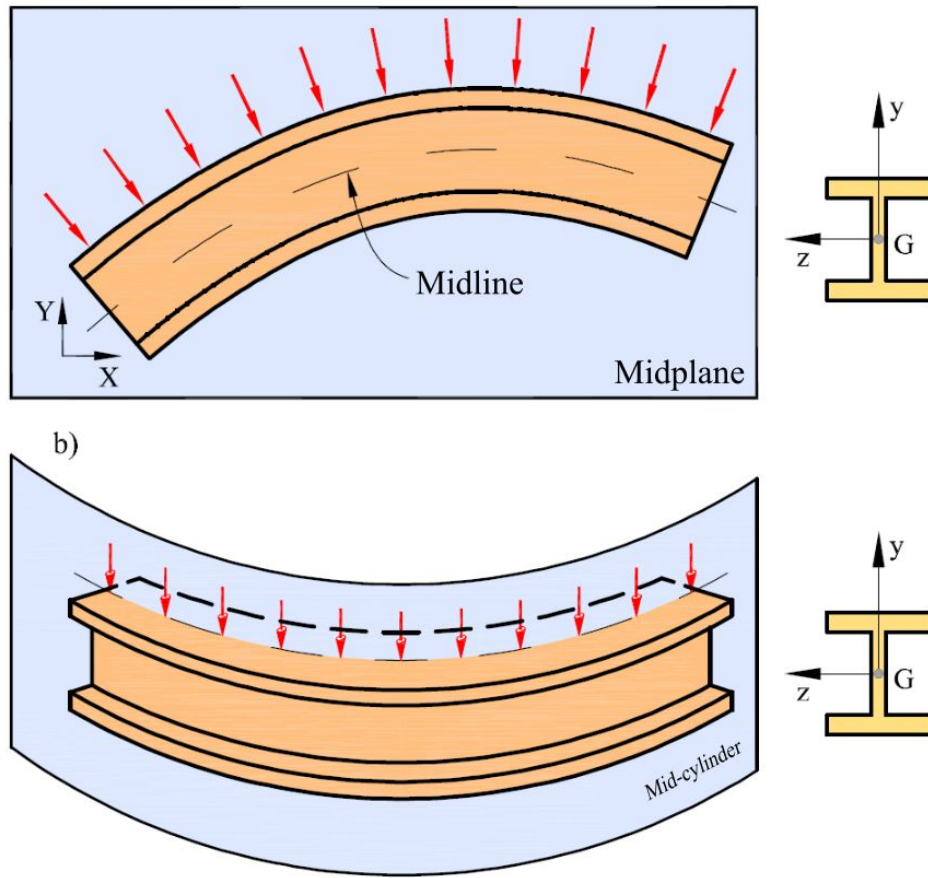


Figure 3.2: (a) mid-plane element (b) mid-cylinder plane element

3.3 Concept of prismatic element

A *prismatic element*, *beam* or *bar*, is the solid body defined by an area S that moving in space so that its center of gravity G runs along a given line l and its plane remains constantly normal to that line. The line l is called *midline* or *axis of the element* and the various positions that S occupies along the axis are called *normal cross-sections* of the element (Figure 3.1).

Prismatic elements can be classified by their geometry, depending on its axis or depending on the shape and variation of the cross-section along the element.

Most of the elements used in civil engineering and in architecture have *straight* axes without curvature. They are also usual *plane* elements, whose axis is a plane curve (with curvature in a single plane).

A *element of mid-plane* is a plane element (straight or curved) in which



Figure 3.3: Prismatic elements: different sections

the cross-section is symmetric with respect to a plane that contains the axis of the element and that is subjected to loads contained in such mid-plane (Figure 3.2a).

A *mid-cylinder plane element* is a plane element (straight or curved) in which the cross-section is symmetric with respect a plane perpendicular to the one that contains the axis of the element and that is subjected to loads perpendicular to the plane of the axis (Figure 3.2b).

If the axis presents curvatures in two perpendicular planes, it defines a *warped* element.

In addition, according to the variation of the section along the element, this can be of *constant section (or inertia) or variable*. Figure 3.3 shows the framed structure of a industrial warehouse in which you can see the combination of elements of constant section (columns and straps) and variable section pieces (lintels).

Depending on the shape of the cross-section of the element, we have *solid* (usually rectangular), *hollow* or *thin-walled* (I-beams, T-beams, H-beams) beams. Figure 3.4 shows three straight elements with a solid rectangular section (Figure 3.4a), an I-beam (Figure 3.4b) and a hollow rectangular (Figure 3.4c).

A *slice* is a differential element that is contained between two infinitely close cross-sections beam (Figure 3.5a). A beam can be seen as a solid generated by a *succession* of differential slices.

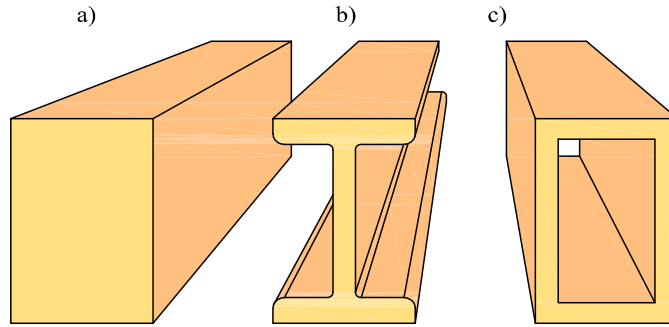


Figure 3.4: Beam: (a) solid, (b) I-beam, (c) hollow

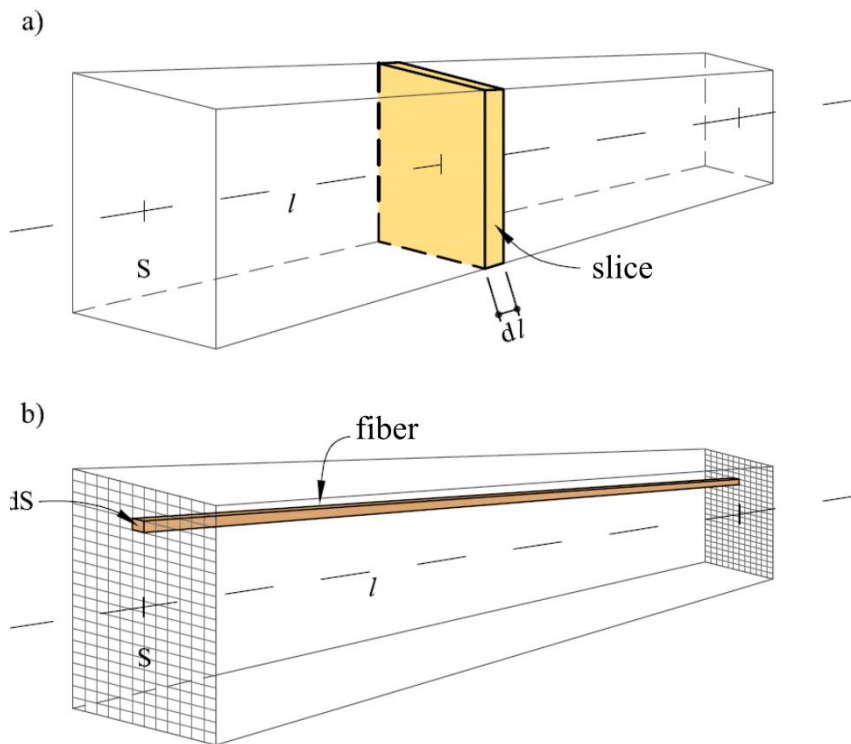


Figure 3.5: Concept of: (a) slice and (b) fiber

A *fiber* is a differential beam element generated by a differential element of the plane area S in the movement that generates the prismatic element

(Figure 3.5b). Therefore, a beam can be seen as a solid formed by a *bundle* of differential fibers. The slice and the fiber are the basic differential elements of study in Strength of Materials.

In the definition of a prismatic element, beam or bar, composite elements made of various materials are also considered, as long as the interfaces between the different materials are *parallel* to the element axis. The definition of slice and fiber in composite pieces are analogous to those corresponding to the prismatic element of homogeneous cross-section. A common example of composite elements are the reinforced concrete beams, where the bars of steel that constitute the longitudinal reinforcement are fibers embedded in the surrounding concrete. Mixed bridge decks also respond to this category, where a concrete bridgehead is placed on top of a base of steel beams or a steel box.

3.4 Concept of Bar Structure

A bar structure is a mechanical system formed by the assembly of prismatic parts. Its purpose is to support the actions that act on it and to transmit the corresponding forces and moments to the supports. Although bar structures are not the only existing structural typology, it is the most commonly used. For this reason, it is the one treated in this Book.

It is usual to classify bar structures according to the *disposition of the axes* of the elements and according to the *type of links* of these.

According to the disposition of the axes of the elements that form the structure, we speak of *planar structures* when the axes of all the pieces are contained in the same plane, or *spatial structures* otherwise.

Spatial structures are much more complex to conceive, design, and build than planar structures. Therefore, it is common that many apparently three-dimensional structures are actually designed as a set of planar structures that work solidarily to achieve the necessary resisting mechanisms.

If the link of the prismatic elements is such that it allows the rotation between the elements, the structure is called *pin-jointed* or *truss structure*, and works fundamentally under axial forces. Pin-jointed structures are represented in an idealized way as follows: the bars are represented by axes of the elements; the links are represented by a circle and the supports are represented taking into account the restricted degrees of freedom: roller or fixed (see Figure 3.6).

If the link of the prismatic elements is in such that completely prevents the relative rotation between the elements, the structure of bars is called a *frame* and works fundamentally in bending and, in some cases, torsion. Bending and torsion are often accompanied by shear forces. Frame structures are idealized so that the beams are represented by their axes and rigid links are idealized by the node where the axes of the bars meet (Figure 3.7).

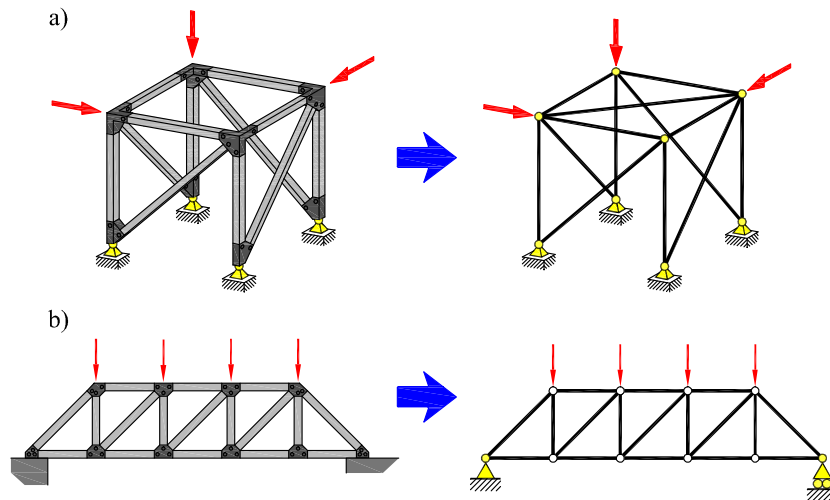


Figure 3.6: Idealization of truss structures: (a) spatial (b) planar

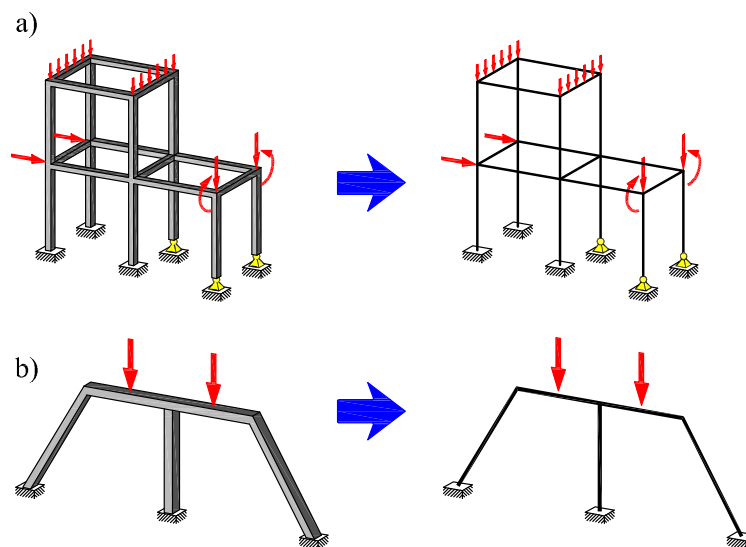


Figure 3.7: Idealization of frame structures: (a) spatial (b) planar

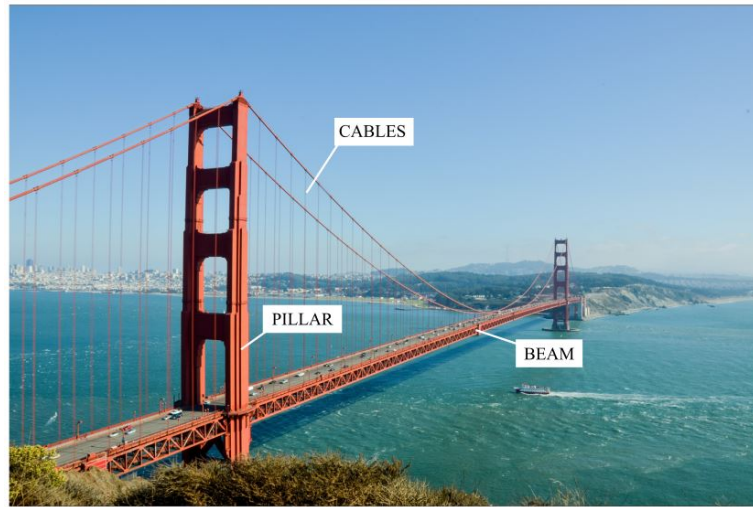


Figure 3.8: Structural elements: beams, pillars, and cables

A *mid-plane structure* is a structure formed by mid-plane elements that are subjected to loads contained in such a mid-plane. A *plane grid* is formed by mid-cylinder elements subjected to loads perpendicular to the mid-plane that contains the element axes.

3.5 Structural elements

The prismatic parts that form structures receive different denominations depending on the function they perform. Thus, a horizontal structural element is a *beam*, generally straight, which supports mainly vertical loads and works fundamentally in bending. A *pillar*, *support* or *column* is a structural element, generally straight and vertical, that resists axial compression loads and, in some cases, also bending moments. The structural element subjected to tensile axial forces is a *cable* or a *brace*; they are often used to bridge large spans, since they are limited only by their weight and anchorage.

Figure 3.8 shows a cable-stayed bridge in which these three basic structural elements are combined.

Arches are plane structural elements with a curved or polygonal axis that support their loads fundamentally in compression. Unlike cables, they must be rigid to preserve their shape, and this generates secondary internal forces (moments and shears) that must be considered in their design.

Figure 3.9 shows a tilt-bridge that combines these elements to support the deck, a beam with a curved axis.



Figure 3.9: Structural elements: arch, cables, and curved beam

3.6 Supports and links in bar structures

3.6.1 Supports

A *support* is a device intended to join a structure to the supporting medium. The supports fulfill the triple function of:

- (a) preventing the rigid solid movements of the structure
- (b) limiting the undergone deformation
- (c) transmitting the loads supported by the structure to the supporting medium

The most common types of support in plane truss and frame structures are described below. Spatial structures also have their supporting devices which are not described here.

Supports in planar truss structures

A cross-section of a plane truss bar has two degrees of freedom, two independent translations. Therefore, there are two types of support, depending on whether they constrain one or both degrees of freedom of the support cross-section. For each restricted degree of freedom, a reaction is transmitted so that movement is prevented.

(a) Simple or roller support

It is a type of support that only restricts the translation movement in one direction of the support cross-section, allowing translation in the perpendicular direction and the rotation of the support cross-section. The reaction

it produces is a force perpendicular to the rolling direction, that is, to the allowed movement; it introduces a *single unknown*, the modulus of the reaction (with known direction) Figure 3.10).

(b) Fixed or hinged support

This type of support completely prevents the translational movement of the support cross-section, but allows rotation.

The reaction it produces is a force of unknown direction and magnitude; *two unknowns* are introduced: the components of the reaction with respect to any two perpendicular axes (Figure 3.11).

3.6.2 Supports in mid-plane truss structures

The most common types of support in mid-plane structures are described below. Since a cross-section of a mid-plane beam has three degrees of freedom, two independent translations in the mid-plane, and one rotation perpendicular to the mid-plane, there are several types of supports, depending if they constrain one, two or all three degrees of freedom of the support cross-section. For each restricted degree of freedom, the support transmits a *reaction* to the structure that prevents movement. These reactions are, equal and in opposite directions to the forces that the structure transmits to the means of support.

(a) Simple or roller support

This type of support only constrains the movement of the support section in one direction, allowing translation in the perpendicular direction and the rotation of the support cross-section (Figure 3.12a). The reaction it produces is a force perpendicular to the rolling direction, that is, to the

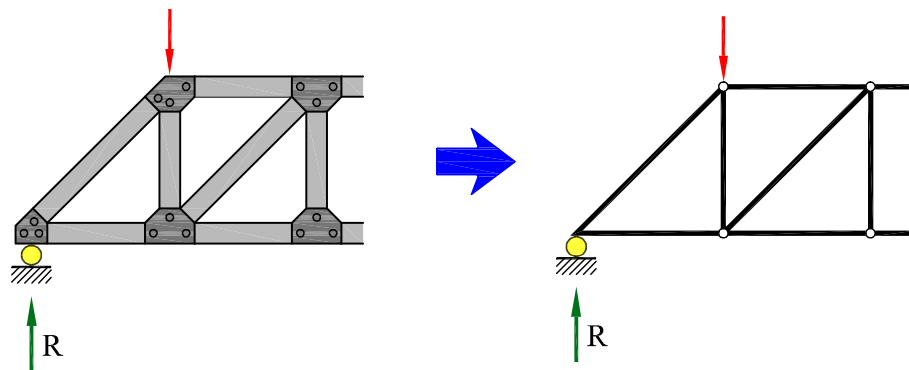


Figure 3.10: Truss structures: Simple or roller support

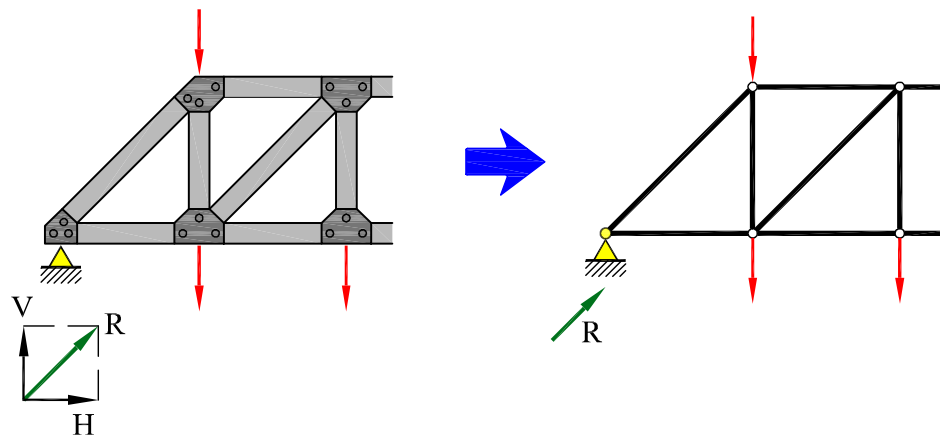


Figure 3.11: Truss structures: Pinned or hinged support

allowed movement. Associated with a simple support, therefore, only one unknown is introduced, the module of the reaction, since its position and direction are known.

(b) Fixed or pinned support

This type of support totally prevents the translation movement of the support section, but allows the rotation (Figure 3.12b). The reaction it produces is a force of unknown direction and magnitude, thus, two unknowns are introduced: the components of the reaction with respect to any two perpendicular axes.

(c) Clamping

This type of support prevents any movement of the support section, both in translation and rotation (Figure 3.12c). The reaction that occurs is a force of unknown position, direction and magnitude; thus, three unknowns are introduced: two components of the reaction and the moment on the support cross-section.

3.6.3 Links

A link is a device intended to join together the different pieces that form a structure. The links fulfill the double function of:

- (a) preventing or limiting the *relative* movements of some pieces with respect to others, and
- (b) transmitting the loads that they support to the others (Figure 3.13).

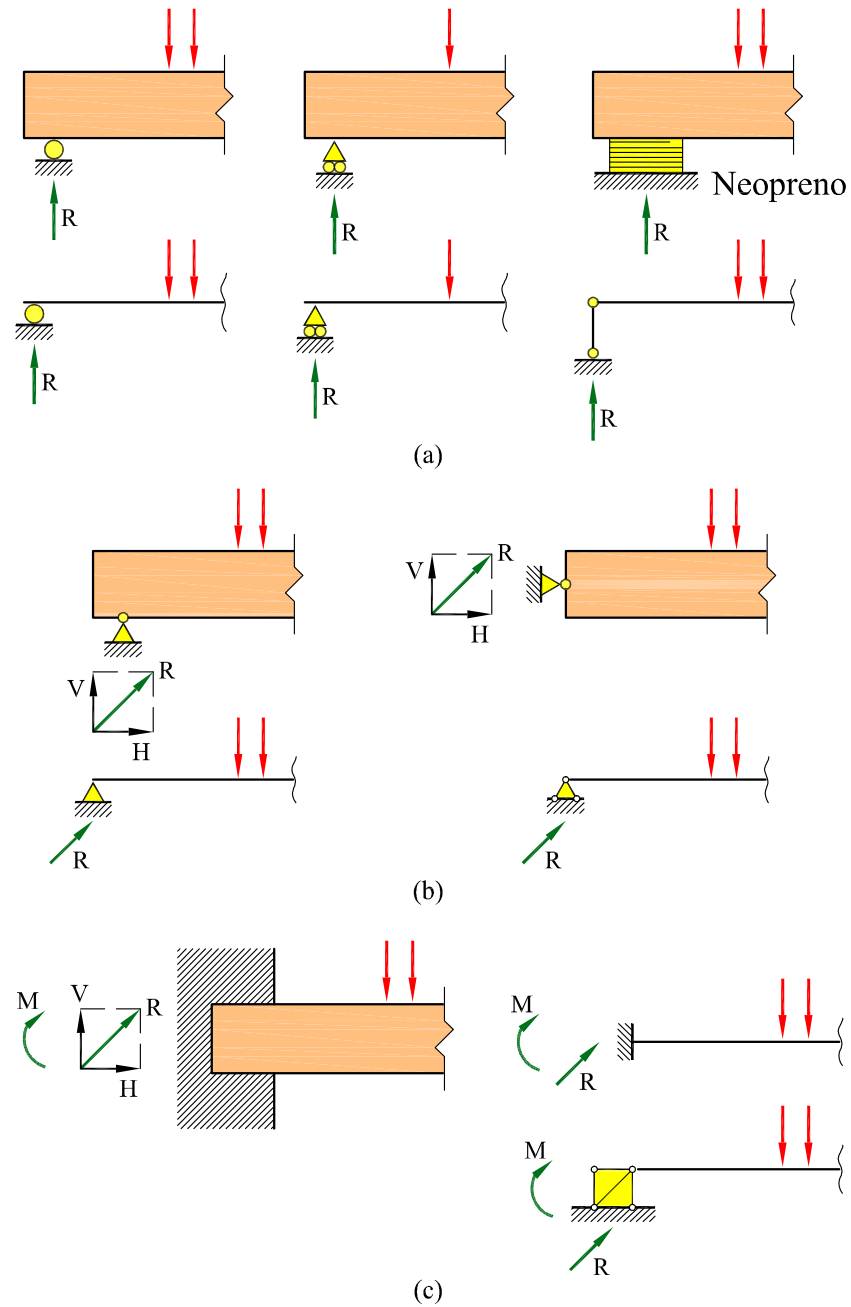


Figure 3.12: Supports in mid-plane structures: (a) simple or roller, (b) fixed or pinned or hinged, and (c) clamped

For each degree of freedom of the link, this transmits an internal force to the concurrent pieces. These forces are equal and in opposite directions in the two pieces that are joined in the link.

The most common types of links in planar structures are described below. Although only links in planar structures are illustrated, they can be generalized to the three-dimensional case.

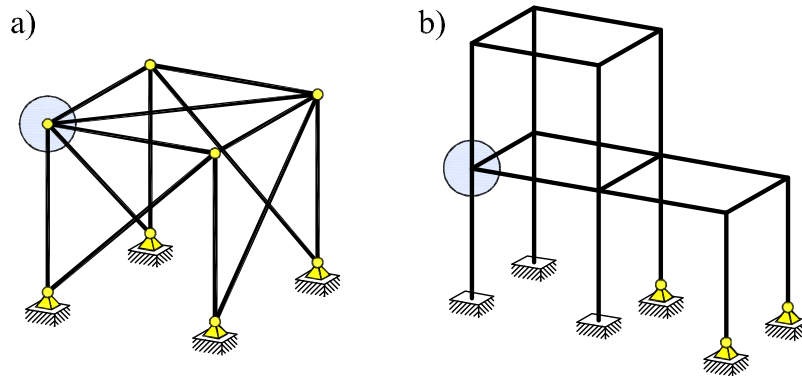


Figure 3.13: Links or nodes: (a) jointed (b) rigid

(a) Roller link

This type of link only constrains relative motion in one direction, allowing translation in the perpendicular direction and rotation of the link section (Figure 3.14a). The reaction it produces is a pair of forces, equal and opposite, perpendicular to the allowed motion. Therefore, only one unknown is introduced: the modulus of the reactions. It is usually represented as a roller between the pieces that slide.

(b) Joint link

This type of link completely prevents the relative translation of the link section, but allows its relative rotation (Figure 3.14b). The reaction it produces is a pair of forces of unknown direction and magnitude. In an internal joint, therefore, two unknowns are introduced: the components of the reaction with respect to any two perpendicular axes. The bending moment is necessarily zero at a joint.

This type of link is the one that joins together the different bars of a truss structure. It can also appear occasionally in other bar structures. It is usually represented with a circle on the node.

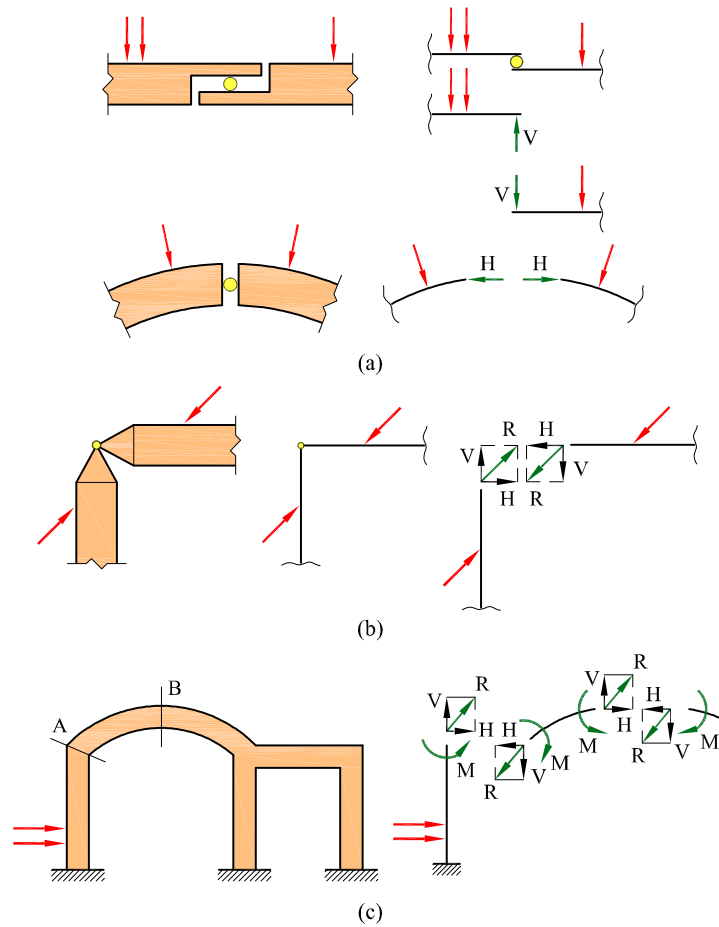


Figure 3.14: Links in mid-plane structures: (a) roller, (b) joint, and (c) fixed links

(c) Rigid link

This type of link prevents any movement of the link section, both in translation and rotation (Figure 3.14c). The reaction it produces is a couple of forces of unknown position, direction and magnitude. In a rigid node, therefore, three unknowns are introduced: two components of the reaction and the moment of the link section.

This type of link is the one that joins together the different bars of a frame structure. It is usually represented as a continuous union between the pieces. The supports and links considered for planar structures can easily be generalized for spatial structures. In this case, the number of degrees of freedom of the section is six (three translations and three rotations), and therefore the number of types of possible supports and links is much greater.

3.7 Principles of Strength of Materials

Strength of Materials is a discipline based on three fundamental principles or hypotheses: the Stiffness Principle, the Superposition Principle and the Saint-Venant Principle. The validity of the results obtained through this theory depends, basically, on the degree of satisfaction of these principles in the specific case in which it is applied.

3.7.1 The Principle of Stiffness

The Principle of Stiffness is a direct consequence of the hypothesis of small displacements applied to the analysis of structures and states that

“the equilibrium equations can be formulated on the undeformed geometry, that is, without considering the movements caused by the load system”.

This hypothesis implies that the movements of the structure due to the loading system are small: the displacements are small, compared to the dimensions of the structure, and the rotations are small (in radians), compared with unity. If this condition is not met, the equilibrium equations must be formulated in the deformed geometry, and the problem ceases to be linear and becomes a *geometrically nonlinear* problem.

In Figure 3.15 the application of the Principle of stiffness is illustrated. Under the hypothesis of small movements (left), the reactions at the left clamping only depend on the initial geometry and the loads ($M = F_1 l$); without it (right), the reactions depend on the deformation of the structure ($M = F_1 (l - \delta) + F_2 f$), and this is unknown *a priori*.

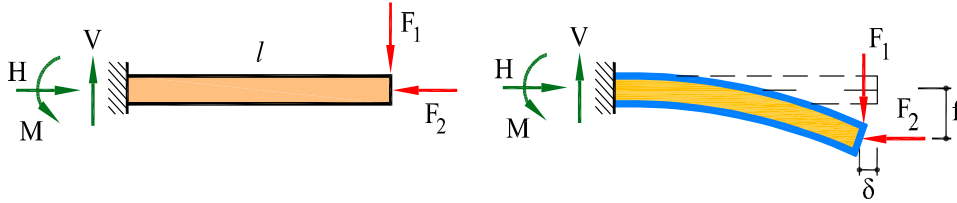


Figure 3.15: Implication of the Principle of Stiffness

3.7.2 The Principle Superposition

As already indicated in Chapter 2, if the hypothesis of small displacements is valid and Hooke's law is satisfied, then the elastic problem is linear. Solving a linear structural problem is much easier than solving a nonlinear one; in addition to other practical difficulties, in a nonlinear problem there is, in general, no guarantee that the solution exists and is unique.

Also, if the problem is linear, the Superposition Principle can be applied. This states that

“the effects that a system of forces causes on a structure are equal to the sum of the effects caused by each of the forces of the system acting separately”.

The Principle of Superposition implies that the reactions, movements, stresses and deformations caused on the structure by the original load state is equal to the sum of the reactions, movements, stresses and deformations caused by the states in which this is decomposed.

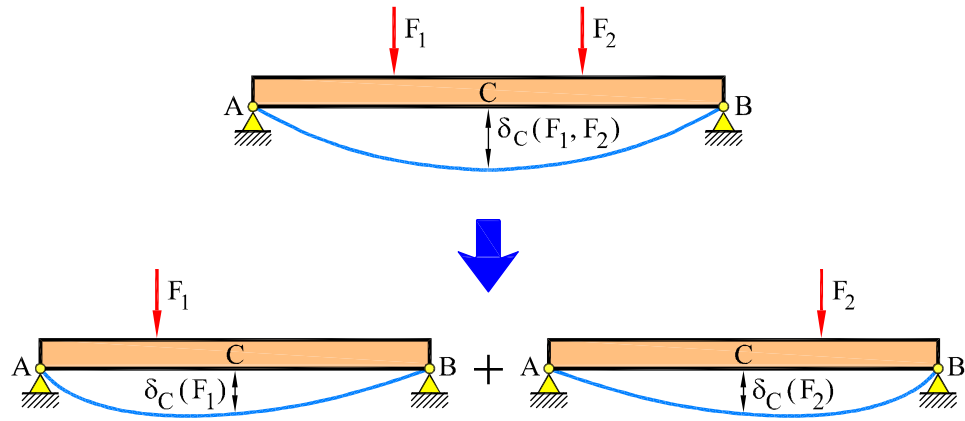


Figure 3.16: Application of the Principle of Superposition

Thus, for example, Figure 3.16 shows a simple-supported beam on which a load system consisting of two point forces acts; this system can be decomposed into the two simpler systems in the lower part of the Figure in order, for example, to calculate the vertical deflection at point C .

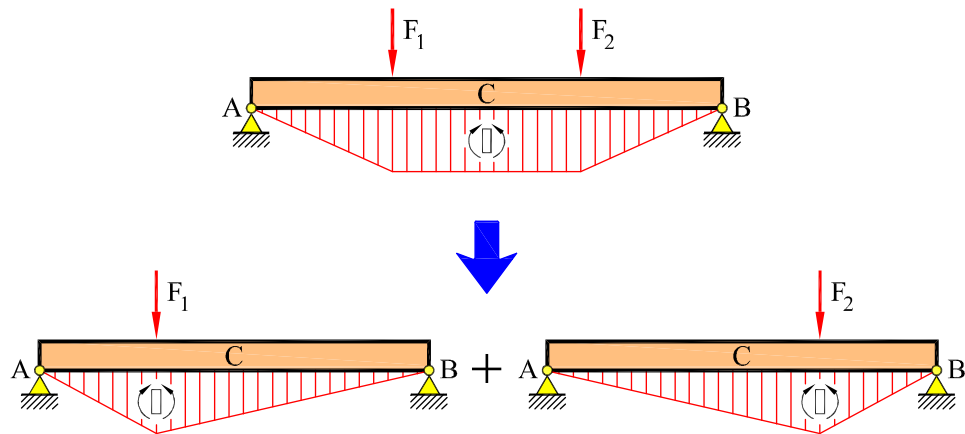


Figure 3.17: Application of the Principle of Superposition

Likewise, in the simple-supported beam of Figure 3.17, the bending mo-

ment due to the load system $F_1 + F_2$ is equal to the sum of the bending moments due to F_1 and F_2 , which act separately.

An immediate consequence of the aforementioned principle is that the final state (stress and strain) of a structure subjected to a system of forces does not depend on the order of application of such forces. This allows, for example, to analyze the case of a complex loading case by breaking it down into simpler cases to study. It is also possible to define and analyze the load states of interest as a combination of previously defined and analyzed load states. Both possibilities are frequently used in Strength of Materials.

3.7.3 The Saint-Venant Principle

The Saint-Venant Principle can be considered the true cornerstone on which Strength of Materials is built, since it establishes that

“in a prismatic element, the stresses that act on a cross-section, sufficiently distant from the points of application of a load system, only depend on the force and moment resulting from the forces located on either side of the considered cross-section.”

This hypothesis implies that local effects (concentrated forces, supports, cross-section variations) only affect a localized area on either side of the sections in that are produced. The precise way in which the loads are applied or the devices used to apply them, as well as the specific construction arrangement of the supporting devices, only affect the stress state of this localized area, of a length approximately equal to the height of the piece. The stress study of the areas affected by local effects is complex and must be done using the Theory of Elasticity.

In Section 3.8 the concept of *internal forces on a cross-section is not defined*, precisely, as the components of the force and moment resulting from the forces located on either side of the cross-section. The calculation of internal forces plays a fundamental role in the Theory of Strength of Materials and Structural analysis, since, according to the Saint-Venant Principle, the stresses in a section only depend on the values of these.

Figure 3.18 shows two examples of cantilevers with a moment applied at their free ends. In the first case, the clamping is achieved by facing the beam against the support and limiting the rotation by means of two separate angles welded to both; the moment at the end is the resultant of a pair of horizontal forces. In the second case, the clamping is achieved by properly embedding the beam into the support; the end moment is the resultant of a couple of forces inclined to the axis of the beam. In both situations, the resultant moment of the acting forces is identical, while the resultant of forces is zero.

Although the realization of the fixing conditions and the applied moments are different, their idealization for calculation purposes is identical, as shown in Figure 3.18. On the other hand, according to the Saint-Venant

Principle, the stresses that appear in the central areas of the pieces depend exclusively on the moment resulting from the acting forces; the extreme zones, on the other hand, are affected by local effects, and the determination of their stress state requires more careful considerations.

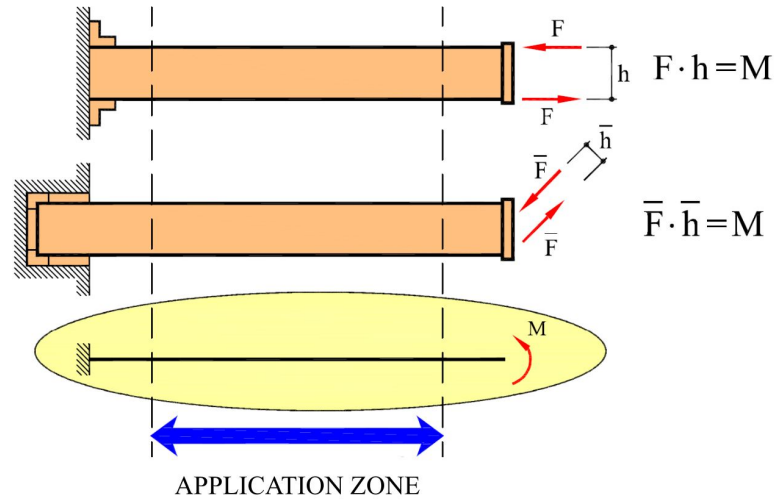


Figure 3.18: Illustration of the Saint-Venant Principle

In engineering practice, the study of the stress state of a structural element is usually done assuming that the Saint-Venant Principle is applicable to all the cross-sections analyzed. The design of the structural elements is usually done based on the results thus obtained. *A posteriori*, the necessary construction provisions are adopted to reduce or adequately resist the local effects in the areas in which the results calculated in this way are not applicable.

3.7.4 Geometrical constraints

In order for the Fundamental Principles to be fulfilled and the results of Strength of Materials to be valid, the structural elements must meet certain conditions:

- *Geometry of the beam axis.* In pieces with a curved axis, the radius of curvature of the axis must be large in relation to the height of the pieces. If this relationship is large enough, the effects of the curvature can be neglected in the analysis of the behavior of the beam slices and use the expressions obtained for pieces with a straight axis.
- *Geometry of the cross-sections:*

- *Dimensions.* The transversal dimensions of the pieces, height and width must be small in relation to their length. This is necessary to fulfil the Saint-Venant Principle and the deformation hypotheses that are commonly used. On the other hand, the geometric characteristics of the cross-sections must ensure that the pieces have the necessary stiffness so that the movements are small. In general, admissible relationships

depth/spam (h/l) are:

$$\begin{aligned} \frac{h}{l} &= \frac{1}{10} \text{ a } \frac{1}{12} && \text{for straight reinforced concrete beams} \\ \frac{h}{l} &= \frac{1}{15} \text{ a } \frac{1}{25} && \text{for straight pre-stressed concrete beams} \\ \frac{h}{l} &= \frac{1}{20} \text{ a } \frac{1}{35} && \text{for straight rolled steel beams} \\ \frac{h}{l} &= \frac{1}{40} \text{ a } \frac{1}{100} && \text{for arches} \end{aligned}$$

- *Variation.* In pieces of variable cross-section, the variation of the transversal dimensions must be gradual and slow. Abrupt variations in the geometric characteristics of the sections produce local effects that invalidate the Saint-Venant Principle

3.8 Internal resultants

In a beam subjected to the action of a system of external loads, any cross-section, S , divides the piece into two parts, (A) and (B) , each located on either side of the considered cross-section, as shown in Figure 3.19.

The system of forces (actions and reactions) acting on the beam must be in static equilibrium. Calling $\mathbf{F}_A$ to the system of forces acting on part (A) of the beam and $\mathbf{F}_B$ to the system of forces that acts on the part (B) , for equilibrium of the beam it must be:

$$\mathbf{F}_A + \mathbf{F}_B = 0 \quad (3.1)$$

Likewise, taking moments about the center of gravity G of section S , it must be:

$$\mathbf{M}_{F_A}^G + \mathbf{M}_{F_B}^G = 0 \quad (3.2)$$

Let us now consider, separately, the equilibrium of the two parts of the piece (A) and (B) . For such an equilibrium to exist, it is necessary that there are forces of interaction that both parts exert on each other, through the surface S .

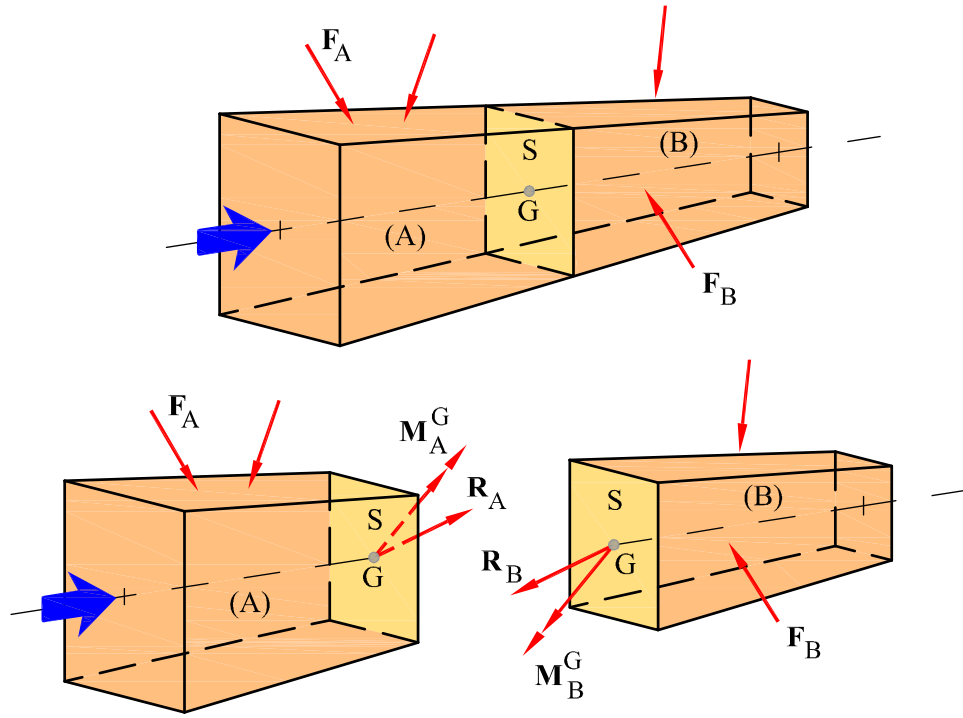


Figure 3.19: Internal forces and moments in a cross-section

Thus, part (B) exerts on part (A) some forces that can be reduced to a resultant $\mathbf{R}_A$ and a resultant moment, about G, $\mathbf{M}_A^G$. Similarly, part (A) exerts on part (B) a resultant force $\mathbf{R}_B$ and a resultant moment $\mathbf{M}_B^G$.

For the equilibrium of part (A) it must be:

$$\begin{aligned}\mathbf{F}_A + \mathbf{R}_A &= 0 \\ \mathbf{M}_{F_A}^G + \mathbf{M}_A^G &= 0\end{aligned}\tag{3.3}$$

For the equilibrium of part (B) it must be:

$$\begin{aligned}\mathbf{F}_B + \mathbf{R}_B &= 0 \\ \mathbf{M}_{F_B}^G + \mathbf{M}_B^G &= 0\end{aligned}\tag{3.4}$$

Adding equations (3.3) y (3.4) and taking into account the global equilibrium (equations (3.1) y (3.2)):

$$\begin{aligned}(\mathbf{F}_A + \mathbf{F}_B) + \mathbf{R}_A + \mathbf{R}_B &= 0 \quad \Rightarrow \quad \mathbf{R}_A = -\mathbf{R}_B \quad (3.5a) \\ (\mathbf{M}_{F_A}^G + \mathbf{M}_{F_B}^G) + \mathbf{M}_A^G + \mathbf{M}_B^G &= 0 \quad \Rightarrow \quad \mathbf{M}_A^G = -\mathbf{M}_B^G \quad (3.5b)\end{aligned}$$

Therefore, the actions that one party exerts on the other are equal and in the opposite direction.

Calling for simplicity

$$\mathbf{R} = \mathbf{R}_A = -\mathbf{R}_B \quad \text{and} \quad \mathbf{M} = \mathbf{M}_A^G = -\mathbf{M}_B^G$$

It will also be

$$\left. \begin{aligned} \mathbf{R} &= -\mathbf{F}_A = \mathbf{F}_B \\ \mathbf{M} &= -\mathbf{M}_{F_A}^G = \mathbf{M}_{F_B}^G \end{aligned} \right\} \quad (3.6)$$

When the resultant force $\mathbf{R}$ and the resulting moment $\mathbf{M}$ are calculated in the form:

$$\mathbf{R} = \mathbf{F}_B \quad \text{and} \quad \mathbf{M} = \mathbf{M}_{F_B}^G \quad (3.7)$$

that is, taking into account the forces that act in “front of” S (part B), the calculation is called “*forward*.”

When $\mathbf{R}$ and $\mathbf{M}$ are calculated in the form

$$\mathbf{R} = -\mathbf{F}_A \quad \text{and} \quad \mathbf{M} = -\mathbf{M}_{F_A}^G \quad (3.8)$$

that is, taking into account the forces that act “behind of” S (part A), the calculation is called “*backward*.” It is clear that both procedures lead to identical results.

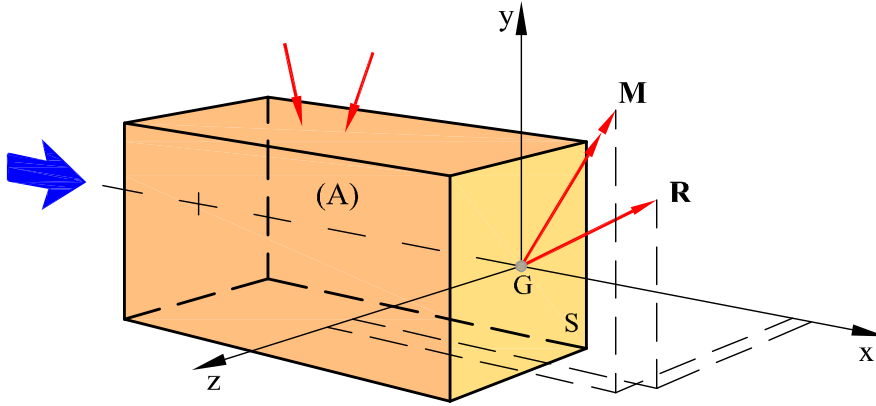


Figure 3.20: Forces and moments resultant in a cross-section

A right-handed Cartesian trihedron xyz , with origin at the center of gravity G of section S , in which the axis x is normal to the section S , oriented from (A) to (B) and tangent to the axis of the beam, and the axes (y, z) are the principal axes of inertia of section S (Figure 3.20) in now defined. Projecting resultant force $\mathbf{R}$ on these axes (Figure 3.21a), the following internal forces are defined:

$$\left. \begin{aligned} &\text{Axial force } N \\ &\text{Shear force } T_y \\ &\text{Shear force } T_z \end{aligned} \right\} \text{ as the projection of } \mathbf{R} \text{ onto axis } \begin{cases} x \\ y \\ z \end{cases}$$

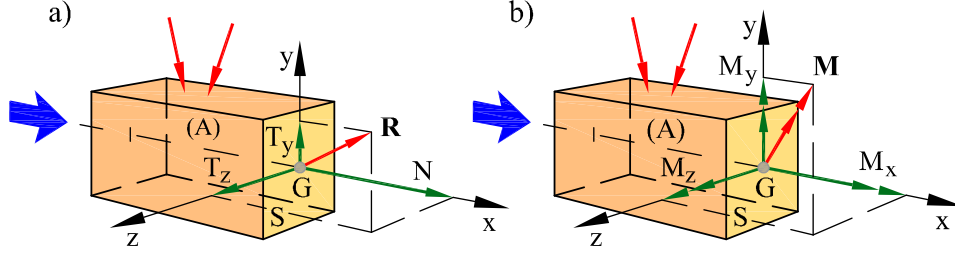


Figure 3.21: Definition of stress in a cross-section

Analogously, projecting the resultant moment $\mathbf{M}$ on the axes (Figure 3.21b), the following internal moments are defined

$$\left. \begin{array}{l} \text{Torsion Moment } M_t \\ \text{Bending Moment } M_y \\ \text{Bending Moment } M_z \end{array} \right\} \text{ as the projection of } \mathbf{M} \text{ onto axis } \left\{ \begin{array}{l} x \\ y \\ z \end{array} \right.$$

3.9 Relationship between internal resultants and stresses

The resultants $\mathbf{R}$ and $\mathbf{M}$, and their respective components, that is, the internal forces on the cross-section, are the static resultants of the *stresses* that act on the section S . As defined in the previous Chapter, these stresses are the interaction forces that one part of the beam exerts on the other, defined per unit area from section S

Thus, as shown in Figure 3.22, on a differential of area dS , of section S , a stress acts $\mathbf{t}$, which can be decomposed into a normal stress, σ_x , according to the axis x , and a tangential stress τ on section S ; in turn the tangential stress can be decomposed, into its Cartesian components along the axes (y, z) , giving the stresses τ_{xy}, τ_{xz} , respectively.

Therefore, $\mathbf{R}$ and $\mathbf{M}$ form are equivalent to the system of differential internal forces $d\mathbf{F} = \mathbf{t} dS$, and the following vector relations must be fulfilled:

$$\mathbf{R} = \int_S \mathbf{t} dS \quad ; \quad \mathbf{M} = \int_S \mathbf{r} \times \mathbf{t} dS \quad (3.9)$$

where $\mathbf{r}$ is the position vector of dS respect to the center of gravity of the cross-section, with components $\mathbf{r}(0, y, z)$.

The above vector relations can be decomposed into the corresponding scalar equations. Thus, the first of the equations (3.9) breaks down into:

$$N = \int_S \sigma_x dS \quad ; \quad T_y = \int_S \tau_{xy} dS \quad ; \quad T_z = \int_S \tau_{xz} dS \quad (3.10)$$

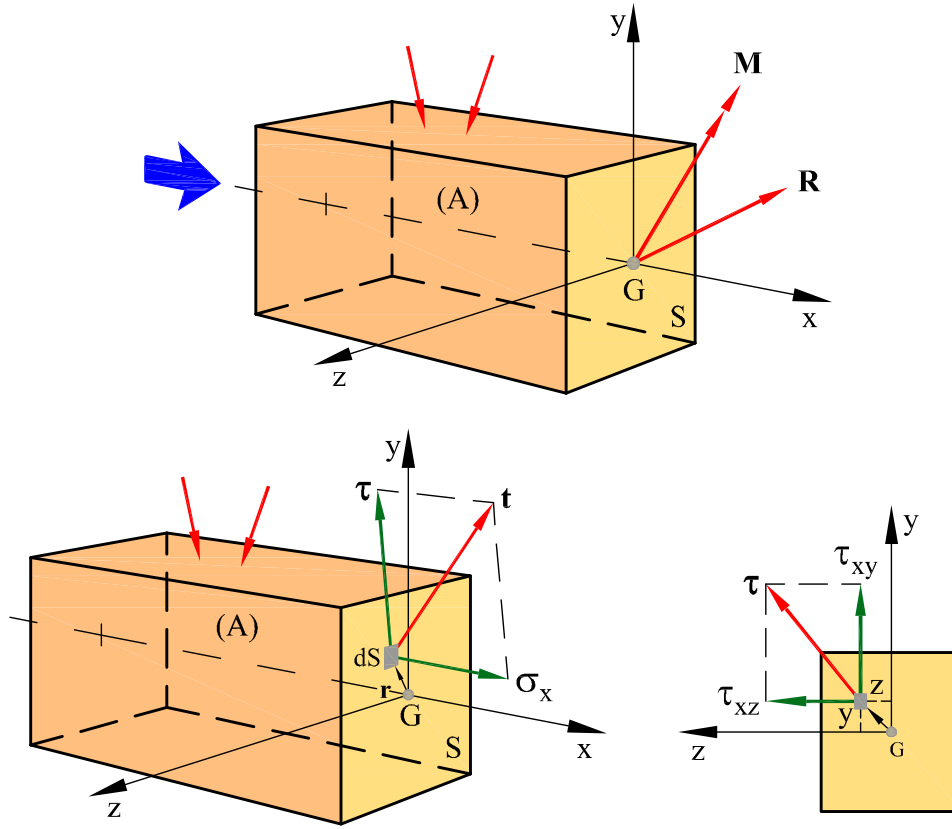


Figure 3.22: Definition of internal forces and stresses in a cross-section

Similarly, the second equation (3.9) is breaks down into:

$$M_t = \int_S (\tau_{xz} y - \tau_{xy} z) dS \quad ; \quad M_y = \int_S \sigma_x z dS \quad ; \quad M_z = - \int_S \sigma_x y dS \quad (3.11)$$

These equations show the relationship between stresses and internal forces acting in a cross-section and will be used often in subsequent Chapters.

Figure 3.23 shows the sign convention used for the stress and moment components. The stresses on the cross-section are positive if they point in the direction of the corresponding positive semiaxes. The moments are positive if, adopting the right-hand criterion, the direction of the moment vectors is positive.

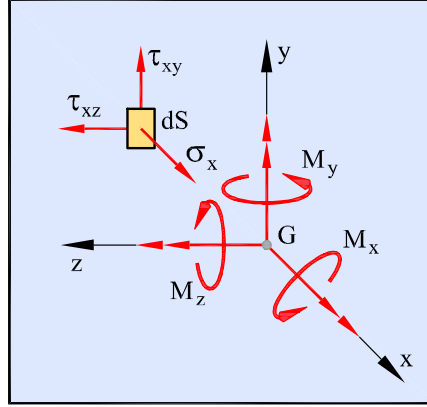


Figure 3.23: Sign convention for stresses and moments.

3.10 Internal resultants in mid-plane elements

Let us consider an element with a mid-plane such as the one shown in Figure 3.24. The mid-plane XY that, it contains the axis of the element. It is the plane of symmetry of the cross-sections and also contains the loads that act on the beam.

The local trihedron xyz of a generic cross-section is defined as follows. The x axis is tangent to the axis of the piece at the center of gravity of section G , and oriented according to the positive direction of the piece axis. The y axis is also in the mid-plane which, due to symmetry, is a principal plane of inertia, and is normal to the x axis, forming with this an angle of $\pi/2$ radians (counter-clockwise). The z axis is normal to the (x, y) axes and such that the trihedron (x, y, z) is right-handed. Thus, the XY plane coincides with the xy plane, and the z axis is perpendicular to it.

As the external forces act only in the XY plane, the resultant $\mathbf{R}$ of the forces that act on one or the other side of the section is contained in such plane and decomposes into an axial force N (along the axis x) and shear force T (along the y axis). Due to symmetry with respect to the mid-plane, T_z is zero. For the same reason, the moment $\mathbf{M}$ resulting from the forces acting on either side of the cross-section is perpendicular to the XY plane and is the bending moment M is its unique component along the z axis. The other two components M_t and M_y are zero due to symmetry with respect to the mid-plane.

When referring to the internal resultants in mid-plane parts, the following sign convention, shown in Figure 3.25 is adopted:

- The axial force N is positive if it is tensile (according to the positive direction of the axis x on the front face) and negative if it is compressive.

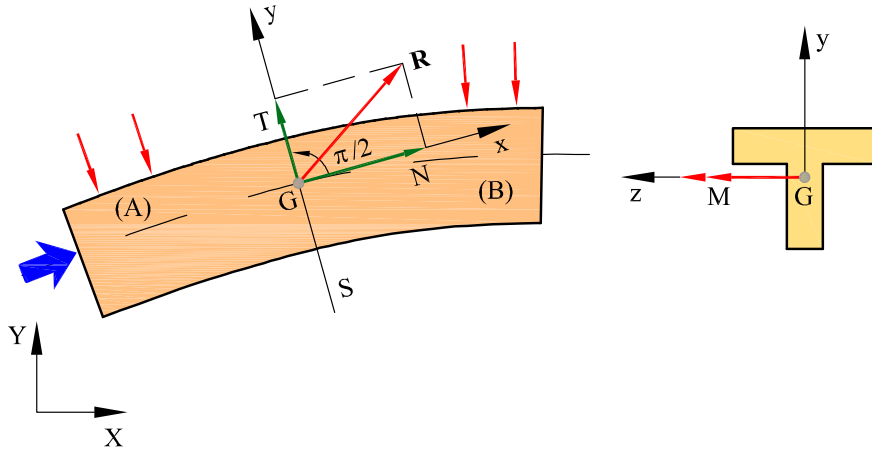


Figure 3.24: Internal forces and moments in mid-plane elements

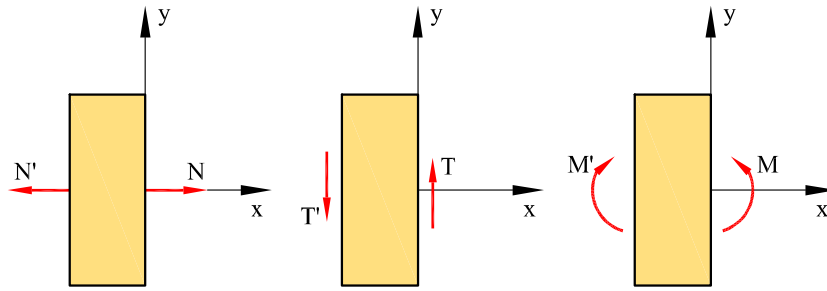


Figure 3.25: Sign convention for internal forces and moments in mid-plane elements

- Shear force T is positive if it tends to rotate the slice counter-clockwise (according to the positive sense of the y axis on the front face); otherwise, it is negative.
- The bending moment M is positive if it is counter-clockwise on the front face (according to the positive direction of the axis z on the front face); otherwise, it is negative.

3.11 Equilibrium equations in straight elements

Let us consider the equilibrium of a differential slice of a straight *mid-plane* piece, on which forces distributed per unit length act, with components (p_x, p_y) according to the coordinate axes, applied to the axis of the slice. As shown in Figure 3.26, it is:

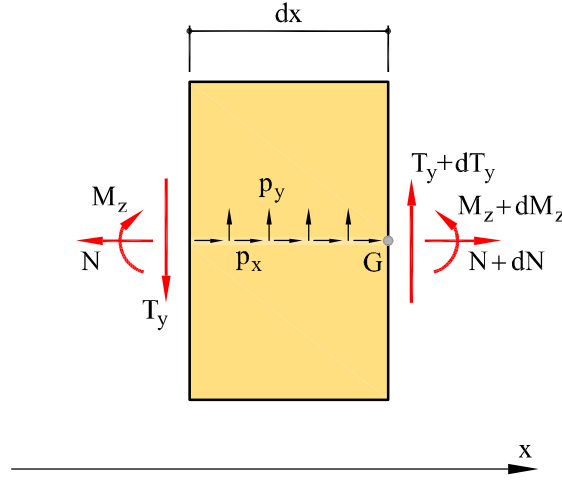


Figure 3.26: Actions on a slice of a straight element

a) For the equilibrium of forces along the axis x :

$$-N + p_x dx + (N + dN) = 0 \quad \Rightarrow \quad p_x = -\frac{dN}{dx} \quad (3.12)$$

b) For the equilibrium of forces along the axis y :

$$-T_y + p_y dx + (T_y + dT_y) = 0 \quad \Rightarrow \quad p_y = -\frac{dT_y}{dx} \quad (3.13)$$

c) For the equilibrium of moments about the center of gravity of the front face:

$$-M_z + T_y dx + (M_z + dM_z) - p_y \frac{dx^2}{2} = 0 \quad \Rightarrow \quad T_y = -\frac{dM_z}{dx} \quad (3.14)$$

where the second-order term due to the distributed forces has been neglected.

These equations establish differential relationships between the load and the internal resultants of a mid-plane straight beam. These relationships can be directly integrated in simple cases to calculate the internal resultants. In more complex cases, these relationships help to calculate or draw these laws.

The generalization of the differential equations of equilibrium of the slice for a general beam with a straight axis (not mid-plane), subjected to forces distributed per unit length of components (p_x, p_y, p_z) according to the local reference trihedron can be obtained analogously, and they are:

$$\begin{aligned} p_x &= -\frac{dN}{dx} & p_y &= -\frac{dT_y}{dx} & p_z &= -\frac{dT_z}{dx} \\ T_y &= -\frac{dM_z}{dx} & T_z &= +\frac{dM_y}{dx} \end{aligned} \quad (3.15)$$

Example 3.11.1

Calculate the internal resultants in the middle cross-section of the beam subjected to the action of the load p uniformly distributed and the forces F and V indicated (Figure 3.27).

Data: $L = 10\text{ m}$, $h = 0.80\text{ m}$, $b = 0.35\text{ m}$, $p = 3\text{ kN/m}$, $F = 30\text{ kN}$, $V = 15\text{ kN}$.

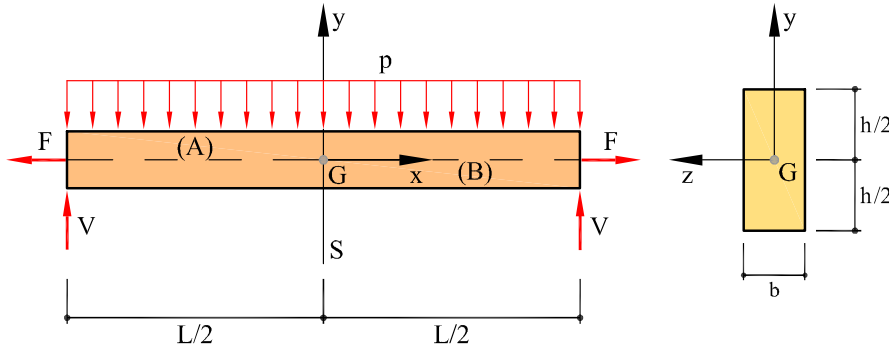


Figure 3.27: Beam of Example 3.11.1

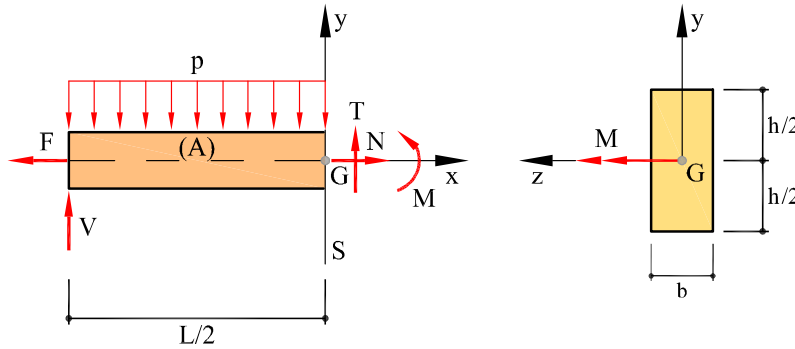


Figure 3.28: Equilibrium of forces and moments

Let N , T and M be the axial and shear forces and bending moment in the middle section of the beam, respectively. This section divides the piece into two parts: (A) on the left and (B) on the right (Figure 3.28). For the equilibrium of (A):

$$\begin{aligned}\sum F_x &= -F + N = 0 \\ \sum F_y &= T + V - \frac{pL}{2} = 0 \\ \sum M_G &= M - V\frac{L}{2} + \frac{pL^2}{8} = 0\end{aligned}$$

Solving:

$$N = 30 \text{ kN} \quad T = -15 + \frac{3 \cdot 10}{2} = 0 \quad M = 15 \cdot 5 - \frac{3 \cdot 10^2}{8} = 37.5 \text{ kN} \cdot \text{m}$$

which are the internal forces in the middle section of the beam.

Example 3.11.2

Calculate the internal resultants in the middle section of the piece subjected to the loads indicated (Figure 3.29).

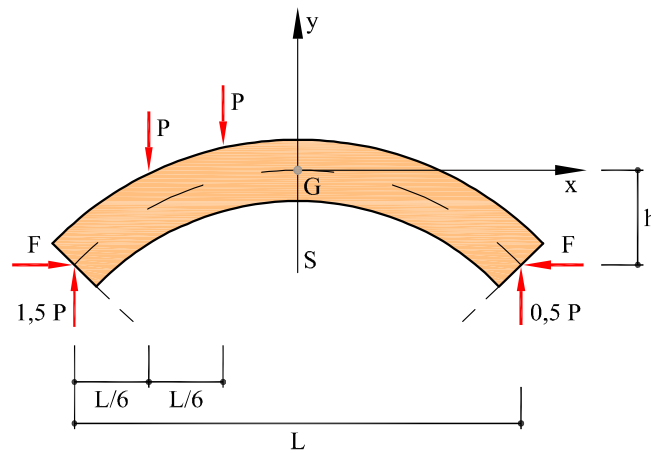


Figure 3.29: Beam of Example 3.11.2

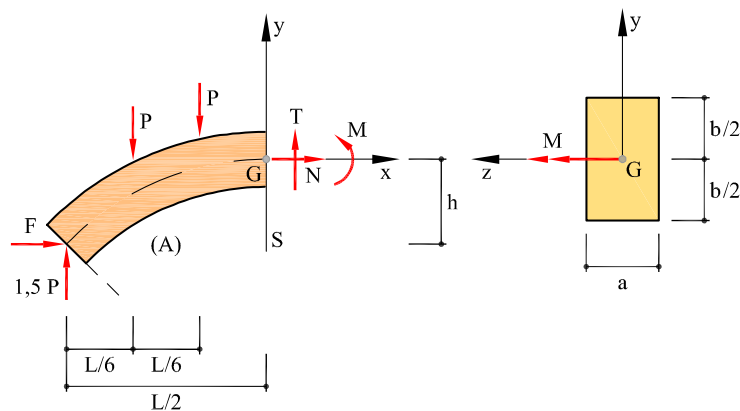


Figure 3.30: Equilibrium of forces and moments

Let N , T and M be the axial and shear forces and bending moment in

the middle section of the beam, respectively. This section divides the piece into two parts: (A) on the left and (B) on the right (Figure 3.30).

For the equilibrium of part (A):

$$\begin{aligned}\sum F_x &= F + N = 0 \\ \sum F_y &= T + 1.5P - 2P = 0 \\ \sum M_G &= M - 1.5P\frac{L}{2} + \frac{2}{6}PL + \frac{1}{6}PL + Fh = 0\end{aligned}$$

Solving:

$$N = -F \qquad T = 0.5P \qquad M = 0.25PL - Fh$$

which are the internal forces in the middle section of the piece.

3.12 Statically indeterminacy

3.12.1 Static equilibrium and solution of structures

The forces (actions and reactions) that act on a structure must be in static equilibrium. This means that they must form a force system with zero resultant and zero resultant moment; therefore, they must satisfy the equations known as *equations of statics*, which, in vector form, can be written as:

$$\sum_i \mathbf{F}_i = \mathbf{0} \quad (3.16a)$$

$$\sum_i \mathbf{M}_i^O = \mathbf{0} \quad (3.16b)$$

where $\mathbf{F}_i$ represents each of the forces acting on the structure, $\mathbf{M}_i^O$ represents the moment of each of the forces with respect to an arbitrary point O and the symbol $\sum_i$ represents the sum over all forces i .

In the case of planar structures loaded in their plane, the previous vector equations are reduced to three scalar equations of the form:

$$\sum_i (F_x)_i = 0 \quad ; \quad \sum_i (F_y)_i = 0 \quad ; \quad \sum_i (M_z)_i^O = 0 \quad (3.17)$$

where the axes x and y are on the plane of the structure and the axis z is perpendicular to them.

The condition of static equilibrium, as expressed by Eq. (3.17) is *necessary* for equilibrium, but it is not *sufficient*. This is because the equilibrium condition and the corresponding equations that express it mathematically must apply not only to the structure considered as a whole, but must be satisfied also for each integral part of it, provided that the forces and moments

that the rest of the structure exerts on the considered part are explicitly considered. In particular, the members that form a bar structure must be in equilibrium, provided that the forces and moments at the end of the bar that the structure exerts on the members are considered. Similarly, the nodes of the structure must be in equilibrium under the action of the forces and moments that act at the ends of the bars that concur in them.

In Structural Mechanics, *solving* a structure calculating the value of the internal resultants that act on each of the cross-sections of all the pieces that form the structure. When this calculation can be performed entirely using only the *equations of statics*, the structure is called statically determined. Otherwise, it is called *statically indeterminate*.

3.12.2 Truss structures

Truss structures of straight elements with loads applied at the nodes work exclusively under axial force, and these axial forces are constant. Such structures are *statically determined* when they can be solved only with the equations of statics.

The number of external reactions (nr) plus the number of axial forces to be determined, one per bar (nb), must be equal to the number of equations of statics that can be established. In planar trusses, two equilibrium equations can be considered per node ($2nn$); therefore, for the structure to be in equilibrium, the following must be fulfilled:

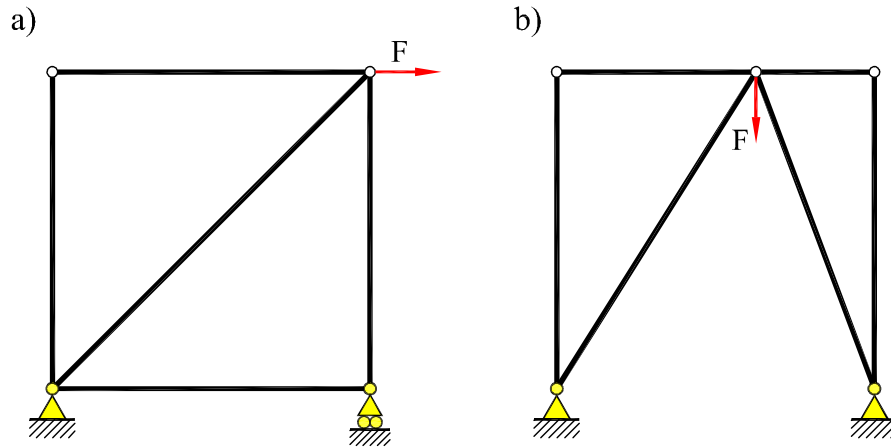


Figure 3.31: Statically determined truss structures

$$nr + nb = 2nn \quad (3.18)$$

Figure 3.31 shows two examples of statically determined truss structures: in structure (a) it can be seen that the number of unknown reactions is

$nr = 3$ and the axial forces to be determined (one per bar) are $nb = 5$, therefore, the number of unknowns of the structure is equal to 8. Since the number of static equations that can be considered is $2nn = 2 \cdot 4 = 8$, the structure can be solved only with the equations of statics, therefore it is statically determined. In structure (b) the number of unknown reactions $nr = 4$ and the number of unknown axial forces is $nb = 6$, the number of equilibrium equations is equal to $2nn = 2 \cdot 5 = 10$, the structure also in this case can be solved only with the conditions of the static and is statically determined.

Trusses are *statically indeterminate* when the equilibrium conditions are not sufficient to solve the structure, that is:

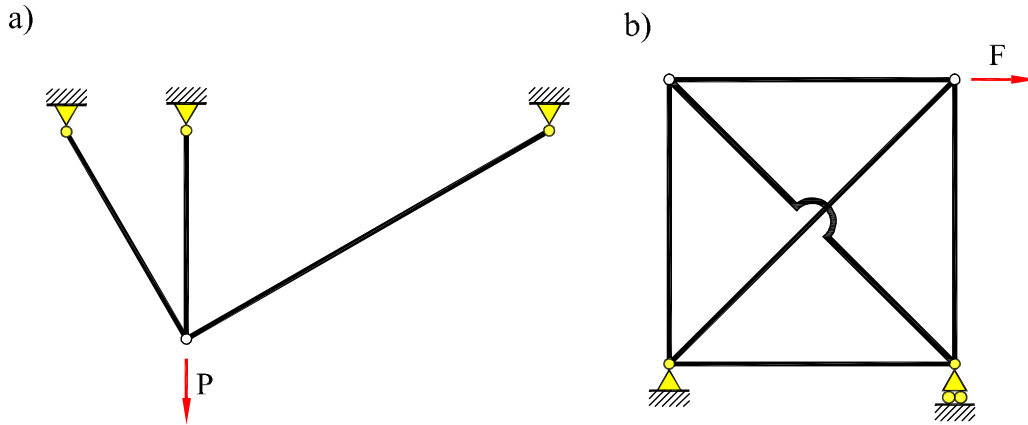


Figure 3.32: Statically indeterminate truss structures

$$nr + nb > 2nn \quad (3.19)$$

In Figure 3.32 two examples of statically indeterminate truss structures can be seen. Thus, in structure (a) the number of reactions is $nr = 6$, the number of bars is $nb = 3$ and the number of equations of statics is $2nn = 2 \cdot 4 = 8$, therefore, the articulated structure (a) is statically indeterminate of degree 1.

In structure (b) the number of reactions is $nr = 3$, the number of unknown axial forces is $nb = 6$ and the number of equilibrium equations is $2nn = 2 \cdot 4 = 8$. Also in this case, the unknowns are greater than the static conditions and the truss structure (b) is statically indeterminate of degree 1.

3.12.3 Frames

Frame structures are formed by prismatic pieces joined by rigid nodes. In these structures, the degree of statically indeterminate is defined regardless

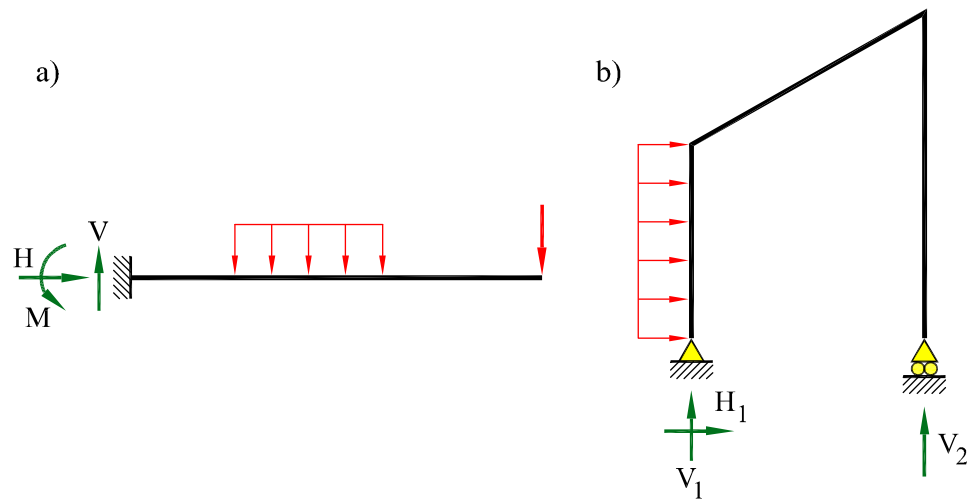


Figure 3.33: Statically determined frame structures

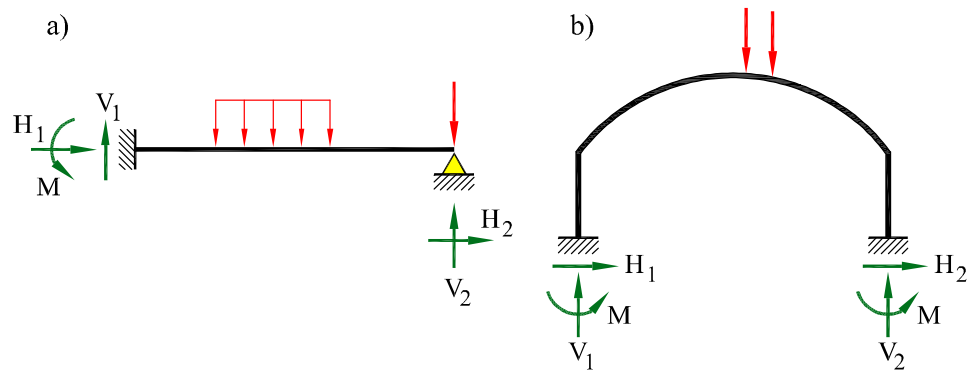


Figure 3.34: Statically indeterminate frame structures

of whether its origin is the indeterminacy of the reactions in the supports or in internal links:

$$h = n - e \quad (3.20)$$

where n is the number of reactions (external or internal) that must be known in order to determine the internal forces acting in any cross-section of the structure, and e is the number of equations of statics.

Once the value of h is calculated, for

- $h = 0$ the structure is statically determined
- $h > 0$ the structure is statically indeterminate of degree h ,
- $h < 0$ the structure is a mechanism of degree h .

Figure 3.33 shows two examples of statically determined frame: in structure (a) it can be seen that the number of unknown reactions is $n = 3$ and the number of equilibrium equations is $e = 3$, therefore, the degree of statical indeterminacy $h = n - e = 3 - 3 = 0$ the structure can be solved only with the equations of the static equilibrium, therefore it is statically determined. In structure (b) the number of unknown reactions $n = 3$ and the number of equilibrium equations is $e = 3$, therefore, the degree of statical indeterminacy is $h = n - e = 3 - 3 = 0$, the structure also in this case can be resolved only with the conditions of the static and it is statically determined.

In Figure 3.34 one can see two examples of statically indeterminate frame structures. In structure (a) the number of reactions is $n = 5$ and the number of equations of statics is $e = 3$; therefore, the frame structure (a) is statically indeterminate of degree $h = n - e = 5 - 3 = 2$.

In structure (b) the number of reactions is $n = 6$, and the number of equilibrium equations is $e = 3$. Also in this case, the unknowns are greater than the static conditions and the frame structure (b) is statically indeterminate of degree $h = 3$.

Chapter 4

Internal resultants

4.1 Introduction

The Saint-Venant Principle (Section 3.7.3) states that, in a prismatic beam, the stresses (and strains) acting on a cross-section sufficiently distant from the points of application of a system of loads depend solely on the force $\mathbf{R}$ and the moment $\mathbf{M}$ resulting from the forces acting on either side of the considered section.

The components of the resulting force $\mathbf{R}$ and moment $\mathbf{M}$ about the reference system of the cross-section define the internal forces and moments acting on the cross-section (Section 3.8).

Therefore, according to the Saint-Venant Principle, stresses (and strains) in a cross-section depend only on the internal forces (and moments) acting on it. As a result, the calculation of these is the starting point for every Strength of Materials computation.

4.2 Internal forces and moments

In statically determined structures, the resultant force and moment, $\mathbf{R}$ and $\mathbf{M}$, on a cross-section S of the structure may be calculated using just the equations of static equilibrium. In accordance with Section 3.8, the internal forces and moments on the cross-section are defined as the projections of these on a right-handed Cartesian trihedron xyz , with origin at the center of gravity G of the cross-section S , in which the x -axis is normal to the section S and tangent to the central axis of the beam and the axes (y, z) are the principal axes of inertia of the section S .

Consequently, by projecting the resultant force $\mathbf{R}$ onto the (x, y, z) axis, the axial force N , shear force T_y , and shear force T_z are defined, respectively.

Similarly, by projecting the resultant moment $\mathbf{M}$ onto the axes (x, y, z) defines the torsional moment M_t , bending moment M_y , and bending moment M_z .

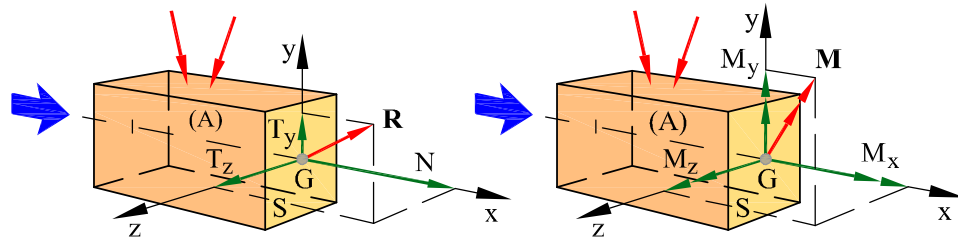


Figure 4.1: Definition of internal forces and moments of in a cross-section

The *internal forces diagrams* are the graphical representation of the variation of these forces and moments along the beam. This representation allows the identification of the cross-section of the beam where internal resultants reach their maximum value. This data is required for designing the beam or, once the beam has been designed, for ensuring that the strength criteria are fulfilled.

Internal resultants diagrams in a statically determined structure are derived as follows:

1. Determination of the support reactions.
2. Determination of the internal resultants variation along each beam.
3. Determination of the maximum values of the internal resultant, at beam and structure levels.

The differential relationships between the applied load and the internal resultants obtained in Section 3.11 are very useful in determining the shape of the internal resultant diagrams in each bar.

4.3 Internal forces in truss structures

Truss structures consist of prismatic bars connected by hinges (Section 3.12.2). It is possible to calculate reaction forces on the supports of statically determined truss structures by applying the equations of statics. In addition, when loads are applied only at the nodes, the internal forces reduce to a constant axial N force in each bar. As a result, the following steps are followed to establish the internal forces diagram:

1. Determination of support reactions.
2. Determination of the axial load N for each bar.

Following this, several examples of the calculation of the axial force diagrams in plane truss structures are developed. The sign convention used to

define the equilibrium of the slice for straight prismatic bar (Section 3.10) must be used.

Example 4.3.1

The truss ABC shown in Figure 4.2 is composed by two bars of the same material. Determine the axial force in the bars for a load P acting at point C .

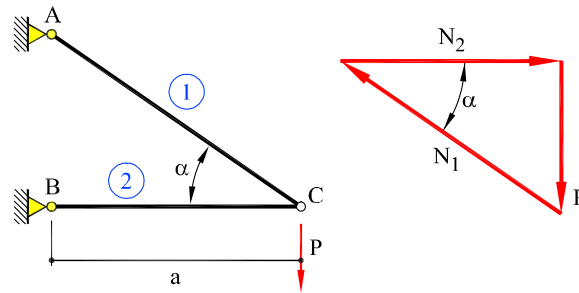


Figure 4.2: Structure of Example 4.3.1 and force equilibrium in C

The structure is statically determined; thus, the axial forces in the bars 1 and 2 are directly determined by imposing equilibrium of forces acting at node C :

$$\begin{aligned} N_1 &= \frac{P}{\sin \alpha} = 80 \text{ kN} && \text{(tensile)} \\ N_2 &= -\frac{P}{\tan \alpha} = -69.3 \text{ kN} && \text{(compression)} \end{aligned}$$

Example 4.3.2

The truss in Figure 4.3 is composed of three identical bars. Determine the axial forces in the bars for the indicated force F acting at C .

Data: $a = 5 \text{ m}$, $F = 20 \text{ kN}$.

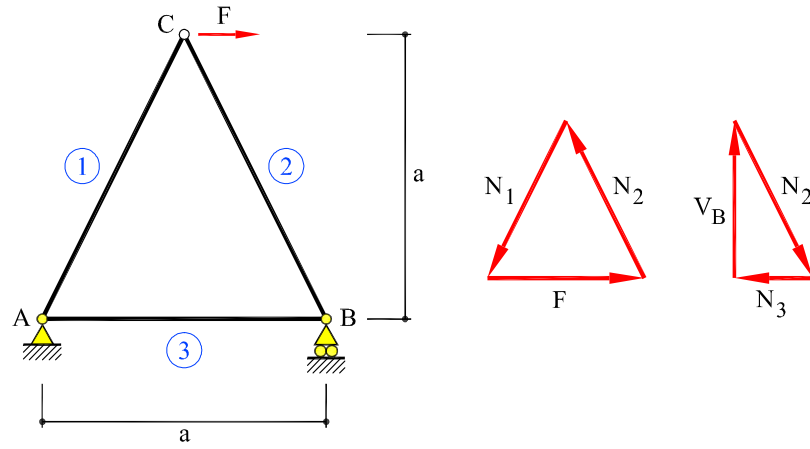


Figure 4.3: Structure of Example 4.3.2 and force equilibrium at nodes C and B

The axial forces can be calculated by taking into account the equilibrium of forces at the nodes.

Considering equilibrium of forces at joint C , the axial forces in bars 1 and 2 can be calculated, and by considering equilibrium of forces at node B , the axial force in bar 3 follows (Figure 4.3):

$$N_1 = 22.4 \text{ kN} \quad N_2 = -22.4 \text{ kN} \quad N_3 = 10 \text{ kN}$$

According to the signs of the calculated axial forces, bars 1 and 3 are under tension while bar 2 is compressed.

Example 4.3.3

The truss showed in Figure 4.4 is composed of five bars of the same material. Determine the corresponding axial forces for a horizontal force F acting at point B .

Data: $a = 4 \text{ m}$, $F = 10 \text{ kN}$.

The structure is statically determined. Force equilibrium is considered to determine the reactions at supports C and D (the directions are shown in Figure 4.4):

$$H_C = F \quad (\leftarrow) \quad V_C = F \quad (\downarrow) \quad V_D = F \quad (\uparrow)$$

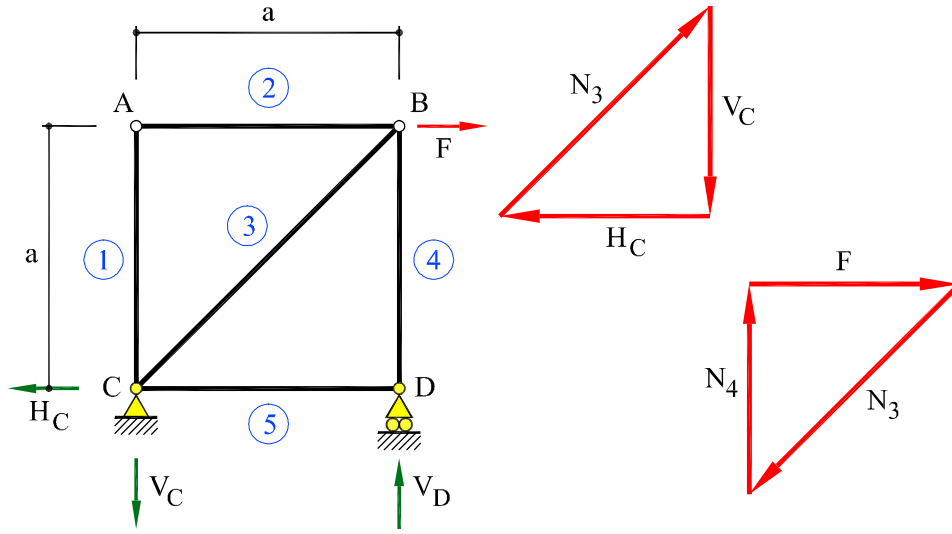


Figure 4.4: Structure of Example 4.3.3 and force equilibrium at nodes C and B

The axial forces can be calculated by taking into account the equilibrium of forces at nodes:

$$\text{node } D \Rightarrow N_4 = -F = 10 \text{ kN} \quad N_5 = 0$$

$$\text{node } C \Rightarrow N_1 = 0 \quad N_3 = \sqrt{2}F = 14.14 \text{ kN}$$

$$\text{node } A \Rightarrow N_2 = 0$$

For the considered load, it is observed that bars 1, 2, and 5 do not carry any load, while bar 3 is under tension and bar 4 is compressed.

Example 4.3.4

The truss showed in Figure 4.5 is composed of five bars of the same material. Determine the corresponding axial forces for a force P acting at node D . Data: $a = 3 \text{ m}$, $P = 40 \text{ kN}$. The structure is statically determined. Force equilibrium is considered to determine the reactions at supports A and B (the directions are shown in Figure 4.5):

$$V_A = P = 40 \text{ kN} \quad (\downarrow) \quad V_B = 2P = 80 \text{ kN} \quad (\uparrow)$$

The axial forces are calculated by taking into account the equilibrium of

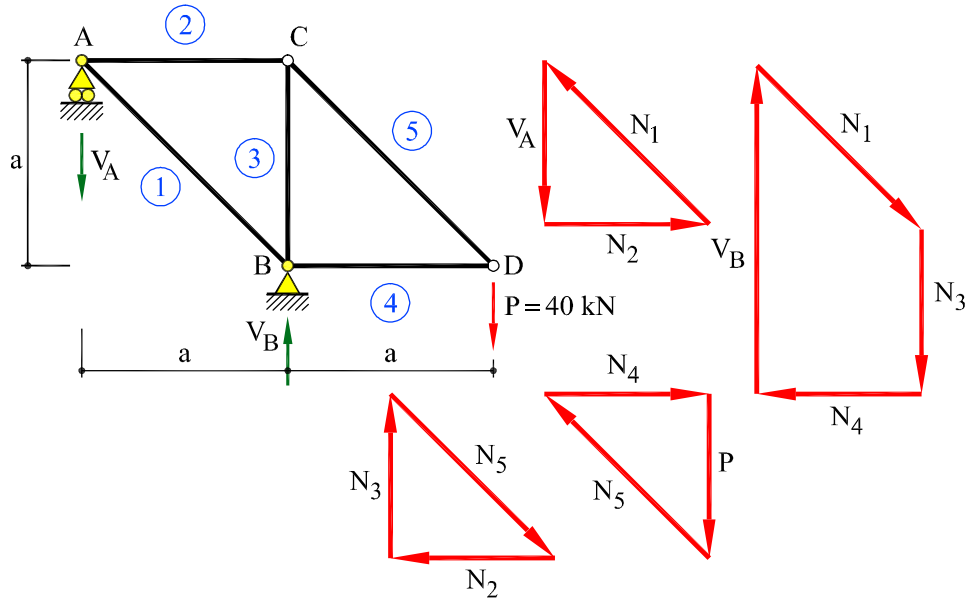


Figure 4.5: Structure of Example 4.3.4 and force equilibrium at nodes

forces at nodes (Figure 4.5):

$$N_1 = -\sqrt{2}P = -56.57 \text{ kN} \quad N_2 = N_4 = P = 40 \text{ kN}$$

$$N_3 = -P = -40 \text{ kN} \quad N_5 = \sqrt{2}P = 56.57 \text{ kN}$$

As can be seen, bars 1 and 3 are in compression and bars 2, 4, and 5 are in tension in the example structure under the specified load.

Example 4.3.5

The truss showed in Figure 4.6 is composed of seven bars of the same material. Determine the corresponding axial forces for the given load system. Data: $a = 5 \text{ m}$, $F = 50 \text{ kN}$.

The structure is statically determined, so the reactions at supports A and E can be determined via force equilibrium (Figure 4.6):

$$V_A = V_E = 1,5F = 75 \text{ kN} \quad (\uparrow)$$

Nodes A and E , as well as nodes B and D , are symmetrical for the considered load state. The equilibrium of forces at nodes is used to calculate the axial

forces (Figure 4.6):

$$N_1 = N_7 = -1.678F = -83.9 \text{ kN} \quad N_2 = N_6 = 0.75F = 37.5 \text{ kN}$$

$$N_3 = N_5 = 0.568F = 28.4 \text{ kN} \quad N_4 = -F = -50 \text{ kN}$$

For the considered load, bars 1, 4, and 7 are compressed while bars 2, 3, 5, and 6 are under tension.

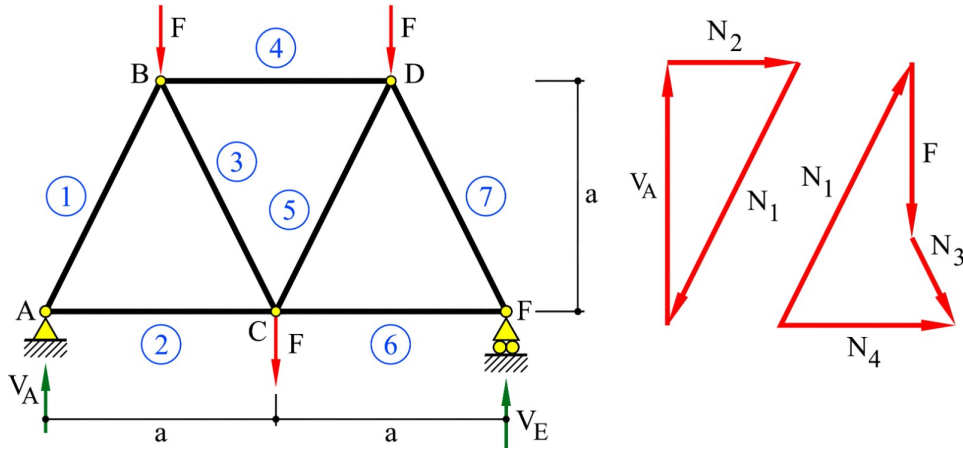


Figure 4.6: Structure of Example 4.3.5 and force equilibrium at joints

Example 4.3.6

The truss showed in Figure 4.7 is composed of seven bars of same material. Determine the axial forces in bars for the given force F .

Data: $a = 5 \text{ m}$, $F = 50 \text{ kN}$.

The structure is statically determined, and the reactions at supports A and B can be determined by force equilibrium (Figure 4.6):

$$H_A = 2F = 100 \text{ kN} (\rightarrow) \quad V_A = F = 50 \text{ kN} (\uparrow) \quad H_B = 2F = 100 \text{ kN} (\leftarrow)$$

The axial forces can be calculated by taking into account the equilibrium of forces at nodes:

$$\text{Node } A \quad N_1 = -50 \text{ kN} \quad N_3 = 0 \quad N_4 = -100 \text{ kN}$$

$$\text{Node } B \quad N_1 = -50 \text{ kN} \quad N_2 = 112 \text{ kN}$$

$$\text{Node } C \quad N_4 = -100 \text{ kN} \quad N_5 = 0 \quad N_7 = -100 \text{ kN}$$

$$\text{Node } E \quad N_6 = 112 \text{ kN} \quad N_7 = -100 \text{ kN}$$

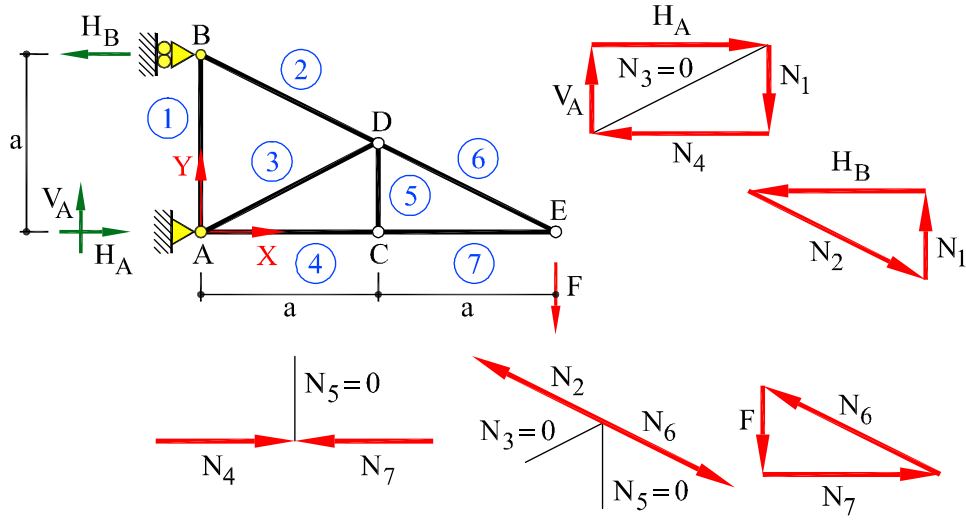


Figure 4.7: Structure of Example 4.3.6 and force equilibrium at joints

For the considered load, bars 3 and 5 do not work, bars 1, 4 and 7 are compressed, and bars 2 and 6 are under tension.

4.4 Internal resultants in midplane structures

Midplane structures are defined by midplane members connected by rigid nodes and so that the mid-plane of all members is the same (Section 3.10). Here, the local reference axes (x, y, z) can be chosen so that the midplane of the member contains the x and y axes, since it is a symmetry plane. Under these conditions, the components $\mathbf{R}$ and $\mathbf{M}$ are reduced to the axial force N , shear force $T = T_y$, and bending moment $M = M_z$, respectively (Figure 4.8).

Consequently, the following steps are taken to determine the internal forces diagram:

1. Calculate the external reactions.
2. Determine the variation of the internal forces on each beam.
3. Identify and calculate the maximum values of the internal forces at the beam and structure levels.

The differential relationships between the applied loads and the internal resultants obtained in Section 3.11 are extremely helpful when the shape of the internal resultants diagrams in each beam is defined.

Following this, some examples of plane straight beam and mid-plane structures are developed. The adopted sign convention for defining the

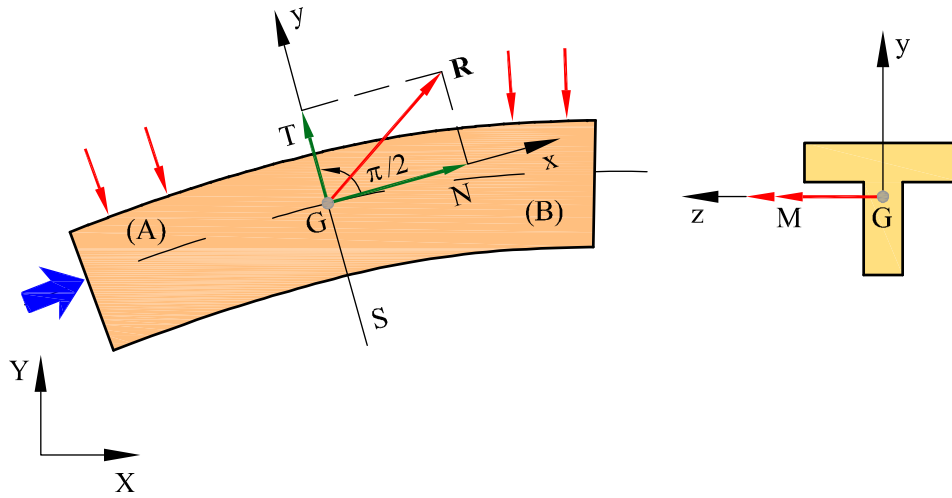


Figure 4.8: Internal forces at midplane structure

equilibrium of the differential beam element of the straight prismatic beam (Section 3.10) is kept.

Example 4.4.1

Calculate and draw the internal force diagrams of the beam ABC in Figure 4.9 subjected to uniformly distributed load p along segment BC .

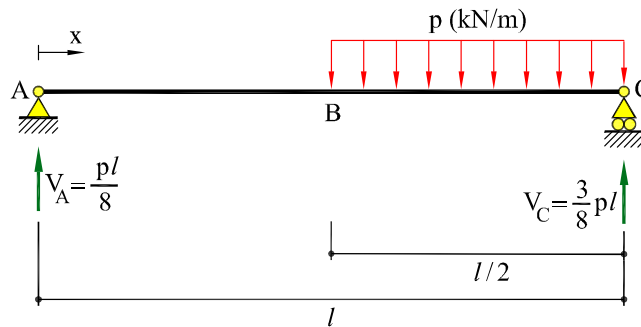


Figure 4.9: Structure of Example 4.4.1

External reactions

Taking the static equilibrium conditions into account (sum of vertical forces equal to zero and sum of moments about support A equal to zero), it is:

$$V_A + V_C = p \frac{l}{2} \quad ; \quad V_C l = p \frac{l}{2} \frac{3l}{4}$$

Therefore, reactions at supports A and C are:

$$V_A = \frac{1}{8} pl \quad (\uparrow) \quad ; \quad V_C = \frac{3}{8} pl \quad (\uparrow)$$

Bending moment diagram

The bending moment diagram is obtained by calculating, for a given cross-section, the moment resulting from the forces applied to either side of the cross-section. In the AB segment, a cross-section set at x distance from support A and calculated backwards from the cross-section yields:

$$M(x) = V_A x = \frac{1}{8} pl x$$

That is, a linear diagram along segment AB , with a maximum moment in section $x = l/2$, and $M_B = pl^2/16$. In segment BC , for a cross-section set at a distance of x from support A , it is:

$$M(x) = V_A x - \frac{p}{2} \left(x - \frac{l}{2} \right)^2 = \frac{pl}{8} \left[-\frac{4x^2}{l} + 5x - l \right]$$

In this instance, the bending moment diagram is parabolic, and the distance at which the maximum value occurs is determined by differentiating with respect to the variable x and equating the derivative to zero.

$$\frac{\partial M}{\partial x} = \frac{pl}{8} \left[-\frac{8x}{l} + 5 \right] = 0 \quad \Rightarrow \quad x = \frac{5l}{8}$$

Resulting in:

$$M_{\max} = M\left(x = \frac{5l}{8}\right) = \frac{9}{128} pl^2$$

Note that the bending moment diagram and its derivative are continuous at point B .

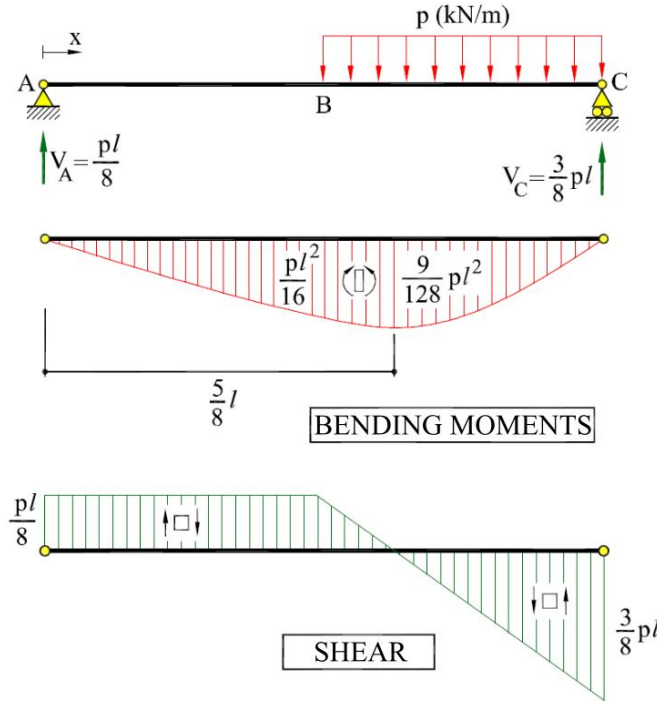


Figure 4.10: Internal forces diagrams for Example 4.4.1

Shear force diagram

Force resultant $\mathbf{R}$ in any cross-section of the beam is calculated by adding all applied forces acting on either side of the cross-section. In this case, the only component of the resultant is the shear force T_y ; therefore, the axial forces are zero over the entire beam.

Along the beam axis, by calculating backwards from the cross-section, the shear diagram is constant in the AB region, and its value is equal to the reaction at support V_A .

In a cross-section set at a distance x from support A and in the BC region, the projection of $\mathbf{R}$ on the cross-section is equal to the sum of the reaction V_A and the sum of the distributed load between point B and the considered cross-section; therefore, the shear diagram along the beam axis is as follows:

$$\begin{aligned} \text{Region } AB : \quad T_y &= V_A = -\frac{1}{8}pl \\ \text{Region } BC : \quad T_y &= V_A + p\left(x - \frac{l}{2}\right) = -\frac{5}{8}pl + px \end{aligned}$$

External reactions, bending moment and shear diagrams are shown in

Figure 4.10, respectively. A constant shear distribution corresponds to a linear bending moment distribution, while a linear shear distribution corresponds to a parabolic bending moment distribution.

Example 4.4.2

The beam $ABCDE$ shown in Figure 4.11 is subjected to a point load P of 200 kN in point B and a uniformly distributed load p of 20 kN/m along the CE span. Calculate and draw the internal resultant diagrams.

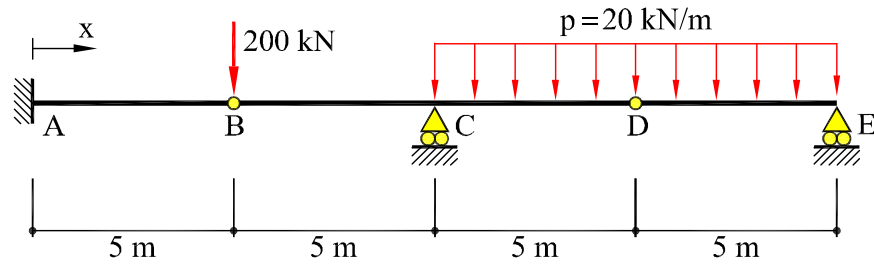


Figure 4.11: Structure of Example 4.4.2

External Reactions

The following equations can be written (see Figure 4.12) by considering the equilibrium equations and conditions (sum of vertical forces equal to zero, sum of bending moments about point A equal to zero, bending moment at joints B and D equal to zero) and expressed in kN and m:

$$\begin{aligned} V_A + V_C + V_E &= 400 \\ M_A + 10V_C + 20V_E &= 4000 \\ M_A - 5V_A &= 0 \\ 5V_E - 250 &= 0 \end{aligned}$$

By solving these, the values of the reactions at the fixed support A as well as at the supports C and E are determined:

$$\begin{aligned} V_A &= 100 \text{ kN } (\uparrow) & ; & & V_C &= 250 \text{ kN } (\uparrow) \\ M_A &= 500 \text{ kN} \cdot \text{m } (\odot) & ; & & V_E &= 50 \text{ kN } (\uparrow) \end{aligned}$$

Bending Moments diagram

In region AB , slicing the structure at a distance x from support A and calculating backwards from the cross-section, the bending moment along the axis of the beam yields the following result:

$$M(x) = -M_A + V_A x = -500 + 100x$$

which corresponds to a linear distribution. For $x = 5$ m, the bending moment is equal $M_B = 0$, as is expected for a joint. Regarding the BC region, the bending moment diagram is:

$$M(x) = -M_A + V_A x - 200(x - 5) = 500 - 100x$$

which is also a linear function. For $x = 10$ m, the moment at support C is $M_C = -500$ kN·m. In the CE region, the bending moment is defined as:

$$\begin{aligned} M(x) &= -M_A + V_A x - 200(x - 5) + V_C (x - 10) - \frac{p}{2}(x - 10)^2 \\ &= -3000 + 350x - 10x^2 \end{aligned}$$

which is a parabolic function with its maximum value located at a distance $x_{\max}$ from the support A , as determined by setting to zero the derivative of the bending moment expression with respect to the variable x . In this example, the maximum moment yields at a distance equal to $x_{\max} = 17.5$ m and $M_{\max}(x = x_{\max}) = 62.5$ kN·m is the corresponding maximum bending moment value.

Shear force diagram

The shear force diagram of the structure is obtained by calculating the resultant forces on either side of the cross-section and determining the component parallel to it. Calculating backwards from the cross-section along the region AB , the resultant is constant and equal to V_A :

$$T_y = V_A = -100 \text{ kN}$$

In region BC , the resultant of forces on either side of the cross-section is also constant. Lastly, in the CE region, using the same methodology, the corresponding shear force is linear:

$$T_y = 100 - 250 + 20(x - 10) = -350 + 20x$$

Note that in each cross-section, the shear force coincides with the derivative of the bending moment with respect to the variable x (sign changed), as it is required by the equilibrium of the differential beam element.

Figure 4.12 shows support reactions and internal resultant diagrams of the beam $ABCDE$.

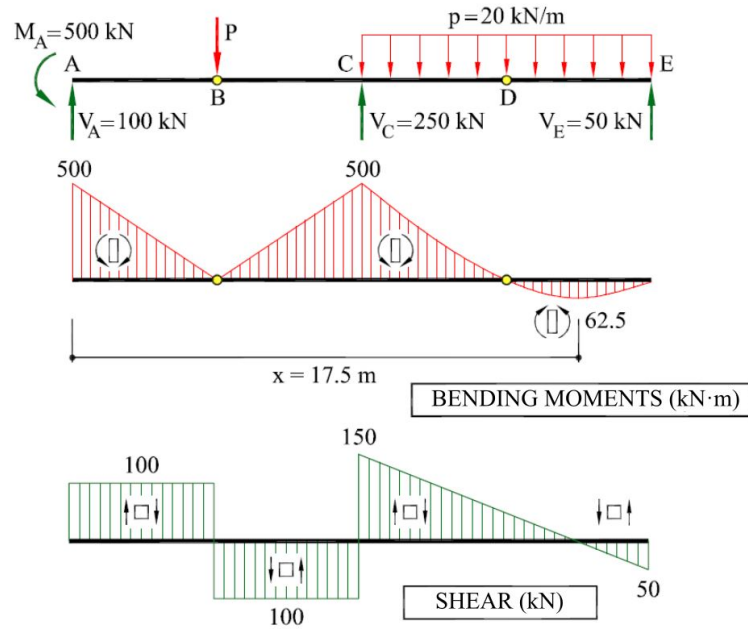


Figure 4.12: Internal resultants diagrams for Example 4.4.2

Example 4.4.3

The structure $ABCD$ in Figure 4.13 has a clamped support at A and is subjected to a vertical load p of 30 kN/m distributed uniformly along the axis of the beam CBD , which is inclined an angle α with respect to the horizontal. Calculate and draw the internal resultants diagrams.

Support Reactions

Considering the equilibrium equation and imposing that the sum of vertical forces and the sum of moments in the fixed support A are both zero, the external reactions of the structure are (see Figure 4.14):

$$\begin{aligned} R_A &= 30 \cdot 7,5 = 225 \text{ kN } (\uparrow) \\ M_A &= 225 \cdot (5 - 3,75) \cos \alpha = 225 \text{ kN} \cdot \text{m } (\curvearrowright) \end{aligned}$$

where α is the slope of the beams BC and BD .

Bending moment diagram

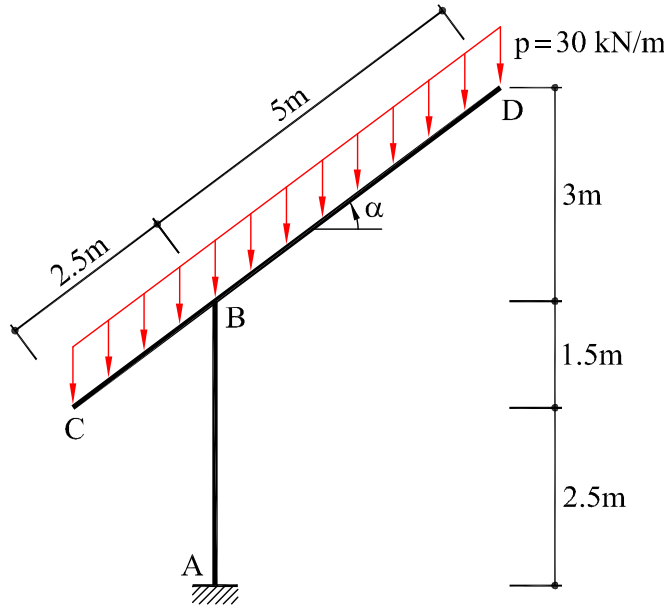


Figure 4.13: Structure of Example 4.4.3

Calculating backwards from the cross-section, the bending moments in the column AB is constant and has a value $M_A = 225 \text{ kN}\cdot\text{m}$. In the beam CB , for a cross-section at a distance s from the free end C (see Figure 4.14b), the bending moment diagram is (see Figure 4.14b):

$$M(x) = -30 s \frac{x}{2} = -15 s x = -15 \frac{x^2}{\cos \alpha}$$

The bending moment function is parabolic, and for $x = 2 \text{ m}$ the value is $M_B = -75 \text{ kN}\cdot\text{m}$, which corresponds to node B . Similarly, in the region BD , the bending diagram for a cross section at distance s from the free end D can be expressed as (see Figure 4.14b):

$$M(x_1) = -30 s_1 \frac{x_1}{2} = -15 s_1 x_1 = -15 \frac{x_1^2}{\cos \alpha}$$

Shear force diagram

Along the longitudinal axis of the vertical beam, the shear force in column AB is zero when calculated. Note that the bending moment function in column AB is constant, so its first derivative is zero. By projecting the resultant of forces onto the cross-section plane, the shear force for regions AB and BD is:

$$\begin{aligned} T_{CB} &= 30 s \cos \alpha &= 30 x \\ T_{BD} &= (30 s - 225) \cos \alpha &= 30 x - 225 \cos \alpha \end{aligned}$$

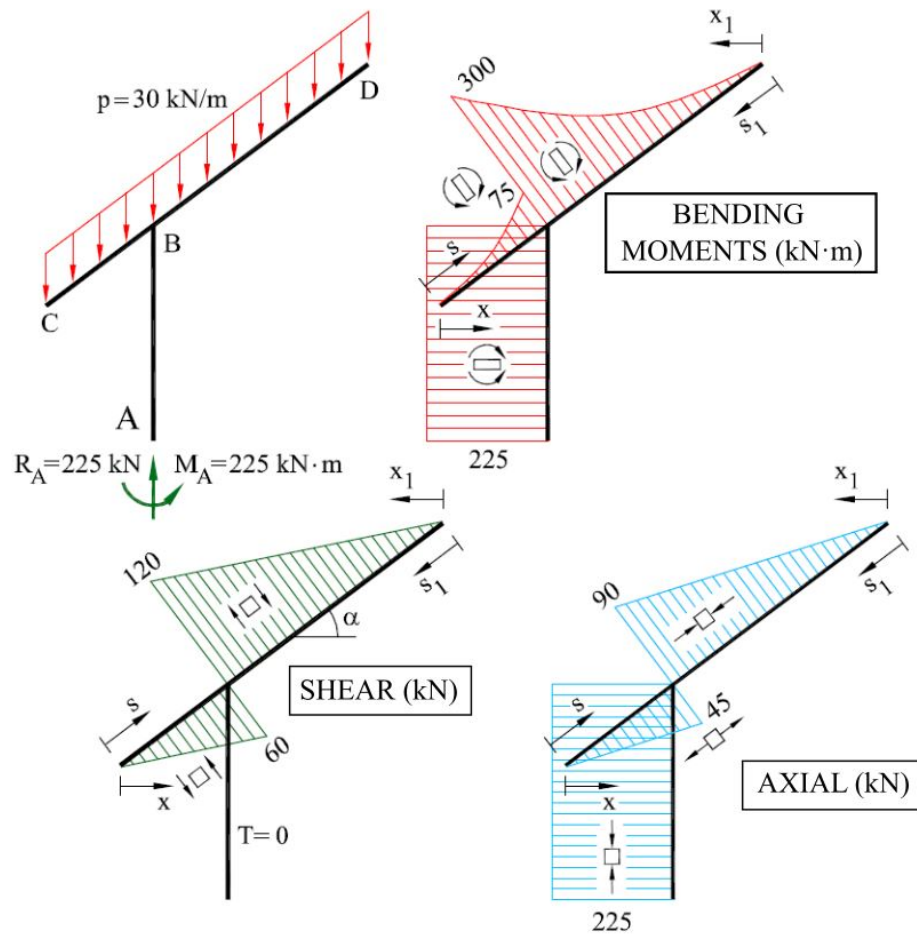


Figure 4.14: Internal resultants diagrams of Example 4.4.3

Both functions are linear, which is consistent with the derivative of parabolic bending moment functions.

Axial force diagram

The axial force along the longitudinal axis of the column AB is compressive, constant, and has a value of $N = 225$ kN.

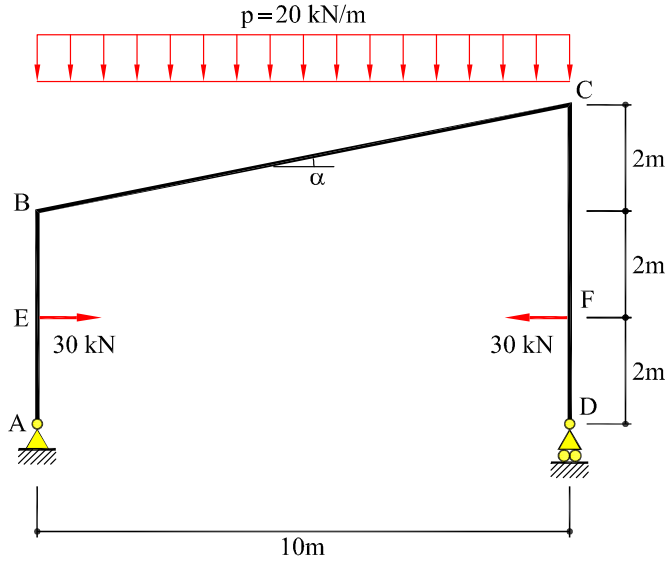


Figure 4.15: Structure of Example 4.4.4

In regions CB and BD , the axial force is obtained by projecting the resultant forces onto the section plane normal as follows:

$$\begin{aligned} N_{CB} &= 30 s \sin \alpha \\ N_{BD} &= (30 s - 225) \sin \alpha \end{aligned}$$

Both functions are linear, as corresponds to a uniform load. Note that the vector composition of the shear and axial forces on each cross-section is equal to the resultant of forces $\mathbf{R}$ on the cross-section, which is a vertical force. In Figure 4.14, the reactions and internal resultant diagram of the structure are showed.

Example 4.4.4

The frame structure in Figure 4.15 is subjected to a 20 kN/m uniformly distributed load p along the horizontal projection of the lintel BC and two horizontal point loads of 30 kN acting on cross-sections E and F (these point loads represent a post-tension cable anchored at nodes E and F). Calculate and draw the internal resultant diagrams.

Support Reactions

Support reactions are computed considering static equilibrium conditions. It is determined that the horizontal reaction H_A is null, and also:

$$\begin{aligned} V_A + V_D &= 200 \\ 10V_D &= 20 \cdot 10 \cdot 5 \end{aligned}$$

and, therefore:

$$V_A = 100 \text{ kN} \quad (\uparrow) \quad ; \quad V_D = 100 \text{ kN} \quad (\uparrow)$$

Bending moments diagram

In the regions AE and DF of both columns, bending moments are zero. Continuing with the column AB , starting from the cross-section E , and calculating considering backwards, the following result is obtained:

$$M(s) = -30s$$

where s indicates the distance from E . This is a linear function, and for $s = 2$ m, the value of the bending moment at cross section B is $M_B = -60 \text{ kN}\cdot\text{m}$.

Similarly, the bending moment function in the DC column is;

$$M(s_1) = -30s_1$$

where s_1 is the distance from F . This is linear function and its value in C cross-section corresponds to $M_C = -120 \text{ kN}\cdot\text{m}$.

At lintel BC , the bending moment function for a cross section x distance from cross-section B corresponds to

$$M(x) = V_A x - 30 \left(2 + \frac{2}{10} x \right) - 20 \frac{x^2}{2} = -60 + 94x - 10x^2$$

a parabolic function whose maximum value is at a distance $x_{\max}$ from cross-section B , which can be determined by setting the derivative of the bending moment function to zero with respect to the variable x . In this case, $x_{\max} = 4,7$ m and the corresponding maximum moment value is $M_{\max}(x = x_{\max}) = 160,9 \text{ kN}\cdot\text{m}$.

Shear force diagram

In the first region AE of column AB , the shear force is zero. Starting at cross-section E , the shear force equals to the horizontal force applied in the E cross-section. Therefore, the shear force is constant and its value is:

$$T_{EB} = 30 \text{ kN}$$

Similarly, the shear force in column DC is zero up to cross-section F . In the FC region of the column, the shear force is constant, with a value equal to the horizontal force applied in cross-section F :

$$T_{FC} = 30 \text{ kN}$$

In region BC , which is inclined at an angle α with respect to the horizontal, the projection of the resultant of the forces on the section plane is equal to:

$$T_{BC} = (-V_A + 20x) \cos \alpha + 30 \sin \alpha$$

which is a linear function with a value of $T_B = -92.17 \text{ kN}$ for the B ($x = 0$) cross-section. Note that the shear force is zero at $x = 4.7 \text{ m}$, as corresponds to maximum bending moment cross section along region BC .

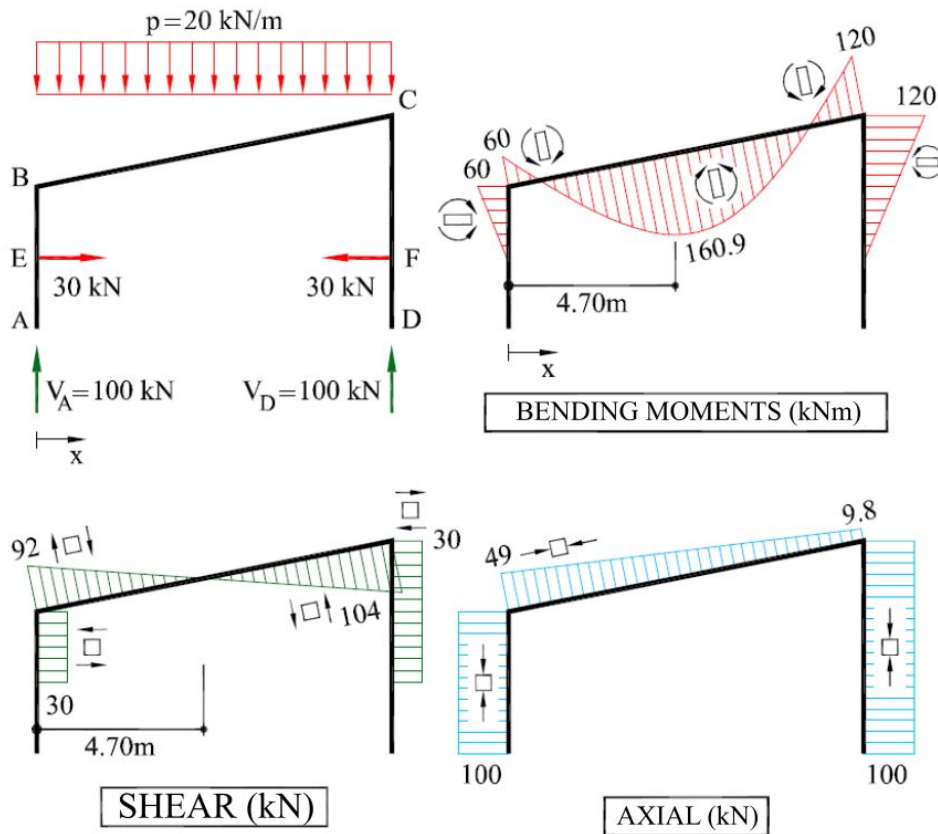


Figure 4.16: Internal resultants diagram of Example 4.4.4

Axial force diagram

In columns AB and CD , the axial forces are compressive, constant, and equal to the respective support reactions. So,

$$N_{AB} = N_{CD} = -100 \text{ kN} \quad (C)$$

In region BC , the axial force is calculated along the longitudinal axis of the beam as:

$$N_{BC} = (-100 + 20x) \sin \alpha - 30 \cos \alpha$$

which is a linear function with its extreme values $N_B = -49 \text{ kN}$ and $N_C = -9,8 \text{ kN}$, respectively, for cross-sections B and C .

In Figure 4.16, the reactions and internal resultants diagrams of the structure are showed.

Example 4.4.5

Calculate and draw the internal resultants diagrams for the structure showed in Figure 4.17, subjected to a horizontal force F acting on cross section B' .

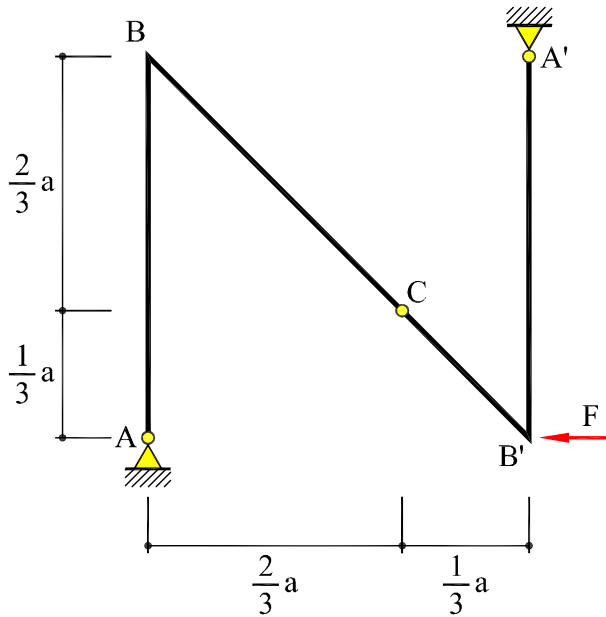


Figure 4.17: Structure of Example 4.4.5

Support reactions

The reactions at the supports are calculated considering the equilibrium conditions of forces and moments (at A) and the condition of zero bending moment at the hinge C (calculated backwards), as shown in Figure 4.18a .

$$\begin{aligned} H_A - H_{A'} - F &= 0 \\ V_A - V_{A'} &= 0 \\ H_{A'} a - V_{A'} a &= 0 \\ V_A \frac{2a}{3} - H_A \frac{a}{3} &= 0 \end{aligned}$$

from where:

$$V_A = F \quad (\uparrow) \quad ; \quad V_{A'} = F \quad (\downarrow) \quad ; \quad H_A = 2F \quad (\rightarrow) \quad ; \quad H_{A'} = F \quad (\leftarrow)$$

Bending moment diagram

For a cross-section in the pillar AB at a x distance from support A , it follows that:

$$M_{AB} = -H_A y = -2Fy$$

which is a linear bending moment function, and for $y = a$, it is $M_B = -2Fa$ corresponding to the value of the bending moment in cross-section B . The bending moment function for a cross-section located at a y distance from cross-section A in the column BA is :

$$M_{B'A'} = H_{A'} (a - y) = F(a - y)$$

a linear bending function whose maximum value is $M_{B'} = Fa$. For a cross section situated in region BB' , at a distance x, y from support A , it is:

$$M_{BB'} = -H_A y + V_A x = F(x - 2y)$$

a linear bending function with its extreme values $M_B = -2Fa$ and $M_{B'} = Fa$.

Shear force diagram

As the bending function is linear in every bar, the shear forces are constant in every bar. The value corresponding to columns is:

$$\begin{aligned} T_{AB} &= H_A = 2F \\ T_{B'A'} &= H_{A'} = F \end{aligned}$$

These values coincide with the horizontal reactions, at bar BB' the value is:

$$T_{BB'} = \frac{\sqrt{2}}{2} (V_A + H_A) = \frac{3\sqrt{2}}{2} F$$

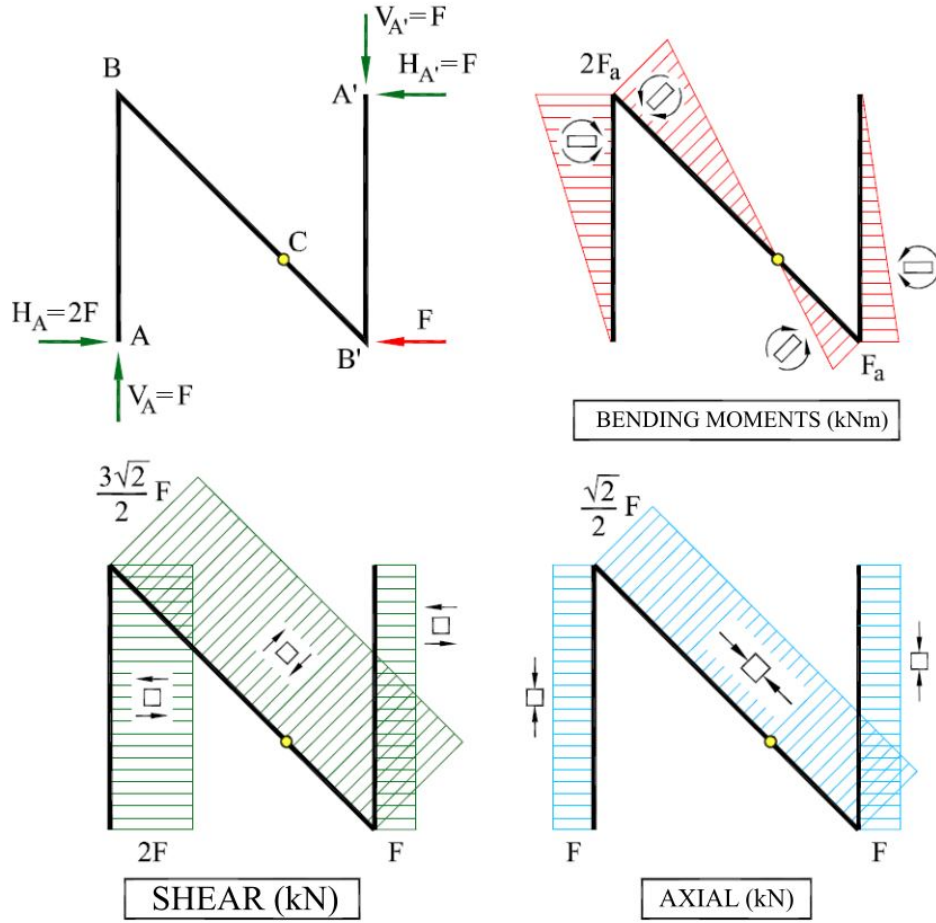


Figure 4.18: Internal resultants diagram of Example 4.4.5

Axial force diagram

Similar to the previous section, the component of $\mathbf{R}$ normal to the cross-sections is determined, and constant axial force values in each bar are obtained:

$$N_{AB} = -F \quad ; \quad N_{BB'} = -\frac{\sqrt{2}}{2} F \quad ; \quad N_{B'A'} = -F$$

In Figure 4.18, the reactions and internal resultants diagrams of the structure are showed.

Example 4.4.6

The statically determined frame structure in Figure 4.19 is subjected to a uniformly distributed load ($p = 20 \text{ kN/m}$) on the BD lintel. Calculate and draw the internal resultants diagrams.

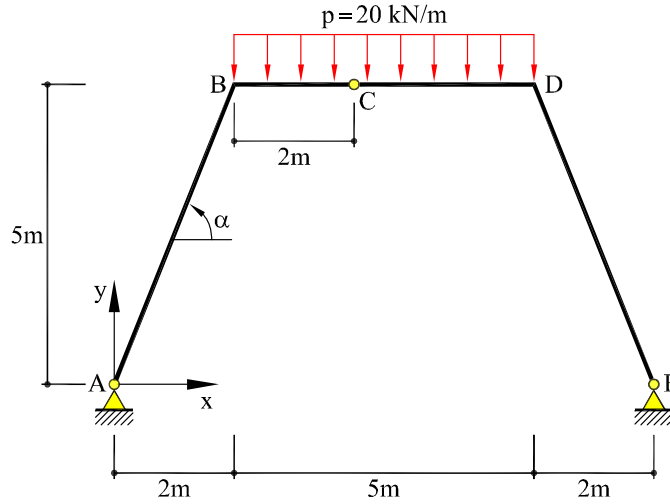


Figure 4.19: Structure of the example 4.4.6

Support reactions

The reactions at the supports are first calculated considering the equilibrium equations and the condition of zero by bending moment at the hinge, as shown in Figure 4.18a:

$$\begin{aligned} H_A - H_E &= 0 \\ V_A + V_E &= 20 \cdot 9 \\ 9V_A &= 20 \cdot 5 \cdot 4.5 \\ 4V_A - 5H_A &= 20 \cdot 2 \cdot 1 \end{aligned}$$

and, therefore, the reactions at supports are:

$$V_A = V_E = 50 \text{ kN} \quad (\uparrow) \quad ; \quad H_A = 32 \text{ kN} \quad (\rightarrow) \quad ; \quad H_E = 32 \text{ kN} \quad (\leftarrow)$$

Note that the reactions are symmetric, despite the asymmetric position of the hinge.

Bending moment diagram

Using the reference axes (x, y) of Figure 4.20, the bending moment corresponding to a generic cross section of region AB is:

$$\begin{aligned} M_{AB} &= V_A x - H_A y \\ &= 50 x - 32 y \end{aligned}$$

a linear function whose the maximum value is $M_B = -60 \text{ kN} \cdot \text{m}$. In region DE , now looking forwards, a symmetrical linear function with respect to the vertical axis that passes through the centre of the lintel is also obtained:

$$\begin{aligned} M_{DE} &= V_E (9 - x) - H_E y \\ &= 50 (9 - x) - 32 y \end{aligned}$$

On the BD lintel, considering the calculating backwards, the corresponding bending moment function can be written as

$$\begin{aligned} M_{BD} &= V_A x - H_A y - \frac{p}{2}(x - 2)^2 \\ &= 50 x - 32 \cdot 5 - \frac{20}{2}(x - 2)^2 = -10 x^2 + 90 x - 200 \end{aligned}$$

a parabolic function with maximum values at the ends $M_B = M_D = -60 \text{ kN} \cdot \text{m}$ and zero value at the hinge. The value of the bending moment at the center is $2,5 \text{ kN} \cdot \text{m}$.

Given the symmetry of the loads and reactions, observe that the bending moments are symmetric about the central vertical axis.

Shear force diagram

Since the bending moment functions for columns AB and DE are linear, the shear forces in these columns are constant. Their values are determined by projecting on the section the reaction components in A and E , respectively:

$$\begin{aligned} T_{AB} &= -V_A \cos \alpha + H_A \sin \alpha = 11 \text{ kN} \\ T_{DE} &= -V_E \cos \alpha + H_E \sin \alpha = 11 \text{ kN} \end{aligned}$$

where α is the angle of the slope of the supports. Calculating backwards, the shear force in the BD region can be expressed as

$$\begin{aligned} T_{BD} &= -V_A + p (x - 2) \\ &= -50 + 20 (x - 2) \end{aligned}$$

This is a linear function whose extreme values correspond to $T_B = -V_A$ and $T_D = V_E$, while the shear force in the centre of the lintel is equal to zero due to symmetry.

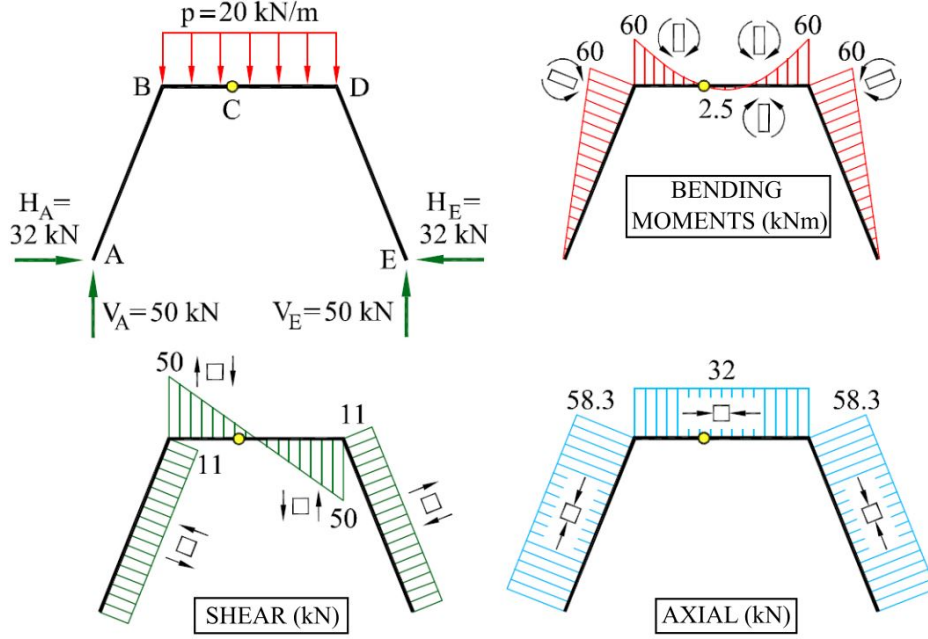


Figure 4.20: Internal resultants diagrams for Example 4.4.6

Axial force diagram

The axial force in each bar is determined using a similar process. At the supports, the horizontal and vertical reactions of the respective supports in their respective directions. Since the projection of the lateral load on the bar is zero, the same procedure is applied to the lintel.

The corresponding axial forces can be expressed as:

$$\begin{aligned} N_{AB} &= N_{DE} = -(V_A \sin \alpha + H_A \cos \alpha) = -58.3 \text{ kN} \\ N_{BD} &= -H_A = -32 \text{ kN} \end{aligned}$$

It is observed that the functions are constant and symmetrical.

In Figure 4.20, the reactions and internal resultants diagrams of the structure are showed.

Chapter 5

Axial Force

5.1 Introduction

A structural element is subjected to *simple traction or compression* when only axial forces act on its cross-sections, that is, forces normal to the cross-sections applied at their respective centers of gravity (see Figure 5.1). Traction stresses are considered positive, and compressive stresses are negative.

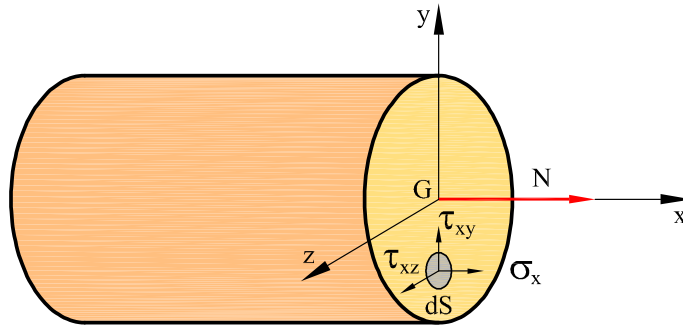


Figure 5.1: Axial force on a cross-section

Given that the different internal results that act on a cross-section are product of the stresses that act on such a section, the following equations must be fulfilled:

$$\begin{aligned} N &= \int_S \sigma_x dS & ; & \quad T_y = \int_S \tau_{xy} dS = 0 \\ T_z &= \int_S \tau_{xz} dS = 0 & ; & \quad M_t = \int_S (\tau_{xz} y - \tau_{xy} z) dS = 0 \quad (5.1) \\ M_y &= \int_S \sigma_x z dS = 0 & ; & \quad M_z = - \int_S \sigma_x y dS = 0 \end{aligned}$$

Obviously, these equations are not enough to determine the distribution of stresses in the cross-section. Therefore, it is necessary to establish simplifying hypotheses relative to the deformation of the element that allows such determination.

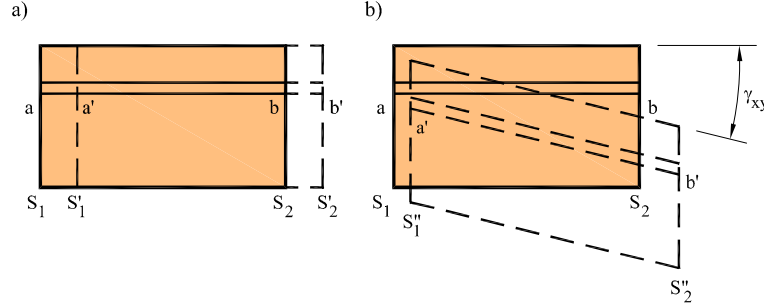


Figure 5.2: Bernoulli's deformation hypothesis

5.2 Axial force in a straight element

In the case of a straight element subjected to an axial force, the following deformation hypothesis is made, which is known as *Bernoulli's hypothesis*: “during the deformation of a straight element subjected to axial force, the cross sections remain plane and parallel to themselves”.

The suitability of this hypothesis can be verified experimentally, as long as the element is long and in the areas where the *Saint-Venant principle* can be applied.

According to this hypothesis, a differential slice bounded by sections S_1 and S_2 can deform according to Figure 5.2a, and transform in $S'_1S'_2$, or, according to Figure 5.2b, and becomes $S''_1S''_2$.

In case (a), there is no distortion of the differential elements of volume; this means $\gamma_{xy} = \gamma_{xz} = 0$ in all the points of the slice and, consequently, $\tau_{xy} = \tau_{xz} = 0$. In case (b), it is $\gamma_{xy} = \text{const}$ and $\gamma_{xz} = \text{const}$, according to the deformation hypothesis and, therefore, $\tau_{xy} = \text{const}$ and $\tau_{xz} = \text{const}$ for all the points of the slice. The equations (5.1) that relate tangential forces and shear stresses are then:

$$T_y = \tau_{xy} \int_S dS = 0 \quad ; \quad T_z = \tau_{xz} \int_S dS = 0 \quad \Rightarrow \quad \tau_{xy} = \tau_{xz} = 0 \quad (5.2)$$

Therefore, the tangential stresses and the distortions are necessarily zero, so that the slice is necessarily deformed according to case (a), and the second, third and fourth integral equations (5.1) hold identically.

In addition, and as shown in Figure 5.3, in the deformation of the slice all the fibers are deformed parallel to the axis of the piece and they suffer

an identical longitudinal deformation, equal to:

$$\varepsilon_x = \frac{du}{dx} \quad (5.3)$$

where du is the elongation (positive for stretching and negative otherwise) of the slice of length dx .

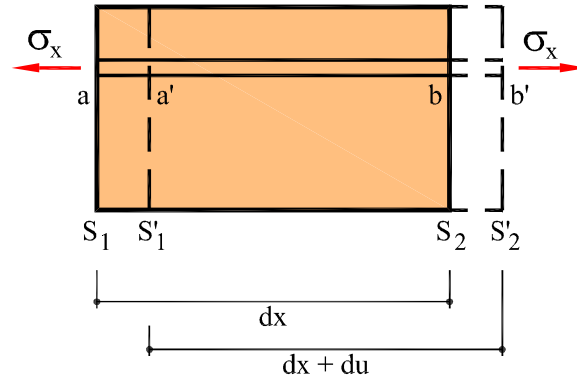


Figure 5.3: Elongations and stresses in the slice

If the material of the element is linearly elastic and Hooke's law is fulfilled, the longitudinal strain ε_x is equal to:

$$\varepsilon_x = \frac{\sigma_x}{E} - \frac{\nu}{E}(\sigma_y + \sigma_z) \quad (5.4)$$

where E is the Young's modulus of the material and ν its Poisson's ratio. Being zero or negligible, in this case, the normal stresses in planes parallel to the axis of the element, σ_y and σ_z , it is:

$$\varepsilon_x = \frac{\sigma_x}{E} \quad \Rightarrow \quad \sigma_x = E \varepsilon_x \quad (5.5)$$

Since the deformations are the same for all the fibers, the stresses are equal

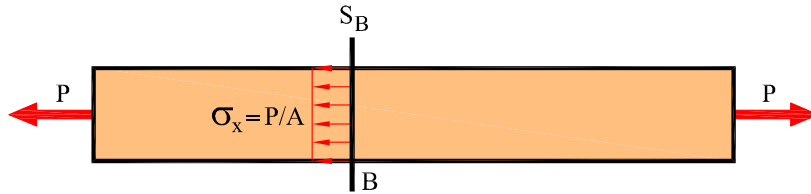


Figure 5.4: Axial force: uniform distribution of stresses

as well; that is, there is a uniform distribution of normal stresses σ_x in the cross-section (Figure 5.4). Therefore, the equations (5.1) can be written:

$$N = \sigma_x \int_S dS = \sigma_x A \quad (5.6a)$$

$$M_y = \sigma_x \int_S z dS \equiv 0 \quad (5.6b)$$

$$M_z = \sigma_x \int_S y dS \equiv 0 \quad (5.6c)$$

since the axes (y, z) , being the principal axes of inertia of the section, pass through the center of gravity of the element. Hereinafter, A is the area of the section. From the equation (5.6a) it results:

$$\sigma_x = \frac{N}{A} \quad \text{and} \quad \varepsilon_x = \frac{N}{EA} \quad (5.7)$$

The longitudinal deformation of the slice is accompanied by a lateral deformation due to the Poisson effect of value::

$$\varepsilon_y = \varepsilon_z = -\nu \varepsilon_x = -\frac{\nu N}{EA} \quad (5.8)$$

It should be noted that the hypothesis of uniform elongation of the fibers and the corresponding uniform distribution of normal stresses is exact when the applied external forces are uniformly distributed on the corresponding cross-sections. If it is not, according to the Saint-Venant Principle, the results obtained are valid for cross-sections far enough from the points where the loads are applied. The distance at which this assumption holds true is approximately equal to the transverse dimensions of the part. Thus, in Figure 5.5 the stress distribution is uniform in a cross-section such as S_B , sufficiently far away from the extremes of the element, but not in one like the S_A , where the local effect of how the loads are applied is still substantial.

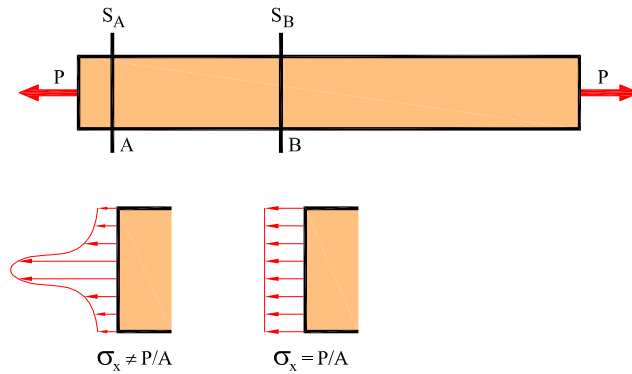


Figure 5.5: Axial force and Saint-Venant's Principle

The variation in the total length of the element, that is, *elongation* or *shortening* of it, is obtained by integration of the longitudinal deformations on all the slices of the element, that is:

$$\Delta l = \int_l \varepsilon_x(x) dx = \int_l \frac{N(x)}{EA(x)} dx \quad (5.9)$$

where it is admitted that both the axial, $N(x)$, and the section of the piece, $A(x)$, can vary along the length l . If the section of the piece A is constant and the axial force is uniform (Figure 5.6), the total elongation of the bar is:

$$\Delta l = \frac{Nl}{EA} \quad (5.10)$$

This expression can be written as $\Delta l = N/k$, with $k = EA/l$ representing the *axial stiffness* of the piece.

The above equations are valid for both compressive and tensile axial forces. In the case of compression, all of the above is valid provided that the piece is not too long in relation to the dimensions of the cross-section, because in that case the equilibrium of the piece can be unstable, a phenomenon known as *buckling*.

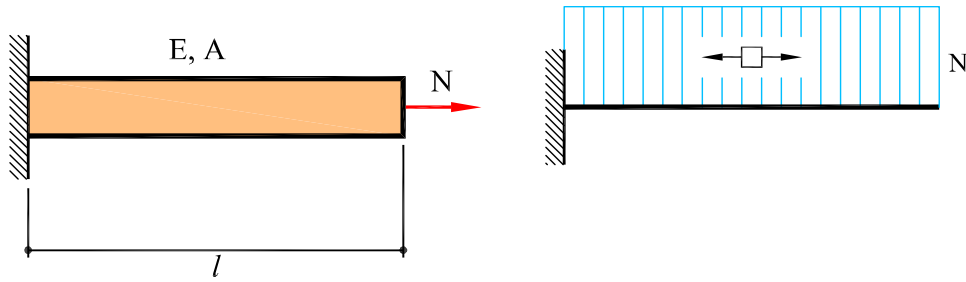


Figure 5.6: Uniform axial distribution

Example 5.2.1

Calculate the total elongation of a bar of constant section A and length l , subjected to the action of its self-weight Q uniformly distributed (Figure 5.7).

Let $\gamma = Q/Al$ be the specific weight of the material. As can be seen in Figure 5.7, the axial force is linear, and is given by the expression:

$$N(x) = \gamma Ax$$

The stresses on each section are obtained simply as :

$$\sigma_x = \frac{N(x)}{A} = \gamma x$$

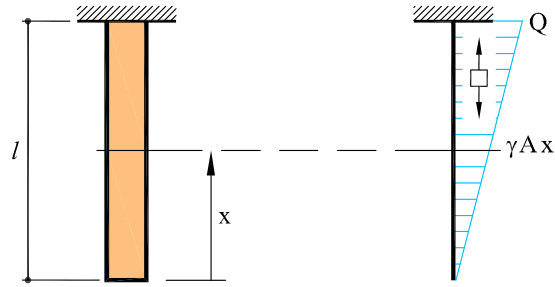


Figure 5.7: Element subjected to its self weight

also a linear function. The total elongation is equal to:

$$\Delta l = \int_l \frac{N(x)}{EA} dx = \int_0^l \frac{\gamma x dx}{E} = \frac{\gamma l^2}{2E} = \frac{Ql}{2EA}$$

Example 5.2.2

Calculate the maximum length that a vertically suspended cable subjected to its self-weight can have (Figure 5.8). Assume an allowable stress of value σ_{all} .

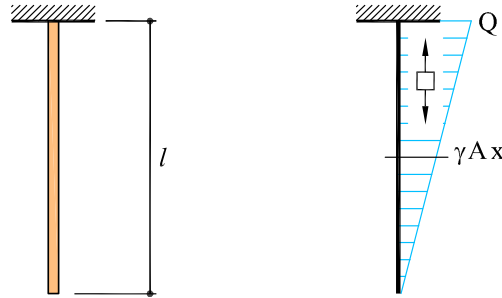


Figure 5.8: Vertically suspended cable

According to the results of the previous example, the maximum stress occurs in the upper section of the cable, which supports a tensile stress

$$\sigma_x(x = l) = \frac{\gamma A l}{A} = \gamma l$$

and it must be

$$\sigma_x \leq \sigma_{\text{all}} \quad \Rightarrow \quad l_{\text{max}} \leq \frac{\sigma_{\text{all}}}{\gamma}$$

A high-strength steel may have a specific weight $\gamma = 75,000 \text{ N/m}^3$ and an allowable stress of $\sigma_{\text{all}} = 1500 \text{ MPa} = 1500 \text{ N/mm}^2$. This results in a maximum length of $l_{\text{max}} \leq 20 \cdot 10^3 \text{ m} = 20 \text{ km}$.

Example 5.2.3

A wood pile of length l , cross-sectional area A , and modulus of elasticity E , is buried and supports a friction load P along its longitudinal perimeter (Figure 5.9a). It is assumed that the friction force f per unit length of the pile is uniformly distributed over the surface of the pile. Calculate: (a) the distribution of stresses along the pile and (b) the shortening δ of the pile as a function of P , l , E y A .

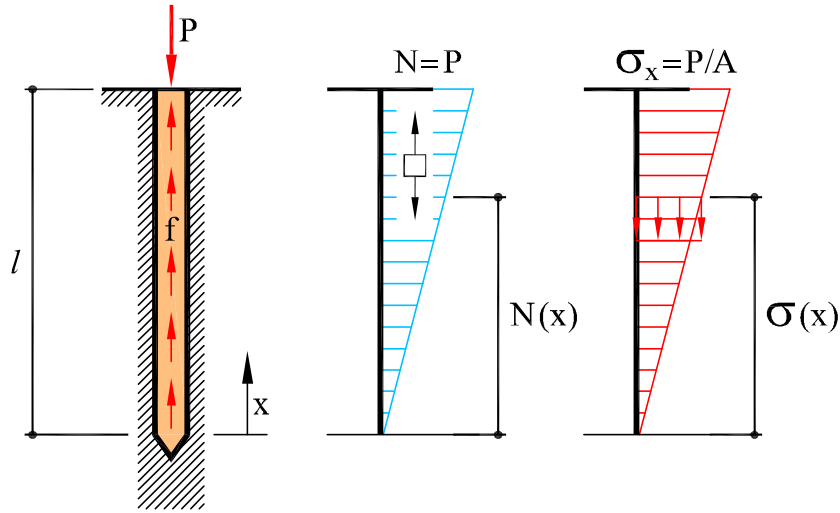


Figure 5.9: Pile, axial forces and stress distribution of Example 5.2.3

(a) *Stress distribution along the pile*

If the load P is supported only by the friction force f , distributed uniformly along the length of the pile, by equilibrium it must be fulfilled:

$$P = f \cdot l$$

and the axial forces in the pile can be expressed by:

$$N(x) = P - f (l - x) = P - \frac{P}{l} (l - x) = P \frac{x}{l}$$

which is a linear function of coordinate x , measured from the bottom of the pile (Figure 5.9).

On the other hand, the normal stress in a generic section located at the distance x is:

$$\sigma_x(x) = \frac{N(x)}{A} = \frac{P}{A} \frac{x}{l}$$

The normal stresses in the pile are compressive and their distribution is linear as a function of x (Figure 5.9).

(b) *Pile shortening*

The pile is subjected to a compression strain that varies linearly with the coordinate x . Therefore, its initial length undergoes a shortening δ , which in turn depends on P , l , E and A , expressed by:

$$\delta = \int_0^l \frac{N(x)}{EA} dx = \int_0^l \frac{P(x/l)}{EA} dx = \frac{Pl}{2EA}$$

Example 5.2.4

The truss structure ABC of Figure 5.10 is formed by two bars of the same material and cross sections of areas A_1 and A_2 , respectively. For a load P acting on node C determine the stresses and elongations in the bars.

Data: $P = 40 \text{ kN}$, $L = 3 \text{ m}$ y $\alpha = 30^\circ$, $A_1 = 5 \text{ cm}^2$, $A_2 = 10 \text{ cm}^2$, $E = 200 \cdot 10^9 \text{ N/m}^2$.

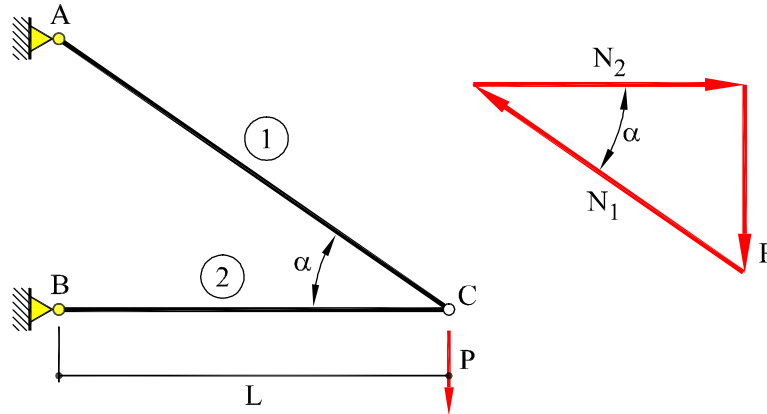


Figure 5.10: Structure of Example 5.2.4 and equilibrium of forces in C

The structure is statically determined and the axial forces in the bars 1 and 2 can be calculated directly by equilibrium of forces in the node C :

$$\begin{aligned} N_1 &= \frac{P}{\sin \alpha} = 80 \text{ kN} && \text{(traction)} \\ N_2 &= -\frac{P}{\tan \alpha} = -69.3 \text{ kN} && \text{(compression)} \end{aligned}$$

The stresses in the bars are:

$$\sigma_1 = \frac{N_1}{A_1} = \frac{80}{5 \cdot 10^{-4}} = 160 \text{ MPa}$$

$$\sigma_2 = \frac{N_2}{A_2} = \frac{-69.3}{10 \cdot 10^{-4}} = -69.3 \text{ MPa}$$

and the corresponding elongations are:

$$\delta_1 = \frac{N_1 l_1}{EA_1} = \frac{80 \cdot 3 / \cos \alpha}{200 \cdot 10^6 \cdot 5 \cdot 10^{-4}} = 0.277 \text{ cm}$$

$$\delta_2 = \frac{N_2 l_2}{EA_2} = \frac{-69.3 \cdot 3}{200 \cdot 10^6 \cdot 10 \cdot 10^{-4}} = -0.104 \text{ cm}$$

Observe that bar 1 is in tension and undergoes elongation, while bar 2 is compressed and shortens, as indicated by the positive and negative signs, respectively.

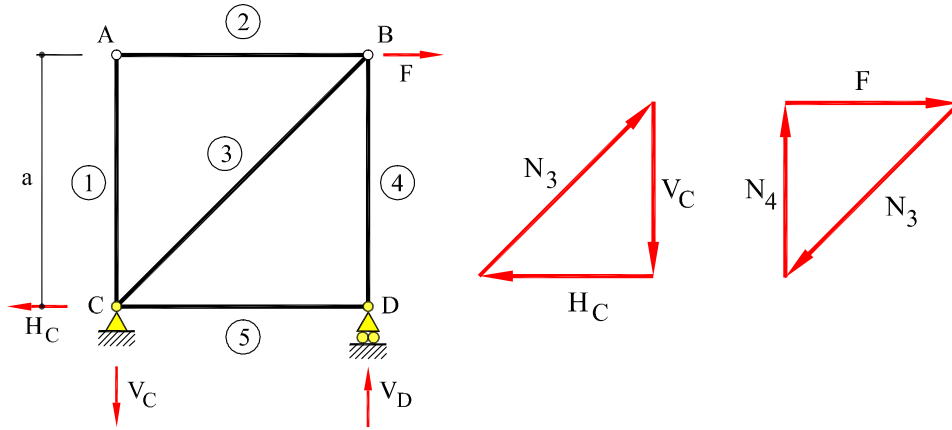


Figure 5.11: Structure from Example 5.2.5 and equilibrium of forces at nodes C and B

Example 5.2.5

The truss structure of Figure 5.11 is formed by five bars of the same material and cross-sections, of area A . For a horizontal force F acting at node B , determine: (a) the axial stresses in the bars and (b) the corresponding elongations.

Data: $F = 10 \text{ kN}$, $a = 4 \text{ m}$, $A = 5 \text{ cm}^2$, $E = 200 \cdot 10^9 \text{ N/m}^2$.

(a) The structure is statically determined and the reactions at supports C and D can be determined by equilibrium of forces and moments:

$$H_C = F \quad (\leftarrow) ; \quad V_C = F \quad (\downarrow) ; \quad V_D = F \quad (\uparrow)$$

Similarly, the axial forces in the bars can be calculated considering the equilibrium of forces at the nodes (Figures 5.11b and (c):

$$\begin{aligned} \text{node } D &\Rightarrow N_4 = -F & ; & \quad N_5 = 0 \\ \text{node } C &\Rightarrow N_1 = 0 & ; & \quad N_3 = \sqrt{2}F \\ \text{node } A &\Rightarrow N_2 = 0 \end{aligned}$$

Bar 3 works in tension and bar 4 in compression. The bars 1, 2 and 5 do not work for the load applied in this case.

(b) The elongation of bar 3 is:

$$\delta_3 = \frac{N_3 L_3}{EA} = \frac{\sqrt{2}F\sqrt{2}a}{EA} = 2 \frac{Fa}{EA} = 0.8 \cdot 10^{-3} \text{m}$$

and the shortening of the bar 4 is:

$$\delta_4 = \frac{N_4 L_4}{EA} = \frac{-Fa}{EA} = -0.4 \cdot 10^{-3} \text{m}$$

Example 5.2.6

For the steel truss of Figure 5.12, the following is requested: (a) find the axial forces in all bars, (b) choose an appropriate single IPN profile and (c) calculate the elongations produced for the chosen profile.

Data: $L = 3 \text{ m}$, $P = 150 \text{ kN}$, $E = 200 \text{ GPa}$, $\sigma_{\text{all}} = 75 \text{ MPa}$.

(a) The structure is statically determinate and the reactions are obtained considering the equilibrium of forces and moments:

$$\begin{aligned} \sum F_H &= H_1 + 150 & = 0 \\ \sum F_V &= V_1 + V_3 & = 0 \\ \sum M_1 &= V_3 L - 150L & = 0 \end{aligned}$$

From the previous equations it is obtained:

$$H_1 = 150 \text{ kN } (\leftarrow) \quad V_1 = 150 \text{ kN } (\downarrow) \quad V_3 = 150 \text{ kN } (\uparrow)$$

In the same way, the axial forces in the bars are obtained considering the equilibrium in the nodes:

$$\begin{aligned} \text{node 1} &\Rightarrow N_{12} = 150 \text{ kN} & ; & \quad N_{13} = 150 \text{ kN} \\ \text{node 2} &\Rightarrow N_{23} = -150\sqrt{2} = -212.5 \text{ kN} \end{aligned}$$

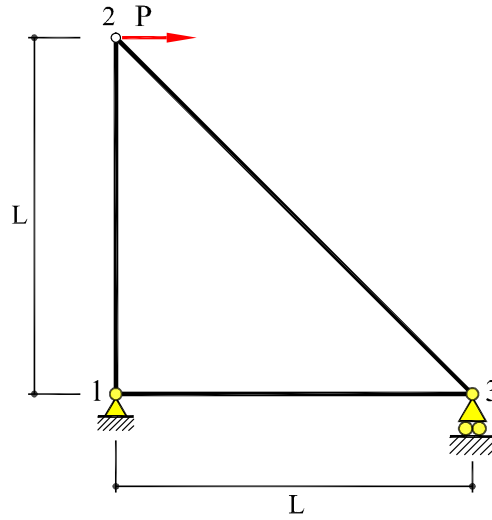


Figure 5.12: Structure of Example 5.2.6

Bars 1 – 2 and 1 – 3 are in tension while bar 2 – 3 is compressed, as indicated by the signs of the respective forces.

(b) Note that the IPN profile should be chosen so that the allowable stress σ_{all} is not exceeded. As the axial force is constant in each bar, the stress is constant in each of the bars. Therefore, the maximum stress will occur in the bar whose axial force is greater, that is, in the 2 – 3 bar:

$$\sigma_{\text{max}} = \frac{N_{\text{max}}}{A} < \sigma_{\text{all}} \quad \Rightarrow \quad A > \frac{N_{\text{max}}}{\sigma_{\text{max}}} = \frac{150\sqrt{2} \text{ kN}}{75 \cdot 10^3 \text{ kN/m}^2} = 28.3 \text{ cm}^2$$

The profile that meets the admissible stress requirement is the IPN200, whose cross-section has an area of 33.5 cm^2 , so this profile is used for all bars.

(c) The deformations that occur in the bars built with an IPN200 profile are the following:

$$\delta_{12} = \delta_{13} = \frac{150 \cdot 3}{2 \cdot 10^8 \cdot 33.5 \cdot 10^{-4}} = 0.67 \cdot 10^{-3} \text{ m}$$

$$\delta_{23} = \frac{-150\sqrt{2} \cdot 3\sqrt{2}}{2 \cdot 10^8 \cdot 33.5 \cdot 10^{-4}} = -1.34 \cdot 10^{-3} \text{ m}$$

Bars 1 – 2 and 1 – 3 are stretched while the 2 – 3 bar is shortened.

Example 5.2.7

For the truss structure in Figure 5.13, subjected to a vertical load of 600 kN, it is requested: (a) calculate the axial forces in the bars. (b) Design the structure with an IPN profile, being the admissible stress of the material σ_{adm} . (c) Calculate the elongation of each bar assuming the entire structure built with the profile chosen in (b).

Data: $P = 600 \text{ kN}$, $a = 4.5 \text{ m}$, $h = 6 \text{ m}$, $\sigma_{\text{all}} = 140 \text{ MPa}$, $E = 200 \text{ GPa}$.

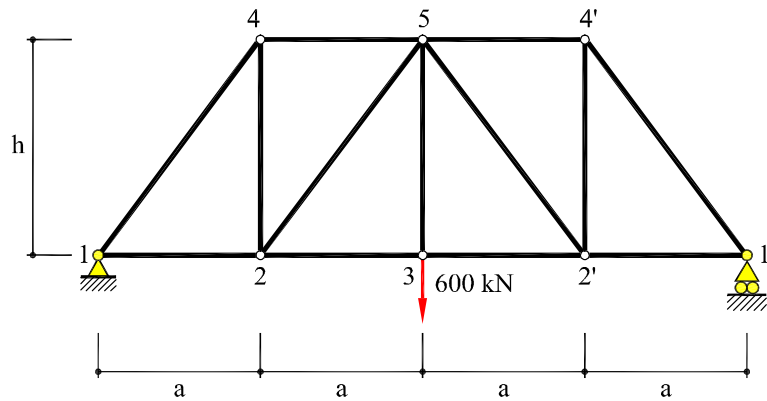


Figure 5.13: Truss of Example 5.2.7

(a) The structure is statically determined and the reactions can be obtained directly from the equilibrium of forces and moments.

$$\begin{aligned}\sum F_H &= H_1 = 0 \\ \sum F_V &= V_1 + V_{1'} - 600 = 0 \\ \sum M_1 &= 18V_{1'} - 600 \cdot 9 = 0\end{aligned}$$

where:

$$H_1 = 0 \quad V_1 = 300 \text{ kN } (\uparrow) \quad V_{1'} = 300 \text{ kN } (\uparrow)$$

The axial forces in the bars are determined from the equilibrium of forces at the nodes. Observe that the structure is symmetric and is loaded symmetrically; this means that the axial forces are also symmetrical:

$$\begin{aligned}\text{node 1} &\Rightarrow N_{12} = 225 \text{ kN (T)} ; N_{14} = -375 \text{ kN (C)} \\ \text{node 4} &\Rightarrow N_{42} = 300 \text{ kN (T)} ; N_{45} = -225 \text{ kN (C)} \\ \text{node 2} &\Rightarrow N_{23} = 450 \text{ kN (T)} ; N_{25} = -375 \text{ kN (C)} \\ \text{node 3} &\Rightarrow N_{35} = 600 \text{ kN (T)} ;\end{aligned}$$

(b) The truss is designed so that the admissible stress σ_{adm} is not exceeded. The bar that supports a greater axial force is bar 35 with a compression force $N = 600$ kN.

The stress in a section of this bar must fulfill:

$$\sigma = \frac{N}{A} \leq \sigma_{\text{adm}} \quad \Rightarrow \quad A \geq \frac{N}{\sigma_{\text{adm}}} = \frac{600}{140 \cdot 10^3} = 42.9 \text{ cm}^2$$

Therefore, the minimum area that the cross-section of the bar requires is 42.9 cm^2 . The smallest profile that exceeds that area is the IPN240 with an area of 46.1 cm^2 .

(c) As the force and the cross-section in all bars are constant, the elongation of each slice is constant throughout the entire bar, and the total elongation of the bar is calculated by:

$$\delta = \frac{Nl}{EA}$$

Assuming that the structure is built with the IPN240 profile the area is 46.1 cm^2 . The length of the horizontal bars is 4.5 m, that of the vertical bars is 6.0 m, while the inclined bars measure 7.5 m. With these data, the elongations of the bars of the structure are:

$$\begin{aligned} \delta_{12} &= \frac{225 \cdot 4.5}{2 \cdot 10^8 \cdot 46.1 \cdot 10^{-4}} = +1.10 \cdot 10^{-3} \text{ m} \\ \delta_{14} &= \frac{-375 \cdot 7.5}{2 \cdot 10^8 \cdot 46.1 \cdot 10^{-4}} = -3.05 \cdot 10^{-3} \text{ m} \\ \delta_{24} &= \frac{300 \cdot 6.0}{2 \cdot 10^8 \cdot 46.1 \cdot 10^{-4}} = +1.95 \cdot 10^{-3} \text{ m} \\ \delta_{23} &= \frac{450 \cdot 4.5}{2 \cdot 10^8 \cdot 46.1 \cdot 10^{-4}} = +2.20 \cdot 10^{-3} \text{ m} \\ \delta_{25} &= \frac{-375 \cdot 7.5}{2 \cdot 10^8 \cdot 46.1 \cdot 10^{-4}} = -3.05 \cdot 10^{-3} \text{ m} \\ \delta_{45} &= \frac{-225 \cdot 4.5}{2 \cdot 10^8 \cdot 46.1 \cdot 10^{-4}} = -1.10 \cdot 10^{-3} \text{ m} \\ \delta_{35} &= \frac{600 \cdot 6.0}{2 \cdot 10^8 \cdot 46.1 \cdot 10^{-4}} = +3.90 \cdot 10^{-3} \text{ m} \end{aligned}$$

Bars 1–2, 2–4, 2–3, and 3–5 are stretched, while bars 1–4, 2–5, and 4–5 are shortened, as indicated by the signs of the respective elongations.

5.3 Cables

Cables are flexible structural elements that work only under traction. The cables are limited only by their self-weight and the way they are anchored. Below are some application examples.

Example 5.3.1

A loading crane, made up of a steel beam AC anchored by a cable BD , supports a load P as shown in Figure 5.14a. The cable has a cross-sectional area $A_c = 481 \text{ mm}^2$ and a modulus of elasticity E_c . Calculate: (a) the stress in the cable when the load P acts and (b) the variation of length of the corresponding cable.

Data: $h = 1.5 \text{ m}$, $l_1 = 3.2 \text{ m}$, $l_2 = 1.6 \text{ m}$, $P = 30 \text{ kN}$, $E_c = 140 \text{ GPa}$.

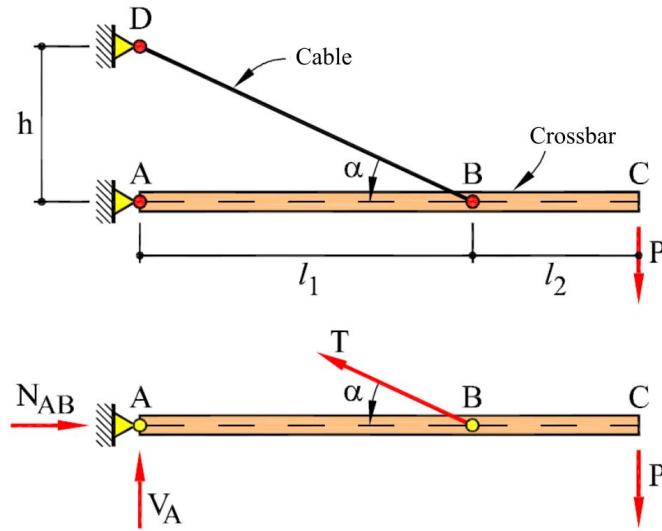


Figure 5.14: Loading crane and equilibrium of forces of the Example 5.3.1

The cable and the beam form an angle α equal to:

$$\tan \alpha = \frac{h}{l_1} = \frac{1.5}{3.2} = 0.4687 \quad \Rightarrow \quad \alpha = 25.11^\circ$$

By equilibrium of moments in A (Figure 5.14):

$$T \sin \alpha = P \left(\frac{l_1 + l_2}{l_1} \right)$$

Therefore, the tensile force in the cable T and the force N_{AB} in the beam are equal to:

$$\begin{aligned} T &= \frac{P}{\sin \alpha} \frac{l_1 + l_2}{l_1} = \frac{30}{0.42436} \cdot 1.5 = +106 \text{ kN} \\ N_{AB} &= -T \cos \alpha = -\frac{30}{0.4687} \cdot 1.5 = -96 \text{ kN} \end{aligned}$$

Note that the cable is pulled (positive axial force) and the beam is compressed between A and B (negative axial force).

(a) *Tension in the cable*

The tension in the cable is uniform, of value:

$$\sigma = \frac{T}{A_c} = \frac{106 \cdot 10^3}{481 \cdot 10^{-6}} = 220 \text{ MPa}$$

(b) *Cable elongation*

Taking into account that $l_{BD} = l_1 / \cos \alpha = 3.53 \text{ m}$, the elongation of the cable is:

$$\Delta l_{BD} = \frac{N_{BD} \cdot l_{BD}}{E_c \cdot A_c} = \frac{106 \cdot 10^3 \cdot 3.53}{140 \cdot 10^9 \cdot 481 \cdot 10^{-6}} = 5.55 \cdot 10^{-3} \text{ m}$$

Example 5.3.2

A steel cable of diameter d , area A_c and length l is used to lift a metal beam that weighs P kN (Figure 5.15). The cable has a modulus of elasticity E_c and its maximum load is $P_{\max}$. Calculate: (a) the tension in the cable to lift the load, (b) its elongation and (c) what is the safety factor of the cable with respect to its maximum load.

Data: $d = 12 \text{ mm}$, $A_c = 76.7 \text{ mm}^2$, $l = 10 \text{ m}$, $E_c = 140 \text{ GPa}$, $P = 20 \text{ kN}$, $P_{\max} = 80 \text{ kN}$.

(a) *Tension in the cable*

When lifting the metal beam, the cable is pulled. If the self-weight of the cable is not taken into account, the axial force is uniform and its value is $N = P$. The tension in the cable is:

$$\sigma = \frac{P}{A} = \frac{20 \cdot 10^3}{76.6 \cdot 10^{-6}} = 261 \text{ MPa}$$

(b) *Cable elongation*

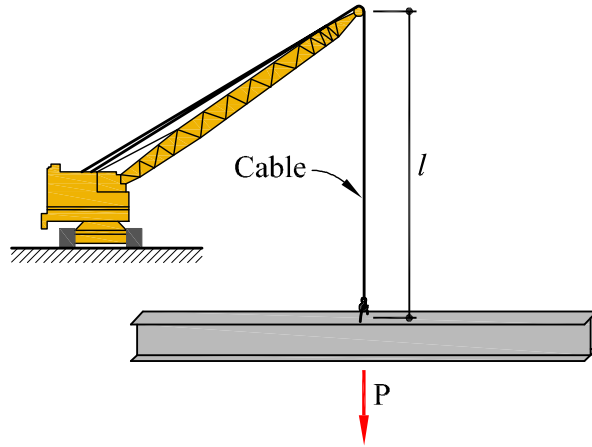


Figure 5.15: Example 5.3.2

Since the axial force in the cable is uniform, the elongation that the cable undergoes when lifting the beam is:

$$\delta = \frac{Pl}{E_c A_c} = \frac{20 \cdot 10^3 \cdot 10}{140 \cdot 10^9 \cdot 76.6 \cdot 10^{-6}} = 1.86 \cdot 10^{-2} \text{ m}$$

(c) *Safety factor with respect to the load $P_{\max}$*

If the maximum load supported by the beam is $P_{\max}$, the safety factor n is:

$$n = \frac{P_{\max}}{P} = \frac{80}{20} = 4$$

Example 5.3.3

A square reinforced concrete plate, with side a and thickness b , is raised by four cables attached to the plate and to an upper hook as shown in Figure 5.16. The area of each cable is $A_c = 173 \text{ mm}^2$. Calculate the tensile stress in the cables due to the weight of the concrete plate and its elongation, taking into account that the specific weight of the concrete is γ .

Data: $h = 1.8 \text{ m}$, $a = 3 \text{ m}$, $b = 0.20 \text{ m}$, $\gamma = 24 \text{ kN/m}^3$, $E_c = 140 \text{ GPa}$.

If γ is the specific weight of the reinforced concrete, the weight of the plate is:

$$P = a^2 b \gamma = 3 \cdot 3 \cdot 0.20 \cdot 24 = 43.2 \text{ kN}$$

According to the geometry of the problem (Figure 5.16), the diagonal length is $d = a\sqrt{2} = 2.21 \text{ m}$ and the length of the cables is $l = \sqrt{h^2 + (d/2)^2} = 2.85 \text{ m}$. On the other hand, $\cos \alpha = h / (d/2) = 0.6315$.

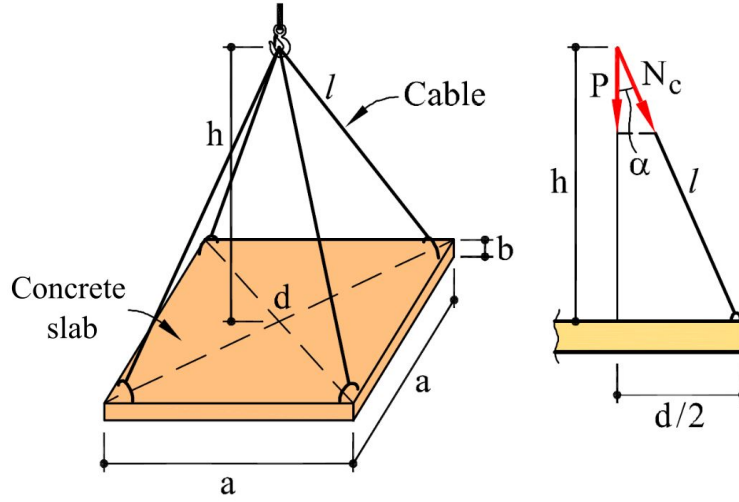


Figure 5.16: Plate of Example 5.3.3

Considering the above, the tensile axial stress in each cable is:

$$N = \frac{1}{4} \frac{P}{\cos \alpha} = \frac{43.2}{4 \cdot 0.6315} = 17.10 \text{ kN}$$

The axial force in the cables is uniform. Therefore, the tension is also uniform and of value:

$$\sigma = \frac{N}{A_c} = \frac{17.10 \cdot 10^3}{173 \cdot 10^{-6}} = 98.8 \text{ MPa}$$

and the corresponding elongation is:

$$\Delta l = \frac{Nl}{E_c A_c} = \frac{17.10 \cdot 10^3 \cdot 2.85}{140 \cdot 10^9 \cdot 173 \cdot 10^{-6}} = 2.12 \cdot 10^{-3} \text{ m}$$

Example 5.3.4

A vertical aluminum post AB , with height h and cross-section A , is braced by four symmetrical steel cables of length l . The anchorage points of the cables form a square whose center is the base of the post (Figure 5.17). Calculate the shortening of the post for a tensile force F in each cable if the modulus of elasticity of aluminum is E_{al} .

Data: $h = 6 \text{ m}$, $A = 110 \text{ cm}^2$, $l = 7.5 \text{ m}$, $F = 15 \text{ kN}$, $E_{al} = 70 \text{ GPa}$.

Each cable forms an angle α with the vertical axis of the post, such that $\cos \alpha = h/l = 0.8$. Therefore, the compression that the cables transmit to the pole is:

$$N = -15 \cdot 0.8 \cdot 4 = -48 \text{ kN}$$

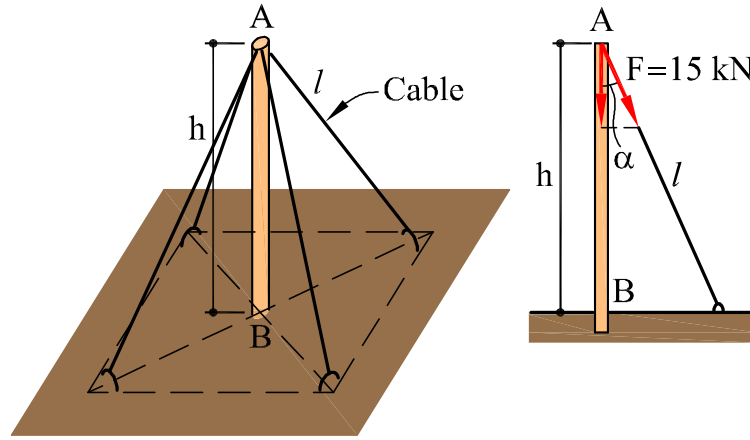


Figure 5.17: Post of Example 5.3.4

and the shortening that this compression produces is:

$$\delta = \frac{N h}{E_{al} A} = \frac{48 \cdot 10^3 \cdot 6}{70 \cdot 10^9 \cdot 110 \cdot 10^{-4}} = 0.374 \cdot 10^{-3} \text{ m}$$

5.4 Axial force and temperature

It is common for the pieces to be subjected concurrently to the action of axial force and to temperature variations. Reviewing Section 2.4, the thermal deformation is:

$$\varepsilon_T = \alpha \Delta T \quad (5.11)$$

Hooke's law including the thermal effect is:

$$\varepsilon_x = \frac{\sigma_x}{E} + \varepsilon_T \quad (5.12)$$

where the stresses σ_y and σ_z have been considered negligible. Then:

$$\sigma_x = E (\varepsilon_x - \varepsilon_T) \quad (5.13)$$

If σ_x is uniform in section, the axial force is:

$$N = \int_S \sigma_x dS = \sigma_x A = EA (\varepsilon_x - \varepsilon_T) \quad (5.14)$$

where:

$$\varepsilon_x = \frac{N}{EA} + \varepsilon_T \quad (5.15)$$

Example 5.4.1

A cantilever beam of length l is formed by a rolled steel profile IPE240. This beam is subjected to a uniform temperature variation $\Delta T > 0$ (Figure 5.18). It is requested: (a) Calculate the axial forces of the cantilever beam. (b) Calculate the elongation suffered by the cantilever beam. (c) What force would have to be applied at the free edge for the beam to recover its initial length? Calculate the new axial force and the stresses in the entire beam. Data $l = 2$ m, $\Delta T = 10^\circ\text{C}$, $\alpha = 10^{-5} \text{ }^\circ\text{C}^{-1}$, $A = 46.1 \text{ cm}^2$, $E = 200 \text{ GPa}$.

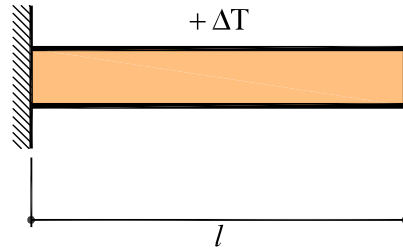


Figure 5.18: Cantilever beam of Example 5.4.1

(a) The cantilever beam is statically determined, so an increase in temperature does not generate reactions or stresses.

(b) Due to the increase in temperature ΔT the beam undergoes a constant dilatation throughout the entire beam of value:

$$\varepsilon_x = \varepsilon_T = \alpha \Delta T$$

In each slice of length dx the incremental length is:

$$du = \varepsilon_x dx = \alpha \Delta T dx$$

Therefore, the displacement at the free edge is:

$$\Delta l = \int_l \varepsilon_x dx = \alpha \Delta T \int_l dx = \alpha \Delta T \cdot l = 0.2 \cdot 10^{-3} \text{ m}$$

(c) The beam subjected to the increase in temperature suffers an elongation $\Delta l = 0.2 \cdot 10^{-3} \text{ m}$, so to recover its initial state it is necessary to apply a uniform compression axial N that causes a shortening of the same value Δl .

$$\Delta l = \frac{Nl}{EA} \implies N = \frac{EA}{l} \Delta l = -92.2 \text{ kN}$$

A force $F = N$ applied at the edge of the beam causes a uniform axial force equal to N throughout the beam. The corresponding stresses are:

$$\sigma_x = \frac{N}{A} = \frac{-92.2 \cdot 10^3}{46.1 \cdot 10^{-4}} = -20 \text{ MPa} \quad (C)$$

The final deformation is:

$$\varepsilon_x = \frac{N}{EA} + \alpha\Delta T = \frac{\sigma_x}{E} + \alpha\Delta T = 0$$

which corresponds to a zero elongation.

Chapter 6

Plane Bending

6.1 Introduction

A beam element is subjected to *pure or uniform bending* when its cross sections are only under bending moment M (Figure 6.1a). The axial force N , shear force T and torsional moment M_t are zero in all cross sections of the beam. When a beam is subjected to *non-uniform bending*, its cross sections are under variable bending moment and, consequently, a shear force is also present (Figure 6.1b). In contrast, a cross section is under *composed bending* when both a bending moment and an axial force act on it (Figure 6.2). Finally, if both bending and torsional moments act on it, the section is said to be under bending-twisting.

6.2 Pure plane bending

Pure bending is the simplest case of bending that can be considered. The results of its study can be applied to the most common cases of non-uniform bending or composed bending, as long as the differences between some cases and others are properly considered. Therefore, the study of pure bending will be undertaken firstly, and then the obtained results will be generalised to the cases of non-uniform and composed bending.

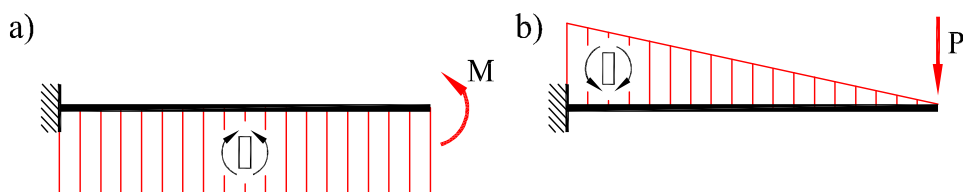


Figure 6.1: Cantilever beam subjected to (a) pure bending y (b) non uniform bending

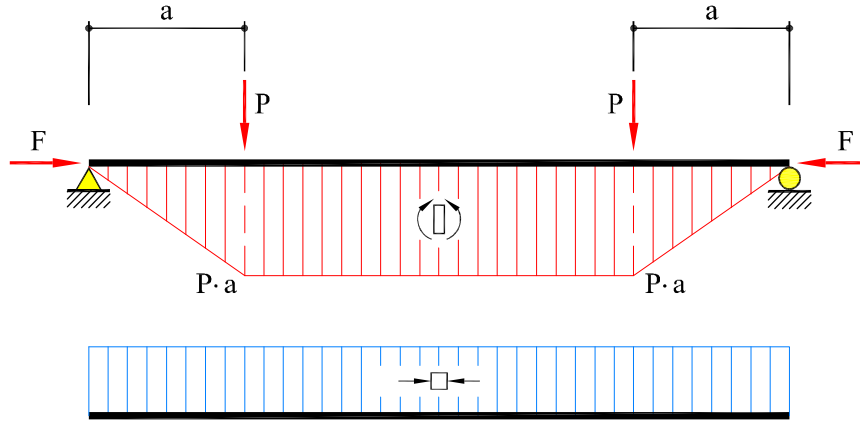


Figure 6.2: Simple supported beam subjected to composed bending

Since the different internal resultants acting on a cross section are the forces and moments resulting from the stresses acting on that cross section, the following equations must be satisfied in the case of *pure bending* with a constant bending moment M_z , as seen in Chapter 3:

$$\begin{aligned}
 N &= \int_S \sigma_x dS = 0 & ; & \quad T_y = \int_S \tau_{xy} dS = 0 \\
 T_z &= \int_S \tau_{xz} dS = 0 & ; & \quad M_t = \int_S (\tau_{xz} y - \tau_{xy} z) dS = 0 \\
 M_y &= \int_S \sigma_x z dS = 0 & ; & \quad M_z = - \int_S \sigma_x y dS
 \end{aligned} \quad (6.1)$$

The stress distribution in a cross section cannot be determined using these integral equations by themselves. Consequently, it is necessary to develop simplifying hypotheses regarding the deformation of the cross section, as was done in the case of the axial force.

6.2.1 Pure bending in midplane beams

An horizontal beam with a constant cross section and a longitudinal symmetry plane subjected to pure bending according to such vertical plane is considered. In this case, the acting bending moment has only one component, M_z ; therefore, the action plane (xy) coincides with the midplane-symmetry plane.

The bending moment is constant throughout the beam under pure bending. Therefore, the deformation of the beam will be identical in all of its differential elements.

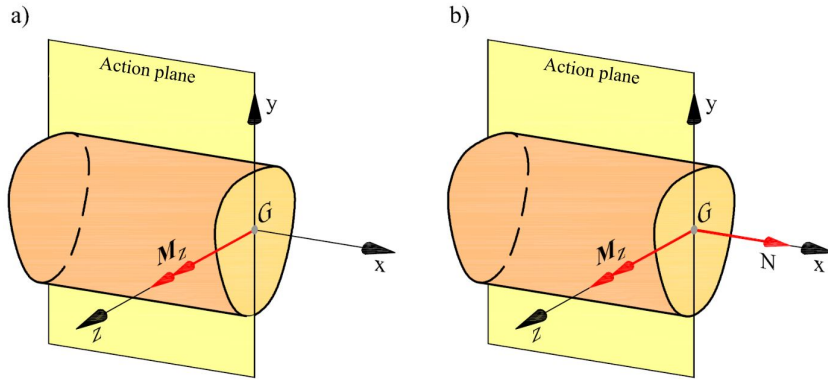


Figure 6.3: Bending in midplane beams: a) Pure bending and b) Composed bending

As a result, the beam's axis deforms into a curve with constant curvature, or an arc of circumference with a centre O that is contained in the symmetry plane of the beam, as shown in Figure 6.4.

Under this conditions and considering the symmetry of the problem with respect to point O , it is that “whose deformed under pure bending cross sections of a differential beam element remain plane and normal to the deformed beam axis in terms of deformation”.

This is Bernoulli's deformation hypothesis for pure bending; its validity can be proven in more general scenarios than those discussed in this section.

The deformation hypothesis states that the longitudinal fibres of the differential beam element are curved and change into arcs of circumference concentric with the deformed beam's axis, as shown in Figure 6.4b. During the deformation process, the length of the fibres can either increase or decrease. They experience compression in the second scenario and longitudinal tensile stress in the first scenario since the axial force on the cross section is null, it is:

$$N = \int_S \sigma_x dS = 0 \quad (6.2)$$

As a result, some of the fibres are compressed and experience negative longitudinal stresses ($\sigma_x > 0$), while other fibres are subjected to tensile with positive longitudinal stresses ($\sigma_x < 0$). As deformation is continuous, the cross section is divided into a tensile and a compressed part along a line where the longitudinal stress is zero ($\sigma_x = 0$). Being the deformation plane, the “neutral line” is a straight line, and due to the symmetry of the problem with respect to the midplane, it is perpendicular to the plane of symmetry. This straight line, contained in the plane of the cross section and perpendicular to the symmetry plane, is the *neutral axis*.

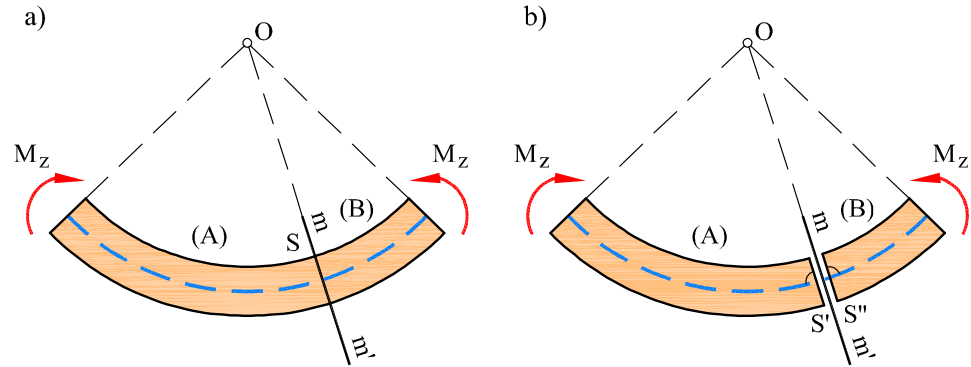


Figure 6.4: Deformed beam axis and deformed plane cross sections of the beam under pure plane bending

For cases of pure bending in straight midplane differential beam elements, the Bernoulli-Navier hypothesis implies that each cross section rotates relative to an adjacent cross section around an axis known as the neutral axis, remaining plane when deformed, and such that the plane of the cross section passes through the centre of curvature of the deformed beam axis.

The bending rotation $d\phi_z$ that occurs in each differential beam element, is small and its corresponding arcs are equal to it tangents in the small displacements hypothesis. Consequently, the deformation of the differential beam element shown in 6.5b is associated with the linearization shown in Figure 6.5c.

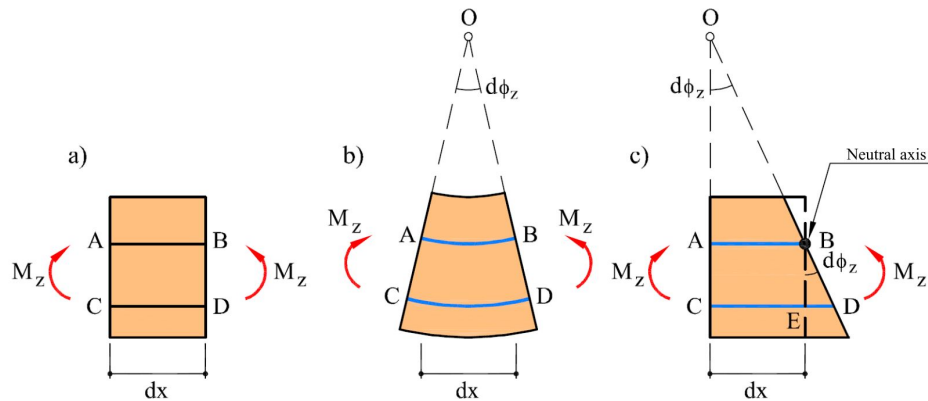


Figure 6.5: Deformation of a differential beam element under plane bending

The deformation of a differential beam element of length dx , as shown

in Figure 6.6, will be analyzed in detail.

In the cross section, the y axis is taken along the plane of symmetry (principal axis of inertia, Figure 6.6b), and the z axis, perpendicular to the y axis, coincides with the neutral axis nn' (Figure 6.6a), whose position is unknown by the time being. Given the similarity of the triangles OAB and BED , it follows that:

$$\frac{\overline{AB}}{\overline{OA}} = \frac{\overline{ED}}{\overline{BE}} \quad \Rightarrow \quad \frac{dx}{\rho_z} = \frac{\varepsilon_x dx}{-y} \quad \Rightarrow \quad \varepsilon_x = -\frac{y}{\rho_z} \quad (6.3)$$

where ρ_z indicates the radius of curvature of the fibres placed on the neutral axis, such as fibre AB , y measures the distance from a generic fibre, such as fibre CD , and the neutral axis, and ε_x represents the strain of one of these fibres. The elongations ε_x of the fibres are proportional to their distance from the neutral axis y ; the same can be said of the stresses according to Hooke's law (and considering that stresses normal to the planes perpendicular to the cross section are negligible compared to the stresses normal to the cross section σ_x , that is, $\sigma_y, \sigma_z \ll \sigma_x$):

$$\sigma_x = E \varepsilon_x = -\frac{E y}{\rho_z} \quad (6.4)$$

As shown in Figures 6.6 and 6.7, the distribution of stresses is linear, which means that all points placed on lines parallel to the neutral axis nn' are subjected to the same stress; and it will be zero for points located on said neutral axis.

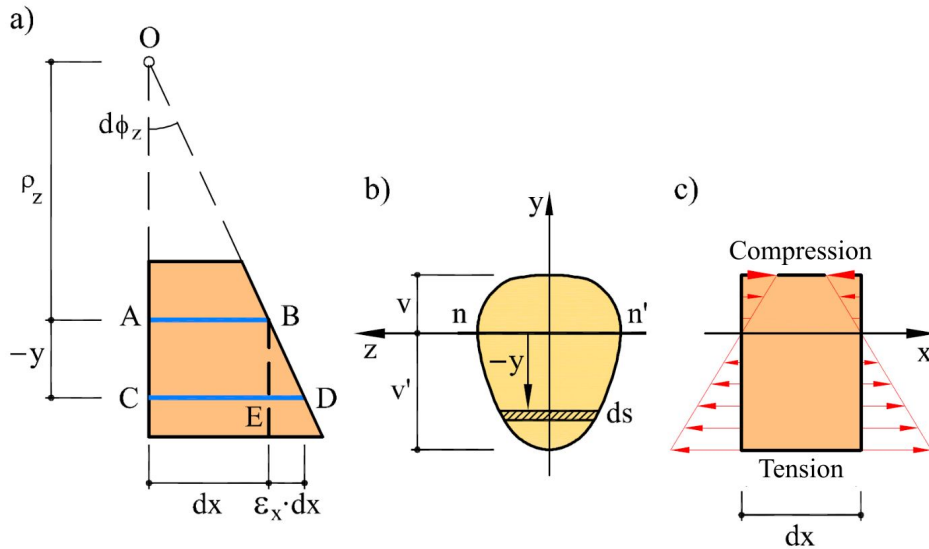


Figure 6.6: Elongation of the fibers and stress distribution

Integrating this stress distribution and assuming that the axial force on the section is zero, it follows:

$$N = \int_S \sigma_x dS = -\frac{E}{\rho_z} \int_S y dS = 0 \Rightarrow \int_S y dS = 0 \quad (6.5)$$

Therefore, the neutral axis passes through the centre of gravity G of the cross section and coincides with its second principal axis of inertia (z axis).

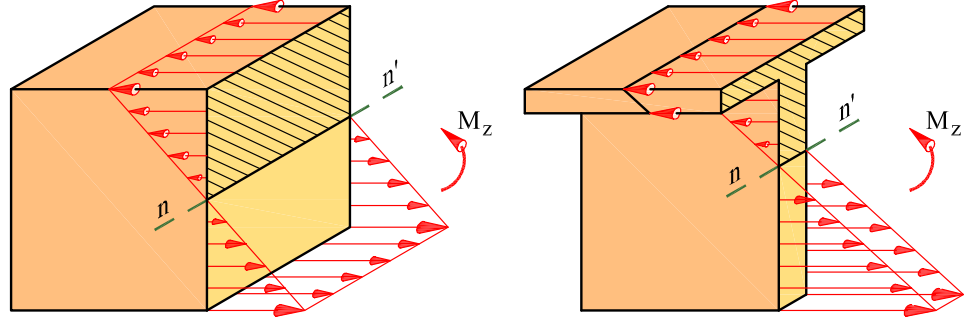


Figure 6.7: Stress distribution under pure bending

For this reason and due to the symmetry with respect to the midplane, it follows that:

$$M_y = \int_S \sigma_x z dS = -\frac{E}{\rho_z} \int_S y z dS \equiv 0 \quad (6.6)$$

Alternatively, the moment due to the stresses about the neutral axis is equal to the acting bending moment:

$$M_z = - \int_S \sigma_x y dS = \frac{E}{\rho_z} \int_S y^2 dS = \frac{E I_z}{\rho_z} \quad (6.7)$$

where $I_z = \int_S y^2 dS$ is the moment of inertia of the cross section about the z axis. Therefore:

$$\frac{1}{\rho_z} = \frac{M_z}{EI_z} \quad (6.8)$$

The obtained expression indicates that the curvature, $\chi_z = 1/\rho_z$, is proportional to the acting bending moment M_z and inversely proportional to the bending stiffness of the cross section, defined by the product EI_z .

In the previous expression, it is also observed that, by definition of curvature $\chi_z = 1/\rho_z = d\phi_z/dx$ (Figure 6.6):

$$d\phi_z = \frac{dx}{\rho_z} = \frac{M_z}{EI_z} dx \quad (6.9)$$

where $d\phi_z$ is the relative angle between the two cross sections that define a differential beam element.

The total relative rotation between the two end cross sections of the beam is obtained by integrating the relative rotations along the beam, which is:

$$\Delta\phi_z = \int_l d\phi_z = \int_l \chi_z dx = \int_0^l \frac{M_z}{EI_z(x)} dx \quad (6.10)$$

where l is the total length of the beam. If the beam has a constant cross section, the total rotation $\Delta\phi_z$ of one end cross section relative to the other is equal to:

$$\Delta\phi_z = \frac{M_z l}{E I_z} \quad (6.11)$$

Finally, eliminating the radius of curvature ρ_z from equations (6.4) and (6.7), the following result is obtained:

$$\sigma_x = -\frac{M_z y}{I_z} \quad (6.12)$$

This is *Navier's law*, which once more demonstrates the linear distribution of the normal bending stresses and allows the calculation of the normal stress at a point of the cross section knowing its distance from the neutral axis. Maximum stresses occurs at the points of the cross section farthest away from the neutral axis, and the maximum compression stress is not equal to the maximum tensile stress unless the distances of the extreme fibres from the centre of gravity are equal (see, for example, the cases shown in Figure 6.7).

Additionally, it is noted that the hypothesis of linear distribution of fibre normal stresses and the linear distribution of fibre strains are both accurate when the external moments are applied precisely in this way to the corresponding end cross sections. If this is not the case, the Saint-Venant Principle states that the obtained results are valid for cross sections that are far enough from the cross sections where the moments are applied. This is verified at a distance approximately equal to the cross-sectional dimensions of the beam.

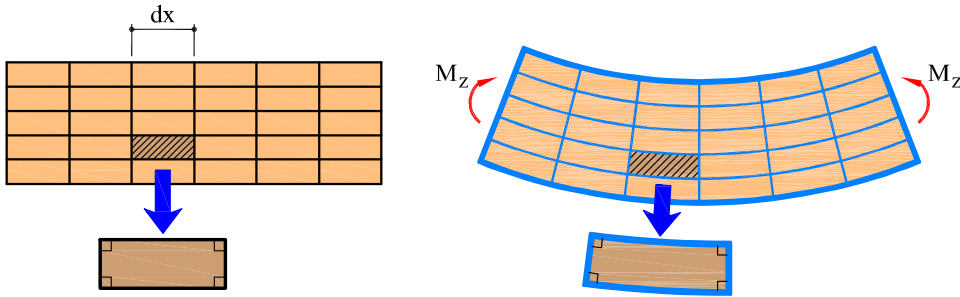


Figure 6.8: Pure bending deformation without angular distortions

As the plane cross sections remain normal to the longitudinal fibres, there is no angular distortion in the differential beam elements, i.e. the right angles are preserved. Thus, distortions γ_{xy} and γ_{xz} are zero everywhere (Figure 6.8), and according to Hooke's law, the shear stresses are also zero, $\tau_{xy} = \tau_{xz} = 0$. Consequently, the equations:

$$T_y = \int_S \tau_{xy} dS \equiv 0 \quad ; \quad T_z = \int_S \tau_{xz} dS \equiv 0 \quad ; \quad M_t = \int_S (\tau_{xz}y - \tau_{xy}z) dS \equiv 0 \quad (6.13)$$

are fulfilled identically.

6.2.2 Pure bending about a principal plane of inertia

Let be considered that the beam is subjected to pure bending by a moment M_z contained in *the principal plane of inertia* defined by the x and y axes, but that this is not a plane of longitudinal symmetry of the beam. In addition, let be assumed that the Bernoulli-Navier hypothesis and Hooke's law are also valid. In this case, the stresses (6.4) are given by equation (6.4). Therefore, the relationship between stresses and strains on the section requires that:

$$M_y = \int_S \sigma_x z dS = -\frac{E}{\rho_z} \int_S y z dS = -\frac{E}{\rho_z} I_{yz} = 0 \quad (6.14)$$

A condition that is satisfied since (y, z) are the principal axes of inertia of the cross section ($I_{yz} = 0$). Consequently, the bending theory developed in the previous section for midplane cross sections will also be applicable to sections without a plane of symmetry, when the bending moment acts in a plane that contains one of the principal axes of inertia, or, in other words, when the action plane coincides with one of the so-called principal bending planes. This result was firstly obtained by Persy, and it was cited by Saint-Venant in 1864.

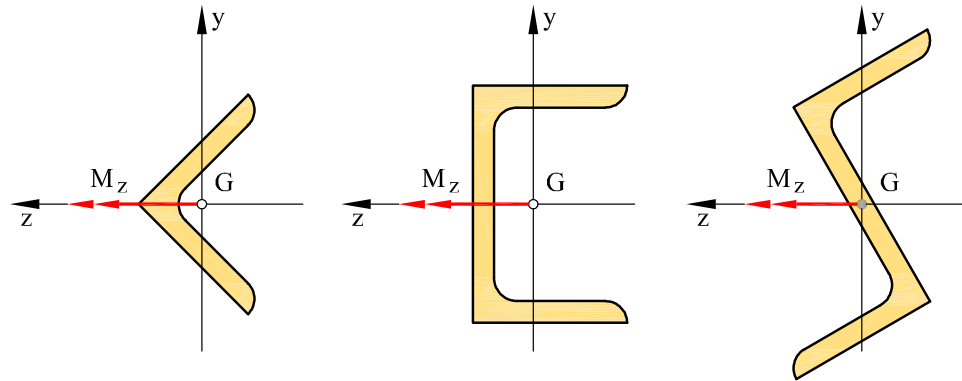


Figure 6.9: Plane bending around principal plane of inertia

The previously studied pure bending theory is characterised by the fact that *the plane in which the bending deformation occurs coincides with the action plane, and the neutral axis is normal to this plane*. Because of this, it is referred to as *pure plane bending* or *Navier's bending*, in honour of the engineer Louis Navier (1785-1842) who established it. Since every cross section has at least two principal planes of inertia, it is always possible to have a beam under plane bending. Figure 6.9 shows scenarios of plane bending on cross sections with different geometries.

Lastly, it is noted that the hypothesis that the cross sections remain plane and normal to the deflection axis after deformation is purely geometric, i.e. it does not establish any conditions regarding the behaviour of the material. The Bernoulli-Navier hypothesis can therefore be extended to materials that are neither homogeneous, isotropic, nor elastic. However, it has been assumed that the beam is homogeneous and that the material is isotropic and linearly elastic in order to obtain the Navier's stress law. In other cases, Navier's law does not apply.

6.2.3 Maximum allowable bending moments

As the stress law in a cross section under pure bending is linear, the fibres farthest from the neutral axis are subjected to the maximum stress values.

Let v and v' be the distances of the section's upper and lower fibres to the z axis, and w and w' will define the distances of the farthest fibres to the left and right of the y axis, respectively (Figure 6.10a and b). Let σ_{all} be the absolute value of the maximum allowable stress of the material. Firstly, this stress is assumed to be equal for tension and compression.

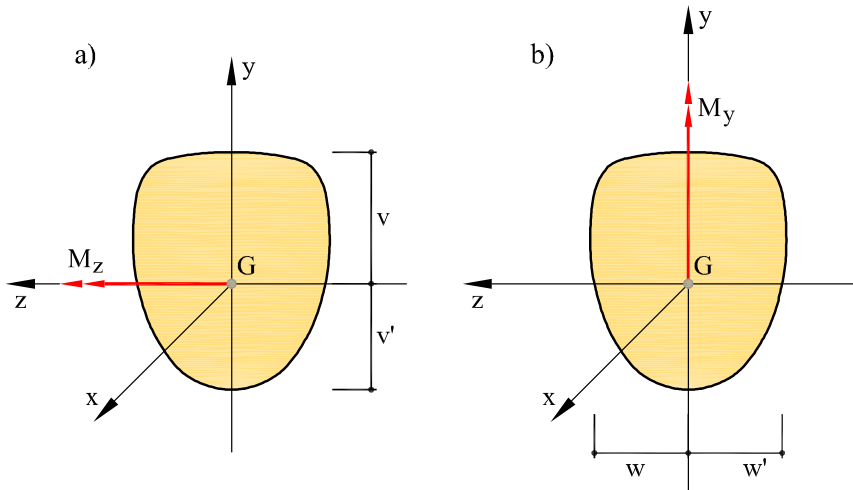


Figure 6.10: Extreme fibre location in plane bending

The maximum bending moment $M_z^{\max}$ supported by a cross-section under pure bending, according to the principal axis of inertia z , is the one that, according to Navier's law, produces a stress equal to σ_{all} in either the upper or lower fibres. Thus, it follows:

$$M_z^{\max} = \frac{\sigma_{\text{all}} I_z}{\max(v, v')} = \min \left(\frac{\sigma_{\text{all}} I_z}{v}, \frac{\sigma_{\text{all}} I_z}{v'} \right) \quad (6.15)$$

Similarly, it is possible to define the maximum bending moment $M_y^{\max}$ supported by a cross-section under pure bending about the principal axis of inertia y , as:

$$M_y^{\max} = \frac{\sigma_{\text{all}} I_y}{\max(w, w')} = \min \left(\frac{\sigma_{\text{all}} I_y}{w}, \frac{\sigma_{\text{all}} I_y}{w'} \right) \quad (6.16)$$

These definitions can be extended to materials with different maximum permissible tensile and compression stresses, σ_{all}^t and σ_{all}^c , respectively.

In this case, it is possible to distinguish between the maximum positive bending moments (tension in fibers v' and w , compression in fibers v y w' , respectively), which are:

$$(M_z^{\max})^+ = \min \left(\frac{\sigma_{\text{all}}^c I_z}{v}, \frac{\sigma_{\text{all}}^t I_z}{v'} \right) \quad ; \quad (M_y^{\max})^+ = \min \left(\frac{\sigma_{\text{all}}^c I_y}{w'}, \frac{\sigma_{\text{all}}^t I_y}{w} \right) \quad (6.17)$$

and the maximum negative bending moments (tension in fibers v y w' , compression in fibers v' y w , respectively), which will be (in absolute value):

$$(M_z^{\max})^- = \min \left(\frac{\sigma_{\text{all}}^c I_z}{v'}, \frac{\sigma_{\text{all}}^t I_z}{v} \right) \quad ; \quad (M_y^{\max})^- = \min \left(\frac{\sigma_{\text{all}}^c I_y}{w}, \frac{\sigma_{\text{all}}^t I_y}{w'} \right) \quad (6.18)$$

6.2.4 Section Modulus

Let the section be subjected to a plane bending by a moment M_z contained in the xy plane, and let σ_{adm} be the allowable stress of the material, equal under tensile all compression. It must be fulfilled that:

$$\sigma_{\text{adm}} \geq \frac{M_z |y|_{\max}}{I_z} = \frac{M_z}{W_z} \quad \text{con} \quad W_z = \frac{I_z}{|y|_{\max}} = \frac{I_z}{\max(v, v')} \quad (6.19)$$

The ratio W_z is known as the section modulus with respect to the z axis and its dimensions are $[L^3]$. Designing a beam subjected to a moment M_z requires that the section modulus satisfies the following condition:

$$W_z \geq \frac{M_z}{\sigma_{\text{adm}}} \quad (6.20)$$

The section moduli can be defined for each principal axis of bending, as geometric properties of the section.

6.2.5 Section Geometrical Efficiency

According to Navier's law, the maximum stresses occur at the fibres farthest from the neutral axis, and they must not exceed the allowable stress. Therefore, the material is not fully exploited since there are points of the section which are subjected to lower stresses than allowable stress value.

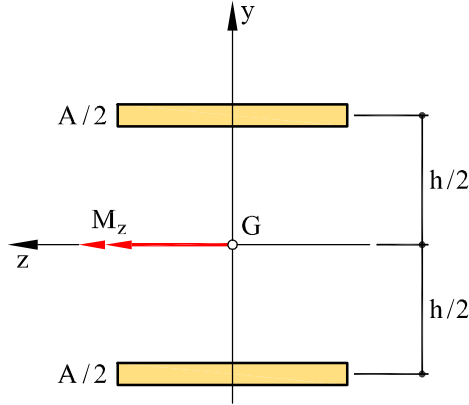


Figure 6.11: Extreme fibres location in plane bending

Therefore, the optimal shape for a cross section under pure bending is the one where most of the area is placed as far away from the neutral axis as possible (Figure 6.11). For a given area A and depth h , the optimal cross section is the one with the greatest section modulus with respect to the bending axis. This maximum value of the section modulus corresponds to an ideal symmetrical section with all the material concentrated in the extreme fibers, at a distance of $h/2$ from the centre of gravity. In this case, the value of the section modulus is (assuming, for example, bending about the z -axis):

$$\begin{aligned} W_z^{\text{opt}} &= \frac{I_z}{|y|_{\max}} \\ &= \frac{2 \frac{A}{2} \left(\frac{h}{2}\right)^2}{\frac{h}{2}} = \frac{Ah}{2} \end{aligned} \quad (6.21)$$

where h is the depth of the beam, that is, the distance between the extreme fibers. For a real section of arbitrary shape with area A and depth h , it is possible to write:

$$W_z = \eta_z W_z^{\text{opt}} = \eta_z \frac{Ah}{2} \quad \text{con} \quad 0 \leq \eta_z \leq 1 \quad (6.22)$$

where η_z is a dimensionless coefficient known as the geometrical efficiency of a cross section around the z axis. Two geometrical efficiency coefficients can

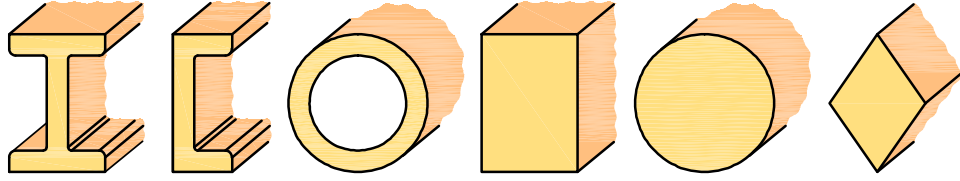


Figure 6.12: The most common sections in Strength of Materials

be defined for the two principal axes of bending, as geometrical properties of the section.

The geometrical efficiency coefficient is always less than one, and the material of the beam is exploited more efficiently as its value (η) along the bending axis increases. The geometrical efficiency coefficient values for some of the most common cross sections (Figure 6.12) are as follows:

$$\begin{aligned}\eta &\simeq \frac{2}{3} \text{ for standard rolled steel I-profiles and W-profiles,} \\ \eta &\simeq \frac{3}{5} \text{ for standard rolled steel U-profiles and rail tracks} \\ \eta &\simeq \frac{1}{2} \text{ for thin-walled circular rings,} \\ \eta &= \frac{1}{3} \text{ for rectangular sections,} \\ \eta &= \frac{1}{4} \text{ for circular sections,} \\ \eta &= \frac{1}{6} \text{ for rhomboidal sections.}\end{aligned}$$

Example 6.2.1

An horizontal beam, with a T shape cross section, is subjected to pure bending in such a way that the upper fiber is under a compressive stress of $\sigma_{\max}^c = 100$ MPa, and the lower fiber is under a tensile stress of $\sigma_{\max}^t = 50$ MPa (Figure 6.13). Determine: (a) the position of the neutral axis, (b) the width b of the beam, (c) the bending moment to which the beam is subjected, and (d) the maximum allowable bending moment if the allowable stress is $\sigma_{\text{all}} = 250$ MPa.

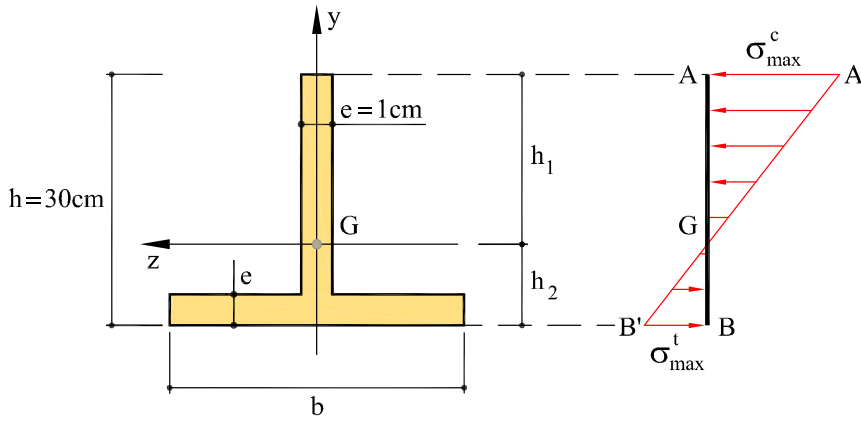


Figure 6.13: Cross section of Example 6.2.1

$$\frac{h_1}{h_2} = \frac{\sigma_{\max}^c}{\sigma_{\max}^t} = \frac{100}{50} = 2$$

(a) Since the stress distribution is linear along the depth of the beam, the position of the neutral axis can be determined directly from the extreme stresses. Based on the similarity of triangles GAA' and GBB' , the following holds:

which leads to the following system of equations

$$\begin{aligned} h_1 + h_2 &= 30 \\ \frac{h_1}{h_2} &= 2 \end{aligned} \quad \Rightarrow \quad \begin{cases} h_1 = 20 \text{ cm} \\ h_2 = 10 \text{ cm} \end{cases}$$

(b) The neutral axis passes through the gravity centre of the cross section; Therefore its first moment of area with respect to the z axis must be zero. Initially, the first moment of area of the web and flange are calculated separately:

$$\begin{aligned} \text{web : } m_{\text{web}}^e &= e [h_1 + (h_2 - e)] \left[\frac{1}{2} (h_1 + (h_2 - e)) - (h_2 - e) \right] \\ &= \frac{e}{2} [h_1^2 - (h_2 - e)^2] \\ \text{flange : } m_{\text{flange}}^e &= -e b (h_2 - \frac{e}{2}) \end{aligned}$$

and by adding both first moments of area:

$$m_{\text{web}}^e + m_{\text{flange}}^e = 0 \quad \Rightarrow \quad \frac{e}{2} [h_1^2 - (h_2 - e)^2] - eb (h_2 - \frac{e}{2}) = 0$$

where:

$$b = \frac{h_1^2 - (h_2 - e)^2}{2(h_2 - \frac{e}{2})} = 16.79 \text{ cm}$$

(c) The external bending moment is equal to the bending moment caused by normal stresses about the neutral axis. Let M_{web} and M_{flange} be the internal bending moments acting on the web and flanges, respectively.

$$\begin{aligned} \text{web} : \quad M_{\text{flange}} &= \left[\frac{1}{2} \sigma_{\text{max}}^c (h_1 e) \right] \frac{2}{3} h_1 + \left[\frac{1}{2} \left(\sigma_{\text{max}}^t \frac{h_2 - e}{h_2} \right) (h_2 - e) e \right] \frac{2}{3} (h_2 - e) \\ \text{flange} : \quad M_{\text{flange}} &= \left[\frac{1}{2} \sigma_{\text{max}}^t (h_2 b) \right] \frac{2}{3} h_2 - \left[\frac{1}{2} \left(\sigma_{\text{max}}^t \frac{h_2 - e}{h_2} \right) (h_2 - e) b \right] \frac{2}{3} (h_2 - e) \end{aligned}$$

$M_{\text{web}} = 14.548 \text{ kN} \cdot \text{m}$ and $M_{\text{flange}} = 7.584 \text{ kN} \cdot \text{m}$ are obtained by substituting values and by adding the both values, the moment of bending acting on the cross section can be obtained:

$$M_z = M_{\text{web}} + M_{\text{flange}} = 22.132 \text{ kN} \cdot \text{m}$$

Alternately, M_z can be calculated by applying Navier's law to any point of the section where the stress is known. First, the moment of inertia of the cross section with respect to the z axis must be calculated:

$$\begin{aligned} I_z &= \frac{1}{12} [h_1 + (h_2 - e)]^3 e + \\ &\quad + [h_1 + (h_2 - e)] e \left[\frac{1}{2} (h_1 + (h_2 - e)) - (h_2 - e) \right]^2 + \\ &\quad + \frac{1}{12} b e^3 + b e \left(h_2 - \frac{e}{2} \right)^2 \\ &= 4426.37 \text{ cm}^4 \end{aligned}$$

By Applying Navier's law, for instance, to the upper fibres, where the maximum compression stress takes place:

$$\sigma_{\text{max}}^c = -\frac{M_z h_1}{I_z} \implies M_z = 22.132 \text{ kN} \cdot \text{m}$$

(d) The maximum allowable moment is the one that produces a stress equal to σ_{adm} in the upper fibre of the cross section, since $h_1 = \max(v, v')$:

$$M_z^{\text{max}} = \min \left(\frac{\sigma_{\text{adm}} I_z}{h_1}, \frac{\sigma_{\text{adm}} I_z}{h_2} \right) = \frac{\sigma_{\text{adm}} I_z}{h_1} = 55.329 \text{ kN} \cdot \text{m}$$

Example 6.2.2

In a beam with a rectangular section of width b and height h (Figure 6.14), determine the maximum bending moment about its two principal axes of inertia if the allowable stress is σ_{adm} .

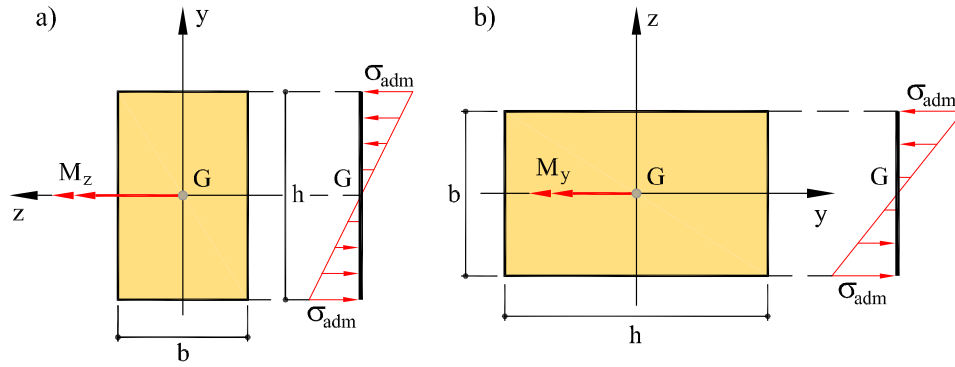


Figure 6.14: Example 6.2.2: A rectangular cross section under plane bending about its two principal inertia planes

(a) The maximum allowable moment M_z^{max} acting in the xy plane is the one that produces a stress equal to σ_{adm} in the extreme fibres of the beam ($y = \pm h/2$). So:

$$\sigma_{\text{adm}} = \frac{M_z^{\text{max}} \frac{h}{2}}{I_z} = \frac{M_z^{\text{max}} \frac{h}{2}}{\frac{1}{12} b h^3}$$

Therefore:

$$M_z^{\text{max}} = \frac{\frac{1}{12} b h^3}{\frac{h}{2}} \sigma_{\text{adm}} = \left(\frac{1}{6} b h^2 \right) \sigma_{\text{adm}} = W_z \sigma_{\text{adm}}$$

Note that $W_z = (1/6) b h^2 = (1/3)(A h/2)$ and therefore, $\eta_z = 1/3$.

(b) Similarly, by exchanging the dimensions b and h , the maximum allowable moment M_y^{max} acting in the xz plane is obtained. Thus:

$$M_y^{\text{max}} = \frac{\frac{1}{12} h b^3}{\frac{b}{2}} \sigma_{\text{adm}} = \left(\frac{1}{6} h b^2 \right) \sigma_{\text{adm}} = W_y \sigma_{\text{adm}}$$

Note that $W_y = (1/6) h b^2 = (1/3)(A b/2)$ and $\eta_y = 1/3$. Therefore relationship between the two maximum moments is:

$$\frac{M_z^{\max}}{M_y^{\max}} = \frac{\left(\frac{1}{6}bh^2\right) \sigma_{\text{adm}}}{\left(\frac{1}{6}hb^2\right) \sigma_{\text{adm}}} = \frac{W_z}{W_y} = \frac{h}{b}$$

Example 6.2.3

Determine the section moduli W_y, W_z as well as the geometrical efficiency coefficients η_y, η_z of the profile section depicted in Figure 6.15, with heights in cm.

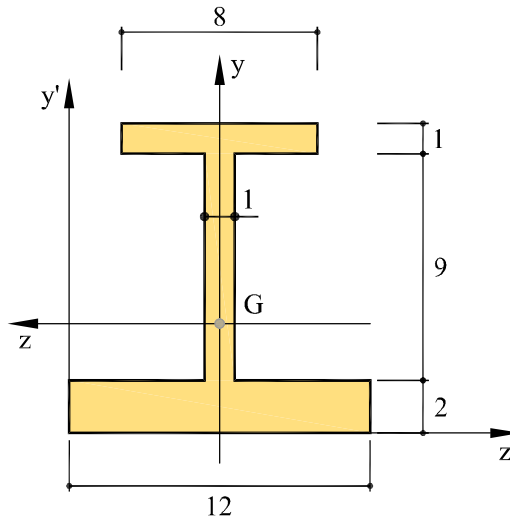


Figure 6.15: Cross section of Example 6.2.3

The profile sectional area is equal to :

$$A = 8 \cdot 1 + 9 \cdot 1 + 12 \cdot 2 = 41 \text{ cm}^2$$

The position of the gravity center is determined by:

$$\begin{aligned} y'_G &= \frac{\sum A_i y_i}{A} = \frac{1}{41} (8 \cdot 11.5 + 9 \cdot 6.5 + 24 \cdot 1) = 4.25 \text{ cm} \\ z'_G &= 6.0 \text{ cm} \quad (\text{symmetry axis}) \end{aligned}$$

The moments of inertia about the axes (y, z) are as follows:

$$\begin{aligned} I_y &= \frac{1}{12} 1 \cdot 8^3 + \frac{1}{12} 9 \cdot 1^3 + \frac{1}{12} 2 \cdot 12^3 \\ &= 331.42 \text{ cm}^4 \\ I_z &= \frac{1}{12} 8 \cdot 1^3 + \frac{1}{12} 1 \cdot 9^3 + \frac{1}{12} 12 \cdot 2^3 \\ &\quad + 1 \cdot 8 \cdot (7.25)^2 + 1 \cdot 9 \cdot (2.25)^2 + 2 \cdot 12 \cdot (3.25)^2 \\ &= 788.98 \text{ cm}^4 \end{aligned}$$

The section modulus and geometrical efficiency coefficients about the axes (y, z) are then equal to:

$$\begin{aligned}
 W_y &= \frac{I_y}{|z|_{\max}} = \frac{331.42}{6} = 55.24 \text{ cm}^3 & ; & \quad \eta_y = \frac{W_y}{\frac{b}{A \frac{1}{2}}} = \frac{55.24}{41 \cdot 6} = 0.225 \\
 W_z &= \frac{I_z}{|y|_{\max}} = \frac{788.98}{7.75} = 101.80 \text{ cm}^3 & ; & \quad \eta_z = \frac{W_z}{\frac{h}{A \frac{1}{2}}} = \frac{101.80}{41 \cdot 6} = 0.414
 \end{aligned}$$

Example 6.2.4

Determine the section modulus of two overlapping rectangular cross-section beams $b \times h$ (total depth of $2h$) based on whether they work (a) jointly or (b) independently.

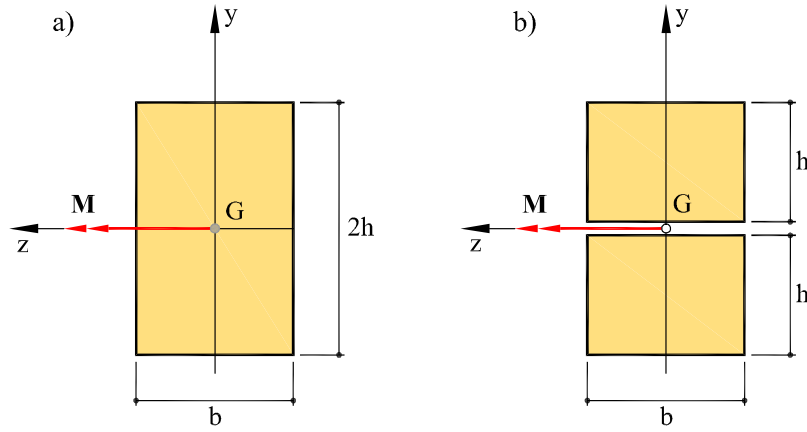


Figure 6.16: Cross sections of Example 6.2.4

For the two beams working jointly, the maximum moment is determined by the moment that produces stresses equal to the allowable stress σ_{adm} in the outer fibres of the composite beam (Figure 6.17a), that is:

$$M_a^{\text{max}} = 2 \cdot \frac{1}{2} \sigma_{\text{adm}} h b \cdot \frac{2}{3} h = \frac{2}{3} b h^2 \sigma_{\text{adm}}$$

so, the section modulus is :

$$W_a = \frac{M_a^{\text{max}}}{\sigma_{\text{adm}}} = \frac{2}{3} b h^2$$

For the two overlapping beams working independently, the maximum moment is double that the one which produces stresses equal to σ_{adm} in the outer fibres of each beam (Figure 6.17b); that is:

$$M_b^{\text{max}} = 2 \left[2 \cdot \frac{1}{2} \sigma_{\text{adm}} \frac{h}{2} b \cdot \frac{2}{3} \frac{h}{2} \right] = \frac{1}{3} b h^2 \sigma_{\text{adm}}$$

so, the section modulus is equal to:

$$W_b = \frac{M_b^{\text{max}}}{\sigma_{\text{adm}}} = \frac{1}{3} b h^2$$

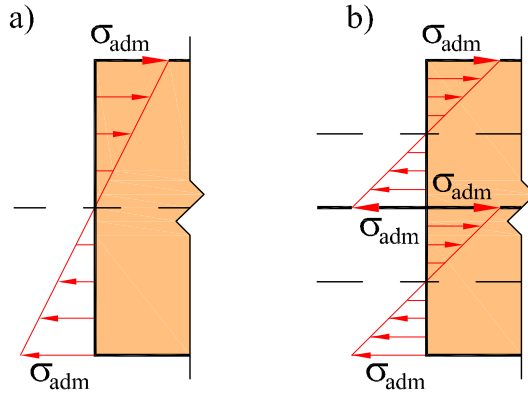


Figure 6.17: Stress distribution for Example 6.2.4

which is double the modulus of each individual beam but half of the modulus of the composite beam.

Figure 6.18 shows how the two beams behave differently under pure bending depending on whether they are working jointly or independently.

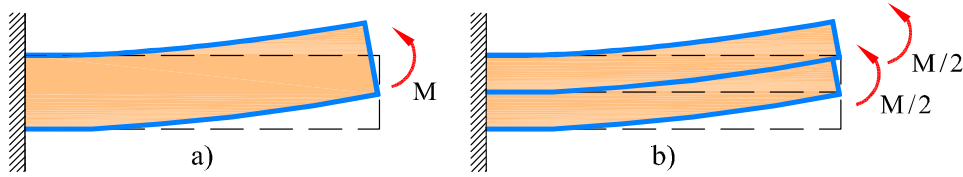


Figure 6.18: Deflections of Example 6.2.4

6.3 Non-uniform bending

Non-uniform bending, is characterized by a variable bending moment, is the most common type of bending in horizontal beams subjected to forces normal to their axis. As shown in Figure 6.19, the non-uniform bending moment requires a shear force acting on the cross sections of the beam for equilibrium of the differential beam elements.

$$T_y = - \frac{dM_z}{dx} \quad (6.23)$$

Therefore, the equations of internal equilibrium are similar to Eqs. (6.1) corresponding to pure bending, with the difference that in non-uniform bending the shear T_y is not zero.

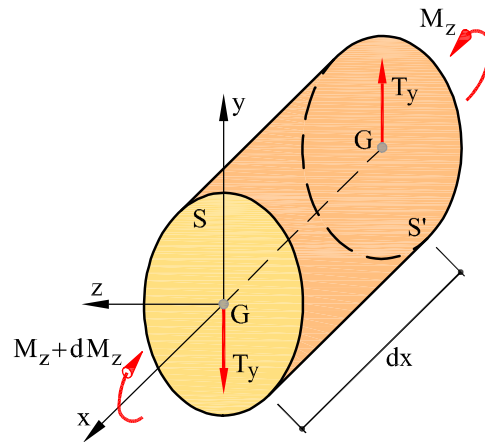


Figure 6.19: Bending moment and shear equilibrium in non-uniform bending

The shear stress leads to the appearance of angular distortions and tangential stresses in the cross section, as will be seen in the corresponding Chapter. In addition to the longitudinal fiber elongation typical of bending, these angular distortions cause a certain deformation of the plane cross sections. Figure 6.20 shows this phenomenon, which is known as *warping* of the cross-sections. The results obtained for pure bending are based on the hypothesis that the plane section remain plane, i.e. without warping. Therefore, the question arises as to whether these results can be extended to a warping case.

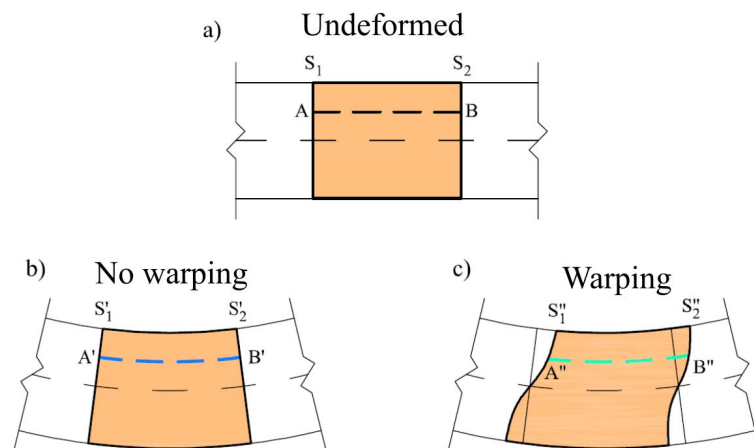


Figure 6.20: Elongation of the fibers in simple bending: (a) undeformed, (b) deformed with out warping (c) deformed with warping

In non-uniform bending, the generalized Bernoulli-Navier hypothesis is adopted; it states that “two infinitely close plane cross sections undergo warping in the deformation, but one of them can be overlapped on the other by means of a translation and a rotation.”

This generalised hypothesis admits the warping of plane sections, but disregards the relative warping between two neighbour sections. This hypothesis is valid since the longitudinal deformations produced by the shear force are usually significantly smaller than those produced by the bending moment. The approximation becomes more accurate as the ratio between the edge and span of the beam decreases, that is, the more slender the beam is.

For slender beams, the effect of warping is negligible, whereas it can be significant for beams of moderate or large depth. If the generalised assumption is accepted, then, as depicted in Figure 6.20, the elongation of a fiber such as AB , which actually deforms at $A'B'$ (with warping), can be considered to be identical to the elongation that would occur if the deformation were as $A'B'$ (without warping). Since all expressions for pure bending are based on the magnitude of the longitudinal elongations of the fibers, it is clear that if the generalised Bernoulli-Navier deformation hypothesis is accepted, all expressions for the case of pure plane bending can be extended to the more general case of non-uniform bending.

Thus, for example, the expression obtained for the stress law (linear distribution) is valid:

$$\sigma_x = -\frac{M_z}{I_z}y \quad (6.24)$$

However, it should be noted that in non-uniform bending the deformation of the axis of the beam is not an arc of circumference, since the curvature changes from section to section as the acting bending moment changes (Figure 6.21). In non-uniform plane bending, the deflection of the midline is a plane curve, but not of constant curvature.

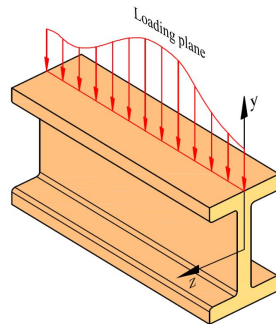


Figure 6.21: Non-uniform bending with constant loading plane

Example 6.3.1

For the cantilever beam subjected to the loads shown in Figure 6.22: (a) determine the maximum bending moment value and location, (b) determine and depict the normal stress distribution and (c) calculate the maximum value that the load P can reach, assuming that the value of the allowable stress is σ_{adm} for both tensile and compression.

Data: $l = 6 \text{ m}$, $P = 100 \text{ kN}$, $a = 40 \text{ cm}$, $b = 30 \text{ cm}$, $h = 50 \text{ cm}$, $e = 1.5 \text{ cm}$, $\sigma_{\text{adm}} = 260 \text{ MPa}$.

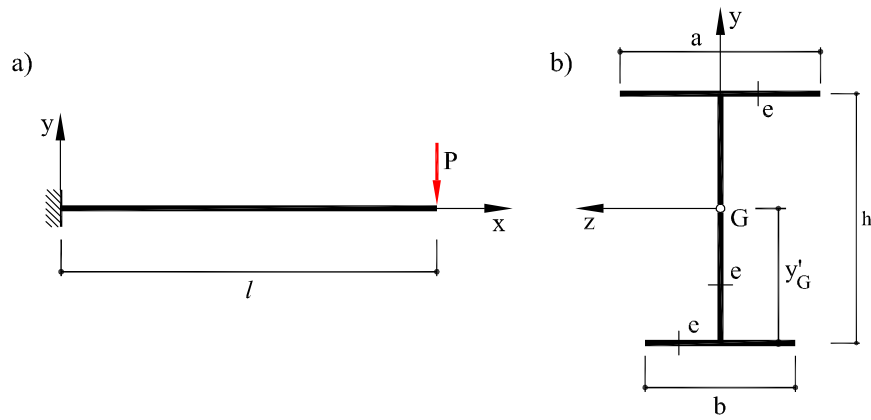


Figure 6.22: Beam and its corresponding cross section of Example 6.3.1

The cross sectional area is equal to:

$$A = (40 + 50 + 30)1,5 = 180 \text{ cm}^2$$

The section is symmetrical about the y axis. Therefore, the centre of gravity is located above it and at a distance from the bottom edge equal to (Figure 6.22):

$$y'_G = \frac{\sum A_i y_i}{A} = \frac{40 \cdot 1,5 \cdot 50 + 50 \cdot 1,5 \cdot 25}{180} = 27 \text{ cm}$$

The principal moment of inertia I_z is:

$$\begin{aligned} I_z &= 40 \cdot 1,5 \cdot 23^2 + 30 \cdot 1,5 \cdot 27^2 + \frac{1}{12} 1,5 \cdot 50^3 + 50 \cdot 1,5 \cdot 2^2 \\ &= 80\,200 \text{ cm}^4 \end{aligned}$$

(a) *Internal resultants diagrams*

The internal resultants diagrams due to the load P are shown in Figure 6.23. It can be seen that the beam is subjected to a non-uniform bending state (bending moment and shear force). The maximum bending moment occurs at the clamped section and its value is:

$$M_z^{\max} = M_z(x = 0) = -P \cdot l = -100 \cdot 6 = -600 \text{ kN} \cdot \text{m}$$

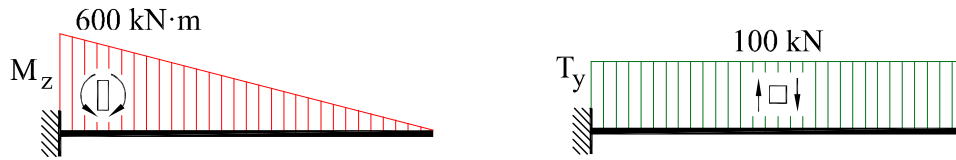


Figure 6.23: Internal resultants diagrams for Example 6.3.1

(b) *Stress distribution and maximum stresses values*

In non-uniform plane bending, the neutral axis coincides with the principal axis z and the maximum stress occurs at the fibers farthest from it, i.e. at the upper and lower fibers (Figure 6.24):

$$\begin{aligned} \sigma_T &= -\frac{M_z}{I_z} y_T = \\ &= -\frac{(-600 \cdot 10^3)}{8.02 \cdot 10^{-4}} (0.23) = +172 \text{ MPa (T)} \\ \sigma_C &= -\frac{M_z}{I_z} y_C = \\ &= -\frac{(-600 \cdot 10^3)}{8.02 \cdot 10^{-4}} (-0.27) = -202 \text{ MPa (C)} \end{aligned}$$

The lowest edge of the section has the maximum absolute stress, compressive. Both stresses are less than the allowable stress, $\sigma_{\text{adm}} = 260 \text{ MPa}$.

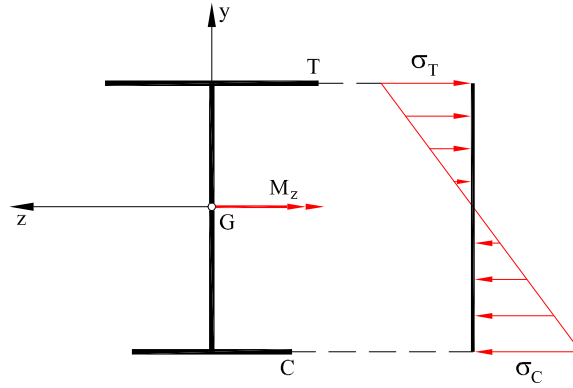


Figure 6.24: Normal stresses distribution of the cross section

(c) *Maximum load value P*

The maximum stress in absolute value occurs at the lower fibers. The maximum value that the load P can reach is determined from the expression:

$$\sigma_{\text{inf}} = -\frac{M_z}{I_z} y_{\text{inf}} \leq \sigma_{\text{adm}} = 260 \text{ MPa}$$

Taking into account that $M_z = -P \cdot 6$, it is:

$$-\frac{(-P \cdot 6 \cdot 10^3)}{8.02 \cdot 10^{-4}} (-0.27) = 260 \text{ MPa} \quad \Rightarrow \quad P_{\text{max}} = 129 \text{ kN}$$

Example 6.3.2

The steel beam with a hollow rectangular cross-section shown in Figure 6.25 is suspended from two points for transport to the construction site. Determine the maximum stress due to self-weight if the specific weight of the material is $\gamma = 78 \text{ kN/m}^3$, the distance between the suspension points is l , and the distance from the suspension points to the ends of the beam is s .

Data: $l = 6 \text{ m}$, $s = 4 \text{ m}$, $a = 20 \text{ cm}$, $b = 40 \text{ cm}$, $e = 1.5 \text{ cm}$.

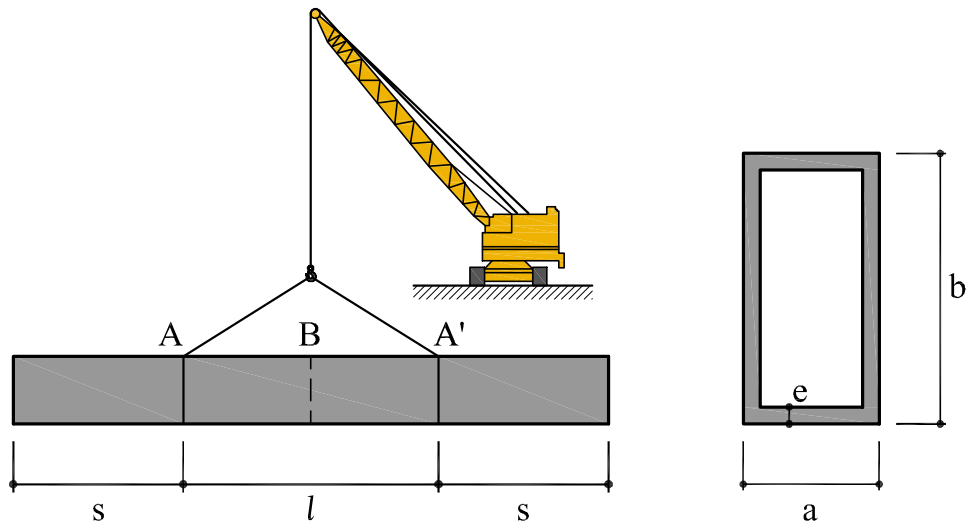


Figure 6.25: Beam and its corresponding cross section for Example 6.3.2

(a) *Geometric characteristics of the cross-section*

The area of the cross section is: $A = (20 \cdot 40) - (17 \cdot 37) = 171 \text{ cm}^2$

The section is doubly symmetric. Therefore, the axes y, z are the principal axes of the section. The principal moment of inertia I_z is:

$$I_z = \frac{20 \cdot 40^3}{12} - \frac{17 \cdot 37^3}{12} = 34908 \text{ cm}^4$$

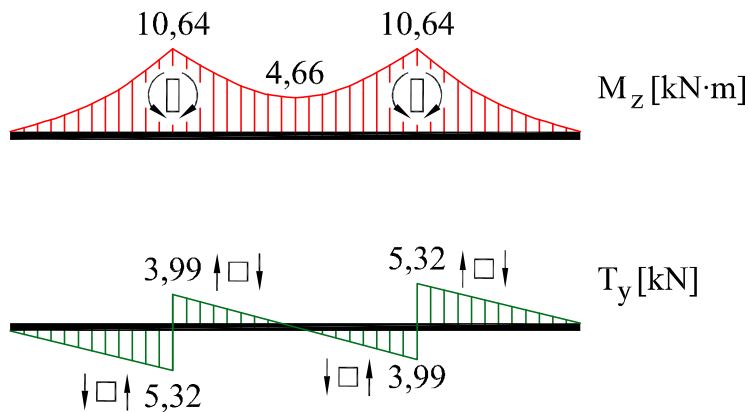


Figure 6.26: Internal resultants diagrams for Example 6.3.2

(b) *Internal resultants diagrams*

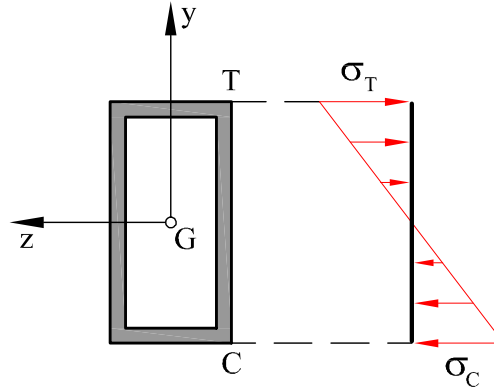


Figure 6.27: Normal stress distribution due to beam transportation

The self-weight of the beam per unit length is:

$$p = \gamma \cdot A = 78 \cdot 171 \cdot 10^{-4} = 1.33 \text{ kN/m}$$

Figure 6.26 shows the bending moment diagram resulting from the uniformly distributed self-weight load. At the cross-sections corresponding to the attachment points A and A' , the bending moment is:

$$M_{A,A'} = -\frac{1.33 \cdot 4^2}{2} = -10.64 \text{ kN}\cdot\text{m}$$

and in the central cross section of the beam (B) the bending moment is:

$$M_B = -10.64 + \frac{1.33 \cdot 6^2}{8} = -4.66 \text{ kN}\cdot\text{m}$$

(c) *Stress distribution and maximum stresses values*

The maximum bending moment occurs in cross section A , and the stress distribution in this section can be seen in Figure 6.27. The maximum stress value occurs at the upper and lower fibers and its value is:

$$\begin{aligned} \sigma_T &= -\frac{M_z}{I_z} y_T = -\frac{(-10.64 \cdot 10^3)}{3.4908 \cdot 10^{-4}} (+20 \cdot 10^{-2}) = +6.1 \text{ MPa} \\ \sigma_C &= -\frac{M_z}{I_z} y_C = -\frac{(-10.64 \cdot 10^3)}{3.4908 \cdot 10^{-4}} (-20 \cdot 10^{-2}) = -6.1 \text{ MPa} \end{aligned}$$

6.4 Composed plane bending

A cross-section is subjected to composed plane bending when the force resultants are reduced to an axial force N and a bending moment M contained in the principal plane of inertia of the cross section (Figure 6.3b).

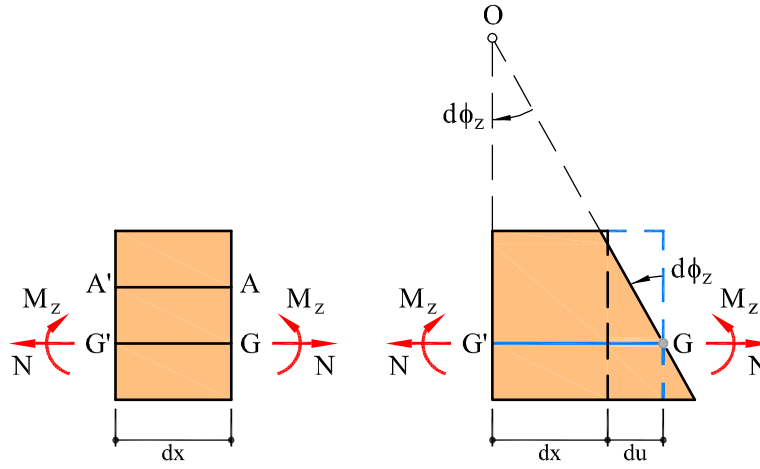


Figure 6.28: Deformation of a differential beam element subjected to composed plane bending.

Consider the case of a beam with a straight axis subjected to composed plane bending. Its cross sections are subjected to an axial force of value N and a bending moment that, for development purposes, is assumed to be contained in the principal plane of inertia containing the x and y axes, resulting in the bending moment vector having only the M_z component. Depending on whether the bending moment M_z is constant or varies along the length of the beam, the shear force T_y may or may not be zero.

Both the axial force and the bending moment produce normal stresses on the cross-sections of the beam, and their distribution can be studied by applying the principle of superposition to the results obtained in the sections corresponding to the studies of the axial force and bending moment, respectively.

As shown in Figure 6.28, the deformation due to axial force follows the Bernoulli hypothesis (plane sections are translated while remaining plane and parallel to themselves), whereas the bending deformation follows the Bernoulli-Navier hypothesis (the plane sections rotate around the neutral axis while remaining plane and perpendicular to the deformation of the beam axis).

By superposition, a section of the differential beam element undergoes a translation du , parallel to the x -axis (due to the axial force), and a rotation $d\phi_z$ around the z -axis (due to the bending force), relative to another plane cross section infinitely close to it.

As can be seen in Figure 6.28, the combination of the two movements, a translation and a rotation, is equivalent to a single rotation about an axis parallel to z , whose position (point O) will be determined later.

The normal stress acting on a generic point of the section is also the sum of the axial and bending stresses. According to deformation hypotheses and

assuming, moreover, that Hooke's law is satisfied, the stress at a point Q placed at a distance y from the z -axis is:

$$\sigma_x = \frac{N}{A} - \frac{M_z y}{I_z} \quad (6.25)$$

Consequently, as depicted in Figure 6.29, the stress distribution is linear on y , and the maximum stresses occur in the fibres of the cross section farthest from the z -axis, and their value are:

$$\begin{aligned} \sigma_v &= \frac{N}{A} - \frac{M_z v}{I_z} \\ \sigma_{v'} &= \frac{N}{A} + \frac{M_z v'}{I_z} \end{aligned} \quad (6.26)$$

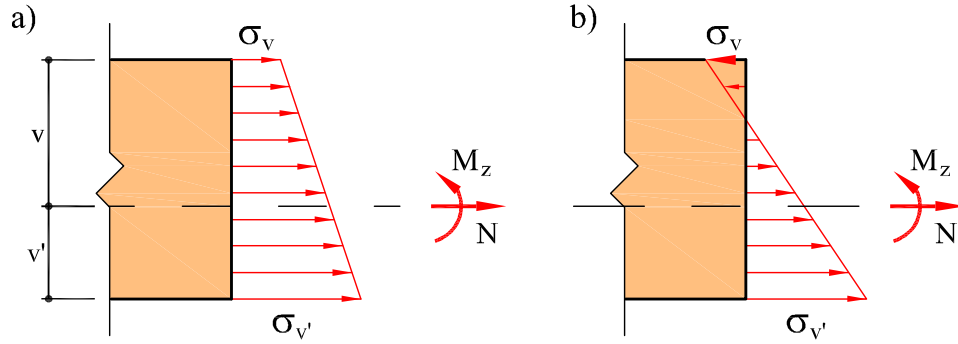


Figure 6.29: Normal stresses distribution under composed plane bending

where v and v' are the distances (absolute value) from the z -axis to the two extreme fibers, respectively. Note that the stresses in the extreme fibers may or may not have the same sign as the axial force, depending on the relative magnitude of the axial and bending stress terms.

Thus, in Figure 6.29a, all the stresses have the same sign as the axial force, whereas in Figure 6.29b, the bending force is large enough to produce stresses with opposite sign to the axial force.

A system of forces composed by a moment M_z and an axial force N applied at the barycenter of cross section G is statically equivalent to the eccentric force N applied at point P at a distance e_y from the barycenter, measured along the y -axis, such that $M_z = -N e_y$ (a positive normal force with a positive eccentricity is statically equivalent to a negative moment M_z). Figure 6.30 describes such scenario.

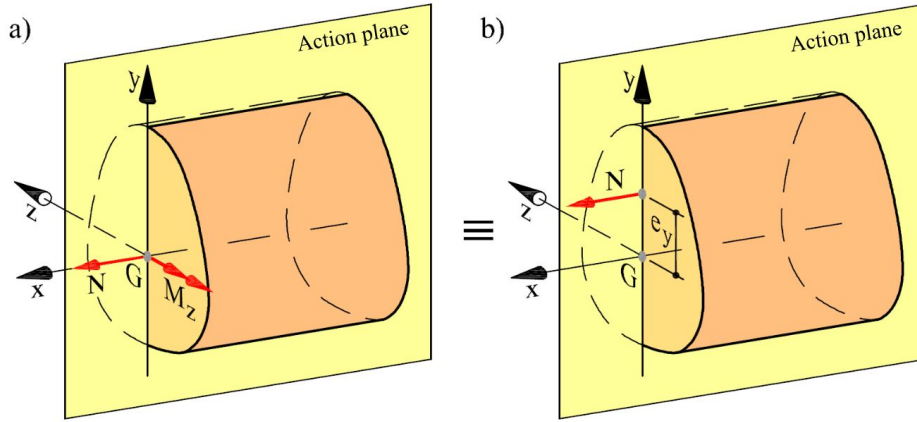


Figure 6.30: Composed plane bending as an eccentric normal force

The point P is the centre of pressure. The eccentricity is the distance e_y between the barycenter and the centre of pressure. Therefore, the stress at a generic point Q of the section placed at a distance y from the principal axis z can be expressed as:

$$\sigma_x = \frac{N}{A} - \frac{M_z y}{I_z} \quad (6.27)$$

$$= \frac{N}{A} + \frac{N e_y y}{I_z} \quad (6.28)$$

$$= \frac{N}{A} \left(1 + \frac{e_y y}{r_z^2} \right) \quad (6.29)$$

where $r_z = \sqrt{I_z/A}$ is the radius of gyration the cross section about the z axis :

According to its definition, the neutral axis is the geometric locus where points are subjected to zero normal stress; so its expression is:

$$1 + \frac{e_y y}{r_z^2} = 0 \quad \Rightarrow \quad y = -\frac{r_z^2}{e_y} \quad (6.30)$$

As can be seen, the neutral axis nn' is a straight line that is perpendicular to the z -axis and parallel to the trace of the action plane, and its position does not depend on the value of the force N , but on the eccentricity e_y . Note that the application point of the normal force and the neutral axis must be on opposite sides with respect the z -axis. Also, the smaller the eccentricity, the greater the ordinate of the neutral axis, i.e. the greater the effect of the axial force in relative terms, the further the neutral axis

is from the z -axis (as shown in Figure 6.29a); conversely, the greater the eccentricity, the smaller the ordinate of the neutral axis, i.e. the greater the effect of the bending in relative terms, the closer the neutral axis is to the z axis (e.g. the case in Figure 6.29b).

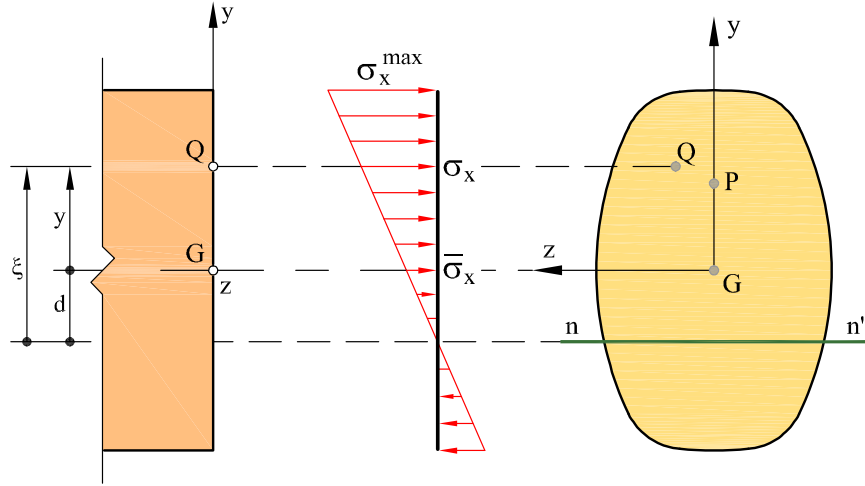


Figure 6.31: Stress distribution under composed plane bending

Defining ξ as the distance between a general point Q and the neutral axis (Figure 6.31), where ξ is positive if point Q and the barycentre G are on the same side of the neutral axis and negative otherwise, equation (6.27c) can be expressed as:

$$\sigma_x = \frac{N}{A} \frac{\xi}{d} \quad (6.31)$$

Thus, looking at Figure 6.31, and referring to $\bar{\sigma}_x = N/A$ as the stress at the barycentre, by similarity of triangles, the following can be concluded:

$$\frac{\sigma_x}{\bar{\sigma}_x} = \frac{\xi}{d} \Rightarrow \sigma_x = \bar{\sigma}_x \frac{\xi}{d} = \frac{N}{A} \frac{\xi}{d} \quad (6.32)$$

The expression (6.31) is alternative to those obtained previously (Eqs. (6.27) and (6.29) and allows direct calculation of stresses in composed bending when the position of the neutral axis is known.

The maximum stress occurs in the fibers farthest away from the neutral axis and its value is:

$$\sigma_x^{\max} = \frac{N}{A} \frac{\xi_{\max}}{d} \quad (6.33)$$

Example 6.4.1

A simple supported beam shown in Figure 6.32 has a rectangular cross section $b \times h$ ($A = bh$, $I_z = (1/12)bh^3$) and is subjected to a pair of compressive forces F , contained in the vertical xy symmetry plane of the beam and applied in the support cross sections, above the beam axis with an eccentricity $e_y = e$. Determine the stress distribution in the cross sections of the beam, their maximum and minimum values, and the maximum eccentricity value for a beam subjected to total compression.

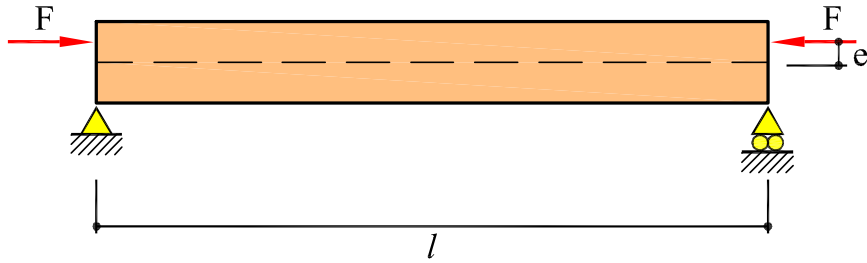


Figure 6.32: Beam of the Example 6.4.1

Considering the external applied loads of the the beam, the internal resultants diagrams are uniform, as shown in Figure 6.33, and can be expressed as :

$$N = -F \quad ; \quad M_z = F e$$

The beam is therefore subjected to plane composed bending. The stress distribution is linear and depends only on the distance of the fibres relative to the z -axis. Thus, the maximum and minimum values of the stresses occur at the extreme fibres ($y = \pm h/2$) and are:

$$\begin{aligned} \sigma_v &= -\frac{F}{bh} - \frac{F e \frac{h}{2}}{\frac{bh^3}{12}} = -\frac{F}{bh} \left(1 + \frac{6e}{h} \right) \\ \sigma_{v'} &= -\frac{F}{bh} + \frac{F e \frac{h}{2}}{\frac{bh^3}{12}} = -\frac{F}{bh} \left(1 - \frac{6e}{h} \right) \end{aligned}$$

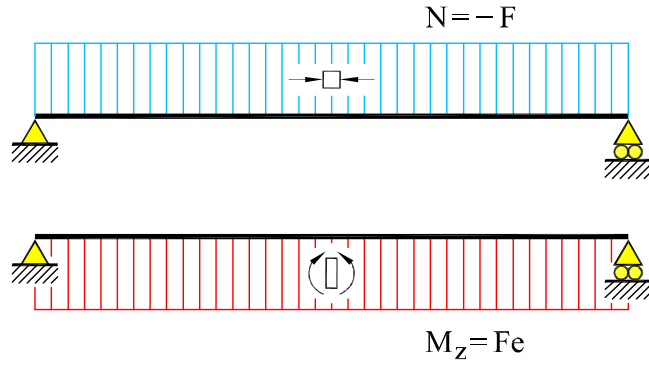
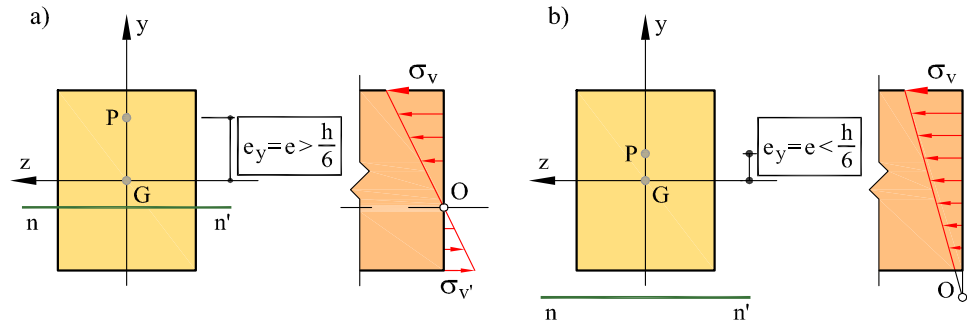


Figure 6.33: Internal resultants diagrams of Example 6.4.1

For compressive load F and positive eccentricity, the stress in the upper fibre, σ_v , is necessarily compressive. The stress in the lower fibre, $\sigma_{v'}$, is compressive if $e < h/6$ and tensile otherwise. Figures 6.34a and b depict the corresponding stress distributions.

The neutral axis is a line parallel to the z -axis, with the equation:

$$y = -\frac{r_z^2}{e_y} = -\frac{I_z}{Ae_y} = -\frac{\frac{1}{12}bh^3}{bhe} = -\frac{1}{12} \frac{h^2}{e}$$

Figure 6.34: Stress distributions according to e_y value

For $e < h/6$, $-y > h/2$, implying that the neutral axis does not intersect the section. For $e > h/6$, $-y < h/2$, implying that the neutral axis does intersect the section.

Example 6.4.2

The simply supported beam in Figure 6.35 has a rectangular cross-section $b \times h$ ($A = bh$, $I_z = (1/12)bh^3$) and is subjected to a uniformly distributed downward vertical load p and a pair of compressive forces F applied at the centres of gravity of the support cross sections. Determine: (a) the curve of pressure of the beam, (b) the minimum value of the forces F so that there are no tensile stresses on the beam, and (c) the stress distribution in sections $x = 0, l/4$ and $l/2$ for this value of F .

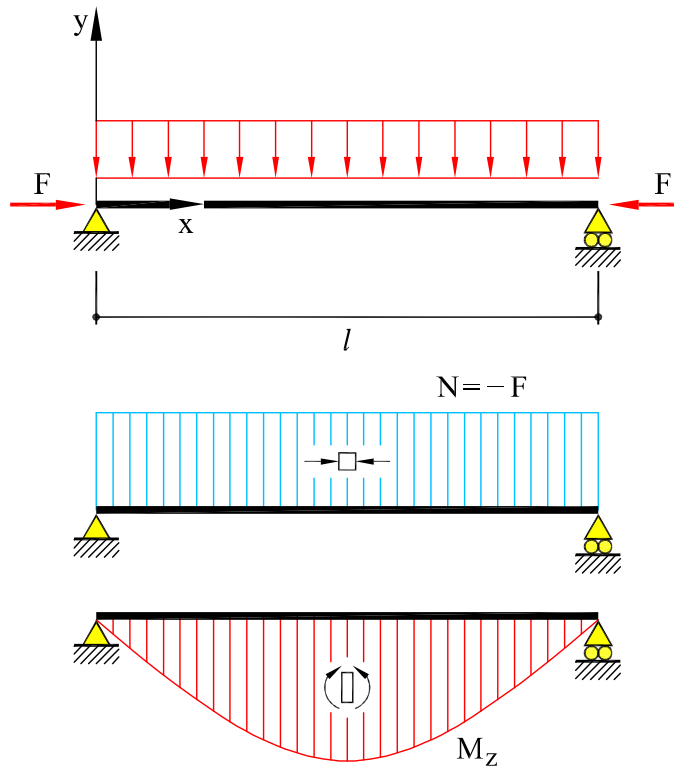


Figure 6.35: Beam of the Example 6.4.2 and its corresponding internal resultants diagrams

(a) *Curve of pressure of the beam*

The internal resultants diagrams, as shown in Figure 6.35, for the external loads applied to the beam are:

$$N(x) = -F \quad ; \quad M_z(x) = \frac{p}{2} x (l - x)$$

The beam is therefore subjected to plane composed bending. The pressure is a curve in the xy plane, whose expression is determined by the position of the centres of pressure of each section. So, it is:

$$e_y(x) = -\frac{M_z(x)}{N(x)} = \frac{p}{2F} x (l - x)$$

Thus, it is a parabola with maximum value $e_y(x = l/2) = \frac{pl^2}{8F}$.

(b) *Minimum value of force F*

The cross section most susceptible to have tensile stresses is the central section, which supports the greatest bending moment, i.e. has the greatest eccentricity. According to the previous Example, if the eccentricity in the central section is less than $h/6$, there is no tension in the lower fibers of the central section.

$$e_y(x = l/2) = pl^2/8F < \frac{h}{6} \implies F > \frac{3pl^2}{4h}$$

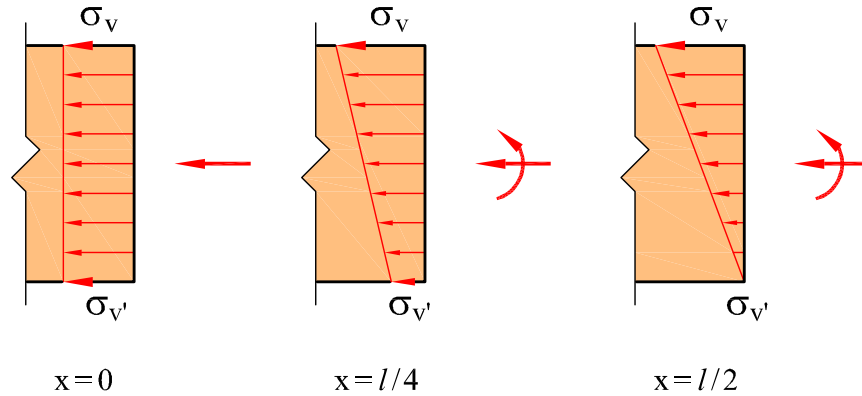


Figure 6.36: Stress distributions in the cross sections $x = 0, l/4$ y $l/2$

(c) *Stress distributions*

Figure 6.36 depicts the stress distributions at cross-sections $x = 0, l/4$ and $l/2$ for $F = 3pl^2/4h$. At section $x = 0$, the bending moment is zero and $e_y = 0$, so the stress distribution is uniform and has a value of :

$$\sigma_v(x = 0) = \sigma_{v'}(x = 0) = -\frac{F}{bh} = -\frac{3pl^2}{4bh^2}$$

At section $x = l/4, M_z = (3/32)pl^2$ and $e_y = (3/32)(pl^2/F) = h/8$ can be determined. The extreme stresses are:

$$\sigma_v(x = l/4) = -\frac{F}{bh} \left(1 + \frac{6e}{h} \right) = -\frac{7}{4} \frac{F}{bh} = -\frac{21}{16} \frac{pl^2}{bh^2}$$

$$\sigma_{v'}(x = l/4) = -\frac{F}{bh} \left(1 - \frac{6e}{h} \right) = -\frac{1}{4} \frac{F}{bh} = -\frac{3}{16} \frac{pl^2}{bh^2}$$

Since $e < h/6$, all stresses are compressive. //

In the section $x = l/2$, $M_z = pl^2/8$ and $e_y = pl^2/8F = h/6$ are established. The extreme stresses are:

$$\sigma_v(x = l/2) = -\frac{F}{bh} \left(1 + \frac{6e}{h} \right) = -2 \frac{F}{bh} = -\frac{3}{2} \frac{pl^2}{bh^2}$$

$$\sigma_{v'}(x = l/2) = -\frac{F}{bh} \left(1 - \frac{6e}{h} \right) = 0$$

and it is checked that there are no tensile stresses at any point on the beam.

Example 6.4.3

A column of rectangular cross-section $b \times h$ ($30 \times 40 \text{ cm}^2$) and length $l = 3 \text{ m}$ supports a slightly inclined compressive force $F = 100 \text{ kN}$. (See Figure 6.37) Determine: (a) Internal resultants diagram (b) the neutral axis position in each section; and (c) the maximum normal stresses in each cross-section.

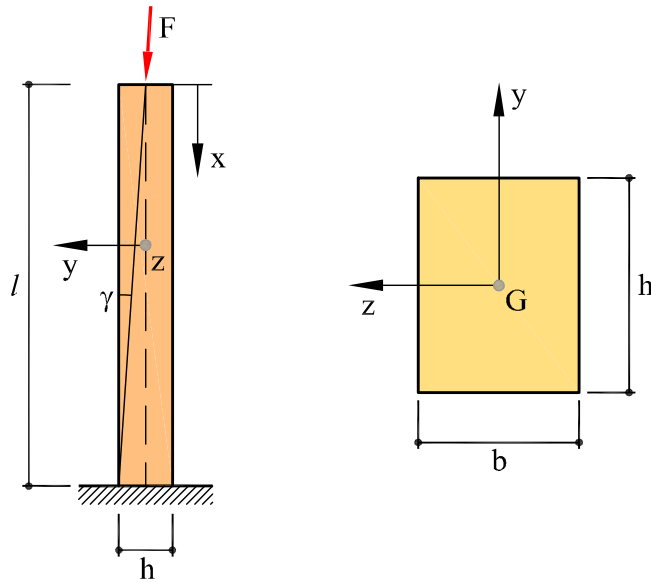


Figure 6.37: Column of Example 6.4.3 and its corresponding cross-section

The geometrical characteristics of the section are:

$$A = 30 \cdot 40 = 1,200 \text{ cm}^2 \quad ; \quad I_z = \frac{1}{12} 30 \cdot 40^3 = 160,000 \text{ cm}^4$$

and, therefore, $r_z^2 = I_z/A = 133.33 \text{ cm}^2$. In addition, the angle of inclination of a force, $\gamma = \arctan(h/2l) = \arctan 1/15 \simeq 1/15$, which can be considered small, is known.

(a) *Internal resultants of the beam*

The cross sections are subjected to plane composed bending moment, with the centre of pressure located on the y -axis of each cross section. The eccentricity of the force as a function of the distance x from the cross section under consideration to the top of the column is:

$$e_y(x) = x \tan \gamma = \frac{h}{2} \frac{x}{l}$$

Therefore, the internal forces diagrams are:

$$\begin{aligned} N(x) &= -F \cos \gamma \simeq -F \\ T_y(x) &= +F \sin \gamma \simeq +F \frac{h}{2l} \\ M_z(x) &= +F \cos \gamma e_y(x) \simeq +F \frac{h}{2} \frac{x}{l} \end{aligned}$$

Along the length of the beam, the axial and shear forces are constant, whereas the bending force increases linearly with x .

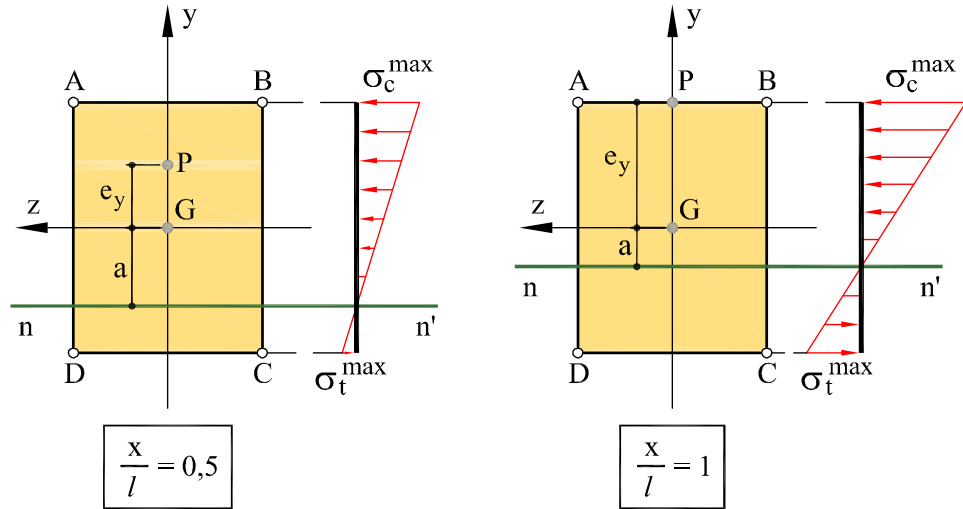


Figure 6.38: Results of Example 6.4.3

(b) *Neutral axis position in each section*

The neutral axis of each section is parallel to each section's z-axis and intersects the y-axis at the ordinate:

$$a(x) = -\frac{r_z^2}{e_y(x)} = -6.66 \frac{l}{x} \text{ cm}$$

Therefore, the neutral axis approaches the centre of gravity as the ratio x/l , increases, as shown in Figure 6.38.

(c) *Maximum stresses in each section*

The stresses at any point on a beam are as follows:

$$\sigma_x = \frac{N}{A} \left(1 + \frac{e_y y}{r_z^2} \right) = \frac{N}{A} \left(1 - \frac{y}{a(x)} \right)$$

Knowing the position of the neutral axis, it is easy to determine that the stress distributions for sections $x/l = 0, 5$ and $x/l = 1$. will be as depicted in Figure 6.38. Maximum compressive stresses occur in the upper fibres (such as *A* or *B*) and its values are:

$$\sigma_c^{\max}(x) = \frac{N}{A} \left(1 + \frac{20}{6.666} \frac{x}{l} \right) = \frac{N}{A} \left(1 + 3 \frac{x}{l} \right)$$

and then $\sigma_c^{\max}(x/l = 0.5) = -2.083 \text{ MPa}$ and $\sigma_c^{\max}(x/l = 1) = -3.333 \text{ MPa}$. The maximum tensile stresses occur in the lower fibres (such as *C* or *D*), and are:

$$\sigma_t^{\max}(x) = \frac{N}{A} \left(1 - \frac{20}{6.666} \frac{x}{l} \right) = \frac{N}{A} \left(1 - 3 \frac{x}{l} \right)$$

and, thus, $\sigma_t^{\max}(x/l = 0.5) = 0.4167 \text{ MPa}$ and $\sigma_t^{\max}(x/l = 1) = 1.667 \text{ MPa}$.

Chapter 7

Skew bending

7.1 Introduction

An element is subjected to *plane bending* when its cross-sections are stressed by a bending moment M contained in a principal plane of flexion. This is the case of mid plane straight beams, which are beams with a vertical longitudinal plane of symmetry bending in such a plane. In the most general case, *skew bending* occurs when a beam is subjected to a bending moment that is applied in a plane that is not a principal plane of inertia of the cross-section. If the acting moment is constant in all the sections of the element, there is *pure skew bending* and in the case that the moment is variable, there is *simple skew bending*, and is accompanied by shear stress.

When an axial force accompanies the action of the bending moment, traction or compression, there is composed skew bending. In this Chapter the study of pure skew bending is addressed, to then generalize the results obtained to the cases of simple skew bending and composed skew bending.

7.2 Pure skew bending

An element is subjected to *pure skew bending* when a uniform bending moment acts on its cross-sections along its length, contained in a plane other than the principal planes of bending of the piece, that is, such that the moment vector does not coincide with any of the principal axes of inertia of the cross-sections, as shown in Figure 7.1.

Given that the different force resultants that act on a section are the forces and moments resulting from the stresses that act on this section, in the case of *pure skew bending* with bending moment M constant of components M_y and M_z , the equations that are fulfilled are:

$$\begin{aligned}
N &= \int_S \sigma_x dS = 0 & ; & \quad T_y = \int_S \tau_{xy} dS = 0 \\
T_z &= \int_S \tau_{xz} dS = 0 & ; & \quad M_t = \int_S (\tau_{xz} y - \tau_{xy} z) dS = 0 \quad (7.1) \\
M_y &= \int_S \sigma_x z dS & ; & \quad M_z = - \int_S \sigma_x y dS
\end{aligned}$$

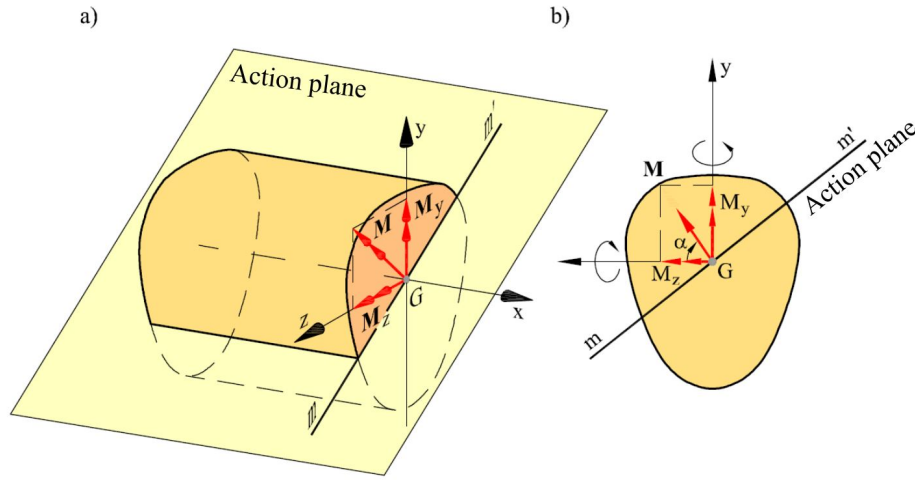


Figure 7.1: Skew bending referred to the principal axes of the cross-section

7.2.1 Stresses in pure skew bending

Let be a straight element subjected to pure skew bending by the action of a pair of moments of magnitude M applied at its ends and contained in a plane called *loading or action plane*, or *with trace*, mm' on the section plane (see Figure 7.1a). Splitting the bending moment vector M (perpendicular to the loading plane) into its two components according to the principal axes of inertia of the section, the components of the moment vector according to such axes are: $M_z = M \cos \alpha$ and $M_y = M \sin \alpha$, where M is the magnitude of the moment and α is the clockwise angle formed by the moment vector with the z axis (see Figure 7.1b).

Next, the effects produced separately by the components M_z and M_y are studied, to then apply the principle of superposition.

The component M_z acts in the plane xy and produces on a point of the section of coordinates (y, z) a tension normal equal to (see Eq. (6.12)):

$$\sigma_x = -\frac{M_z}{I_z} y \quad (7.2)$$

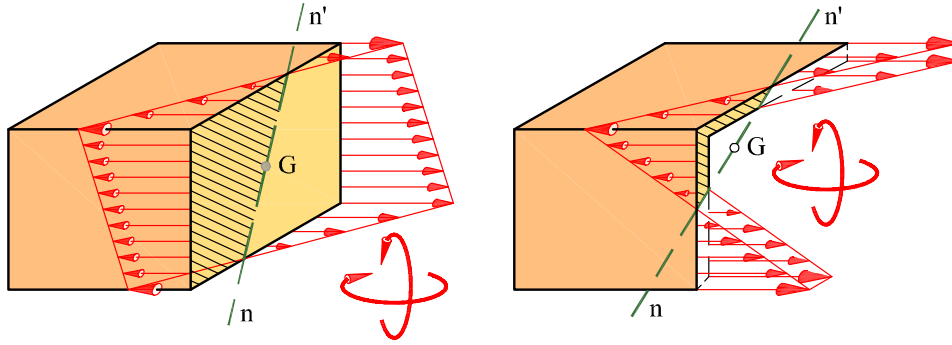


Figure 7.2: Stress distribution in skew bending

according to the sign conventions established for moments and stresses (that is, positive moments M_z compress the fibers with coordinate $y > 0$). Similarly, the component M_y acts in the plane xz and produces on a point of the section of coordinates (y, z) , a normal stress of value:

$$\sigma_x = \frac{M_y}{I_y} z \quad (7.3)$$

(that is, positive moments M_y pull the fibers with coordinate $z > 0$).

Adding the effects of the two components of the bending moment M it is:

$$\sigma_x = -\frac{M_z}{I_z} y + \frac{M_y}{I_y} z \quad (7.4a)$$

$$= M \left(-\frac{\cos \alpha}{I_z} y + \frac{\sin \alpha}{I_y} z \right) \quad (7.4b)$$

Note that the stress distribution obtained is linear in y, z . The stress distribution is linear and can be represented as shown in Figure 7.2.

The equation of the neutral axis is obtained as the locus of the points where the normal stress is zero, that is:

$$-\frac{M_z}{I_z} y + \frac{M_y}{I_y} z = 0 \quad \text{or} \quad -\frac{\cos \alpha}{I_z} y + \frac{\sin \alpha}{I_y} z = 0 \quad (7.5)$$

This is the equation of a line that passes through the center of gravity of the section and has a slope:

$$\tan \beta = \frac{y}{z} = \frac{M_y}{M_z} \frac{I_z}{I_y} = \frac{\sin \alpha}{\cos \alpha} \frac{I_z}{I_y} = \tan \alpha \frac{I_z}{I_y} \quad (7.6)$$

where α and β are the clockwise angles formed by the vector M and the neutral axis with the z axis, respectively (see Figure 7.3). Observe that the neutral axis is *not normal*, in general, to the trace of the loading plane

mm' , since the normal line to the plane mm' has a slope, with respect to the z axis, $\tan \alpha = y/z = M_y/M_z$, while the neutral axis has slope $\tan \beta = y/z = (M_y/M_z) (I_z/I_y)$. The skew angle θ is the angle formed by the moment vector with respect to the neutral axis.

The angles α and β only coincide ($\theta = 0$) if:

1. One of the components, M_y or M_z , is null; that is, if the moment is in a principal plane of inertia, or if
2. $I_z/I_y = 1$; that is, when all the directions are principal of inertia, and therefore, so is the direction mm' .

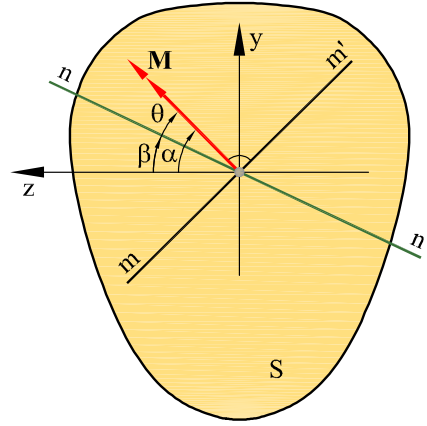


Figure 7.3: Angles in skewed bending

7.2.2 Deformations in pure skew bending

Similarly to the stress study, the deformation of a straight beam subjected to pure skew bending can be studied by decomposing the acting bending moment into its components according to the principal axes of inertia and applying the principle of superposition. The differentials of the bending angles that a differential slice of width dx , undergoes due to the moments M_z and M_y , are, respectively :

$$d\phi_y = \frac{M_y}{EI_y} dx \quad ; \quad d\phi_z = \frac{M_z}{EI_z} dx \quad (7.7)$$

where $d\phi_y$ is a differential rotation contained in the xz plane, and $d\phi_z$ is a differential rotation contained in the xy plane, as shown in the Figure 7.4.

The vectorial composition of these two differential rotations is equivalent to a differential rotation $d\phi$ of components $(d\phi_y, d\phi_z)$ according to the

the acting moment; since this is constant in skew pure bending, and for a beam of uniform section, the curvature is identical for all the sections and the deformed axis of the beam is an arc of circumference.

Therefore, in pure skew bending, and according to the Bernoulli-Navier hypothesis, the cross-sections remain plane in deformation, rotating around the neutral axis and keeping perpendicular to the deformed axis.

Consequently, and as shown in Figure 7.4, the deformed axis of the piece is an arc of circumference contained in the plane normal to the neutral axis that, in general, *does not coincide* with the loading plane (the one that contains the acting bending moment). Hence the name *skewed bending* for this type of solicitation.

Example 7.2.1

A rectangular cross-section of width $b = 30$ cm and depth $h = 40$ cm is subjected to the action of a skew bending moment contained in the plane of one of the diagonals of the cross-section and of value $M = 50$ kN·m. Calculate: (a) the direction of the neutral axis, (b) the maximum stresses, and (c) the curvature of the axis of the slice, if the maximum elastic modulus is $E = 20$ GPa.

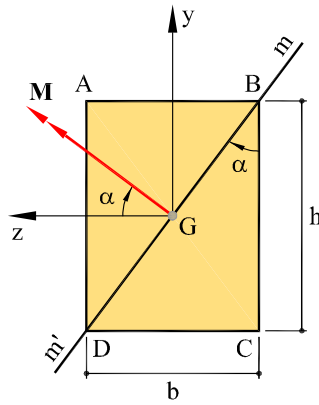


Figure 7.5: Cross-section of Example 7.2.1

(a) On the one hand, the moments of inertia of the section with respect to its principal axes of inertia are:

$$I_y = 9 \cdot 10^{-4} \text{ m}^4 \quad I_z = 16 \cdot 10^{-4} \text{ m}^4$$

On the other hand, and assuming that the moment is contained in the diagonal plane BD of the rectangle and that it has the sign of Figure 7.5,

the components of the moment bending with respect to the axes (y, z) are:

$$M_y = M \sin \alpha = 50 \cdot \frac{3}{5} = 30 \text{ kN}\cdot\text{m}$$

$$M_z = M \cos \alpha = 50 \cdot \frac{4}{5} = 40 \text{ kN}\cdot\text{m}$$

Therefore, the slope of the neutral axis is:

$$\tan \beta = \frac{y}{z} = \frac{M_y I_z}{M_z I_y} = \tan \alpha \frac{I_z}{I_y} = \frac{3}{4} \frac{16 \cdot 10^4}{9 \cdot 10^4} = \frac{4}{3}$$

As can be seen in Figure 7.6, in this case the neutral axis coincides with the other diagonal of the rectangle, the line AC .

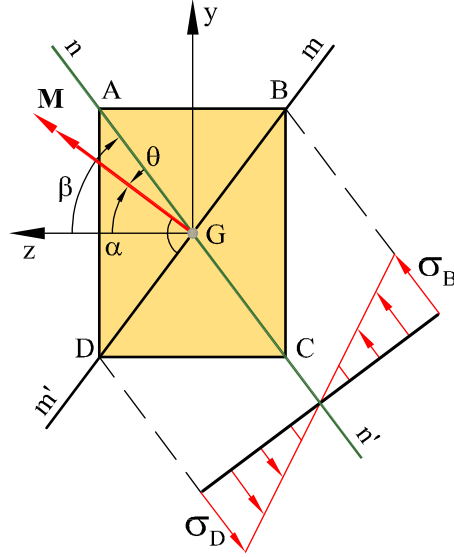


Figure 7.6: Solution of Example 7.2.1

(b) Given the direction of the neutral axis, the maximum stresses occur at points B and D , which correspond to the maximum compression and traction respectively:

$$\begin{aligned} \sigma_B &= -\frac{M_z}{I_z} y_B + \frac{M_y}{I_y} z_B \\ &= \left[-\frac{40 \cdot 10^3}{16 \cdot 10^{-4}} 0.20 + \frac{30 \cdot 10^3}{9 \cdot 10^{-4}} (-0.15) \right] \text{ Pa} = -10 \text{ MPa} \\ \sigma_D &= -\frac{M_z}{I_z} y_D + \frac{M_y}{I_y} z_D \\ &= \left[-\frac{40 \cdot 10^3}{16 \cdot 10^{-4}} (-0.20) + \frac{30 \cdot 10^3}{9 \cdot 10^{-4}} 0.15 \right] \text{ Pa} = +10 \text{ MPa} \end{aligned}$$

The corresponding stress distribution can be seen in Figure 7.6.

(c) The curvature of the deformed axis can be calculated as:

$$\begin{aligned}\chi &= \frac{1}{E} \sqrt{\frac{M_y^2}{I_y^2} + \frac{M_z^2}{I_z^2}} \\ &= \frac{1}{20 \cdot 10^9} \sqrt{\frac{(30 \cdot 10^3)^2}{(9 \cdot 10^{-4})^2} + \frac{(40 \cdot 10^3)^2}{(16 \cdot 10^{-4})^2}} \text{ m}^{-1} = 2.0833 \cdot 10^{-3} \text{ m}^{-1}\end{aligned}$$

which corresponds to a radius of curvature of $\rho = 480 \text{ m}$.

Example 7.2.2

The thin-walled cross-section of Figure 7.7 is subjected to the action of a skew bending moment M . Determine: (a) Calculate and draw the distribution of normal stresses, showing their maximum values and (b) Calculate the maximum value of the moment M that supports the section if the allowable stress of the material is σ_{all} .

Data: $M = 350 \text{ kN}\cdot\text{m}$, $\alpha = 30^\circ$, $\sigma_{\text{all}} = 260 \text{ MPa}$, $a = 40 \text{ cm}$, $b = 30 \text{ cm}$, $h = 50 \text{ cm}$, $e = 1.5 \text{ cm}$.

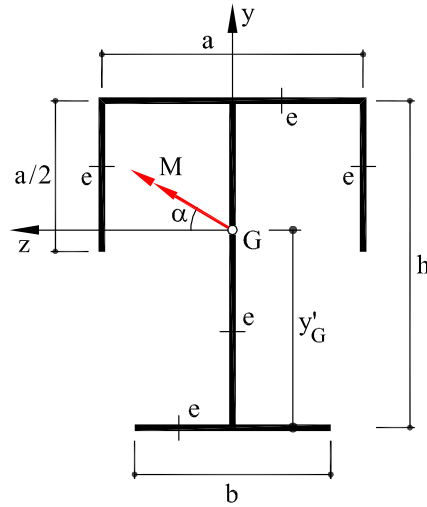


Figure 7.7: Cross-section of Example 7.2.2

The cross-section is symmetric about the y axis. Therefore, the center of gravity is located on this vertical axis. Being symmetric with respect to the y axis, the (y, z) axes are the principal axes of inertia of the section.

The geometric characteristics of the cross-section are (see Example 6.2.6):

Area of the cross-section: $A = 240 \text{ cm}^2$,

Position of the center of gravity with respect to the bottom edge: $y'_G = 30.3 \text{ cm}$,

Principal moments of inertia: $I_y = 35400 \text{ cm}^4$, $I_z = 89977 \text{ cm}^4$.

(a) *Distribution of normal stresses*

Taking into account the slope α of the moment M , the components of the bending moment with respect to the axes y, z are:

$$M_y = M \sin \alpha = 350 \frac{1}{2} = 175 \text{ kN} \cdot \text{m} \quad M_z = M \cos \alpha = 350 \frac{\sqrt{3}}{2} = 303 \text{ kN} \cdot \text{m}$$

Therefore, the slope of the neutral axis is:

$$\tan \beta = \frac{M_y I_z}{M_z I_y} = \tan \alpha \frac{I_z}{I_y} = 0.5774 \frac{8.9977 \cdot 10^4}{3.54 \cdot 10^4} = 1.467 \Rightarrow \beta = 55.73^\circ$$

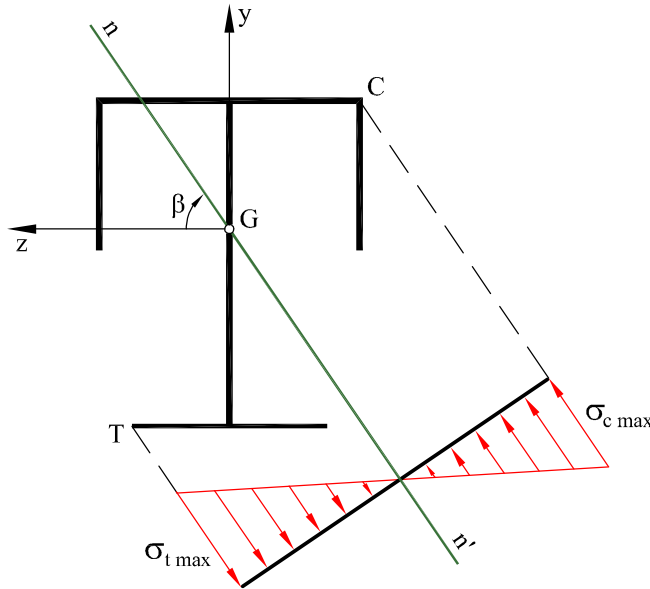


Figure 7.8: Solution of Example 6.5.2

The neutral axis is the line that delimits the region of tension and compression in the cross-section (Figure 7.8). The normal stress is proportional to the distance from the neutral axis. Therefore, the points of maximum tension are those farthest away from this axis. In this case, the farthest points are: $C(19.7; -20.0)$ and $T(-30.3; 15.0)$ (expressed in cm), and the

stresses that result in them are:

$$\begin{aligned}
 \sigma_C^{\max} &= -\frac{M \cos \alpha}{I_z} y_C + \frac{M \sin \alpha}{I_y} z_C = \\
 &= \left[-\frac{303 \cdot 10^3}{8.9977 \cdot 10^{-4}} (0.197) + \frac{175 \cdot 10^3}{3.54 \cdot 10^{-4}} (-0.20) \right] \text{ Pa} = -165 \text{ MPa (C)} \\
 \sigma_T^{\max} &= -\frac{M \cos \alpha}{I_z} y_T + \frac{M \sin \alpha}{I_y} z_T = \\
 &= \left[-\frac{303 \cdot 10^3}{8.9977 \cdot 10^{-4}} (-0.303) + \frac{175 \cdot 10^3}{3.54 \cdot 10^{-4}} (0.15) \right] \text{ Pa} = 176 \text{ MPa (T)}
 \end{aligned}$$

(b) *Maximum allowable moment M*

To calculate the maximum allowable moment M and assuming that the value of the allowable stress σ_{all} is valid both for tension as for compression, it must be taken into account that the maximum tension (176 MPa) at point T . Therefore:

$$\sigma_T^{\max} = M_{\max} \left[-\frac{\cos \alpha}{I_z} (-0.303) + \frac{\sin \alpha}{I_y} (0.15) \right] < \sigma_{\text{all}} = 260 \text{ MPa m}$$

and:

$$M_{\max} = 516 \text{ kN}. \quad (7.10)$$

7.3 Simple skew bending

As in the case of plane simple bending (Section 6.3), in simple skew bending, the acting moment is not constant and it is accompanied by shear forces:

$$T_y = -\frac{dM_z}{dx} \quad T_z = \frac{dM_y}{dx} \quad (7.11)$$

The shear forces give rise to the appearance of angular distortions and tangential stresses in the beam. As a consequence of the existence of these angular distortions, in addition to the elongation of the longitudinal fibers typical of bending, a certain deformation of the planar cross-sections occurs. This phenomenon is explained in Section 6.3 for the case of plane bending; it is known as *warping* and the considerations exposed there can be extended to the case of skewed bending.

Since all the expressions obtained for pure bending are based on the magnitude of longitudinal elongations of the fibers, it follows that if the generalized Bernoulli-Navier hypothesis of deformation is admitted, all the expressions found for the case of pure bending can be extended to the more general case of simple bending.

Example 7.3.1

For the straight beam of Figure 7.9 subjected to the indicated loads: (a) determine the value and location of the maximum bending moments; (b) calculate and draw the distribution of normal stresses and (c) calculate the maximum value that the load P_y can reach if the load P_z remains constant, assuming that the allowable stress value σ_{all} is valid for both tension and compression.

Data: $l = 5$ m, $P_y = 90$ kN, $P_z = 15$ kN, $a = 40$ cm, $b = 30$ cm, $h = 50$ cm, $e = 1.5$ cm, $\sigma_{\text{all}} = 260$ MPa, $E = 210$ GPa.

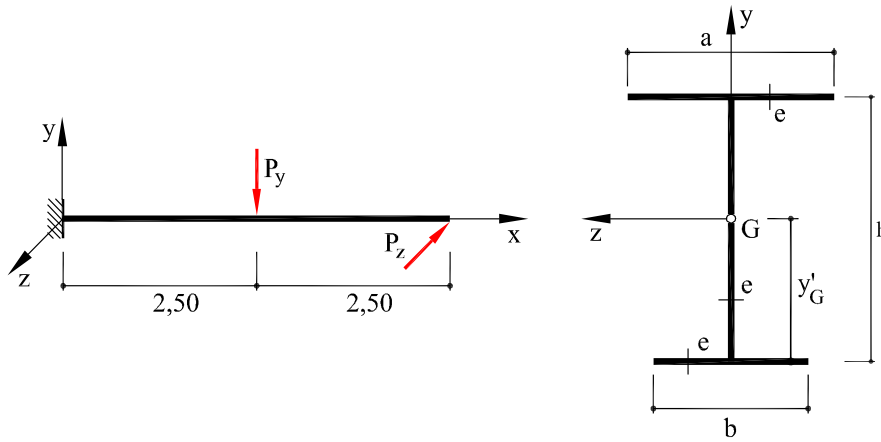


Figure 7.9: Beam and cross section of Example 7.3.1

The cross-section is symmetric about the y axis. Therefore, the axes (y, z) are the principal axes of inertia of the section.

The geometric characteristics of the section are: Area of the cross-section: $A = 180 \text{ cm}^2$, Position of the center of gravity from the bottom edge: $y'_G = 27 \text{ cm}$, Principal moments of inertia: $I_y = 11389 \text{ cm}^4$, $I_z = 80200 \text{ cm}^4$.

(a) The stresses due to the loads P_y and P_z are shown in Figure 7.10. From these it is concluded that the beam is subjected to a state of simple bending (bending moment plus shear). The bending moment in a generic cross-section has components along both axes (y, z) . Consequently, the sections are subjected to a state of skew bending. The maximum moment occurs in the clamping section and its components according to the principal axes are:

$$M_y = 15 \cdot 5 = 75 \text{ kN} \cdot \text{m}$$

$$M_z = -90 \cdot 2.5 = -225 \text{ kN} \cdot \text{m}$$

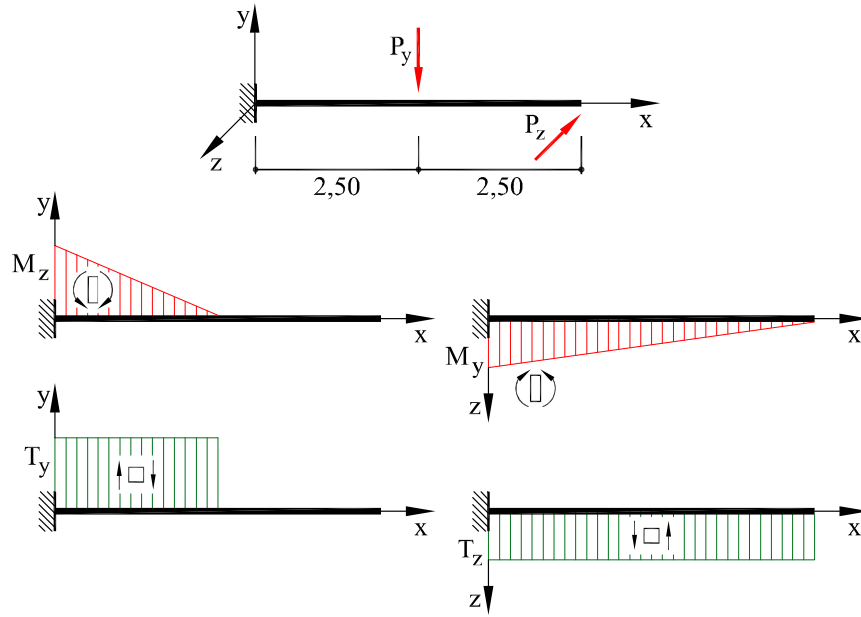


Figure 7.10: Bending moments and shear forces of Example 7.3.1

(b) The slope of the neutral axis is:

$$\tan \beta = \frac{M_y I_z}{M_z I_y} = \frac{75 \cdot 10^3}{(-225 \cdot 10^3)} \frac{8.02 \cdot 10^{-4}}{1.14 \cdot 10^{-4}} = -2.35 \quad \Rightarrow \quad \beta = -66.95^\circ$$

The points farthest from the neutral axis are $A(0.23 \text{ m}; 0.20 \text{ m})$ and $B(-0.27 \text{ m}; -0.15 \text{ m})$ (Figure 7.11); in them the maximum stresses occur:

$$\begin{aligned} \sigma_A &= -\frac{M_z}{I_z} y_A + \frac{M_y}{I_y} z_A = \\ &= \left[-\frac{(-225 \cdot 10^3)}{8.02 \cdot 10^{-4}} (0.23) + \frac{75 \cdot 10^3}{1.14 \cdot 10^{-4}} (0.20) \right] \text{ Pa} \\ &= +197 \text{ MPa (T)} \\ \sigma_B &= -\frac{M_z}{I_z} y_B + \frac{M_y}{I_y} z_B = \\ &= \left[-\frac{(-225 \cdot 10^3)}{8.02 \cdot 10^{-4}} (-0.27) + \frac{75 \cdot 10^3}{1.14 \cdot 10^{-4}} (-0.15) \right] \text{ Pa} \\ &= -175 \text{ MPa (C)} \end{aligned}$$

The maximum axial tensile stress occurs in A , and this is also the absolute

maximum stress value in the cross-section. The maximum compression stress occurs in B . It is verified that both stresses are less than the allowable stress of the material $\sigma_{\text{all}} = 260 \text{ MPa}$.

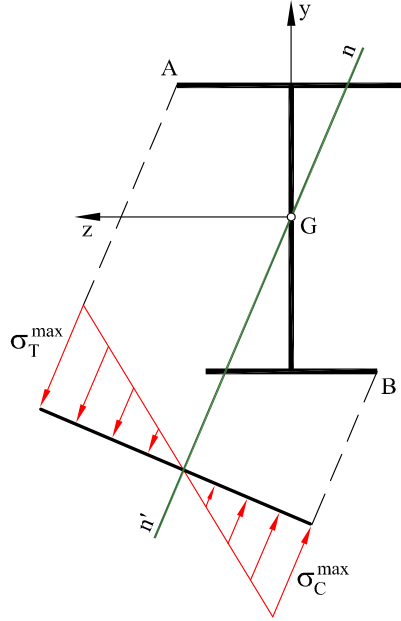


Figure 7.11: Distribution of normal stresses

(c) At point A of the section, the maximum stress in absolute value is produced. The maximum value that the charge P_y can reach is determined from the expression:

$$\sigma_A = -\frac{M_z}{I_z}y_A + \frac{M_y}{I_y}z_A \leq \sigma_{\text{adm}} = 260 \text{ MPa}$$

Taking into account that $M_z = -P_y \cdot 2.5$, it follows:

$$\left[-\frac{(-P_y^{\text{max}} \cdot 2.5 \cdot 10^3)}{8.02 \cdot 10^{-4}} (0.23) + \frac{75 \cdot 10^3}{1.14 \cdot 10^{-4}} (0.20) \right] \text{ Pa} = 260 \text{ MPa} \quad P_y^{\text{max}} = 178 \text{ kN}$$

Example 7.3.2

It is required to choose the cross-section of some prismatic beam (Figure 7.12) to use as straps on a roof with a slope $\alpha = 20^\circ$. There is a standard IPN80 profile (width $a = 42$ mm, depth $b = 80$ mm, $I_z = 77.8$ cm⁴, $I_y = 6.29$ cm⁴) and a rectangular hollow section (width $a = 40$ mm, depth $b = 80$ mm, thickness $e = 2$ mm, $I_z = 38.97$ cm⁴, $I_y = 13.12$ cm⁴). Determine: (a) which of them is the most resistant, if both are subjected to downward vertical load and (b) the value of α so that both sections resist the same bending moment.

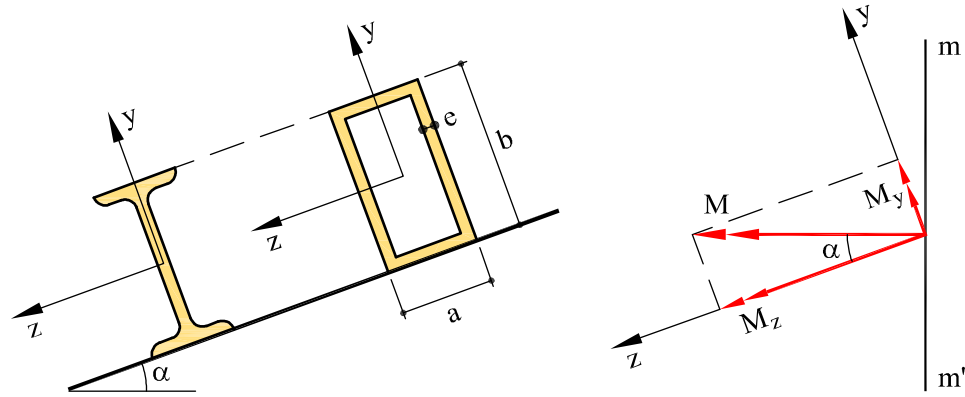


Figure 7.12: Cross-sections of Example 7.3.2

(a) Due to symmetry, both cross-sections have the same principal directions of inertia. The components of the vertical bending moment M on these principal directions (y, z) are:

$$\begin{aligned} M_z &= M \cos \alpha = 0.940 M \\ M_y &= M \sin \alpha = 0.342 M \end{aligned}$$

IPN 80 profile: The slope of the neutral axis for a moment M is:

$$\tan \beta = \frac{y}{z} = \frac{\sin \alpha}{\cos \alpha} \frac{I_z}{I_y} = \frac{0.342}{0.940} \frac{77.8}{6.29} = 4.50 \quad \Rightarrow \quad \beta = 77^\circ 28' 16''$$

The maximum stress occurs at the points farthest away from the neutral axis, that is, at the points C y D (Figure 7.13), and its value can be calcu-

lated, for a moment M [kN·m] as:

$$\begin{aligned}
 \sigma_C &= -\frac{M_z}{I_z} y_C + \frac{M_y}{I_y} z_C \\
 &= \left[-\frac{0.940 \cdot 10^3 M}{77.8 \cdot 10^{-8}} 4.0 \cdot 10^{-2} + \frac{0.342 \cdot 10^3 M}{6.29 \cdot 10^{-8}} (-2.1 \cdot 10^{-2}) \right] \text{ Pa} \\
 &= -162.5 M \text{ MPa} \quad (\text{maximum compression})
 \end{aligned}$$

The stress at point D has the same absolute value, but it has the opposite sign, that is, it corresponds to the maximum tension.

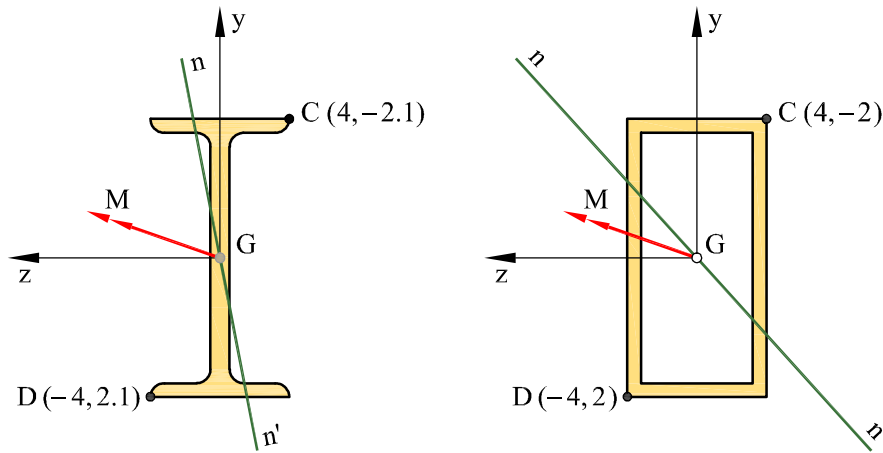


Figure 7.13: Neutral axes and points of maximum stress in Example 7.3.2

Hollow rectangular cross-section: The slope of the neutral axis, in this case, is equal to:

$$\tan \beta = \frac{y}{z} = \frac{\sin \alpha}{\cos \alpha} \frac{I_z}{I_y} = \frac{0.342}{0.940} \frac{38.97}{13.12} = 1.08 \Rightarrow \beta = 47^\circ 12' 9''$$

The maximum stress occurs at the vertices C and D of the cross-section (Figure 7.13) and its value, for a moment M [kN.m], is equal to

$$\begin{aligned}
 \sigma_C &= -\frac{M_z}{I_z} y_C + \frac{M_y}{I_y} z_C \\
 &= \left[-\frac{0.940 \cdot 10^3 M}{38.97 \cdot 10^{-8}} 4 \cdot 10^{-2} + \frac{0.342 \cdot 10^3 M}{13.12 \cdot 10^{-8}} (-2 \cdot 10^{-2}) \right] \text{ Pa} \\
 &= -148.6 M \text{ MPa} \quad (\text{maximum compression})
 \end{aligned}$$

The stress at the vertex D is equal in absolute value, but it is tensile.

From the values of the maximum stresses obtained, it can be seen that, for the load considered, the tubular section is *more resistant* than the IPN 80 profile, with a difference of approximately 10 %. (b) For an arbitrary angle α , it is:

$$\begin{aligned}\sigma &= -\frac{M_z}{I_z}y + \frac{M_y}{I_y}z \\ &= M \left(-\frac{\cos \alpha}{I_z}y + \frac{\sin \alpha}{I_y}z \right)\end{aligned}$$

Maximum stress in the IPN 80 profile:

$$\sigma_{\max}^{\text{IPN}} = M \left[-\frac{\cos \alpha}{77.8}(4) + \frac{\sin \alpha}{6.29}(-2.1) \right] \cdot 10^9$$

Maximum stress in the hollow section:

$$\sigma_{\max}^{\text{hollow}} = M \left[-\frac{\cos \alpha}{38.97}(4) + \frac{\sin \alpha}{13.12}(-2) \right] \cdot 10^9$$

Equating:

$$\frac{4}{77.8}\cos \alpha + \frac{2.1}{6.29}\sin \alpha = \frac{4}{38.97}\cos \alpha + \frac{2}{13.12}\sin \alpha$$

where

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{0.051230}{0.181424} = 0.2823 \quad \Rightarrow \quad \alpha = 15^\circ 46' 6''$$

Obviously, for angles less than $\alpha = 15^\circ 46' 6''$, the IPN is more resistant, since it has a greater W_z section modulus. For angles greater than $\alpha = 15^\circ 46' 6''$, the tubular section is more resistant, since it has a greater W_y section modulus.

7.4 Skew composed bending

A beam is subjected to *skew composed bending* when a system of stresses acts on its cross-sections that is reduced to an axial force N and a bending moment M with components M_y and M_z , contained in a plane *other than* the principal bending planes of the beam, that is, such that the moment vector *does not coincide* with any of the principal axes of inertia of the sections (Figure 7.14a).

In a similar way to what is described for plane composed bending, the deformation of the slice can be deduced by applying the superposition principle. Thus, the deformation due to the axial force occurs according to Bernoulli's hypothesis (plane cross-sections are translated remaining plane

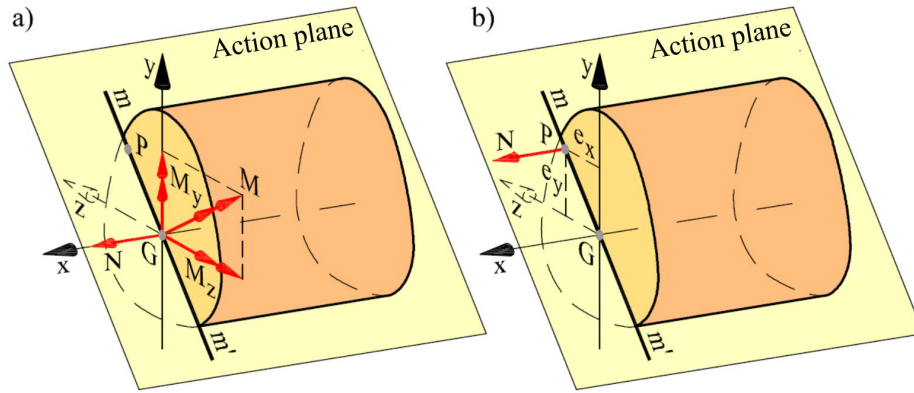


Figure 7.14: Skew composed bending as eccentric normal force

and parallel to themselves), while bending deformation occurs according to the hypothesis of Bernoulli-Navier (plane cross-sections rotate around the neutral axis, remaining plane and perpendicular to the deformed axis). By superposition of effects, a section of the slice undergoes a translation du , parallel to the axis x (due to the axial force), a rotation $d\phi_z$, about the z axis (due to bending M_z), and a rotation $d\phi_y$, about the axis y (due to moment M_y), relative to another cross-section infinitely close. The composition of these displacements is equivalent to a single rotation around the *neutral* axis, whose position will be determined later.

The value of the normal stresses at a generic point of the section, with coordinates (y, z) with respect to the principal axes, can also be obtained applying the superposition principle. In this case, the stresses obtained for the axial force are superimposed on those corresponding to skewed bending, in the form:

$$\sigma_x = \frac{N}{A} - \frac{M_z y}{I_z} + \frac{M_y z}{I_y} \quad (7.12)$$

Observe that, according to the Bernoulli-Navier deformation hypothesis and Hooke's law, the stress distribution is linear in the coordinates y, z .

On the other hand, the axial force N , applied at the center of gravity of the section G , plus the moment M are a system of resultants statically equivalent to a single eccentric force of value N applied at the center of pressures P (Figure 7.14b), whose position with respect to the principal axes $P(e_y, e_z)$ is given by the relations:

$$M_y = N e_z \quad ; \quad M_z = -N e_y \quad (7.13)$$

or,

$$e_z = \frac{M_y}{N} \quad ; \quad e_y = -\frac{M_z}{N} \quad (7.14)$$

In skewed composed bending, the center of pressure P is not located on any of the principal axes. The direction $\overline{GP}$ (or mm' in Figure 7.14), orthogonal to the moment vector, is the trace of the loading plane. According to the above, the stress at a generic point of the section can also be expressed as:

$$\sigma_x = \frac{N}{A} - \frac{M_z y}{I_z} + \frac{M_y z}{I_y} \quad (7.15a)$$

$$= \frac{N}{A} + \frac{N e_y y}{I_z} + \frac{N e_z z}{I_y} \quad (7.15b)$$

$$= \frac{N}{A} \left(1 + \frac{e_y y}{r_z^2} + \frac{e_z z}{r_y^2} \right) \quad (7.15c)$$

where $r_z = \sqrt{I_z/A}$ and $r_y = \sqrt{I_y/A}$ are the radius of gyration the cross-section with respect to the axes (y, z) , respectively.

By definition, the *neutral axis* is the locus of points that have zero normal stress. Therefore, setting to zero the expression for the stresses, the equation of the neutral axis is obtained:

$$1 + \frac{e_y y}{r_z^2} + \frac{e_z z}{r_y^2} = 0 \quad (7.16)$$

This is the equation of a line contained in the plane yz . Such a line *does not pass* through the center of gravity of the section and, in general, *is not perpendicular* to the trace of the loading plane (direction mm').

If the points of intersection of the neutral axis with the axes are labelled A and B (y, z) , respectively, we have:

$$\text{for } z = 0 \Rightarrow y = a = -\frac{r_z^2}{e_y} \quad \text{being } a \text{ the ordinate of point } A$$

$$\text{for } y = 0 \Rightarrow z = b = -\frac{r_y^2}{e_z} \quad \text{being } b \text{ the abscissa of point } B$$

It should be noted that if the center of pressure P approaches the center of gravity of the section G , that is, the eccentricity decreases, the bending moment has less relative importance and the neutral axis moves away from the section, all of it being subjected to tension or compression, depending on the sign of the normal force N . Indeed, for $e_y, e_z \rightarrow 0$, $a, b \rightarrow \infty$. Conversely, if P moves away from G , that is, the eccentricity increases, the bending has greater relative importance, the neutral axis approaches the center of gravity of the section and intersects the section dividing it into two zones, one in tension and the other in compression ($e_y, e_z \rightarrow \infty$, $a, b \rightarrow 0$).

It can be shown that if the center of pressure moves along a line r of equation:

$$\frac{y}{a} + \frac{z}{b} = 1$$

the neutral axes corresponding to the successive positions are lines that rotate around the coordinate point $R(y_R, z_R)$:

$$y_R = -\frac{r^2 z}{a} \quad z_R = -\frac{r^2 y}{b}$$

Example 7.4.1

A pillar of rectangular section $b \times h$ ($30 \times 40 \text{ cm}^2$) and length $l = 2.5 \text{ m}$ supports a slightly inclined compression force $F = 100 \text{ kN}$, as shown in Figure 7.15. Calculate: (a) the force resultants in the piece; (b) position and direction of the neutral axis in each cross-section; (c) maximum stresses in each cross-section.

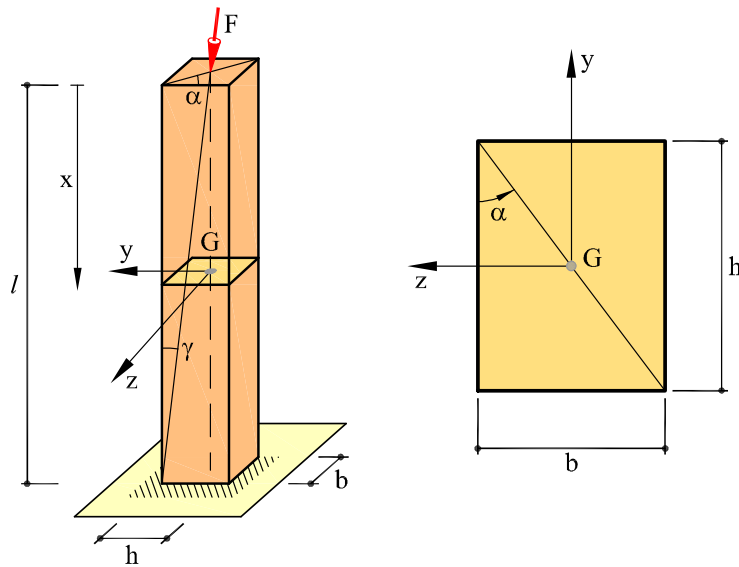


Figure 7.15: Pillar and cross-section of Example 7.4.1

The geometric characteristics of the section are:

$$A = 30 \cdot 40 = 1200 \text{ cm}^2$$

$$\begin{aligned}
I_y &= \frac{1}{12} 40 \cdot 30^3 = 90\,000 \text{ cm}^4 & ; & & r_y^2 &= \frac{I_y}{A} = 75.00 \text{ cm}^2 \\
I_z &= \frac{1}{12} 30 \cdot 40^3 = 160\,000 \text{ cm}^4 & ; & & r_z^2 &= \frac{I_z}{A} = 133.33 \text{ cm}^2
\end{aligned}$$

The angles α and γ , with $d^2 = h^2 + b^2$, are:

$$\alpha = \arctan \frac{b}{h} = \arctan \frac{3}{4} \quad ; \quad \gamma = \arctan \frac{d/2}{l} = \arctan \frac{1}{10} \simeq \frac{1}{10}$$

The eccentricities of the load as a function of the distance x from the cross-section considered to the upper edge of the pillar are:

$$e_y(x) = \frac{h}{2} \frac{x}{l} \quad ; \quad e_z(x) = \frac{b}{2} \frac{x}{l}$$

(a) *Force resultants*

The force resultants, as a function of the eccentricities in each section, are:

$$\begin{aligned}
N(x) &= -F \cos \gamma \simeq -F \\
T_y(x) &= +F \sin \gamma \cos \alpha \simeq +0.08F \\
T_z(x) &= +F \sin \gamma \sin \alpha \simeq +0.06F
\end{aligned}$$

$$\begin{aligned}
M_y(x) &= -F \cos \gamma e_z(x) \simeq -F \frac{b}{2} \frac{x}{l} \\
M_z(x) &= +F \cos \gamma e_y(x) \simeq +F \frac{h}{2} \frac{x}{l}
\end{aligned}$$

The axial and shear forces are constant throughout the pillar, while the bending forces grow linearly with x .

(b) *Position and direction of the neutral axis in each cross-section*

The neutral axis of each section intersects the axes (y, z) of the section at points A and B , of coordinates

$$a(x) = -\frac{r_z^2}{e_y(x)} = -6.66 \frac{l}{x} \text{ cm} \quad ; \quad b(x) = -\frac{r_y^2}{e_z(x)} = -5.00 \frac{l}{x} \text{ cm}$$

Therefore, the neutral axis approaches the center of gravity as the ratio x/l increases, as shown in Figure 7.16.

(c) *Maximum stresses in each cross-section*

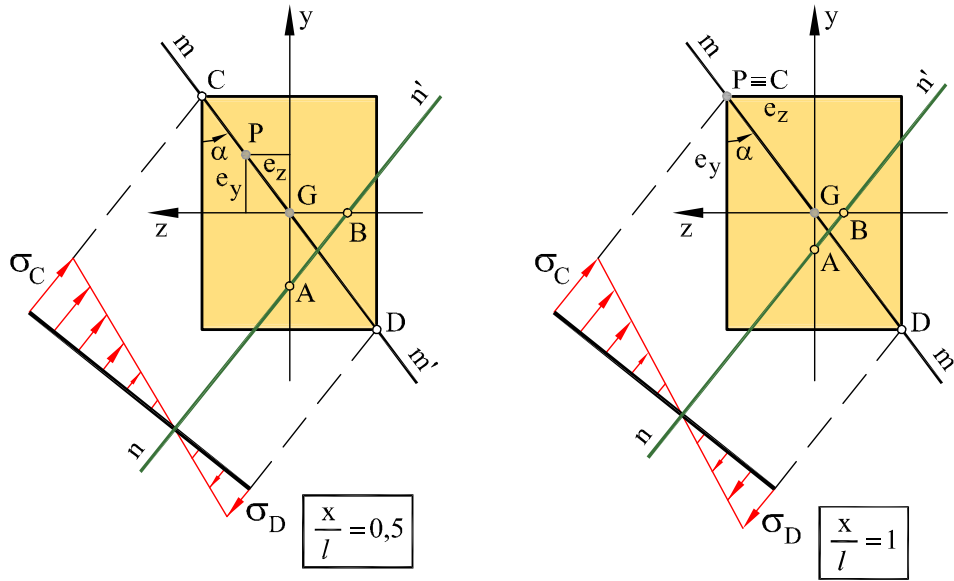


Figure 7.16: Solution of Example 7.4.1

The stresses at any point of the pillar are:

$$\begin{aligned}\sigma_x &= \frac{N}{A} \left(1 + \frac{e_y y}{r_z^2} + \frac{e_z z}{r_y^2} \right) \\ &= \frac{N}{A} \left(1 - \frac{y}{a(x)} - \frac{z}{b(x)} \right)\end{aligned}$$

Knowing the position of the neutral axis, it follows that the stress distributions will be like those drawn in Figure 7.16 for sections $x/l = 0.5$ and $x/l = 1$.

The maximum compression occurs at point C , upper left corner, and is:

$$\sigma_C(x) = \frac{N}{A} \left(1 + \left[\frac{20}{6.66} + \frac{15}{5.0} \right] \frac{x}{l} \right) = \frac{N}{A} \left(1 + 6.0 \frac{x}{l} \right)$$

then, $\sigma_C(x/l = 0.5) = -3.333$ MPa y $\sigma_C(x/l = 1) = -5.833$ MPa.

The maximum tension occurs at point D , lower right corner, and is:

$$\sigma_D(x) = \frac{N}{A} \left(1 - \left[\frac{20}{6.66} + \frac{15}{5.0} \right] \frac{x}{l} \right) = \frac{N}{A} \left(1 - 6.0 \frac{x}{l} \right)$$

then, $\sigma_D(x/l = 0.5) = 1.667$ MPa y $\sigma_D(x/l = 1) = 4.167$ MPa.

7.5 Central core of a cross-section

The *central core* of a cross-section is defined as the locus of points such that, taken as the centers of pressure in eccentric tension or compression, produce normal stresses of the same sign at all points of the section (Figure 7.17).

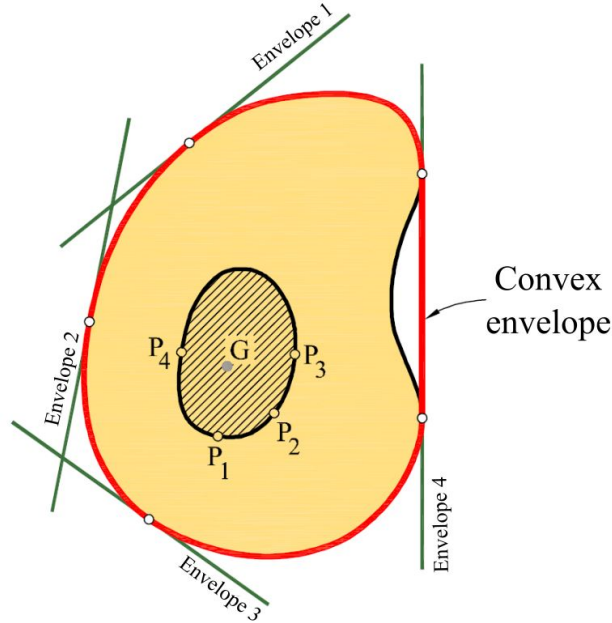


Figure 7.17: Central core of a cross-section

To determine the central core the following properties are used:

- a) for the cross-section to be subjected to normal stresses of the same sign, it is necessary and sufficient that the neutral axis *does not intersect* the section.
- b) the center of pressures and the neutral axis are related by Eq. (7.16).

Consequently, the central core of a cross-section is formed by the interior points of a closed curve, the locus of the points which, taken as centers of pressure, correspond to neutral axes that are tangent lines to the convex envelope of the cross-section.

The concept of central core of a cross-section can be visualized more easily if it is taken into account that, as shown in Figure 7.17:

1. *the central core is a closed enclosure.* If it were not, it would necessarily include points infinitely far from the center of gravity, which is impossible, since centers of pressure far enough from the barycenter

correspond to sufficiently close neutral axes to this so that they will intersect the cross-section.

2. *the central core includes the centroid of the cross-section as an interior point*, since neutral axes correspond to centers of pressure sufficiently close to the center of gravity far enough away from it so as not to intersect the cross-section.
3. *the central core is inside the convex envelope of the cross-section*, since the centers of pressure are the points at which the resultant of the normal stresses acting over the section acts; if these are all of the same sign (traction or compression), their resultant must necessarily pass through a point inside the convex envelope.
4. *the central core is a connected and convex enclosure*, given that the *convex enclosure* of the section is as well.

Both the concept and the definition and calculation method of the central cores are due to J.A.C. Bresse, who included them in his book on Strength of Materials, published in 1859. Next, the central core of some particular sections is determined.

7.5.1 Circular cross-section

In a circular cross-section of radius R any axis that passes through the center of the circle is a principal axis of inertia and the corresponding radius of gyration is

$$r^2 = \frac{I}{A} = \frac{\pi R^4/4}{\pi R^2} = \left(\frac{R}{2}\right)^2 \quad (7.18)$$

By radial symmetry, the central core of a circular cross-section is a concentric circle (see Figure 7.18). The radius of the central core is determined by imposing that the neutral axis is tangent to the cross-section. Take, for instance, the case of tangency on the lower fiber of the circle, of equation $y = -R$. Identifying this equation with the general expression of the neutral axis in composed bending about the axis z ,

$$1 + \frac{e_y y}{r_z^2} = 0 \quad (7.19)$$

it is:

$$1 + \frac{e(-R)}{(R/2)^2} = 0 \quad \Rightarrow \quad e = \frac{R}{4} \quad (7.20)$$

where e is the radius of the circle which corresponds to the central core of the circular cross-section of radius R .

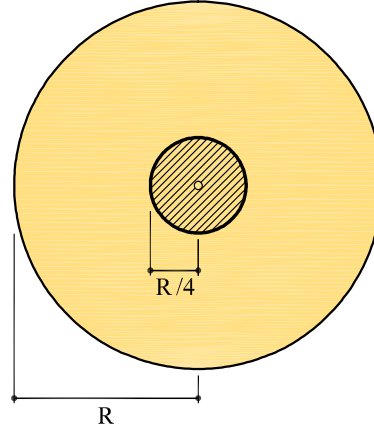


Figure 7.18: Central core of a circular cross-section

7.5.2 Circular crown cross-section

In a circular crown cross-section of radii R_e and R_i , exterior and interior, respectively, any axis that passes through the center of the circle is a principal axis of inertia and the corresponding radius of gyration is

$$r^2 = \frac{I}{A} = \frac{\pi (R_e^4 - R_i^4)/4}{\pi (R_e^2 - R_i^2)} = \frac{R_e^2 + R_i^2}{4} \quad (7.21)$$

By radial symmetry, the central core of a circular crown cross-section is a concentric circle. Applying a process similar to that used for the solid circular cross-section, we calculate the radius of the central core by imposing that the neutral axis is tangent to the section; for example, in the inferior fiber of the circle, of equation $y = -R_e$. Identifying this equation with the general expression of neutral axis in straight composed bending around the axis z , it is

$$1 + \frac{e(-R_e)}{\frac{R_e^2 + R_i^2}{4}} = 0 \quad \Rightarrow \quad e = \frac{R_e^2 + R_i^2}{4R_e} > \frac{R_e}{4} \quad (7.22)$$

where e is the radius of the circle that corresponds to the central core of the circular crown cross-section.

7.5.3 Rectangular cross-section

The convex enclosure of a rectangular cross-section of dimensions $b \times h$ is the section itself. The radii of gyration about the principal axes of inertia are $r_z^2 = h^2/12$ and $r_y^2 = b^2/12$. Let us first consider a neutral axis coincident with the side $\overline{CD}$, of equation $y = -h/2$; the corresponding center of pressure is the point M , and the equation of the neutral axis is:

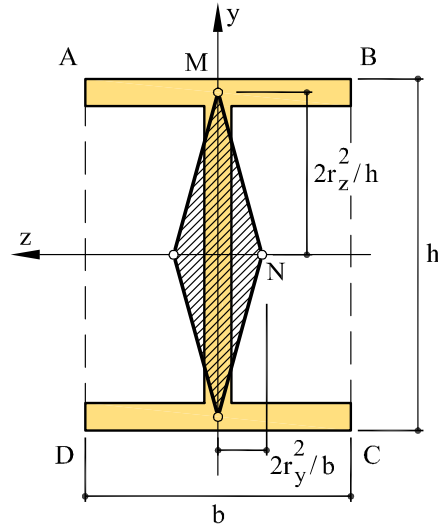


Figure 7.19: Central core of wide flange W-profile

$$1 + \frac{e_M(-\frac{h}{2})}{r_z^2} = 0 \quad \Rightarrow \quad e_M = \frac{h}{6} \quad (7.23)$$

Similarly, for the neutral axis coincident with the side $\overline{AD}$, of equation $z = b/2$, the center of pressure is point N with eccentricity $e_N = -b/6$.

When the center of pressure moves along the segment $\overline{MN}$, the corresponding neutral axes rotate around point D , since in D the neutral axes corresponding to the centers of pressure M and N intersect. Due to double symmetry with respect to the axes (y, z) , the central core is a rhombus with diagonals $h/3$ and $b/3$, respectively, as seen in Figure 7.20.

7.5.4 Wide flange W-profile

The convex enclosure of a symmetrical wide flange W-profile is a rectangle, as shown in Figure 7.19. Therefore, the central core is a rhombus as in the previous case, but the lengths of its diagonals are given by the radii of gyration of the profile.

The position of point M is obtained by considering that side $\overline{CD}$, equation $y = -h/2$ is the neutral axis corresponding to which M is the center of pressure:

$$1 + \frac{e_M(-\frac{h}{2})}{r_z^2} = 0 \quad \Rightarrow \quad e_M = \frac{2r_z^2}{h} \quad (7.24)$$

and, for point N and its corresponding neutral axis $\overline{AD}$, equation $z = b/2$,

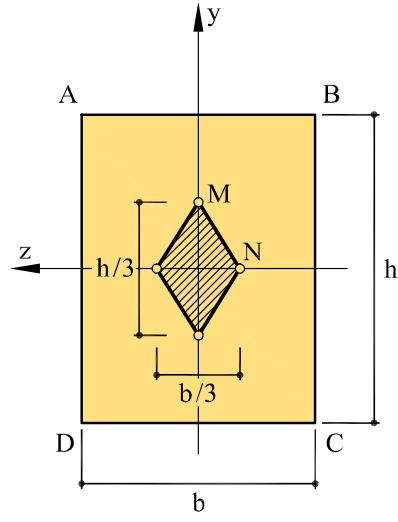


Figure 7.20: Central core of a rectangular cross-section

is

$$1 + \frac{e_N \frac{b}{2}}{r_y^2} = 0 \quad \Rightarrow \quad e_N = -\frac{2r_y^2}{b} \quad (7.25)$$

Chapter 8

Shear Force

8.1 Introduction

A cross-section of a beam is subjected to shear when it is under a shear force, which is the resultant of forces parallel to plane of the cross section.

The action of a shear force $\mathbf{T}$ on the cross-section implies the existence of a shear stress distribution on the plane of the cross-section satisfying the integral relationships:

$$T_y = \int_S \tau_{xy} dS \quad ; \quad T_z = \int_S \tau_{xz} dS \quad (8.1)$$

where T_y, T_z are the shear force components and τ_{xy}, τ_{xz} are the tangential stress components.

The previous equations are not enough to determine the shear stress distribution and, as in previous Chapters, hypotheses about the deformation of the differential beam element must be considered. Different deformation hypotheses lead to different results. The so-called elementary shear theory, based on a very simple deformation hypothesis, is considered first. Then, the Collignon shear theory based on the Saint-Venant bending deformation hypothesis (generalised Bernoulli Navier hypothesis) is studied.

8.2 Elementary shear theory

The simplest hypothesis for the deformation of a differential beam element subjected to shear, is that “the cross-sections remain plane and move parallel to themselves in the direction of the shear force.”

In such a case, a cross section such as S_2 will have a relative displacement of value dv , relative to an infinitesimally close cross section S_1 in the direction of the acting shear in a differential beam element of length dx as shown in Figure 8.1.

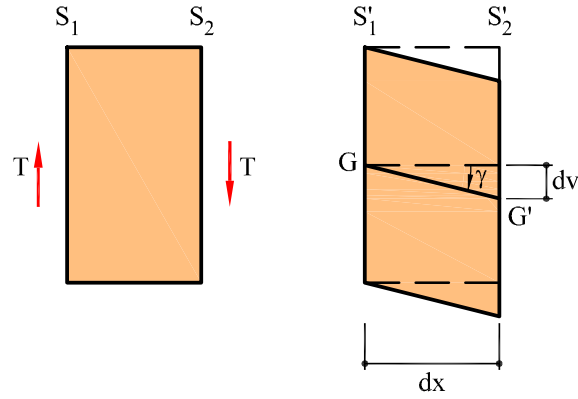


Figure 8.1: Deformation hypothesis of the Elementary Shear Theory

The angular distortion produced at any point on the differential beam element is:

$$\gamma = \frac{dv}{dx} \quad (8.2)$$

Therefore, the differential beam element is subjected to a uniform state of distortion of value γ . If Hooke's law holds, the tangential stress acting at any cross-sectional point is:

$$\tau = G\gamma \quad (8.3)$$

where G is the shear modulus of the beam material.

As the resultant of these stresses must equal the shear stress, the tangential stress distribution is uniform with the same direction as the shear stress and its value is:

$$\tau = \frac{T}{A} \quad (8.4)$$

where A is the cross sectional area. The angular distortion is:

$$\gamma = \frac{T}{GA} \quad (8.5)$$

For the analysis of riveted, screwed or weld bead joints (see Figure 8.2), it is common to adopt the elementary shear theory.

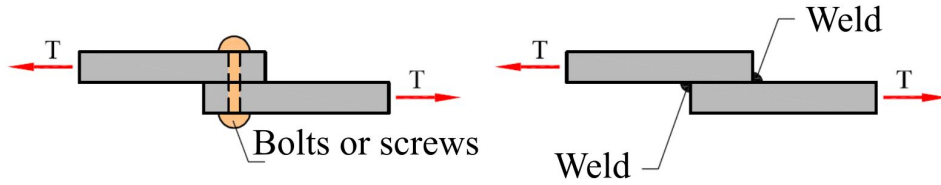


Figure 8.2: Applications of elementary shear theory: welded or riveted joints

Example 8.2.1

Three steel plates, each 1,6 cm thick, are joined by two rivets of diameter $d = 1,3$ cm each, as shown in Figure 8.3. If the value of the load T is 44,5 kN, calculate: (a) the average tangential stress on the rivets, and (b) the maximum load T supported by the joint if the allowable tangential stress on the rivets is $\tau^* = 220$ MPa. Neglect the friction between plates.

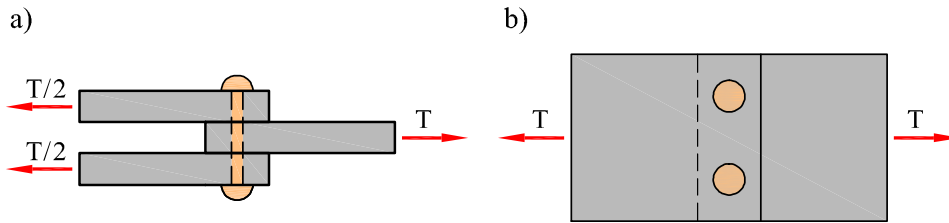


Figure 8.3: Joints between plates for the Example 8.2.1: a) Cross section b) Plan view.

(a) The plates are joined by two rivets; therefore, the force T is supported by four cross sections of these, and the mean tangential stress in the rivets is:

$$\tau_{med} = \frac{T}{4A} = \frac{T}{4(\pi d^2/4)} = \frac{44.5 \cdot 10^3}{4 \cdot (\pi \cdot 1.3^2/4) \cdot 10^{-4}} \text{ Pa} = 83.8 \text{ MPa}$$

(b) If the maximum allowable tangential stress in the rivets is $\tau^* = 220$ MPa, then the maximum force T is:

$$T_{\max} = 220 \cdot 10^6 \cdot 4A = 220 \cdot 10^6 \cdot 4 \cdot \frac{\pi \cdot 1.3^2}{4} \cdot 10^{-4} = 116.8 \text{ kN}$$

Example 8.2.2

The support of a machine consists of a neoprene pad of thickness h and a steel plate of dimension $a \times b$ (Figure 8.4). For a horizontal force T , calculate: (a) the mean tangential stress in the neoprene and (b) the horizontal displacement u of the steel plate, if the shear modulus of stiffness of the neoprene is G_n . Neglect the friction between the plate and the neoprene.

Data: $a = 200$ mm; $b = 150$ mm; $h = 12$ mm; $T_y = 8$ kN

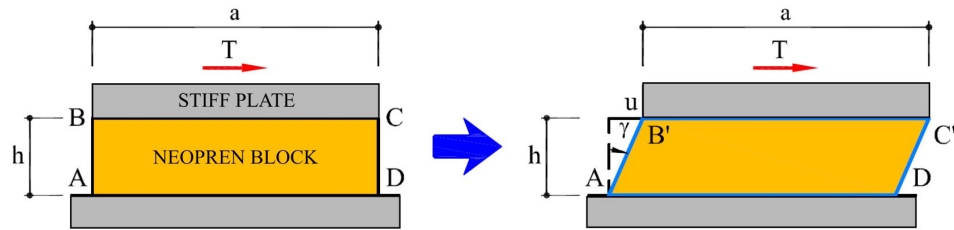


Figure 8.4: Support of the Example 8.2.2

(a) The mean tangential stress τ in the neoprene is:

$$\tau = \frac{T}{ab} = \frac{8 \cdot 10^3}{0.2 \cdot 0.15} \text{ Pa} = 0.267 \text{ MPa}$$

(b) The average distortion γ in the neoprene is

$$\gamma = \frac{\tau}{G_n} = \frac{0.267 \cdot 10^6}{0.9 \cdot 10^6} = 0.297 \text{ rad}$$

Consequently, the horizontal displacement u of the steel plate is given by:

$$u = h \cdot \gamma = 12 \cdot 0.297 = 3.56 \text{ mm}$$

8.3 Collignon shear theory

Consider a prismatic plane beam of arbitrary cross-section constant along the beam's length, subjected to non-uniform bending in a principal plane of inertia (plane bending). Let be assumed for development purposes that a variable bending moment M_z and a shear force T_y act on a differential beam element, both contained in the principal plane of inertia xy (Figure 8.5).

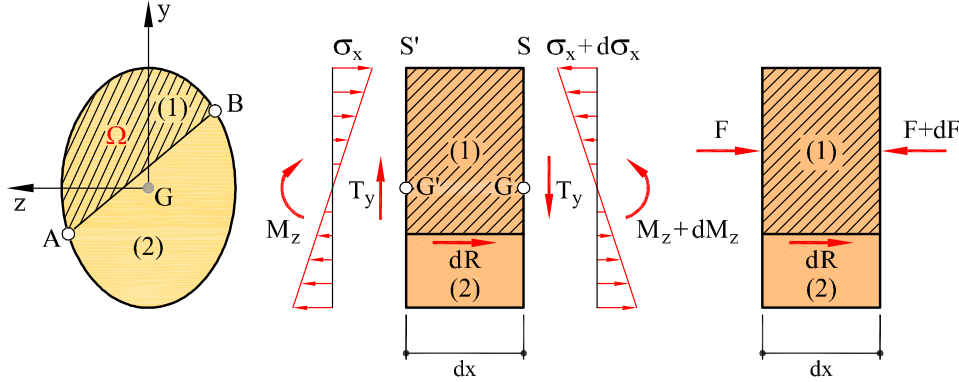


Figure 8.5: Collignon shear theory

From the equilibrium of the differential beam element with two cross-sections S (frontal) and S' (dorsal) delimited by a distance dx , the following relationship is derived:

$$T_y = \frac{dM_z}{dx} \quad (8.6)$$

Let be considered now the differential beam element divided by a plane surface $ABA'B'$ parallel to the axis of the beam, which contains the line AB on the plane of cross-section S and the line $A'B'$ on the plane of section S' .

Let the equilibrium of forces along the x -axis acting on the shaded region of the differential beam element (1) which is placed above the cut surface be examined. According to Navier's stress law, normal stresses due to M_z act on face S' , whose resultant force F on the part of section (1) bounded by line $A'B'$ is:

$$F = \int_{\Omega} \sigma_x d\Omega = \int_{\Omega} -\frac{M_z}{I_z} y d\Omega = -\frac{M_z}{I_z} m_z^e(\Omega) \quad (8.7)$$

where Ω is the part of the cross-section over which this resultant is evaluated (shaded area in Figure 8.5) and $m_z^e(\Omega)$ is the first moment of area of Ω with respect to the z -axis.

Similarly, normal stresses act on face S as a result of the acting moment $M_z + dM_z$, whose force resultant $F + dF$ is equal to:

$$F + dF = - \int_{\Omega} \frac{M_z + dM_z}{I_z} y d\Omega = -\frac{M_z + dM_z}{I_z} m_z^e(\Omega) \quad (8.8)$$

The difference between both normal forces is:

$$dF = -\frac{dM_z}{I_z} m_z^e(\Omega) \quad (8.9)$$

$$= -\frac{T_y dx}{I_z} m_z^e(\Omega) \quad (8.10)$$

where $dM_z = T_y dx$ has been considered.

Note that the previous expression is independent of the value of the bending moment M_z and only depends on dM_z , that is, the variation of the bending moment and, consequently, the shear T_y . As the z -axis passes through the barycenter of the section, it is important to keep in mind that the first moment of area of the striped area above line AB is equal to and has the opposite sign of the first moment of area of the area below it.

Consider instead the tangential stresses that the region of the differential beam element (2) placed below the the cutting plane exerts on the upper region (1). By referring to dR as the result of these interfacial shear stresses, it is possible to write that

$$dR = dx \int_{AB} \tau dl \quad (8.11)$$

$$\simeq dx \tau_{\text{med}} l_{AB} \quad (8.12)$$

where l_{AB} is the length of the segment $\overline{AB}$ and τ_{med} is the average shear stress on the surface $ABA'B'$.

The equilibrium of forces along the x -axis of the differential element (1) requires that $dR = |dF|$ (Figure 8.5). Equating equations (8.10) and (8.12) leads to :

$$\tau_{\text{med}} = \frac{T_y m_z^e(\Omega)}{I_z l_{AB}} \quad (8.13)$$

The above expression is known as Collignon's formula (1877), although this approximate method of evaluating shear stresses in beams had already been used by Jourawski in 1844 to calculate timber beams and beams reinforced with rivets, and published by him in 1856. Note that this formula does not give the value of the intersicial shear stress at a particular point, but the average value of the intersicial shear stress along the contact surface.

Considering the equilibrium of forces along the x -axis acting on the lower part of the differential beam element (2) below the cutting surface, it is deduced that the interfacial force that the upper region (1) exerts on it is also equal to dR and acts in the opposite direction to the force that acts on the upper region (Figure 8.6a). Interfacial stresses on the cutting surface of the lower region (2) are obviously of equal magnitude and opposite direction to those on the cutting surface of the upper region (1).

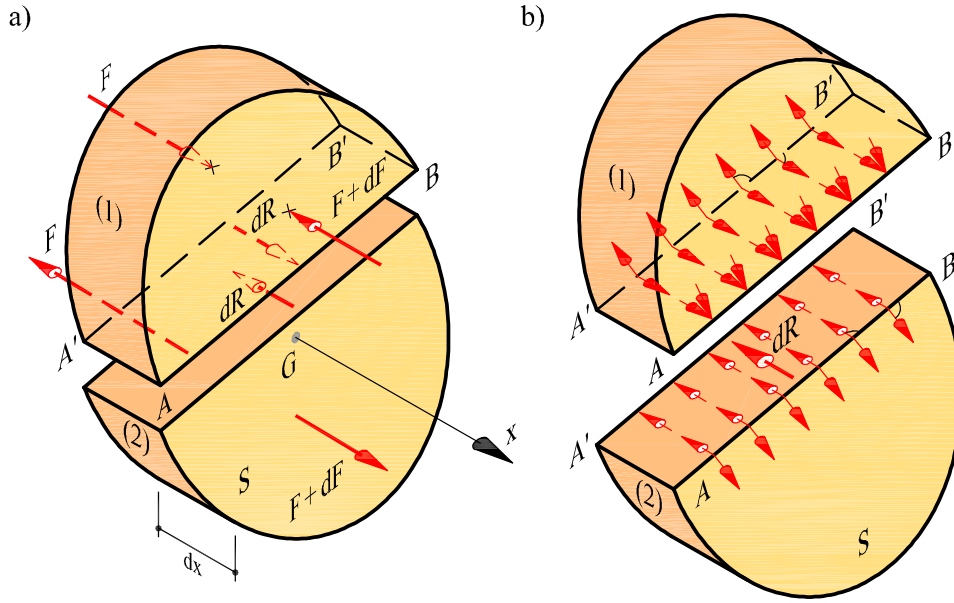


Figure 8.6: Interfacial and tangential stresses

In order to satisfy equilibrium regarding to dF , by the principle of reciprocity of the tangential stresses, as well as the interfacial stresses, parallel to the axis of the beam which appear on the surface $ABA'B'$, stresses contained in the plane of S (and of S') appear, whose normal component to line AB (and $A'B'$) will be equal at each point to the corresponding tangential stress contained in the plane surface $ABA'B'$.

These interfacial stresses on the S and S' cross sections are depicted in Figure 8.6b. Observe that, by the principle of reciprocity, both will be directed toward the line (in this case, toward AB) or away from it (as at $A'B'$). The average interfacial shear stress acting on the surface $ABA'B'$ and the average tangential stress contained in the section S (and S'), along $\overline{AB}$ (and $\overline{A'B'}$) and normal to this line can be determined using Collignon's formula (8.13), since both modulus are equal. The direction of these shear stresses can be determined without ambiguity based on the same analysis.

It is deduced from the fundamental formula that the tangential stresses are zero at the points farthest from the z -axis, since for both it is verified that the corresponding first moment of area $m_z^e(\Omega)$ is zero: at the upper point because Ω area is zero, and at the lower point because it is equal to the cross section Ω area, and the first moment of area of the section with respect to a barycentric axis such as the z -axis is zero.

A simple experiment can be used to point out the importance of interfacial shear stresses. Let us consider a simply supported beam, subjected to non-uniform bending, such as the one shown in Figure 8.7a, let us imagine

a longitudinal cut of the beam along a horizontal plane AA' that divides it into two elements: (1) an element above the cutting plane and (2) an element below the cutting plane. Interfacial stresses appear along the surface AA' when the beam is bent, preventing element (1) from sliding on element (2) and causing the beam to behave as a solid element, as shown in Figure 8.7b. If the AA' cut is real, and parts (1) and (2) are simply superimposed (without friction), these interfacial stresses do not appear, and the assembly behaves totally differently, as shown in Figure 8.7c.

If two superimposed beams are joined in such a way that they work together, slippage of one beam over the other must be avoided and the necessary interfacial forces must be allowed to develop in order to significantly increase the bending strength (see Example 6.2.5). To achieve this, the elements of the beams must be joined using wedges, bolts, or any other device capable of transmitting and resisting the required interfacial forces.

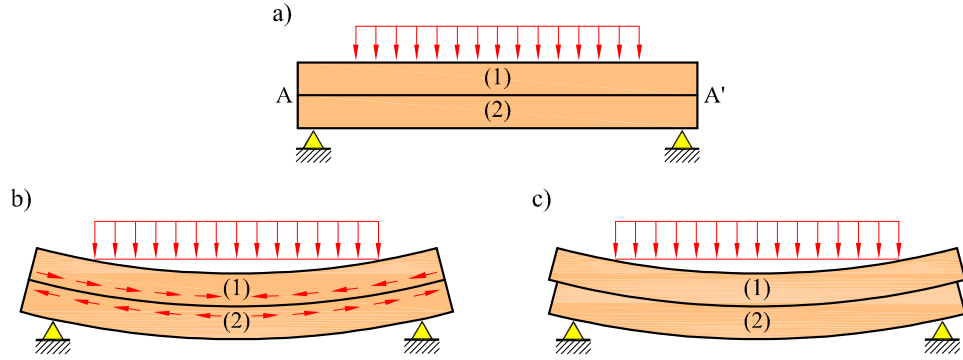


Figure 8.7: Beam deformation under non-uniform bending : (b) with interfacial shear stresses and (c) without interfacial shear stresses

Example 8.3.1

A uniformly distributed load of $p = 3 \text{ kN/m}$ is applied to the simply supported beam with a span of $l = 6 \text{ m}$ as shown in Figure 8.8. Three independent $40 \times 10 \text{ cm}^2$ pieces joined by connectors used to construct the beam as shown in Figure 8.8. Calculate the diameter of these connectors, given that they are spaced $d = 15 \text{ cm}$ from each other and that their allowable tangential stress is $\tau_{\text{all}} = 200 \text{ MPa}$.

The maximum shear stress in the beam is equal to the vertical reactions at the cross sections at the supports:

$$T_{\text{max}} = \frac{pl}{2} = \frac{3 \cdot 6}{2} = 9 \text{ kN}$$

The moment of inertia of the cross-section with respect to the horizontal barycentric axis is:

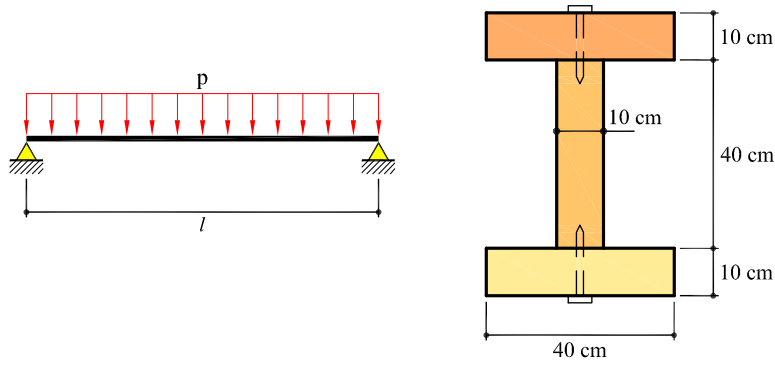


Figure 8.8: Composite simply supported beam Example 8.3.1

$$\begin{aligned}
 I &= \left[\frac{1}{12} 10 \cdot 40^3 + 2 \cdot \left(\frac{1}{12} 40 \cdot 10^3 + 40 \cdot 10 \cdot 25^2 \right) \right] \text{ cm}^4 \\
 &= 56 \cdot 10^{-4} \text{ m}^4
 \end{aligned}$$

According to Collignon's formula, the mean interfacial stress on the contact surfaces between the flanges and the web of the section is:

$$\tau_{\text{med}} = \frac{T_{\text{max}} m^e(\Omega)}{I b}$$

where $m^e(\Omega)$ is the first moment of area of the flange with respect to the horizontal barycentric axis and b is the width of the contact surface, so:

$$\begin{aligned}
 m^e(\Omega) &= 10 \cdot 40 \cdot 25 \text{ cm}^3 = 0.01 \text{ m}^3 \\
 b &= 0.1 \text{ m}
 \end{aligned}$$

Thus, $\tau_{\text{med}} = 0,161 \text{ MPa}$. Each connector in the supporting area must withstand an inter-facial force with a value equal to:

$$R = \tau_{\text{med}} b d = 0.161 \cdot 10^6 \cdot 0.1 \cdot 0.15 \text{ N} = 2.42 \text{ kN}$$

Assuming now that the elementary shear theory is applicable to the connector, the inter-facial stress in each of them is:

$$\tau_{\text{con}} = \frac{R}{A_{\text{con}}} \leq \tau_{\text{all}}$$

which the area of each connector is calculated as:

$$A_{\text{con}} \geq \frac{R}{\tau_{\text{all}}} = \frac{2.42 \cdot 10^3}{200 \cdot 10^6} \text{ m}^2 = 12.1 \text{ mm}^2$$

A minimum diameter of $\phi = 4$ mm is required if the connectors have a circular cross-section.

The previous calculations determine the minimum diameter of the connectors in those cross sections close to the supports, where shear stress is maximum. Since the variation of shear stresses along a beam is linear, with the lowest values close to centre of the span, it is possible to increase the spacing between connectors as shear stresses decreases. Thus, for example, in the middle three metres of the beam, the shear is less than $T = 4,5$ kN and the spacing can be increased to $d = 30$ cm without exceeding the allowable stress in the connectors.

8.4 Rectangular cross section

Using Collignon's theory, the shear stress τ at any point on the plane cross section subjected to a vertical shear stress of magnitude T for a beam with a rectangular cross-section of dimensions $b \times h$ is considered.

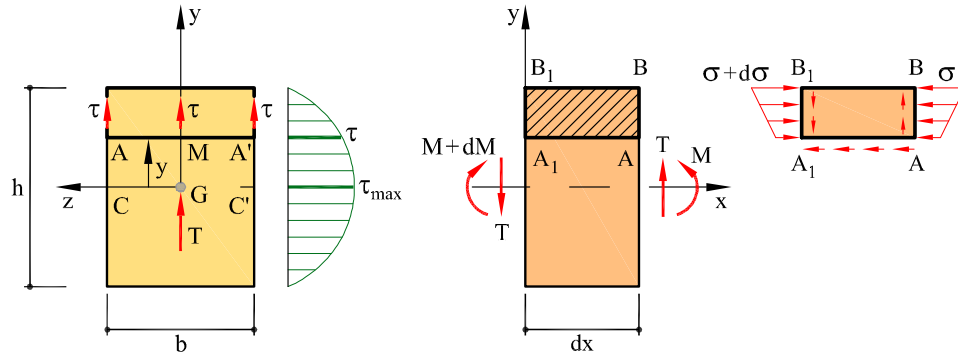


Figure 8.9: Tangential and inter-facial stresses of the rectangular cross section

According to the principle of reciprocity of shear stresses, at points such as A and A' on the cross-section contour (Figure 8.9), the shear stresses must be directed along the cross-section contour, thus, they are vertical.

By symmetry, it must also be vertical at point M . As a result, and as a simplified hypothesis, shear stresses at any point along the line AA' are parallel to the shear force T , so they are vertical and their distribution is uniform along the line AA' . The direction of shear stresses coincides with that of the shear force. Then, the value at each point corresponds to the mean value established by the fundamental formula applied to the region of the cross section above the line AA' .

For this case, $T_y = T$, $I_z = bh^3/12$ y $l = b$. The first moment of area of the region above the line AA' is calculated as a function of the y ordinate

of the cut and is given by

$$m_z^e(y) = b \left(\frac{h}{2} - y \right) \frac{1}{2} \left(\frac{h}{2} + y \right) = \frac{b}{2} \left(\frac{h^2}{4} - y^2 \right) \quad (8.14)$$

Consequently, substituting the preceding values into Collignon's formula:

$$\tau(y) \simeq \tau_{\text{med}}(y) = \frac{3}{2} \frac{T}{bh} \left[1 - \left(\frac{2y}{h} \right)^2 \right] \quad (8.15)$$

It follows from this expression that the distribution of shear stresses in a rectangular section subjected to vertical shear force is parabolic in terms of the ordinate y , is zero in the extreme fibres ($y = \pm h/2$), and has its maximum value at the points on the z -axis ($y = 0$) where it is equal to:

$$\tau_{\text{max}} = \frac{3}{2} \frac{T}{bh} = \frac{3}{2} \frac{T}{A} \quad (8.16)$$

Consequently, the maximum shear stress is 50% greater than that obtained by assuming a uniform stress distribution, as would be obtained by applying elementary shear theory (Jourawski, 1844). It should also be noted that the following integral relationships are satisfied:

$$\begin{aligned} T_y &= \int_S \tau_{xy} dS = \int_{-\frac{h}{2}}^{+\frac{h}{2}} \frac{3}{2} \frac{T}{bh} \left[1 - \left(\frac{2y}{h} \right)^2 \right] b dy = T \\ T_z &= \int_S \tau_{xz} dS \equiv 0 \end{aligned} \quad (8.17)$$

The values provided by the expression (8.15) are not exact, but are extremely approximated, especially when h/b is large. Calculating the exact values for the points on the z axis ($y = 0$) using the Theory of Elasticity shows, for example, that the distribution is not uniform along the axis, with a maximum at points C and C' (Figure 8.9) and a minimum at point G . Table 8.1 provides the correction factors to be applied to the approximated maximum value. Observe how close the factor is to one when $h/b > 1$.

	$h/b = 2$	$h/b = 1$	$h/b = 1/2$	$h/b = 1/4$
$z = 0$	0.983	0.940	0.856	0.805
$z = \pm b/2$	1.033	1.126	1.396	1.988

Table 8.1: Correction values for tangential stresses along $y = 0$ axis

8.5 Thin-walled cross-sections

8.5.1 Open thin-walled cross section

For open thin-walled cross sections, Collignon's shear theory allows the distribution of shear stresses due to shear force to be determined very accurately, as it is reasonable to assume that shear stresses are distributed uniformly throughout the thickness.

Wide Flange (W) and I cross sections

Consider the case of a symmetric Wide Flange (W-profile) or I section subjected to increasing vertical shear force T (Figure 8.10a). The vertical components of the tangential stress, τ_{xy} , are computed first and then the horizontal components, τ_{xz} .

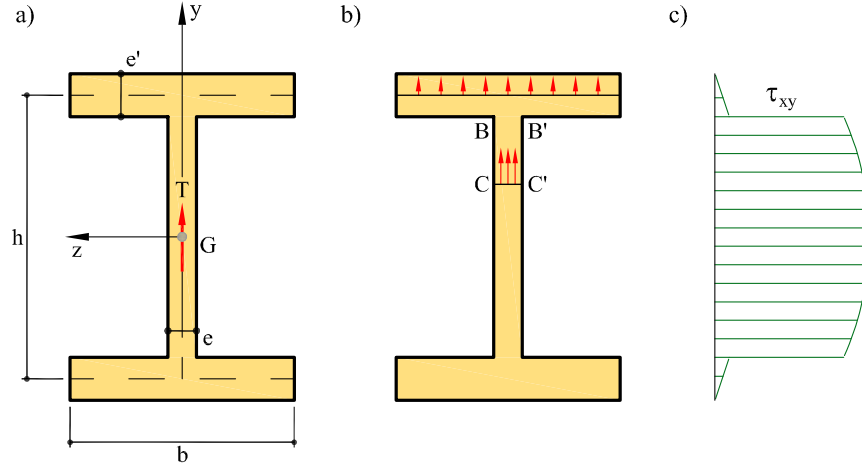


Figure 8.10: Wide Flange(W) or I cross section under shear stress along the web. Vertical tangential stresses

The vertical component of the tangential stress τ_{xy} is calculated by applying Collignon's formula to horizontal cuts of the cross section, as shown in Figure 8.10b. Thus,

$$\tau_{xy} = \frac{T}{I_z} \frac{m_z^e}{l} \quad (8.18)$$

where l is the “width” of the cut. Therefore, when moving from the flange to the web, the value of τ_{xy} undergoes a significant discontinuity, as l changes from $l = b$ in the flange to $l = e$ in the web (see Figure 8.10b). Since the thickness is small, $e \ll b$, the vertical shear stress in the flanges is negligible compared to that in the web, and the web supports almost the vertical

shear stress T . In the web, both τ_{xy} and $m_z^e(y)$ change as a function of the y -coordinate. This variation is symmetric and parabolic (Figure 8.10c):

$$m_z^e(y) = be' \frac{h}{2} + \left(\frac{h}{2} - y \right) e \frac{1}{2} \left(\frac{h}{2} + y \right) = be' \frac{h}{2} + \frac{e}{2} \left(\frac{h^2}{4} - y^2 \right) \quad (8.19)$$

The minimum vertical stresses in the web (Figure 8.10b) are given at the web end points B and B' for $y = \pm h/2$, and they are

$$\tau_{xy}^{\min} = \frac{T}{I_z e} \left(be' \frac{h}{2} \right) \quad (8.20)$$

while the maximum value occurs at the height of the barycentre, G , for $y = 0$, and is

$$\tau_{xy}^{\max} = \frac{T}{I_z e} \left(be' \frac{h}{2} + \frac{e}{2} \frac{h^2}{4} \right) \quad (8.21)$$

For standard W and I cross-sections, the difference between the minimum (in B and B') and maximum values (in G) of the parabola is small. In this instance, the shear stress distribution in the web can be considered uniform, with a value of:

$$\tau_{xy} \simeq \tau_{xy}^{\text{med}} = \frac{T}{\Omega_{\text{web}}} \quad (8.22)$$

where $\Omega_{\text{web}} = he$ is the area of the web. By default, this approximation results in maximum value errors of less than 10% for standard cross-sections. Obviously the direction of the shear stresses τ_{xy} and the shear force T coincide.

Therefore, the vertical τ_{xy} stresses in the flanges are negligible, but the horizontal τ_{xz} components are not; they can be calculated similarly by applying Collignon's formula to vertical cross-sectional cuts in the flanges (Figure 8.11). Therefore:

$$\tau_{xz} = \frac{T m_z^e(s)}{I_z e'} \quad (8.23)$$

where $m_z^e(s) = 'h/2$ is the first moment of the area of the shaded region $AABB$ about the z -axis.

The stresses along BB are assumed to be uniform and parallel as a result of the thin thickness. Considering that T , I_z and e' are constant and that $m_z^e(s)$ varies linearly with s , the tangential stresses distribution in the flanges is linear, and zero at the ends. The direction of τ_{xz} is determined explicitly by equilibrium as shown in Figures 8.11 and 8.12.

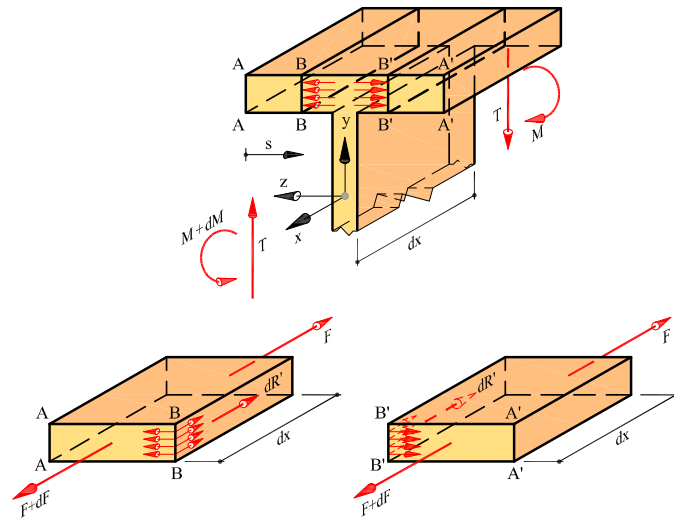


Figure 8.11: Wide Flange (W) or I cross section subjected to shear force. Horizontal tangential stresses

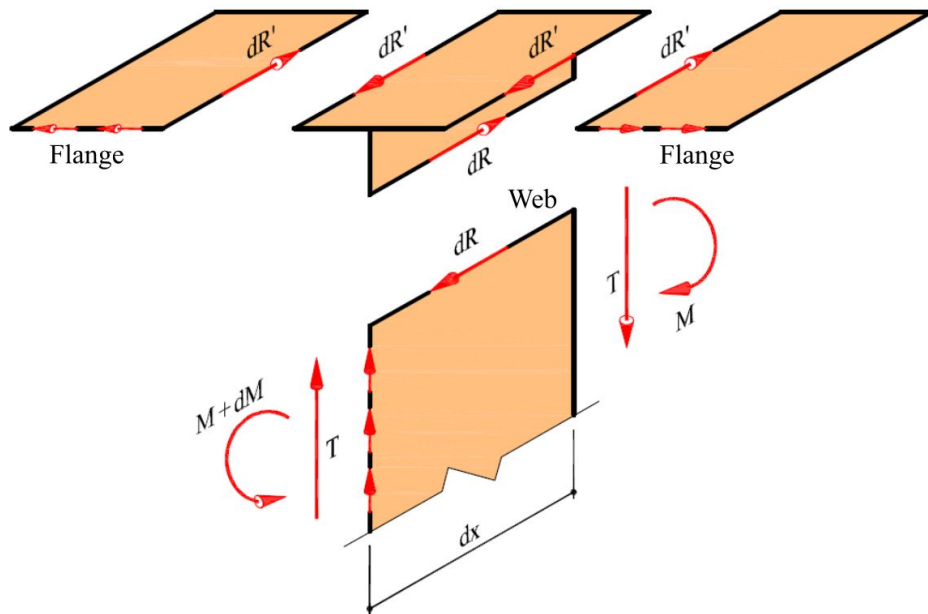


Figure 8.12: Wide Flange (W) or I cross section subjected to shear stress. Interfacial and tangential stresses

Figure 8.12 depicts the differential interfacial forces at the intersection

of the upper flanges and the web. By equilibrium of forces along the x -axis, it must be that:

$$dR = 2dR' \quad (8.24)$$

where dR is the web top interfacial force and dR' is each of the interfacial forces at the flanges, which are equal to each other due to cross section symmetry and antisymmetry of the loading with respect to the y -axis. It follows that

$$\tau_{\text{web}} e dx = 2 \tau_{\text{flange}} e' dx \quad (8.25)$$

where τ_{web} and τ_{flange} indicate the shear stresses in the web and flange, respectively, at the joining point. Therefore, by equilibrium, the following relationship between the two values must be fulfilled:

$$\frac{\tau_{\text{web}}}{\tau_{\text{flange}}} = \frac{2e'}{e} \quad (8.26)$$

The final distribution of shear stresses (with their corresponding directions) is shown in Figure 8.13. Because of the symmetry of the cross-section and the antisymmetry of the shear forces with respect to the z -axis, the shear stresses in the lower flange are antisymmetric to those in the upper flange. According to the approximations considered, the direction of the shear stress at a cross section point depends on the shape of the cross section and not on the shear stress direction, as stated at the beginning of this Chapter.

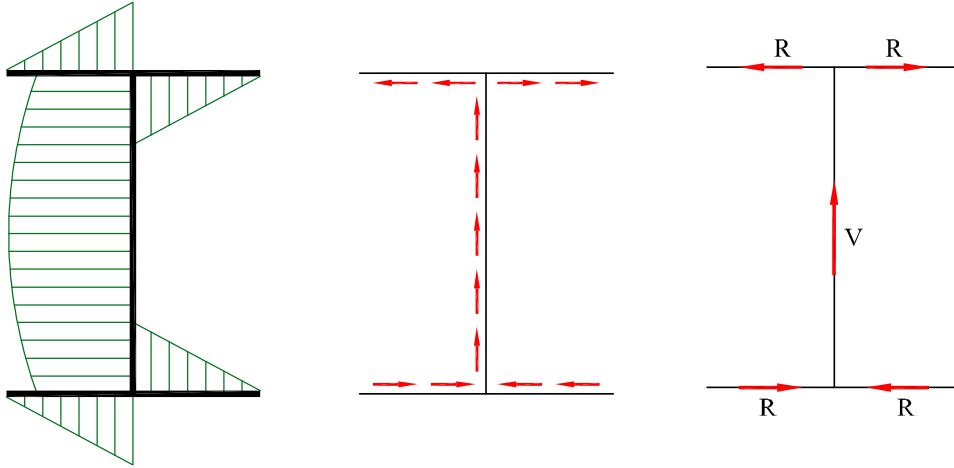


Figure 8.13: Tangential stress and interfacial stress distribution of W or I cross section under shear force

Furthermore, it can be observed that the stress distribution satisfies the equations of internal equilibrium. Let R and V be the resultants of the

shear stresses in the flanges and web, respectively; their values can be easily calculated by finding the area of the shear stress diagrams and multiplying by the corresponding thickness. In the web, the resultant V is:

$$V = \frac{T}{I_z e} \left(be' \frac{h}{2} + \frac{2}{3} \frac{e h^2}{4} \right) h e \quad (8.27)$$

$$= \frac{T}{I_z} \left(\frac{be' h^2}{2} + \frac{eh^3}{12} \right) \quad (8.28)$$

$$= T \quad (8.29)$$

since the moment of inertia of the profile, neglecting the moment of inertia of the flanges, is equal to:

$$I_z \simeq \frac{be' h^2}{2} + \frac{eh^3}{12} \quad (8.30)$$

On the other hand, it is obvious from symmetry that the total resultant of the horizontal tangential stresses in the wings is zero, which corresponds to the fact that $T_z = 0$.

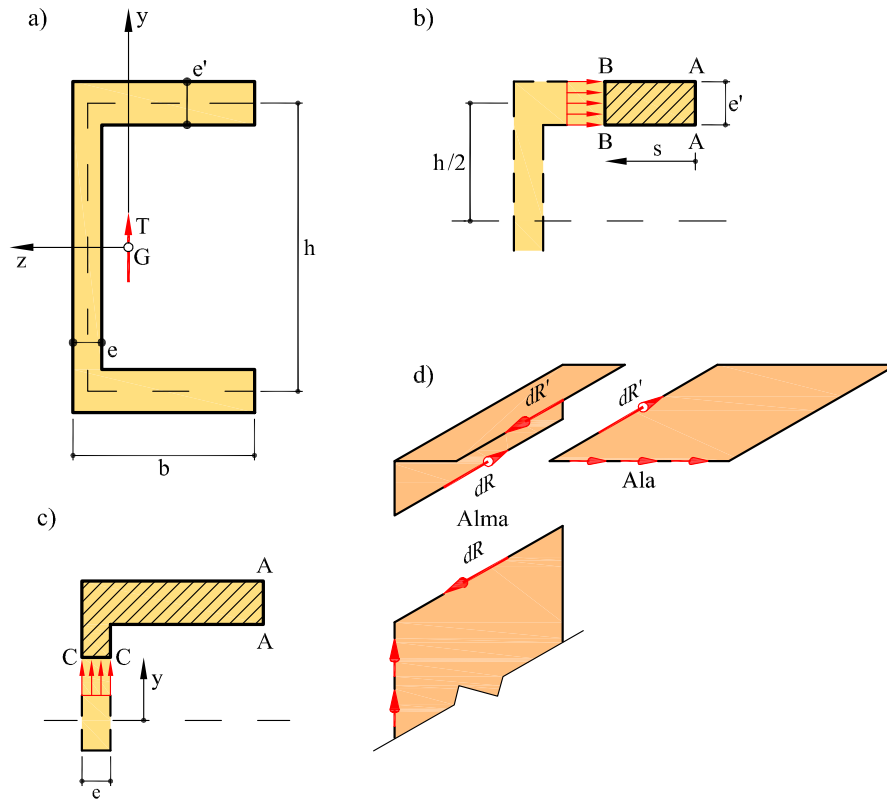


Figure 8.14: U cross section under shear force

U cross-sections

Consider a U-shaped cross section that is subjected to a shear force T in a direction parallel to its web and in an upward direction. Let h be the depth of the profile and b the width of the flanges, e' the thickness of the flanges and e the thickness of the web. Using the same hypotheses as in the previous section, the tangential stress due to shear T at point B of the top flange (Figure 8.14b) can be calculated as follows:

$$\tau(s) = \frac{T m_z^e(s)}{I_z e'} \quad (8.31)$$

with $m_z^e(s) = se'h/2$ for $e' \ll h$. Consequently:

$$\tau_{xz}(s) = \frac{T sh}{I_z 2} \quad (8.32)$$

It is observed that the shear stresses along the flange vary linearly from a zero value at the A end to a maximum value at the web joint equal to

$$\tau(s=b) = \frac{T bh}{I_z 2} \quad (8.33)$$

The direction of shear stresses is determined by the equilibrium between the normal forces acting on the shaped area and the interfacial forces, as showed in Figure 8.14d. For the lower flange, the same expression is obtained, but the interfacial and tangential stresses have the opposite direction of those of the upper flange. In order to calculate the shear stresses in the web, for example at a point C above the z -axis, the static moment of area is:

$$m_z^e(y) = be' \frac{h}{2} + e \left(\frac{h}{2} - y \right) \frac{1}{2} \left(\frac{h}{2} + y \right) \quad (8.34)$$

and then:

$$\tau(y) = \frac{Tbe'h}{2I_z e} + \frac{Th^2}{8I_z} \left[1 - \left(\frac{2y}{h} \right)^2 \right] \quad (8.35)$$

According to the previous expression, the variation of shear stresses in the web is symmetric and parabolic, with a maximum value on the z -axis ($y=0$) equal to:

$$\tau_{xy}^{\max} = \tau_{xy}(y=0) = \frac{Tbe'h}{2I_z e} + \frac{Th^2}{8I_z} \quad (8.36)$$

Consider R and V to be the resultants of the shear stresses in the flanges and web, respectively (Figure 8.15); their values can be easily calculated by computing the area of the shear stress diagrams and multiplying it by the corresponding thickness. The resultant V in the web is:

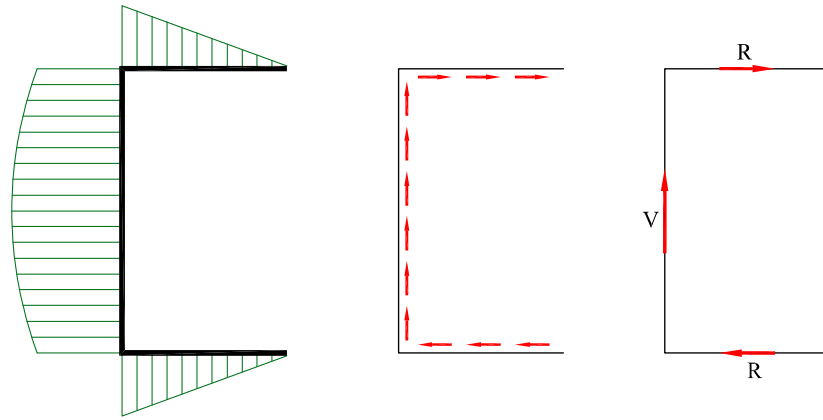


Figure 8.15: Tangential stresses distribution of U cross section under shear force

$$V = \frac{Tbe'h^2}{2I_z} + \frac{Teh^3}{12I_z} \quad (8.37)$$

$$= \frac{T}{I_z} \left(\frac{be'h^2}{2} + \frac{eh^3}{12} \right) \quad (8.38)$$

$$= T \quad (8.39)$$

since the moment of inertia of the profile, neglecting the moment of inertia of the flanges, is equal to:

$$I_z \simeq \frac{be'h^2}{2} + \frac{eh^3}{12} \quad (8.40)$$

It can be seen that the web takes the entire shear force, with negligible contribution from the flanges, which, however, are subjected to horizontal tangential stresses. The resultants of these stresses in the flanges, R , must be equal and in opposite directions, since $T_z = 0$.

8.5.2 Closed thin-walled cross sections

The general problem of determining the distribution of shear stresses produced by a shear force in a closed thin-walled section is statically undetermined since it is not possible to determine the value of the shear stresses, along a single cut through the thickness, using only equilibrium considerations.

Symmetric closed thin-walled cross sections

In the particular case where closed thin-walled the cross-section is symmetric about the shear stress axis, the shear stress distribution can be easily

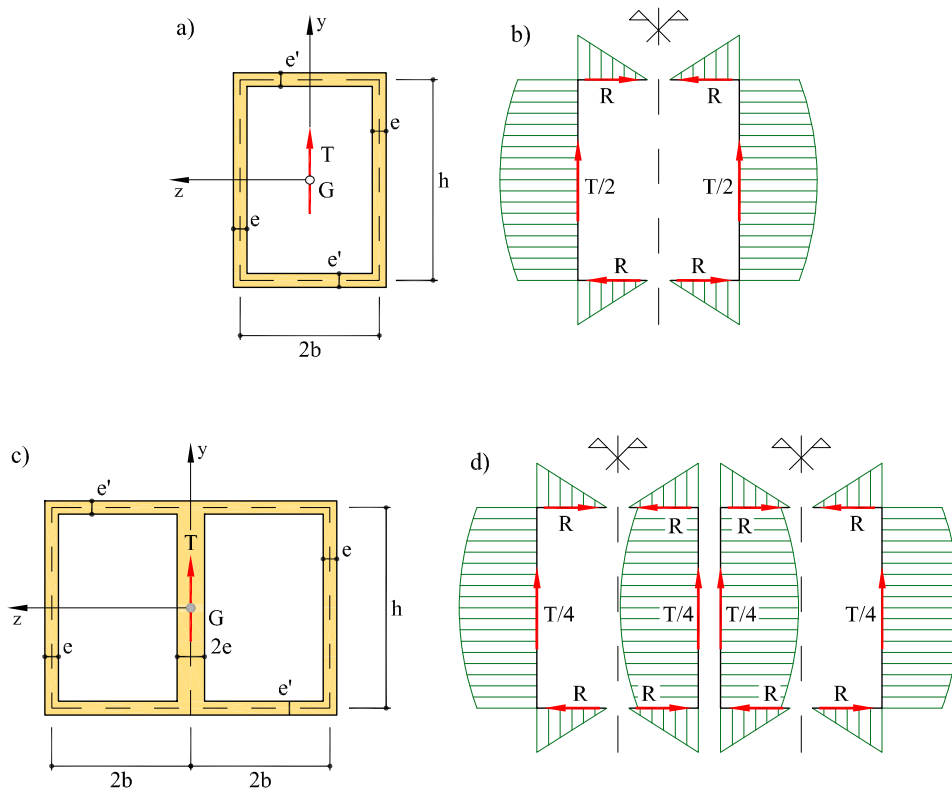


Figure 8.16: Symmetric closed thin-walled cross-sections subjected to shear stress according to its symmetry axis

determined using Collignon's formula. In this case, due to symmetry, the interfacial shear stresses on the plane of symmetry must be zero. The tangential stresses in the cut of the cross-section along the axis of symmetry must be zero, reciprocally. Therefore, for calculation, the section can be considered as two open half-sections, working side by side.

Consider the rectangular tubular section in Figure 8.16a, which has a depth h and a width $2b$ and is subjected to an upward shear force of value T .

Given the symmetry of the cross section and the loading, it can be analysed as two U-shaped sections of depth h and width b working side by side and carrying a shear force of $T/2$ (See Figure 8.16). The stress distribution in each is similar to that shown in Figure 8.15. It is worth noting that the shear stresses along the axis of symmetry are zero.

Similarly, the two-cell rectangular tubular section of depth h and width $4b$ in Figure 8.16c subjected to an upward shear force of T can be considered. The central web is double the thickness of the side webs.

Given the symmetry of the section and the loading, it can be analysed as

four U -sections, of depth h and width b , working side by side and supporting a shear force $T/4$ each (shown in Figure 8.16d).

8.6 Shear center

All previous developments have implicitly assumed that the action line of the resultant shear force passes through the centre of gravity of the cross section. In fact, the equilibrium approach used to derive Collignon's formula did not take into account the position of this shear force on the section plane and therefore, the corresponding tangential stress distribution (mean values and directions) corresponding to non uniform bending does not consider it either. However, it is necessary to consider whether the resultant of this distribution passes through the centre of gravity of the cross-section. In general, this is not the case, and thus the concept of shear centre, as the acting point of the resultant of the shear forces due to non uniform bending, must be introduced.

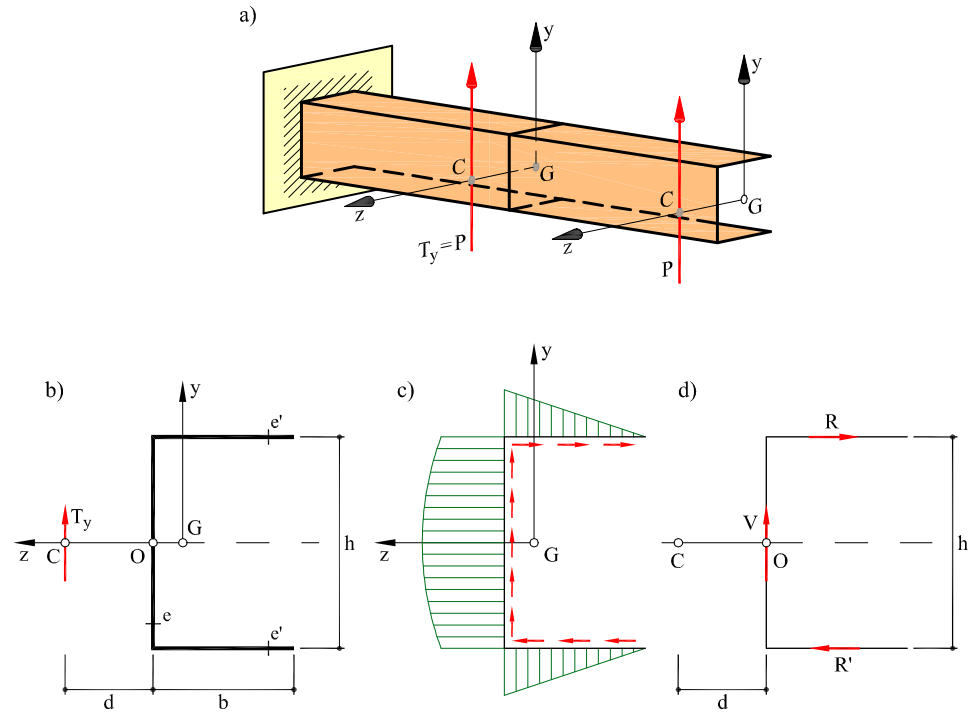


Figure 8.17: Shear center of a U cross-section

Consider a cantilever with a U cross-section, symmetrical about the z -axis (principal inertia), subjected to a constant shear force of value T_y along the length of the beam, vertical and upwards, whose action line passes

through a point C placed at a distance d from point O , the web's centre of gravity. Let h be the section's depth, b the width of the flanges, e' the thickness of the flanges, and e the thickness of the web (Figures 8.17 a and b).

Figure 8.17c depicts the tangential stress distribution that results from a vertical shear force on such a profile, as studied in Section 8.5.1. It has a linear distribution of horizontal stresses across the flanges and a parabolic distribution of vertical stresses across the web.

Let R and R' be the force resultants on the upper and lower flanges, respectively, and V is the force resultant on the web. To satisfy the equations of internal equilibrium, the system formed by the forces R , R' and V (Figure 8.17d) must be statically equivalent to the system formed by shear force acting on point C (Figure 8.17b). For this, it must be fulfilled that:

- (a) The resultants R and R' of the tangential stresses in the flanges must be equal and in opposite directions, since $T_z = 0$. Their value can be calculated as:

$$R = R' = \frac{1}{2} \frac{T_y (be' \frac{h}{2})}{I_z e'} be' = \frac{T_y b^2 h e'}{4 I_z} \quad (8.41)$$

- (b) The resultant V of the tangential stresses on the web equals the shear force T_y , $V = T_y$, as checked in Section 8.5.1.

- (c) Furthermore, since the internal resultants on the section are equal to the forces and the moments resulting from the stresses, the moment of the stresses at any point must be equal to that shear force T_y . Considering moments respect to point O it should be :

$$T_y d = R \frac{h}{2} + R' \frac{h}{2} = R h \quad (8.42)$$

where:

$$d = \frac{R h}{T_y} = \frac{b^2 h^2 e'}{4 I_z} \quad (8.43)$$

Therefore, the calculated tangential stress system is statically equivalent to the shear force T_y only if T_y passes through point C . It should also be noted that the position of point C is determined purely by the geometry of the cross-section, rather than by the value of the acting shear force T_y . As a result, it is a geometrical characteristic of the section.

As a general proposition, the resultant of the tangential stresses due to non-uniform bending (i.e. the shear force) always passes through a specific point in the plane of the cross-section. This point is the shear centre of the cross-section and is characteristic of the cross-section geometry. The

concept of the shear centre and the method for its calculation are due to R. Maillart (1922).

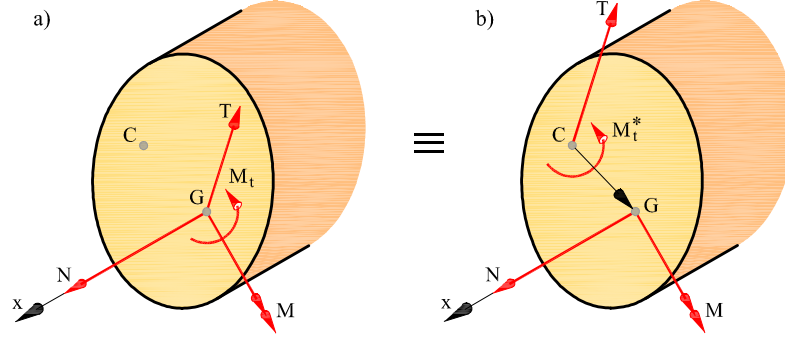


Figure 8.18: Stress reduction to the center of shear stresses

The introduction of the concept of the shear centre requires to reconsider the definition of the torsional moment reduced to the centre of gravity of the section, as introduced in Chapter 3. Consider a cross-section subjected to a system of internal forces consisting of a normal force $\mathbf{N}$, a shear force $\mathbf{T}$, a bending moment $\mathbf{M}$ and a torsional moment $\mathbf{M}_t$, reduced to the centre of gravity of the section, as shown in Figure 8.18. The force system formed by $\mathbf{T}$ and $\mathbf{M}_t$, reduced to the centre of gravity G (Figure 8.18a), is statically equivalent to the system formed by $\mathbf{T}$ and $\mathbf{M}_t^*$, reduced to the shear centre C (Figure 8.18b), with:

$$\mathbf{M}_t^* = \mathbf{M}_t + \overrightarrow{CG} \times \text{textbf T} \quad (8.44)$$

$$\text{textbf T} \quad (8.45)$$

$$= \int_S \mathbf{r} \times \boldsymbol{\tau} \, dS + \overrightarrow{CG} \times \int_S \boldsymbol{\tau} \, dS \quad (8.46)$$

$$= \int_S (\mathbf{r} + \overrightarrow{CG}) \times \boldsymbol{\tau} \, dS \quad (8.47)$$

$$= \int_S \mathbf{r}_c \times \boldsymbol{\tau} \, dS \quad (8.48)$$

where $\boldsymbol{\tau}$ is the tangential stress, $\mathbf{r} = \mathbf{r}(y, z)$ is the position vector of a generic point of the section with respect to G , and $\mathbf{r}_c = \mathbf{r}_c(y - y_c, z - z_c)$ is the corresponding position vector with respect to the centre C , of coordinates $C(y_c, z_c)$ (Figure 8.19).

$\mathbf{M}_t^*$ is the resultant moment of the shear stresses respect the shear center. Therefore, the existence or nonexistence of the torsional phenomenon, as will be studied in the following Chapter, and the associated tangential stresses is dependent on the value of the moment $\mathbf{M}_t^*$, reduced to the shear center,

and not on the value of M_t (in modulus), reduced to the centre of gravity. As a result, M_t^* is also known as the real torsional moment, and the shear centre is also known as the torsion centre.

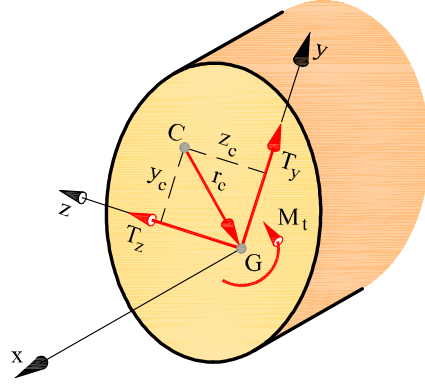


Figure 8.19: Coordinates of the shear center

The reduction of the torsional moment to the shear center can then result in three scenarios:

- (a) $T = 0$ and $M_t^* \neq 0$. The section is subjected to only torsion or torsion and pure bending (there is no shear force), and the tangential stresses are due to the torsional moment.
- (b) $T \neq 0$ and $M_t^* = 0$. The section is only subjected to non-uniform bending, and the tangential stresses are those caused by T passing through the shear center.
- (c) $T \neq 0$ y $M_t^* \neq 0$. The cross-section is subjected to bending-torsion and the total tangential stresses will be those due to bending (statically equivalent to the shear force T passing through shear center C) and torsion (equivalent to the torsional moment M_t^*).

In order to define the real torsional moment acting on the cross-section, the position of the centre of shear center must be known. Particularly, it is necessary to determine in which cross sections this does not coincide with the centre of gravity, as these may be subjected to concurrent bending and torsion inadvertently.

As an example, let us see what happens to the U -section cantilever with which the discussion was started, depending on where the action line of its end load (and, therefore, of the shear force) passes through. In Figure 8.20 a the shear force passes through the shear centre C and, in the lack of a torsional moment $M_t^* = M_t = 0$, the section is subjected to bending only (for this reason, the shear centre is also called the bending centre). In

Figure 8.20b the shear does not pass through C and therefore the section is subjected to bending-torsion, with an real torsional moment $M_t^* = T d$, anti-clockwise, which causes a torsional rotation in the sign indicated in the figure. Figure 8.20c demonstrates the same phenomenon, but with a clockwise torsional moment, opposite to that in the previous figure. This would be the case, for example, due to self-weight loads, since the line of action of T would then pass through the centre of gravity of the cross-section. As a result, a U -beam is subjected to torsion due to the action of its self-weight along its principal plane of inertia.

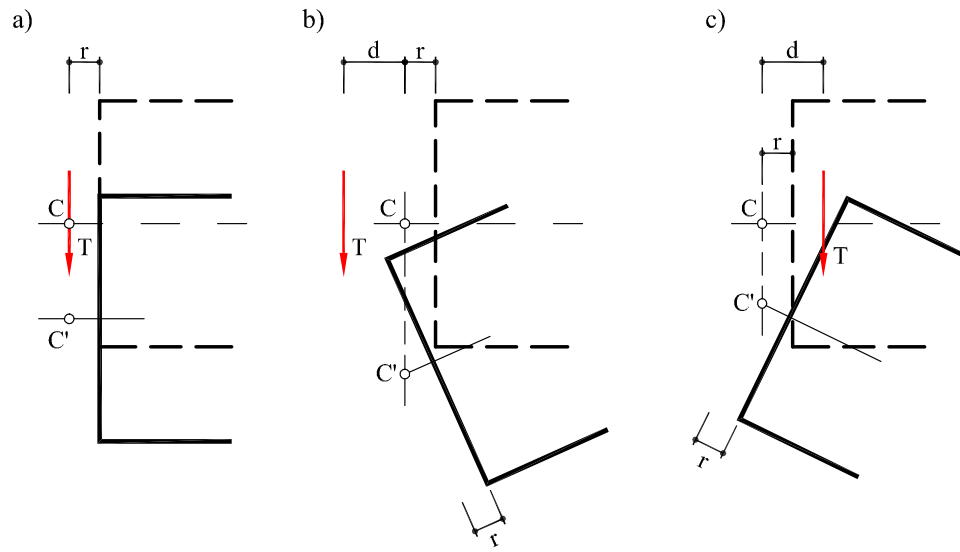


Figure 8.20: Cantiliever beam with U cross-section subjected to shear forces according different action lines

8.6.1 Position of the shear centre

The position of the shear centre C is determined directly from its definition; that is, its position is such that a shear force of any direction applied at C is statically equivalent to the system of forces resulting from the tangential stresses due to that shear force. The moment of a shear force in any direction, assumed to be applied at C , with respect to an arbitrary point O of the cross-section, must be equal to the moment resulting from the tangential stresses due to that shear force, with respect to point O (see Figure 8.21). This is:

$$\overrightarrow{OC} \times T = \int_S \mathbf{r}_o \times \boldsymbol{\tau} dS \quad (8.49)$$

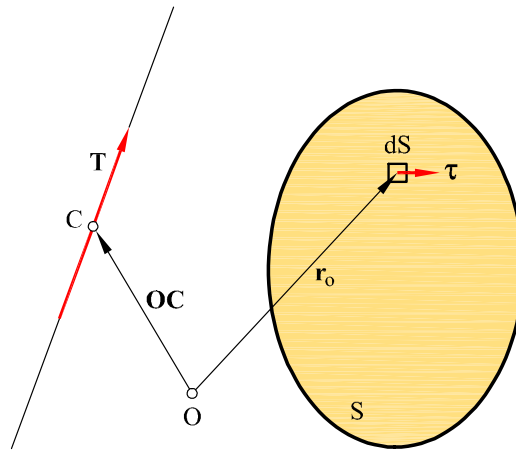


Figure 8.21: Determination of the location of the shear center

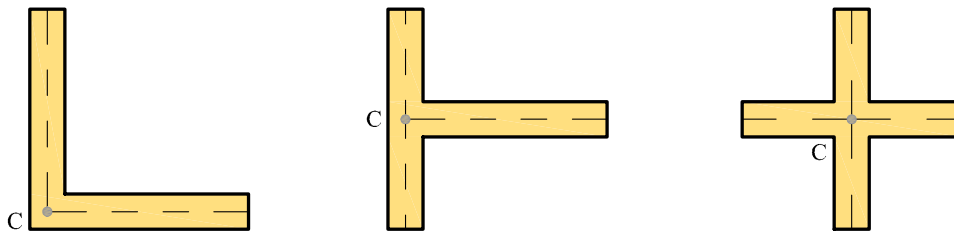


Figure 8.22: Position of the shear center in some particular cross-sections

In general, this is a scalar equation with two unknowns: the two coordinates of point C in the plane of the cross-section. Therefore, to solve it, it has to be stated twice, for shear forces $\mathbf{T}$ acting in two different directions. It is common to state it for shear forces acting along the two principal axes of inertia of the cross-section, as this simplifies the two cases to be studied to two plane shear problems (not skewed).

When the cross-section has a symmetry axis, the shear centre is on that axis, which reduces the unknown coordinates of the shear centre to one. This is the case, for example, of the U -section studied above, where the position of the shear stresses centre has been calculated by means of Eq. (8.49). If the section has two or more axes of symmetry, the shear stresses centre must be at the intersection of those axes and coincides with center of gravity.

This is the case, for example, of rectangular or circular cross-sections, as well as symmetrical W - and I - cross-sections. In open thin-walled cross-sections composed of straight sections joined at a point, such as those shown in Figure 8.22, this point is necessarily the shear centre, since the sum of

moments with respect to that point of the tangential stresses produced on each section by an arbitrary shear force is zero.

Example 8.6.1

The cantilever shown in Figure 8.23 is designed for a uniformly distributed downward vertical load of $p = 5 \text{ kN/m}$, assuming that the normal stresses do not exceed the allowable stress value, $\sigma_{\text{adm}} = 250 \text{ MPa}$. a) Determine the normal stress distribution and asses that the allowable stress is not exceeded. (b) Draw the shear stress distribution and calculate the maximum value in the beam. (c) Determine and draw the location of the shear centre.

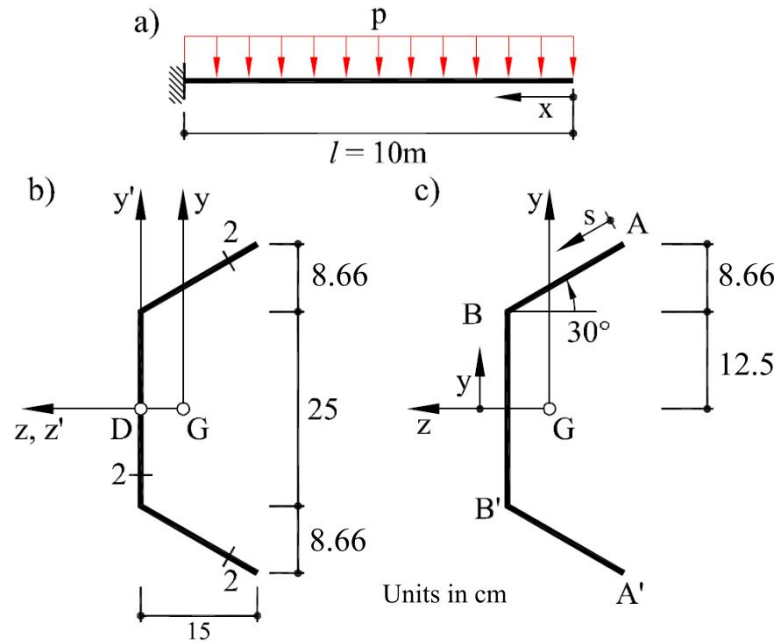


Figure 8.23: Cantilever beam in Example 8.6.1

(a) The bending moment and shear forces laws due to load p are:

$$M(x) = \frac{1}{2}px^2 \quad ; \quad T(x) = px$$

with maximum values in the clamping section ($x = 10 \text{ m}$) of value:

$$M_{\text{max}} = 5 \cdot \frac{10^2}{2} = 250 \text{ kN} \cdot \text{m} \quad T_{\text{max}} = 5 \cdot 10 = 50 \text{ kN}$$

The maximum normal stresses in the clamping section can be calculated by the following formula:

$$\sigma_{\max} = \frac{M_{\max}}{I_z} z_{\max} \quad (8.50)$$

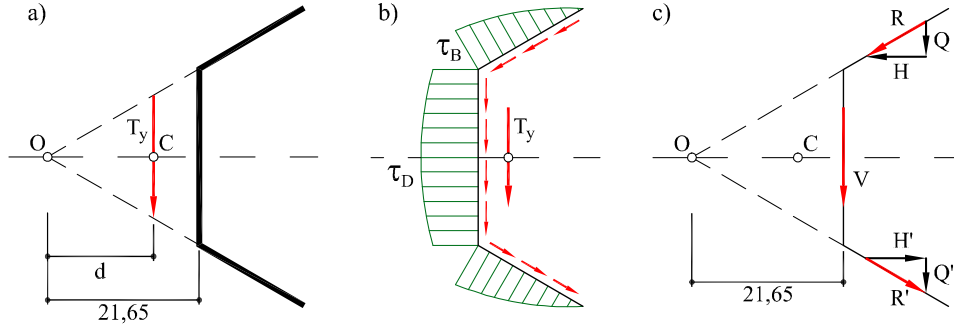


Figure 8.24: Position of the shear center in Example 8.6.1

(b) The tangential stress distribution is obtained by applying Collignon's formula:

$$\tau(s) = \frac{T_y m_z^e(s)}{I_z e}$$

In the upper flange, the value of the first moment of area m_z^e , as a function of the s -coordinate (Figure 8.23b) is equal to:

$$m_z^e(s) = e s (12.5 + 8.667 - \frac{s}{2} \sin 30^\circ) = 42.33 s - 0.5 s^2 \text{ cm}^3$$

Therefore, at point A the tangential stress is zero, $\tau_A = 0$. At point B ($s = 17,32, \text{ cm}$) the value of the first moment of area is equal to $m_z^e = 583,2 \text{ cm}^3$ and, consequently, the tangential stress at that point is:

$$\tau_B = \frac{T_y \cdot 583,2 \cdot 10^{-6}}{2,2669 \cdot 10^{-4} \cdot 2 \cdot 10^{-2}} = 128,6 T_y \text{ Pa}$$

with T_y in Newtons. By replacing the value of the shear stress T_y in the previous expression with $T_{\max} = 50 \text{ kN}$, $\tau_B = 6,43 \text{ MPa}$ is obtained. The stresses in the lower flange are $\tau_A = \tau_{A'} = 0$ and $\tau_B = \tau_{B'} = 6,43 \text{ MPa}$. due to the symmetry of the section and the anti symmetry of the loading. The signs of the stresses in the flanges can be deduced from simple equilibrium considerations, as shown in Figure 8.24b.

In the web, the static moment about the z -axis is calculated in terms of the y -coordinate of the cut (Figure 8.23b):

$$m_z^e(y) = 583,2 + (12,5 - y) e \frac{1}{2} (12,5 + y) = 739,45 - y^2 \text{ cm}^3$$

and the maximum value for point D ($y = 0$ cm) is $m_z^e(y = 0) = 739.5 \text{ cm}^3$; the tangential stress at that point is:

$$\tau_D = \frac{T_y \cdot 739.45 \cdot 10^{-6}}{2.2669 \cdot 10^{-4} \cdot 2 \cdot 10^{-2}} = 163.1 T_y \text{ Pa}$$

with T_y in Newtons. For a shear force $T_{\max} = 50 \text{ kN}$, it is equal to $\tau_D = 8.155 \text{ MPa}$.

Note the small value of the tangential stresses due to shear compared to the normal stresses due to the bending moment.

(c) Because the cross-section is symmetrical about the z -axis, the shear centre C must necessarily be located on the z -axis, and its position on the z -axis must be calculated. To do this a vertical shear passing through the shear centre is applied, as shown in Figure 8.24a. The corresponding tangential stress distribution is known because it coincides with the one calculated in the section (b) of this example. The shear stress resultants in the flanges R and R' as well as the web V (Figure 8.24b) are:

$$\begin{aligned} R &= \int_{s=0}^{s=17.32} \frac{T_y m_z^e(s)}{I_z e} e ds = \frac{T_y}{I_z} \int_0^{17.32} (42.32 s - 0.5 s^2) ds = 0.2419 T_y \\ Q &= R \sin 30^\circ = 0.1210 T_y \\ V &= 2 \int_{y=0}^{y=12.5} \frac{T_y m_z^e(y)}{I_z e} e dy = \frac{2T_y}{I_z} \int_0^{12.5} (739.45 - y^2) dy = 0.7580 T_y \end{aligned}$$

Note that $V + 2Q = T_y$. Equating the moments with respect point O of the systems of forces formed by, on the one hand, T_y in Figure 8.24a, and, on the other hand, the resultants R, R' and V in Figure 8.24c

$$T_y d = V \cdot 21.65 \quad \Rightarrow \quad d = 21.65 \frac{V}{T_y} = 16.41 \text{ cm}$$

which gives the position of the centre of shear forces C on the axis of symmetry of the section.

Chapter 9

Torsion

9.1 Introduction

A prismatic element is subjected to *simple torsion* when the resultant moment acting on its cross-sections has only component along the longitudinal x axis of the element, that is, a *torsion*, or *twisting moment* or *torque* M_t (Figure 9.1a). If the acting torque is constant throughout the element, the state is of *pure torque*.

The determination of the torque must be done with respect to the shear center C of the cross-section, in order to be able to determine the *real torque* that acts on it (see Section 8.6 dedicated to *shear center*). In this Chapter, it will always be considered that the torque M_t is the *real torque* and, therefore, the corresponding qualifier is omitted.

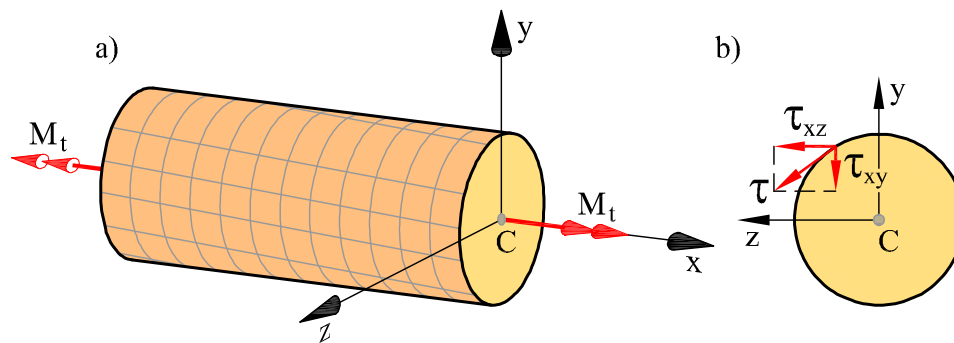


Figure 9.1: Bar subjected to torsion

Given that the different force resultants that act on a cross-section are the forces and moments resulting from the stresses that act on that section, in the case of *pure torsion* with constant torque M_t , the following equations

must be satisfied:

$$\begin{aligned}
 N &= \int_S \sigma_x dS = 0 & ; & \quad T_y = \int_S \tau_{xy} dS = 0 \\
 T_z &= \int_S \tau_{xz} dS = 0 & ; & \quad M_t = \int_S (\tau_{xz} y - \tau_{xy} z) dS \\
 M_y &= \int_S \sigma_x z dS = 0 & ; & \quad M_z = - \int_S \sigma_x y dS = 0
 \end{aligned} \tag{9.1}$$

where the distances y, z are measured with respect to the shear center C (Figure 9.1). The action of a torque on the section implies the appearance of tangential stresses on the section (Figure 9.1b), in such a way that the integral relations:

$$\mathbf{T} = \int_S \boldsymbol{\tau} dS = 0 \quad ; \quad \mathbf{M}_t = \int_S \mathbf{r}_c \times \boldsymbol{\tau} dS \tag{9.2}$$

are fulfilled, where $\mathbf{r}_c$ is the position vector with respect to the shear center. In the cases studied in this Chapter, the longitudinal stresses are null, $\sigma_x \equiv 0$. Non-uniform torsion, not discussed here, is a notorious case in which the normal stresses are not zero.

As in previous Chapters, these integral equations are not enough to determine the distribution of tangential stresses, so it is necessary to make hypotheses about the deformation of the differential slice, whose validity must be supported by experience.

The general study of prismatic elements subjected to torsion is a complex subject that escapes from the fundamental hypotheses on which Strength of Materials is based. The main difficulty is that, in general, cross-sections subjected to torsion do not remain plane, but suffer deformation out of their plane, a phenomenon known as *warping*. Based on this fact, a distinction is made between *uniform torsion*, when the relative warping is negligible, and non-uniform *non-uniform* torsion, when relative warping must be considered.

Uniform torsion is studied by Coulomb's theory, valid for cross-sections in which there is no warping, and Saint-Venant's theory, more general, valid for situations without relative warping. Both theories are compatible with the general approach of Strength of Materials. On the contrary, when the relative warping is relevant, *the Saint-Venant principle* is not fulfilled, that is, the stresses in a cross-section do not depend exclusively on the value of the torque on it. In these situations the study remains, strictly speaking, outside the field of application of Strength of Materials. The problem can be solved with a suitable *ad-hoc* approach, known as the *theory of non-uniform torsion*, or by the general approach of the Theory of Elasticity.

In this Chapter, simple torque on cylindrical bars with a circular cross-section will be studied in detail, through Coulomb's theory. The Saint-

Venant theory for uniform torsion will also be introduced, and the appropriate analogies will be described to consider the torsion of elements of arbitrary cross-section, obtaining results of practical interest for the study of thin-walled profiles, both open and closed. In summary, in what follows we will only treat the case of uniform torsion, leaving treatment of non-uniform torsion situations for specialized books.

9.2 Coulomb theory

9.2.1 Circular sections

Let us consider a straight prismatic element of constant circular section, subjected to *torsion* under the action of two moments M_t , equal and in opposite directions, applied at its ends cross-sections.

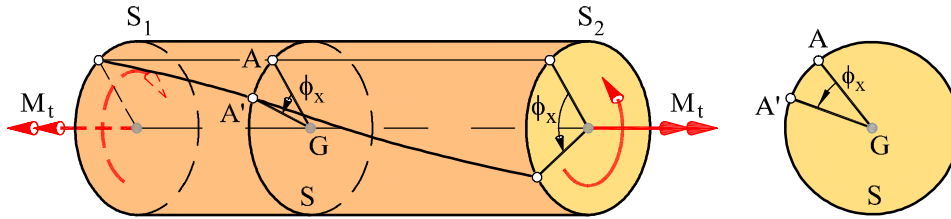


Figure 9.2: Element of circular cross-section subjected to pure torque

Geometrical considerations based on the symmetry of the element and the loading (Figure 9.2), make it possible to ensure that in the torsional deformation:

- the plane cross-sections rotate around their center of gravity, due to axial symmetry with respect to the axis of the element.
- plane cross-sections remain circular and plane. Indeed, the cross-sections must remain circular due to axial symmetry with respect to the axis of the element. In addition, they must remain plane due to the symmetry of the loading with respect to any cross-section.
- the radii of the cross-sections remain straight, due to the symmetry of the solicitation with respect to any cross-section.
- the angle between any two radii of the cross-section remains unchanged, due to axial symmetry with respect to the axis of the element.

In summary, the deformation by torsion of straight elements of circular cross-section can be described by the hypothesis of C. A. de Coulomb (1784), which can be stated:

“In a straight element with a circular cross-section subjected to pure torsion, the deformation is such that each cross-section rotates around its center without deforming in its plane and without warping, as if it were a rigid disk. The longitudinal fibers of the bar are deformed into helices”.

Under the hypothesis of small displacements and deformations twist, the rotation due to torsion are small enough and the curvature of the helical arc of the deformed fibers can be assumed to be negligible, that is, these can be assumed rectilinear.

The Coulomb deformation hypothesis is purely geometric, it can be verified experimentally and agrees with the results of the Theory of Elasticity, even when the torque is variable along the axis of the part, that is, in situations of *simple torque*.

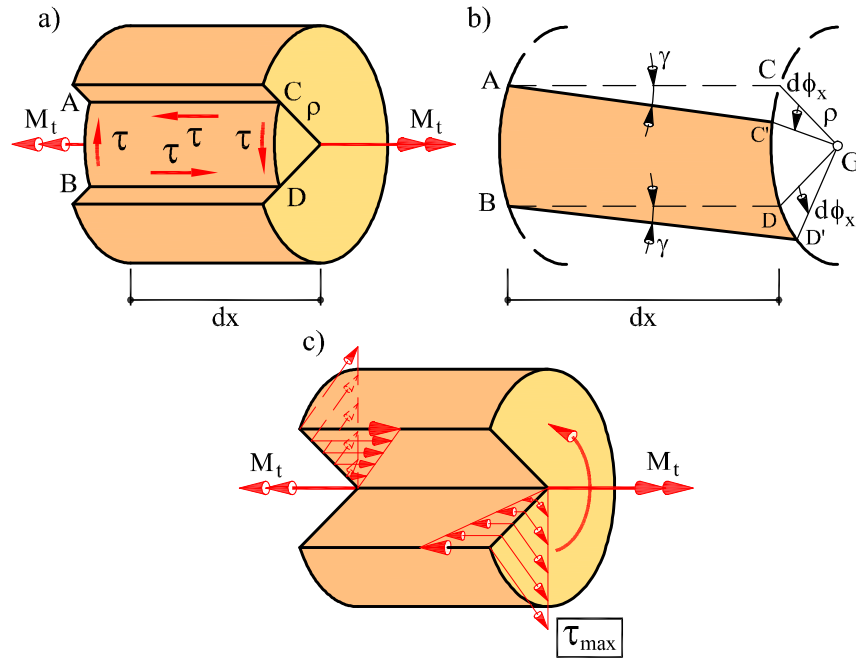


Figure 9.3: Slice subjected to Coulomb torsion

Let us analyze a differential slice of length dx of a prismatic element of circular section subjected to simple *torque*, as shown in Figure 9.3a. Let us consider in it the differential element $ABCD$, located on a cylinder of radius ρ , concentric with the axis of the element; this element deforms to torsion according to the “parallelogram” $ABC'D'$ of Figure 9.3b, where AC' and BD' are helix segments of infinitesimal length, so they can be considered rectilinear.

The angular deformation of the element is:

$$\gamma \simeq \tan \gamma = \frac{\overline{CC'}}{\overline{AC}} = \frac{\rho d\phi_x}{dx} = \rho \theta \quad (9.3)$$

where ϕ_x is the twist angle, and $\theta = d\phi_x/dx$ is the twist angle per unit of length. Note that ϕ_x is dimensionless, and is measured in radians [rad]. In contrast, the twist angle per unit of length, θ , is measured in [rad/L]. Therefore, it follows that the distortions are proportional to ρ :

$$\gamma = \rho \theta \quad (9.4)$$

According to Hooke's law, the shear stresses in the element $ABCD$ are proportional to the distortions, in the form:

$$\tau = G \gamma = G \rho \theta \quad (9.5)$$

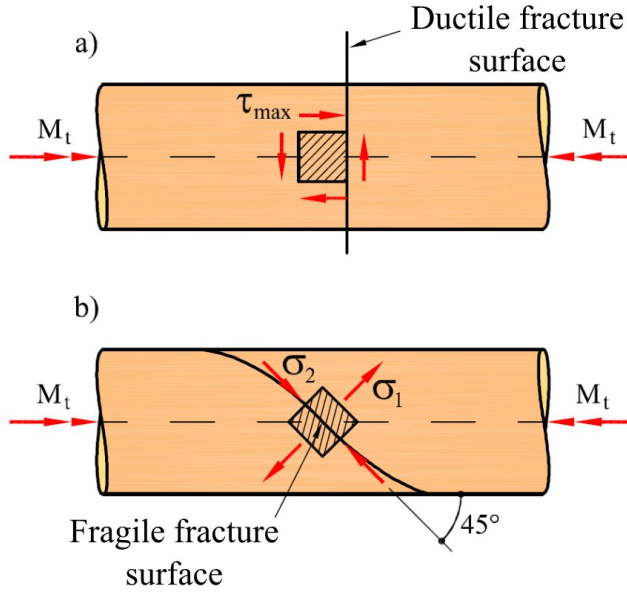


Figure 9.4: Failure of bar element subjected to Coulomb torsion: (a) ductile failure, (b) rigid failure

where G is the modulus of shear stiffness of the material. These tangential stresses have the directions and signs shown in Figures 9.3 a and c. It is observed that the shear stresses at a point of the section are perpendicular to the radius passing through that point, and that they vary linearly with the distance ρ from the center of the cross-section; therefore, the maximum

values are given in its the contour.

The torsional moment resulting from the tangential stresses must be equal to the acting torque M_t ; therefore:

$$M_t = \int_S \rho \tau dS = \int_S G \theta \rho^2 dS = G \theta \int_S \rho^2 dS = G \theta I_p \quad (9.6)$$

where I_p is the polar moment of inertia of the circular section. For a circle of radius r , it is $I_p = \pi r^4/2$, and the twist angle per unit of length θ is:

$$\theta = \frac{M_t}{G I_p} \quad (9.7)$$

The product $G I_p$ is called *torsional stiffness* of the cross-section. Substituting this value in the expression (9.5) of the tangential stresses, we obtain:

$$\tau = \frac{M_t \rho}{I_p} \quad (9.8)$$

Therefore, in a circular cross-section the value of the maximum tangential stress is given in the outer circumference and its value is given by the expression:

$$\tau_{\max} = \frac{M_t r}{I_p} = \frac{2M_t}{A r} \quad (9.9)$$

It is interesting to note that the differential elements such as the one in Figure 9.4 are subjected to pure shear. To this state of stress correspond principal stresses $\sigma_1 = -\sigma_2 = \tau_{\max}$ in directions at $\pm 45^\circ$ with the longitudinal axis of the element.

If the material of the bar has tensile and compressive strengths of similar values, the torsion failure occurs in a plane perpendicular to the axis of the element, when the maximum tangential stress reaches the failure value. This type of failure is common in most metals and is generally of a ductile nature. On the contrary, if the strength of material of the element to tension is much smaller than to compression, the failure occurs due to tension, appearing helical cracks at 45° with the axis of the element. This type of failure is generally brittle and occurs in materials such as cast iron, concrete, and rocks.

The total twist rotation of the bar due to torsion, between its ends, is:

$$\phi_x = \int_l \theta dx = \int_0^l \frac{M_t(x)}{G I_p(x)} dx \quad (9.10)$$

where l is the total length of the bar. In the particular case of pure torsion and a bar of constant cross-section, the total torsion twist is:

$$\phi_x = \frac{M_t l}{G I_p} \quad (9.11)$$

9.2.2 Hollow circular cross-section

Consider a hollow circular section, with r_e and r_i as the outer and inner radii, respectively, as shown in Figure 9.5. The geometric considerations and the expressions obtained in the two previous Sections are, also applicable in this case. Consequently, the distribution of stresses in a tubular section is represented in Figure 9.5. The polar moment of inertia I_p and the torsion modulus of the hollow section are:

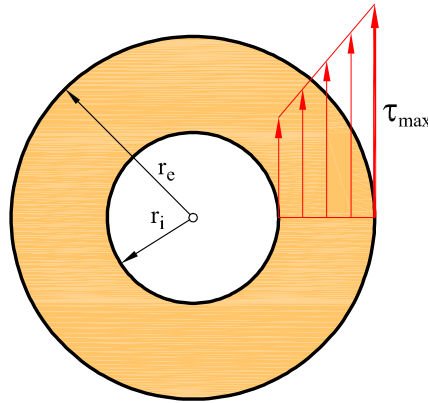


Figure 9.5: Hollow circular cross-section

$$I_p = \frac{\pi (r_e^4 - r_i^4)}{2} \quad (9.12)$$

In addition, if the thickness of the section is small in relation to the outer radius, it can be approximated:

$$I_p \simeq 2\pi r_m^3 e = A r_m^2 \quad (9.13)$$

with $A = 2\pi r_m e$, equal to the area of the cross-section. In this way, in a hollow circular section of small thickness, the tangential stress due to the torque M_t can be considered *uniform* throughout the section and of value:

$$\tau_{med} = \frac{M_t}{A r_m} \quad (9.14)$$

Thus, it can be seen that the material is very well exploited in thin-walled tubular sections under torsion, since all the material is subjected to practically uniform stress.

9.2.3 Closed thin-walled cross-sections

Consider a slice of an element with a closed thin-walled cross-section subjected to uniform torsion and consider the equilibrium of a differential element like the one in Figure 9.6. Since the thickness is small, the distribution

of tangential stresses through the thickness is almost uniform.

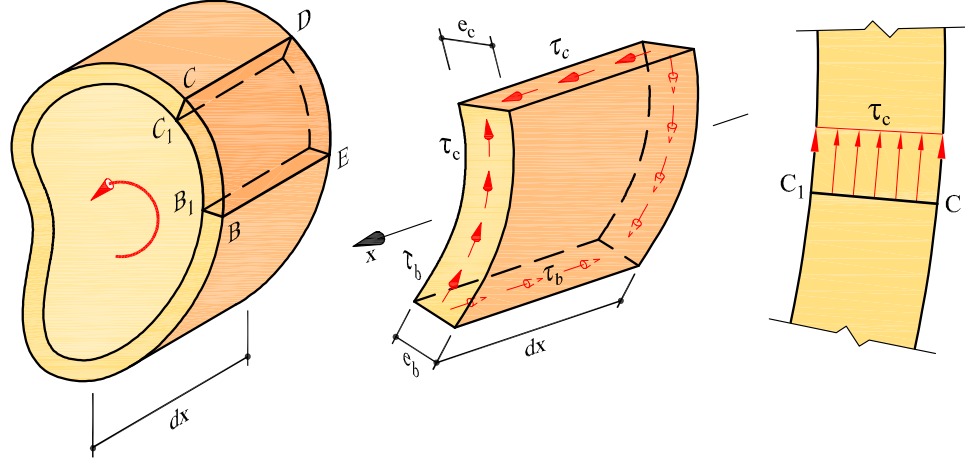


Figure 9.6: Closed thin-walled cross-section

By equilibrium of forces along the x axis, it must be:

$$\tau_b e_b dx - \tau_c e_c dx = 0 \quad (9.15)$$

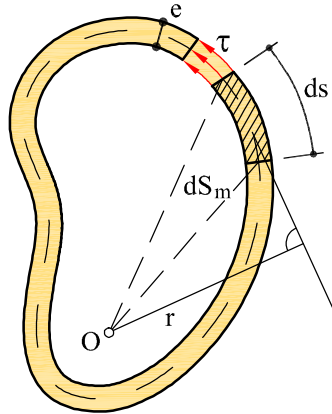


Figure 9.7: Area enclosed by the midline

where e is the thickness at a given point of the cross-section. From the previous expression, it follows that:

$$\tau_b e_b = \tau_c e_c = q = cte \quad (9.16)$$

This means that *the flux of shear stresses through the thickness, $q = \tau e$, is*

constant throughout the cross-section (*Bredt's hypothesis*).

To find the stress on each cut of the section, the equality between the moment resulting from the stresses and the acting torque is established, with respect to any interior point O . It is:

$$M_t = \int_S r \tau dS = \oint_s r \tau e ds = \oint_s q r ds = q \oint_s r ds = q \oint_{S_m} 2 dS_m = 2qS_m \quad (9.17)$$

where r is the distance to the point O (arbitrary) of the tangent at a given point on the midline, s is the arc length measured along the midline, and S_m is the area enclosed by the midline of the section (Figure 9.7).

Therefore, the stress flow q is equal to:

$$q = \tau e = \frac{M_t}{2S_m} \quad (9.18)$$

This expression is known as *Bredt's first formula*.

Using energy considerations, it can be deduced that in closed thin-walled cross-sections the torsion twist per unit length, θ , is:

$$\theta = \frac{1}{2GS_m} \oint_s \frac{q}{e} ds \quad (9.19)$$

and by substituting $\tau = q/e = M_t/(2S_me)$, it is:

$$\theta = \frac{M_t}{G 4 S_m^2 / \oint_s \frac{ds}{e}} = \frac{M_t}{GI_t} \quad (9.20)$$

This equation is known as *Bredt's second formula*, and it defines the value of the torsional inertia in this type of cross-section:

$$I_t = \frac{4 S_m^2}{\oint_s \frac{ds}{e}} \quad (9.21)$$

where the integral $\oint_s \frac{ds}{e}$ is the perimeter of the midsection divided by the thickness. In the case of polygonal midlines this term is equal to $\sum_i \frac{l_i}{e_i}$, where l_i is the length of each side of the polygon and e_i the corresponding thickness.

Example 9.2.3.1

Calculate the torsional inertia and the maximum torque that can be resisted by an element with a *closed* circular ring section of radius r and thickness e (Figure 9.8) if the allowable tangential stress is τ_{adm} .

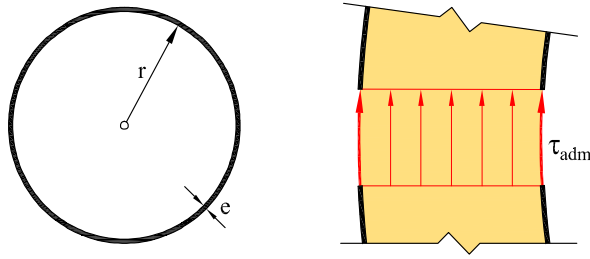


Figure 9.8: Closed circular ring

The torsional inertia of the closed cross-section is equal to (see Eq. (9.21) and Eq. (9.13a)):

$$I_t = \frac{4S_m^2}{\oint \frac{ds}{e}} = \frac{4(\pi r^2)^2}{\frac{2\pi r}{e}} = 2\pi r^3 e = Ar^2 = I_p$$

and comparing with the torsional inertia of the same open cross-section (see Example 9.3.5.1) the relationship is:

$$\frac{(I_t)_{\text{ce}}}{(I_t)_{\text{ab}}} = \frac{Ar^2}{\frac{1}{3}Ae^2} = 3 \left(\frac{r}{e} \right)^2$$

The allowable torque, according to the Eq. (9.18) is:

$$M_t^{\text{max}} = 2S_m \tau_{\text{adm}} e = 2\pi r^2 e \tau_{\text{adm}} = Ar \tau_{\text{adm}}$$

and all the points of the section are subjected to a stress τ_{adm} , as the thickness is uniform.

Example 9.2.3.2

Calculate the length a of the cross-section of Figure 9.9, subjected to a torque $M_t = 150 \text{ kN}\cdot\text{m}$, considering that the allowable tangential stress is $\tau_{\text{adm}} = 100 \text{ MPa}$. Calculate the angle of torsion per unit length if the modulus of shear stiffness is $G = 100 \text{ GPa}$.

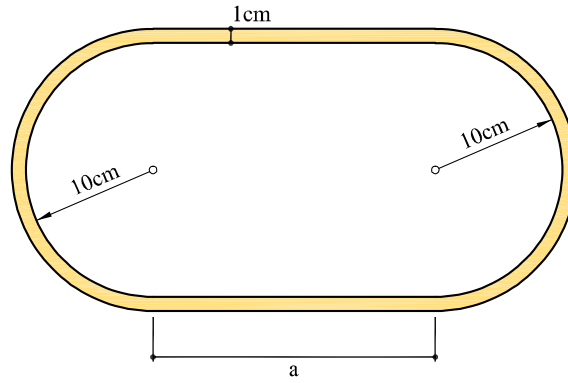


Figure 9.9: Cross-section of Example 9.2.3.2

The value of the tangential stress is uniform throughout the cross-section and is:

$$\begin{aligned}\tau &= \frac{M_t}{2S_m e} \\ &= \frac{150 \cdot 10^3}{2(a \cdot 21.0 + \pi \cdot 10.5^2) \cdot 10^{-4} \cdot 10^{-2}} \text{ Pa} = \frac{1.5 \cdot 10^5}{42.0a + 692.72} \text{ MPa}\end{aligned}$$

Setting $\tau < \tau_{\text{adm}} = 100 \text{ MPa}$ gives $a > 19.22 \text{ cm}$. The torsional inertia of the section for $a = 20 \text{ cm}$ is:

$$\begin{aligned}I_t &= \frac{4S_m^2}{\oint \frac{ds}{e}} = \frac{4S_m^2}{\sum \frac{l_i}{e_i}} \\ &= \frac{4 \cdot (20.0 \cdot 21.0 + \pi \cdot 10.5^2)^2}{\frac{2 \cdot (20.0 + \pi \cdot 10.5)}{1.0}} \text{ cm}^4 = 22\,168.14 \text{ cm}^4\end{aligned}$$

Then, the angle of torsion per unit length is:

$$\begin{aligned}\theta &= \frac{M_t}{GI_t} \\ &= \frac{150 \cdot 10^3}{100 \cdot 10^9 \cdot 2.2168 \cdot 10^{-4}} \text{ rad/m} \\ &= 6.766 \cdot 10^{-3} \text{ rad/m} = 0.387^\circ \text{ m}^{-1}\end{aligned}$$

9.3 Saint-Venant torsion

9.3.1 General formulation

As shown above, the solution of the torsion problem of an element with a circular section is obtained by assuming as hypothesis of deformation that the

plane cross-sections of the element remain plane and rotate without deforming. This hypothesis, formulated by Coulomb, was later applied by Navier to the case of a prismatic element of rectangular cross-section, reaching the following *erroneous* conclusion: for a given torque M_t , the angle of torsion is inversely proportional to the polar moment of inertia with respect to the centroid of the section and the maximum values of the tangential stress occur at the points farthest from the centroid.

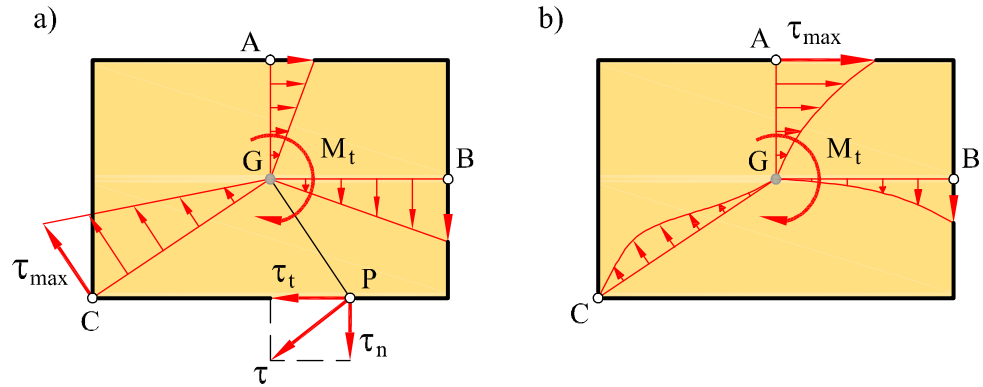


Figure 9.10: Distribution of shear torsion stresses in a rectangular cross-section: (a) Coulomb's theory, (b) Saint-Venant's theory

On the other hand, in experimental tests carried out with elements of rectangular cross-section it is verified that:

- (a) the angular distortion is maximum at the midpoints of the longest sides of the rectangle, which are not the farthest points from the center of gravity of the section, while in the corners the distortion is null. This implies that the real distribution of shear stresses in a rectangular cross-section is as shown in Figure 9.10b, and
- (b) the cross sections of the element *do not remain plane* when are subjected to torsion, but rather warp.

These experimental evidence clearly indicate that Coulomb's theory is not adequate to represent the phenomenon of torsion in pieces of non-circular cross-section.

The exact solution of the *uniform torque* problem in prismatic elements of arbitrary shape was obtained by Saint-Venant (1853), assuming that the deformation is *uniform*, that is, identical for all cross-sections of the element and consisting of:

- (a) a *rigid twist rotation* of the cross-sections in their plane, and

(b) a *warping* of the sections out of its plane.

The Reciprocity Theorems prove that the center of rotation of the cross-sections in their plane is, precisely, the shear center, whose position can be calculated as shown in Chapter 8. For this reason, the shear center is also called *center of torsion* of the cross-section. In uniform torsion, the locus of the torsion centers is called *axis of torsion*, and is an axis parallel to the axis of the element.

Figure 9.11 shows the deformation in pure torque of two prismatic elements, one with a circular cross-section and the other of rectangular one. The piece with a circular section deforms according to the Coulomb hypothesis, that is, without warping, and the cross-sections remain plane. On the contrary, the piece with a rectangular cross-section is deformed according to the Saint-Venant hypothesis, and the cross-sections rotate in their plane and they warp.

Obviously, Coulomb's theory is a particular case of Saint-Venant's theory in which warping is zero.

In Saint-Venant's theory, both the twist torsion per unit length, θ , and the warping are uniform throughout the element. Assuming that the warping is uniform for all cross-sections simplifies the statement of the problem, since in this way normal stresses due to the warping phenomenon do not appear. However, it is common for an element subjected to torsion to have some cross-section forced to remain fixed, as is the case of clamped sections that cannot warp during torsion. This restriction of the warping of a cross-section gives rise to the appearance of non-uniform torsion throughout the piece.

The influence that non-uniform warping has on the behavior of a part subjected to torsion may be important when the cross-sections are open and thin-walled, but it is negligible in the other types of sections.

The complete study of the problem of uniform torsion (or Saint-Venant's torsion) is done through the Theory of Elasticity.

The solution of the elastic problem of uniform torsion in sections of arbitrary shape is a complex mathematical problem, which is often solved by numerical methods, such as finite differences, finite elements, or boundary elements.

9.3.2 Membrane analogy

As an alternative to solving the elastic problem of uniform torsion by analytical or numerical methods, the solution can be approached by the *membrane analogy*. Let us imagine a homogeneous elastic membrane whose edges are fixed on a contour equal to that of the cross-section of the bar to be studied, subjected to uniform traction at its edges and normal pressure also uniform (Figure 9.12a). If the cross-section is multiply connected, that is, it has interior holes, these are covered with rigid plates of negligible weight (Figure 9.12b).

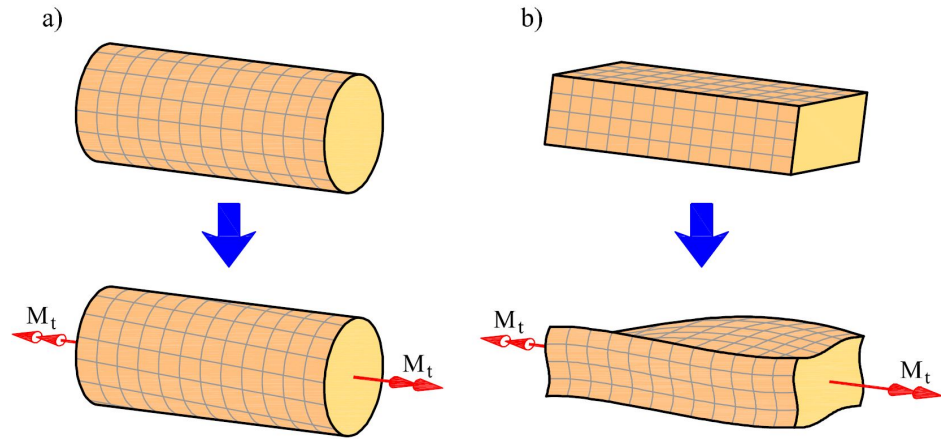


Figure 9.11: Deformation by torsion: (a) element of circular section (b) element of rectangular section

Let q be the normal pressure per unit area acting on the membrane and Q the uniform traction per unit length acting on its contour. It can be shown that the membrane deformation problem and the uniform torsion elastic problem have an analogous analytical formulation. Therefore, representing the deformed surface of the membrane by level curves, several important conclusions regarding the distribution of tangential stresses in the torsion problem can be deduced:

(a) the tangential stress at a generic point of the cross-section has the direction of the tangent to the curve that passes by this point (Figure 9.13). In particular, the tangential stress in the contour is parallel to it.

(b) the magnitude of the tangential stress τ at a generic point of the cross-section is analogous to the slope of the membrane at this point (Figure 9.13). Therefore, the tangential stress will be greater in the areas of the cross-section where the membrane presents a greater slope and the closer the level curves are to each other.

(c) the torque acting on the cross-section is *analogous* at twice the volume enclosed by the deformed elastic membrane.

Therefore, the membrane analogy provides a qualitative and quantitative description of the distribution of shear stresses that appear in a cross-section subjected to uniform torsion.

Thus, for example, in a rectangular cross-section, the vertices A_1, B_1, C_1, D_1 have zero tangential stress, since the slope of the membrane is zero at these points, and the stress trajectories are like those in Figure 9.14 a; the maximum stresses occur at the midpoints of the longest sides, E_1 and F_1 , where the slope of the membrane is maximum. Due to similar considerations, it

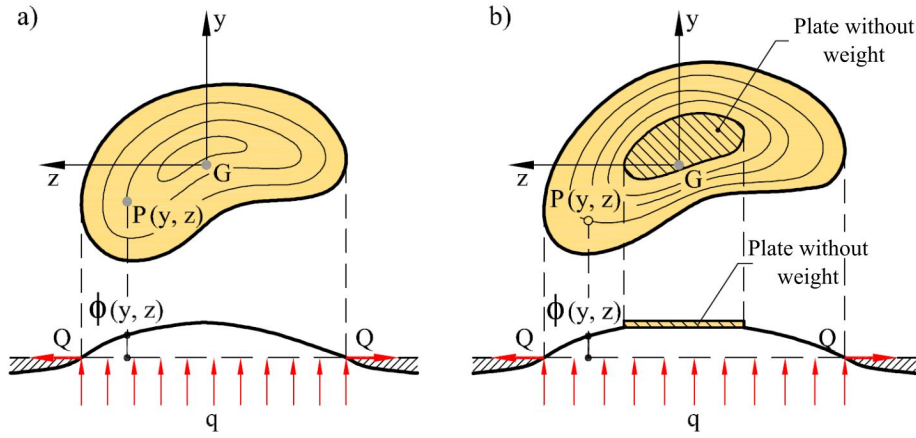


Figure 9.12: Membrane analogy: (a) connected domain (b) “holed ” domain

can be deduced that in a cross-section like the one in Figure 9.14b, the stress is zero in the salient angles (A_2, B_2, C_2, D_2, E_2) and infinite in the reentrant angles (F_2). In practice, this type of singularity is avoided by rounding off the reentrant angles. Figure 9.14c shows the stress trajectories for a hollow circular section; due to central symmetry, these turn out to be concentric circles, a result that coincides with Coulomb’s theory.

9.3.3 Rectangular cross-section

The solution of the problem of uniform torsion in elements of rectangular cross-section can be obtained analytically by solving the corresponding elastic problem on a rectangular domain. However, the mathematical methods to be used are beyond the level used in this book. Through the membrane analogy a qualitative image of the stress trajectories is obtained, that is, of the direction of the tangential stresses, which are like those shown in the Figure 9.14 a.

b/c	1	1.5	2	3	6	10	∞
α	0.208	0.231	0.246	0.267	0.299	0.312	0.333
β	0.141	0.196	0.229	0.263	0.299	0.312	0.333

Table 9.1: Coefficients for rectangular cross-sections

For a rectangular cross-section with sides $b \times c$, where b is the longest side, the maximum tangential stress occurs at the midpoints of the longest sides

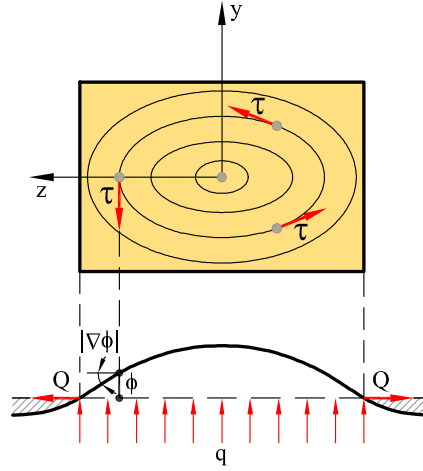


Figure 9.13: Membrane analogy in a rectangular cross-section

lengths of the rectangle and is:

$$\tau_{\max} = \frac{M_t}{\alpha b c^2} \quad (9.22)$$

where α is a value that depends on the ratio $b/c > 1$. The twist of torsion per unit length is

$$\theta = \frac{M_t}{G \beta b c^3} \quad (9.23)$$

where β also depends on the ratio b/c . Table 9.1 provides values of the coefficients α and β for various values of this relationship. Observe that both α and β tend to be $1/3$ when the rectangle is long and narrow.

9.3.4 Narrow rectangular cross-sections

The membrane analogy allows to obtain a very simple solution of the problem of the uniform torsion of a prismatic element of narrow rectangular cross-section of dimensions $b \times c$, with $b \gg c$ (Figure 9.15).

Indeed, if the rectangle is long enough, the effect of the edges of this on the deformation of the membrane in areas far enough away from them can be neglected. This is equivalent to say that, except in the areas close to the extremes, the membrane is deformed in the form of parabolic cylinder (Figure 9.15a). In this case:

$$\tau_{\max} = \frac{M_t c}{\frac{1}{3} b c^3} = \frac{M_t c}{I_t} \quad \text{and} \quad \theta = \frac{M_t}{G \frac{1}{3} b c^3} = \frac{M_t}{G I_t} \quad (9.24)$$

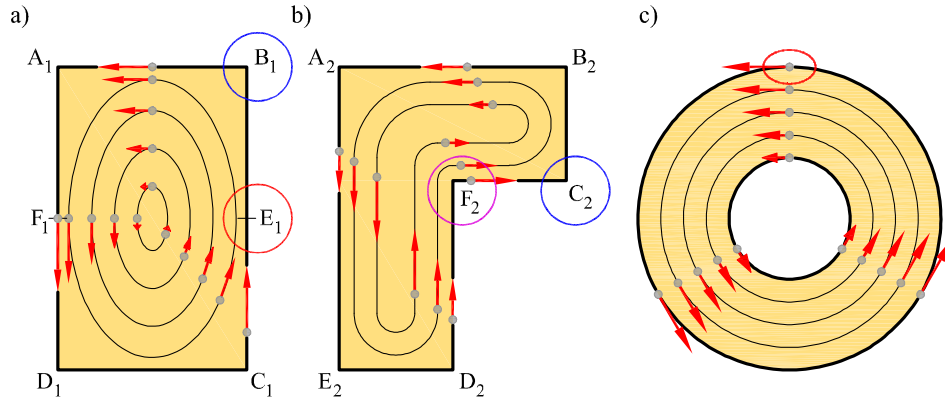


Figure 9.14: Stress distribution on cross-sections of different geometries

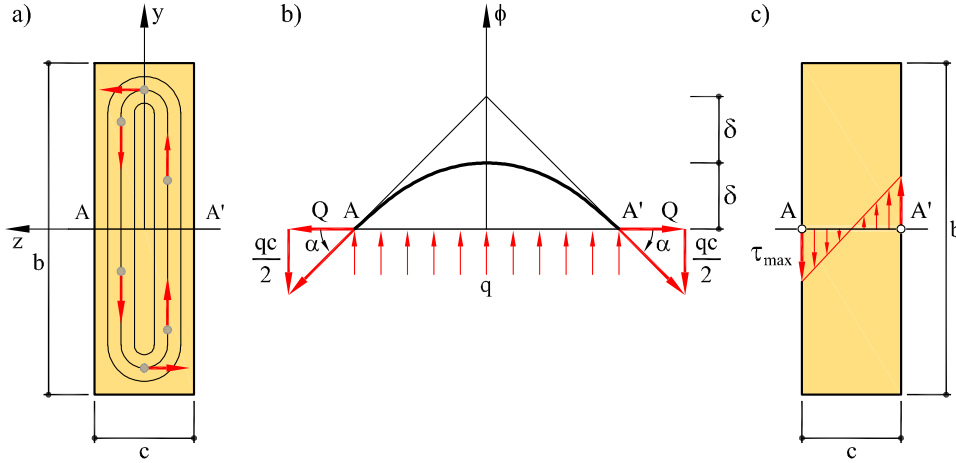


Figure 9.15: Membrane analogy in a narrow rectangular cross-section

where $I_t = bc^3/3$ is the *torsional inertia of the cross-section*.

Furthermore, in the parabola the slope varies linearly; therefore, the distribution of tangential stresses, along a section AA' , is linear:

$$\tau(z) = \tau_{\max} \frac{z}{c/2} = \frac{M_t 2z}{I_t} \quad (9.25)$$

with the maximum values at the points A and A' of the contour.

The previous expressions are so much more approximate the larger is the relationship b/c . For $b/c > 10$ they can be considered as exact.

9.3.5 Open rolled profiles

Expressions (9.24) and (9.25), obtained for narrow rectangular cross-sections, can be applied approximately to thin-walled open sections of uniform thickness that are not rectangular but can be “rectified” to this geometry. In them, the magnitude “ b ” is taken as the “rectified” length of the midline of the section and the thickness “ e ” as the other dimension (Figures 9.16a and 9.16b). The argument for this generalization is assuming that the deformed membrane would be, approximately, a parabolic cylinder of length b .

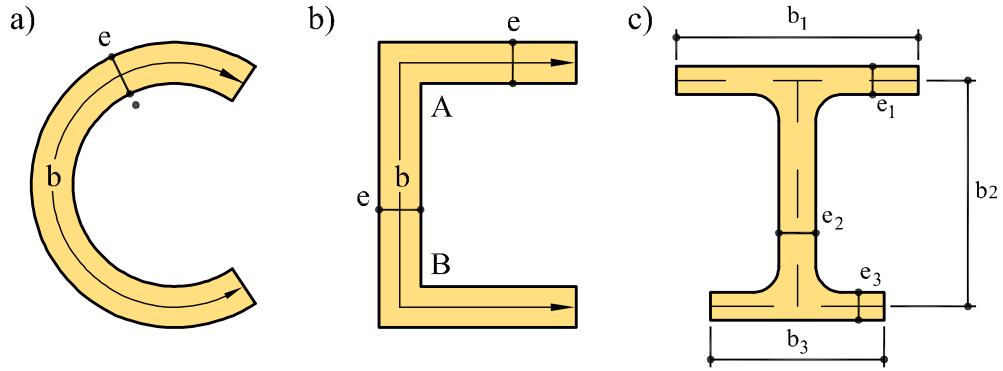


Figure 9.16: Open thin-walled profiles: a) and b) of uniform thickness c) with variable thicknesses

In the case of “branched” rolled profiles, in which the lengths and the thicknesses of the “branches” that compose them are different, but it is fulfilled that $b_i/e_i \gg 1$ (Figure 9.16c), the following approximation is considered adequate.

Let m_i be the part of the torque absorbed by the i -th “branch” and let θ be the angle of torsion per unit of length of the cross-section, assumed to be the same for all “branches” (uniform torsion). Considering that each branch takes a fraction of the total torque proportional to its own torsional stiffness, m_i , such that $\sum_i m_i = M_t$, it is:

$$\theta = \frac{m_i}{G \frac{1}{3} b_i e_i^3} = \frac{M_t}{G \sum_i \frac{1}{3} b_i e_i^3} \quad (9.26)$$

Therefore:

$$I_t = \sum_{i=1}^n (I_t)_i = \sum_{i=1}^n \frac{1}{3} b_i e_i^3 \quad (9.27)$$

is defined as the torsion inertia of the cross-section (sum of those corresponding to the various branches). Furthermore, according to the membrane analogy, the maximum stress in each “branch” occurs along its contour, and

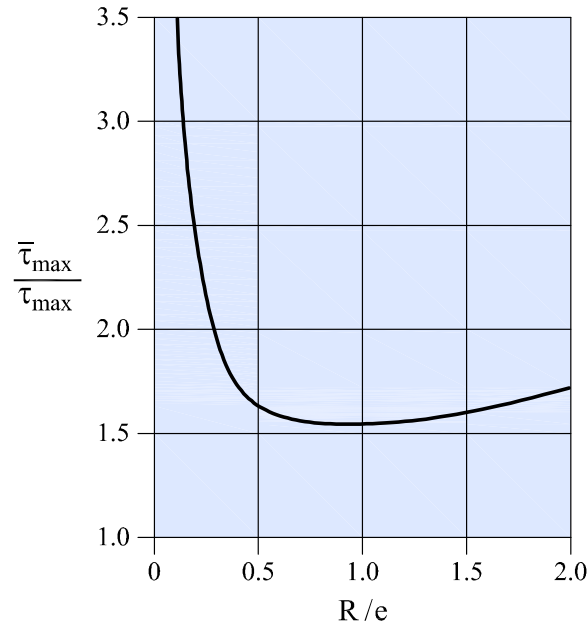


Figure 9.17: Effect of stress concentration in rounder reentrant angles as a function of ratio radius/thickness

is:

$$(\tau_{\max})_i = G \theta e_i = \frac{M_t e_i}{I_t} \quad (9.28)$$

This approximation neglects the local effects that occur at the reentrant angles of the cross-section, such as angles *A* and *B* of Figure 9.16b. However, the membrane analogy predicts that in these angles, the stresses tend to infinity. In practice, these singularities are avoided by rounding these angles. It can be shown by the Theory of Elasticity that the relation between the real maximum stress, $\bar{\tau}_{\max}$, and the stress predicted by the expressions above, $\tau_{\max}$, at these angles is a function of the relationship R/e between the radius of curvature R of the rounding and the thickness e of the element, as shown in Figure 9.17.

Example 9.3.5.1

Calculate the torsional inertia and the maximum torque that it can resist a bar of open circular ring cross-section of radius r and thickness e , for an allowable shear stress τ_{adm} .

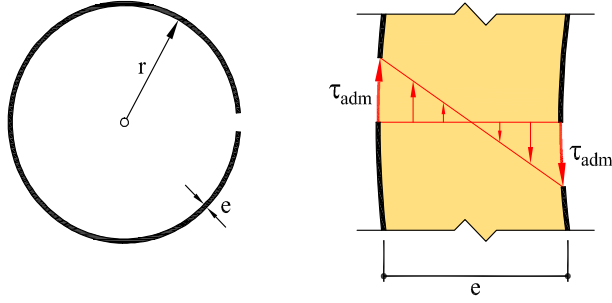


Figure 9.18: Open circular ring

The torsional inertia of the cross-section is:

$$I_t = \frac{1}{3} b e^3 = \frac{1}{3} 2\pi r e^3 = \frac{1}{3} A e^2$$

and therefore, according to the Eq. (9.24b),

$$M_t^{\max} = \frac{I_t \tau_{\text{adm}}}{e} = \frac{1}{3} A e \tau_{\text{adm}}$$

and the stress τ_{adm} is reached only along the contour of the section. Comparing this result with the one obtained in Example 9.2.3.1 for the *closed* thin-walled cross-section of equal area, there is a relationship:

$$\frac{(M_t^{\max})_{\text{ce}}}{(M_t^{\max})_{\text{ab}}} = \frac{A r \tau_{\text{adm}}}{\frac{1}{3} A e \tau_{\text{adm}}} = 3 \left(\frac{r}{e} \right)$$

Both results demonstrate the superiority of closed cross-sections compared to open ones when subjected to uniform torque.

Example 9.3.5.2

The cantilever of Figure 9.19 is subjected to a downward vertical uniformly distributed load $p = 5 \text{ kN/m}$ (see Example 8.8.1.1. of Chapter 8). If the plane of action of the vertical loads is the one that contains the centers of

gravity of the cross-sections: (a) Calculate the actual torque acting on each cross-section. (b) Calculate the maximum value of the tangential stresses due to torsion. Compare with the results obtained in section (b) of Example 8.8.2.1. and comment. (c) Calculate the twist angle at the end of the cantilever and comment on the result.

Data: $G = 80 \text{ GPa}$.

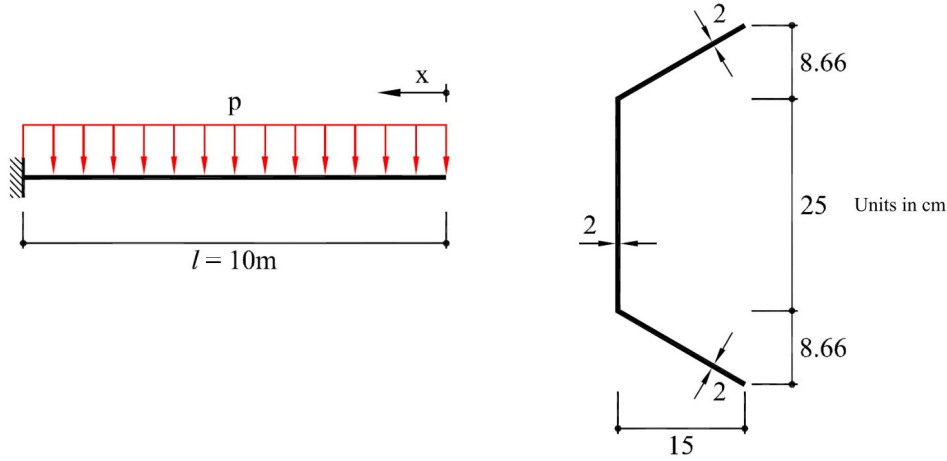


Figure 9.19: Cantilever of Example 9.3.5.2

(a) As calculated in Example 8.8.1.1 of Chapter 8, the shear center of section C is located on the axis of symmetry of the cross-section, to the left of the web, at 9.596 cm from the center of gravity G . When the loads act in the plane that contains the centers of gravity of the cross-sections, and since the shear center does not coincide with the center of gravity, a real torque is produced in each section:

$$M_t = T_y \cdot \overline{CG} = 9.596 \cdot 10^{-2} T_y \text{ kN} \cdot \text{m}$$

with the shear force T_y in kN.

(b) As this is an open thin-walled profile, the maximum tangential stresses due to the torque M_t can be calculated using the expression:

$$\tau = \frac{M_t e}{I_t}$$

where e is the thickness of the profile. The torsional inertia of the cross-section is equal to:

$$I_t = \frac{1}{3} b e^3 = \frac{1}{3} (25 + 2 \cdot 17.32) \cdot 2^3 = 159.04 \text{ cm}^4$$

where b is the length of the midline of the profile. Therefore, the value of the maximum shear stress is

$$\tau_{\max} = \frac{9.596 \cdot 10^{-2} T_y \cdot 10^3 \cdot 2 \cdot 10^{-2}}{159.04 \cdot 10^{-8}} \text{ Pa} = 1.207 T_y \text{ MPa}$$

with T_y in kN. For the maximum value of the shear force $T_{\max} = 50$ kN at the clamping, the maximum shear stress due to torsion is $\tau_{\max} = 60.35$ MPa. If this result is compared with that obtained in section (b) of the Example 8.8.1.1, it is observed that the maximum shear stress due to torsion is approximately seven times greater than the maximum shear stress due to the shear force.

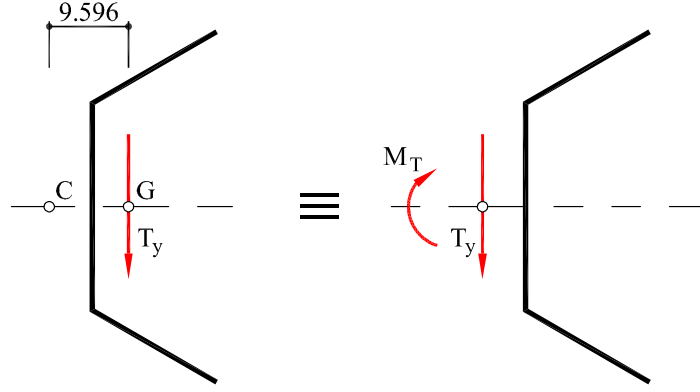


Figure 9.20: Location of the shear center and real torque

(c) The twist of torsion per unit length is equal to:

$$\theta = \frac{M_t}{GI_t} = \frac{\overline{CG} \cdot T_y}{GI_t} = \frac{\overline{CG} \cdot px}{GI_t} \text{ rad/m}$$

where $\overline{CG} = 9,596 \cdot 10^{-2}$ m, $p = 5$ kN/m y $GI_t = 127232$ N·m². Therefore, the value of the twist at the end of the cantilever is equal to:

$$\phi_x = \int_l \theta dx = \frac{\overline{CG} p}{GI_t} \int_0^l x dx = \frac{\overline{CG} pl^2}{2GI_t} = 0.189 \text{ rad} = 10.81^\circ$$

As can be seen, the value of the torsion twist at the end of the cantilever is very high, and is outside the hypotheses of small displacements. The section should be redefined to limit the torsion twist.

Example 9.3.5.3

The hollow rectangular section of Figure 9.21a is open at the midpoint of its lower base, by means of a cut of negligible width. The cross-section is subjected to a shear force parallel to the y axis and applied in the vertical left web. Calculate: (a) the real torque acting on each cross-section, (b) the maximum value of the tangential stresses due to the torque. (c) The maximum shear stress that occurs in the cross-section, (d) the maximum value of the shear T_y that can be supported by the cross-section if the allowable shear stress is τ_{adm} .

Data: $b = 20$ cm, $h = 40$ cm, $e = 1$ cm, $d_{CG} = 42.86$ cm, $T_y = 20$ kN, $\tau_{\text{adm}} = 150$ MPa, $G = 100$ GPa.

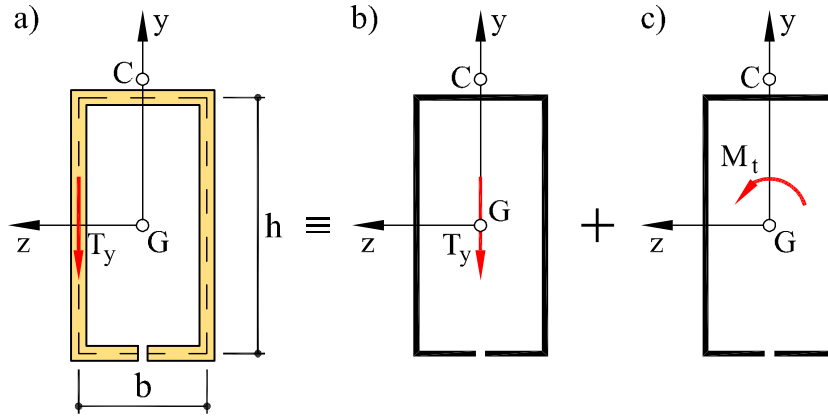


Figure 9.21: a) Cross-section of Example 9.3.5.3 b) reduction of the shear force T_y to the shear center c) real torque

(a) By reducing the shear force T_y (Figure 9.21b) to the shear center C , the real torque on the cross-section, with T_y expressed in Newtons is:

$$M_t = 10 \cdot 10^{-2} T_y \text{ N} \cdot \text{m}$$

and its direction is counterclockwise, as shown in the Figure 9.21c.

(b) The cross-section is open and thin-walled. The maximum shear stresses due to a torque M_t can be calculated by the expression:

$$\tau_{\text{max}}(M_t) = \frac{M_t e}{I_t}$$

where e is the thickness. The torsional inertia of the open cross-section is equal to:

$$I_t = \sum \frac{1}{3} b_i e_i^3 = \frac{1}{3} (2 \cdot 20 + 2 \cdot 40) \cdot 1^3 = 40.0 \text{ cm}^4$$

The maximum value of the tangential stresses due to the torsion moment is:

$$\tau_{\max}(M_t) = \frac{10 \cdot 10^{-2} T_y \cdot 1 \cdot 10^{-2}}{40 \cdot 10^{-8}} \text{ Pa} = 2500 T_y \text{ Pa}$$

with T_y expressed in Newtons. For a shear force $T_y = 20 \text{ kN}$, the maximum shear stress due to torsion is $\tau_{\max} = 50 \text{ MPa}$. As the thickness is constant throughout the cross-section, the value of the maximum tangential stress is the same throughout the cross-section and occurs in its contours (Figure 9.22b).

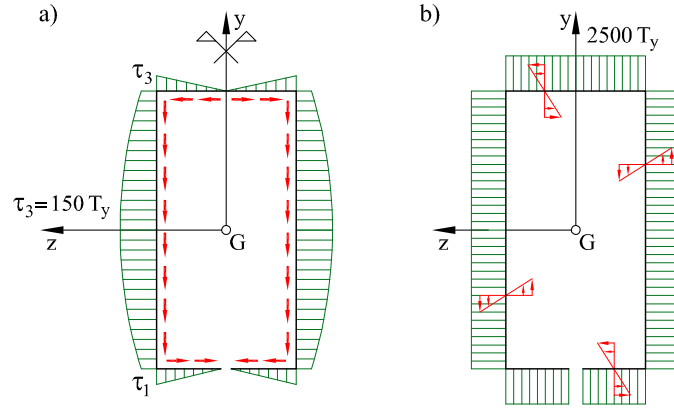


Figure 9.22: Distribution of tangential stresses due to T_y and M_t

(c) The maximum total tangential stress that the cross-section supports is obtained by the superposition of the tangential stresses due to the shear T_y acting in the shear center and those corresponding to the real torsion moment.

The tangential stresses due to shear $T_y = 20 \text{ kN}$, applied at the shear center (Figure 9.22a) have been calculated in Example 8.5.1.2 of Chapter 8. The maximum value is given at the midpoint of the vertical webs and its value is:

$$\tau_{\max}(T_y) = 150 T_y \text{ Pa} = 3.0 \text{ MPa}$$

Therefore, the total maximum tangential stress occurs at the midpoint of the vertical webs and its value is:

$$\tau_{\max} = \tau_{\max}(T_y) + \tau_{\max}(M_t) = (150 T_y + 2500 T_y) \text{ Pa} = 53 \text{ MPa}$$

(d) The maximum value of the shear T_y that the cross-section can support for an allowable stress $\tau_{\text{adm}} = 150 \text{ MPa}$ is:

$$\tau_{\max} = 2650 T_y \text{ Pa} \leq \tau_{\text{adm}} \quad \Rightarrow \quad T_y = \frac{150 \cdot 10^6}{2650} \text{ N} = 56.60 \text{ kN}$$

Appendix A

Geometric characteristics of the section

A.1 Axial first moment of area of a cross section

Let S be a homogeneous section lying in a plane with dextrorotatory Cartesian axes (y, z) , as shown in Figure A.1. The first moment of area of the cross section with respect to the axes (y, z) are defined as the integrals:

$$m_y = \int_S z \, dS \quad ; \quad m_z = \int_S y \, dS \quad (\text{A.1})$$

where dS is a differential area element and the distances from it to the coordinate axes are (y, z) .

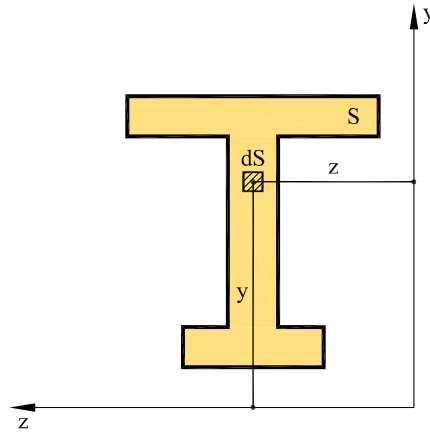


Figure A.1: Axial static moments of a section

The first moment of area of a cross section with respect to an axis can be positive, negative or zero, depending on the distance and position of the

section with respect to that axis. The dimensions of the static moment are L^3 , therefore, in the *SI* they are measured in m^3 .

Gravity centre

The centre of gravity or barycentre of the section S is defined as the point G of coordinates (y_G, z_G) satisfying the relations:

$$y_G = \frac{\int_S y dS}{\int_S dS} \quad ; \quad z_G = \frac{\int_S z dS}{\int_S dS} \quad (\text{A.2})$$

According to equations (1.1) and (1.2), the first moment of area of the section S with respect to the (y, z) axes can also be expressed as :

$$m_y = z_G A \quad ; \quad m_z = y_G A \quad (\text{A.3})$$

where $A = \int_S dS$ is the cross section area. These equations also show that the first moment of area of a section with respect to an axis is zero if and only if that axis passes through the barycentre of the cross section.

It is obvious that if the section S has a symmetry axis, the static moment with respect to this axis is zero. In fact, for a section with the y -axis as the axis of symmetry, there is another identical element dS with negative abscissa z for each element dS with positive abscissa z , so $m_y = 0$. As a result of the equations (A.2), $z_G = 0$, indicating that the section's centre of gravity is located on the axis of symmetry.

Following the same procedure, it is demonstrated that if the section has two axes of symmetry, the barycentre coincides with the point of intersection of these axes. Considering the preceding, it is possible to easily determine the centre of gravity of some cross sections, such as:

- *rectangle or parallelogram*: the barycentre coincides with the point of intersection of the medians or diagonals (see Figure A.2). Each of the medians is the axis of symmetry of the section, and the static moment of the section with respect to them is zero.
- *triangle*: the barycentre is located at the point of intersection of the three medians (Figure 1.4). The medians are not axes of symmetry, but the area can be said to be "symmetrically" placed on either side.

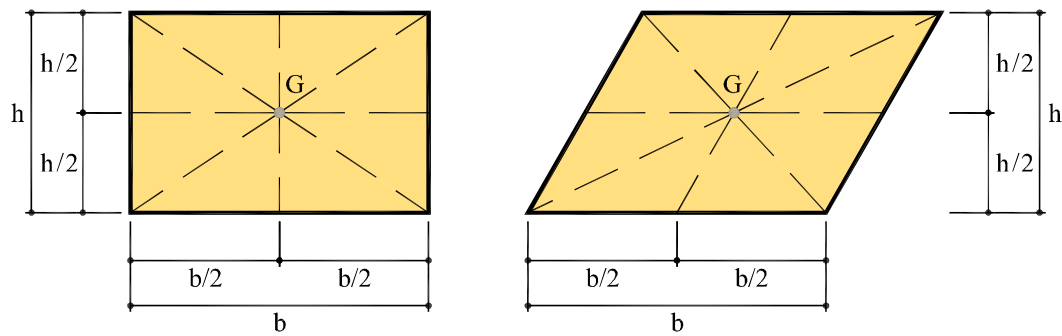


Figure A.2: Barycentre: rectangle and parallelogram

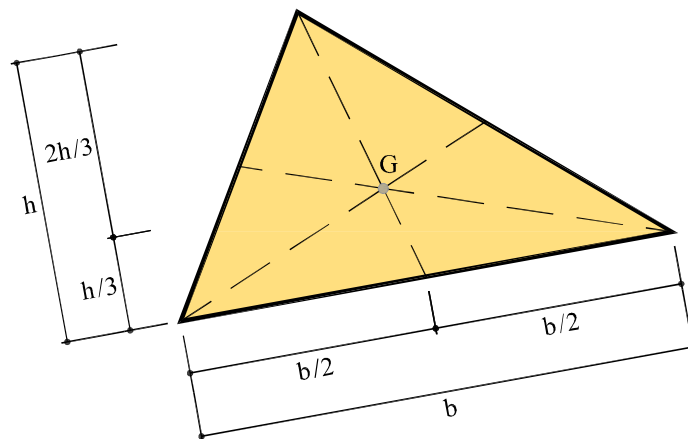


Figure A.3: Barycentre of a triangle

Example A.1.1

Calculate, for the cross section depicted in Figure A.4: (a) the static moment about the y' -axis , (b) the static moment about the z' -axis , and (c) the position of the centre of gravity G of the cross section .

Data: $b = 15 \text{ cm}$, $h = 30 \text{ cm}$, $e = 4 \text{ cm}$.

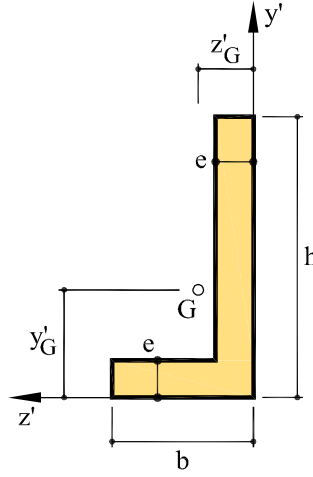


Figure A.4: Cross section of Example A.1.1

(a) Splitting the section into two rectangles: 15×4 cm $\times$ cm at the bottom and 4×26 cm $\times$ cm at the top, the static moment of the section about y' -axis is:

$$m_{y'} = \int_S z' dS = 15 \cdot 4 \cdot 7.5 + 4 \cdot 26 \cdot 2 = 658 \text{ cm}^3$$

(b) Similarly, for the static moment of the section about z' -axis, the following is obtained:

(c) By Applying Eq. (A.2) and taking into account that:

$$\int_S dS = A = 15 \cdot 4 + 4 \cdot 26 = 164 \text{ cm}^2$$

the position of the centre of gravity of the section is defined by:

$$\begin{aligned} y'_G &= \frac{m_{z'}}{A} = \frac{1888}{164} = 11.51 \text{ cm} \\ z'_G &= \frac{m'_{y'}}{A} = \frac{658}{164} = 4.01 \text{ cm} \end{aligned}$$

which are their coordinates with respect to the axes y', z' .

A.2 Moments of inertia of a cross-section

A.2.1 Axial moment of inertia and rotation radius

Let S be a homogeneous section contained in a plane on which some Cartesian dextrorotating axes are located (y, z) . The following integrals are defined as the moments of inertia of the section about the axes:

$$I_y = \int_S z^2 dS \quad ; \quad I_z = \int_S y^2 dS \quad (\text{A.4})$$

where dS is a differential area element and the distances from it to the coordinate axes are (y, z) (Figure A.5). The moment of inertia of a cross-section with respect to an axis is always positive. The dimensions of the moments of inertia are L^4 so, they are measured in m^4 in the SI.

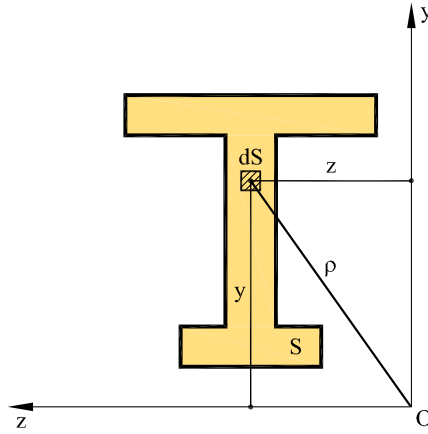


Figure A.5: Moments of inertia of cross section

The rotation radius of a cross-section with respect to an axis is the square root of the ratio between the moment of inertia and the area of the cross-section.

It is possible to write the following for the (y, z) axes:

$$r_y^2 = \frac{I_y}{A} \quad ; \quad r_z^2 = \frac{I_z}{A} \quad (\text{A.5})$$

A.2.2 Second polar moment of area

Let S be a section contained in a plane and O be a point in that plane (Figure A.5). The second polar moment of area I_O of the section S with respect to the point O is defined as the integral:

$$I_o = \int_S \rho^2 dS \quad (\text{A.6})$$

where dS is a differential element of area and ρ is the distance between it and point O . The polar moment, like the axial moment of inertia, is always positive and has the dimensions $[L^4]$.

If two orthogonal axes are drawn through point O (Figure A.5), and the distances y, z of a differential of area dS to these axes are measured, the following expression can be written:

$$\rho^2 = y^2 + z^2 \quad (\text{A.7})$$

and therefore:

$$I_o = \int_S \rho^2 dS \quad (\text{A.8})$$

$$= \int_S (y^2 + z^2) dS \quad (\text{A.9})$$

$$= \int_S y^2 dS + \int_S z^2 dS \quad (\text{A.10})$$

$$= I_z + I_y \quad (\text{A.11})$$

As a result, the polar moment of inertia with respect to a point O equals the sum of the moments of inertia with respect to two orthogonal axes that pass through that point. The sum of axial moments of inertia is also an invariant for any pair of orthogonal axes passing through the same point, as the previous property implies.

A.2.3 Product of inertia

The product of inertia of the surface S with respect to the axes (y, z) is defined as the integral.

$$I_{yz} = \int_S yz dS \quad (\text{A.12})$$

The product of inertia of a section can be positive, negative or zero. Its dimensions are $[L^4]$.

If a section S has an axis of symmetry, the product of inertia of the section with respect to this axis and another axis perpendicular to it is zero. In fact, by symmetry, each differential element of area dS with positive abscissa z corresponds to another identical element with negative z abscissa, whose ordinate y has the same value and sign, so that the products of inertia of these differential elements cancel each other out.

Example A.1.2

For the rectangular section with base b and height h (Figure A.6), calculate:
 (a) the moment of inertia with respect to the z' axis passing through its base and (b) the moment of inertia with respect to the barycentric axis z_o .

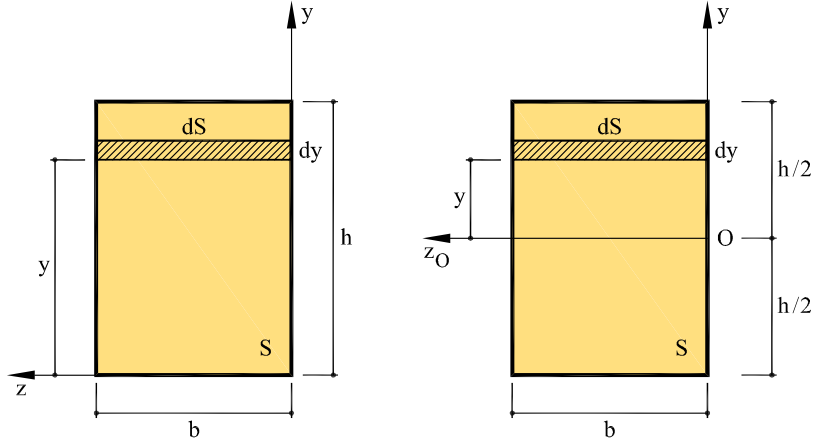


Figure A.6: Rectangular cross section of Example A.1.2

(a) The moment of inertia of the rectangular cross-section with respect to the z -axis is calculated using Eq. (A.4). Since the section width is constant, the differential area can be expressed as $dS = b \, dy$. Integrating between $y = 0$ and $y = h$ yields:

$$I_z = \int_S y^2 dS = \int_0^h y^2 b \, dy = b \left. \frac{y^3}{3} \right|_0^h = \frac{bh^3}{3} \quad (\text{A.13})$$

(b) Applying once again Eq. (A.4) to calculate the inertia of the section with respect to the barycentric axis z_o , integrating between $y = -h/2$ and $y = +h/2$ yields:

$$I_z = \int_S y^2 dS = \int_{-h/2}^{+h/2} y^2 b \, dy = b \left. \frac{y^3}{3} \right|_{-h/2}^{+h/2} = \frac{bh^3}{12} \quad (\text{A.14})$$

These results are often used for sections that can be decomposed into a sum of rectangles.

Example A.1.3

For the section in Figure A.7 calculate: (a) the moment of inertia about the y' -axis, (b) the moment of inertia about the z' -axis, (c) the polar moment of inertia about the point O and (d) the product of inertia of the section.

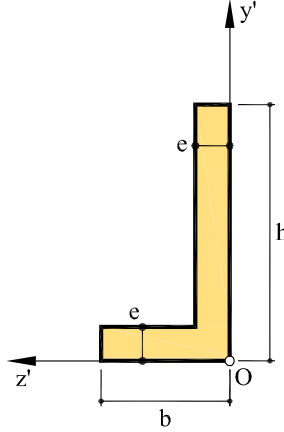


Figure A.7: Cross section for Example A.1.3

(a) The moment of inertia of the cross-section with respect to the y' -axis is obtained by decomposing the cross-section into rectangles, $b \times e = 15 \times 4 \text{ cm} \times \text{cm}$ at the base and $e \times (h - e) = 4 \times 26 \text{ cm} \times \text{cm}$ at the vertical flange, using Eq. (A.13):

$$\begin{aligned} I_{y'} &= \frac{eb^3}{3} + \frac{(h-e)e^3}{3} \\ &= \frac{4 \cdot 15^3}{3} + \frac{26 \cdot 4^3}{3} = 5\,054.67 \text{ cm}^4 \end{aligned}$$

(b) Similarly, by decomposing the cross section into rectangles, $(b - e) \times e = 11 \times 4 \text{ cm} \times \text{cm}$ at the base and $e \times h = 4 \times 30 \text{ cm} \times \text{cm}$ at the vertical flange, and applying Eq. (A.13), the moment of inertia with about the z' axis is:

$$\begin{aligned} I_{z'} &= \frac{(b-e)e^3}{3} + \frac{eh^3}{3} \\ &= \frac{11 \cdot 4^3}{3} + \frac{4 \cdot 30^3}{3} = 36\,234.67 \text{ cm}^4 \end{aligned}$$

(c) The polar moment of inertia of the section is calculated by applying Eq. (A.8a):

$$\begin{aligned}
I_o &= \int_S \rho^2 dS = I_{z'} + I_{y'} \\
&= 5054.67 + 36234.67 = 41\,289.34 \text{ cm}^4
\end{aligned}$$

(d) Finally, by applying Eq. (A.12), the product of inertia of the section is equal to:

$$\begin{aligned}
I_{yz} &= \int_S yz dS \\
&= (15 \cdot 4) \cdot 7.5 \cdot 2 + (4 \cdot 26) \cdot 2 \cdot 17 = 4\,436 \text{ cm}^4
\end{aligned}$$

A.3 Translation of reference axes

A.3.1 Axial moment of inertia

Let z_o be a barycentric axis of the section S and let z be an axis parallel to it at a distance d_y from it (Figure A.8). The moment of inertia I_z can be calculated as:

$$I_z = \int_S y^2 dS \quad (\text{A.15})$$

$$= \int_S (y_o + d_y)^2 dS \quad (\text{A.16})$$

$$= \int_S y_o^2 dS + 2d_y \int_S y_o dS + d_y^2 \int_S dS \quad (\text{A.17})$$

$$= I_{z_o} + d_y^2 A \quad (\text{A.18})$$

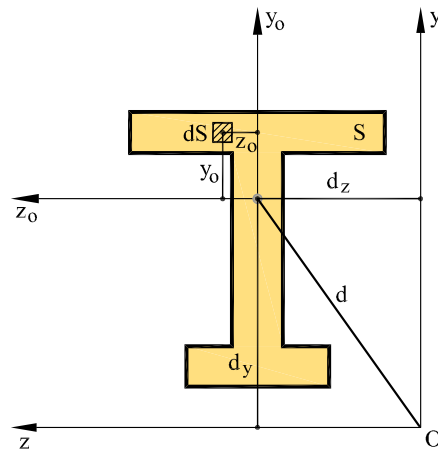


Figure A.8: Translation of reference axes

where the second integral in Eq. (A.17c) is zero due to the fact that z_o is a barycentric axis. As a result, the moment of inertia of a section with respect to an arbitrary axis is the sum of the moment of inertia with respect to the previous axis and the area of the section multiplied by the square of the distance between both axes (Steiner's Theorem).

$$I_z = I_{z_o} + d_y^2 A \quad (\text{A.19})$$

By dividing this equation by the cross section area:

$$r_z^2 = r_{z_o}^2 + d_y^2 \quad (\text{A.20})$$

This expression gives the radius of inertia of the section with respect to an axis as a function of the radius of inertia with respect to a parallel barycentric axis.

A.3.2 Second polar moment of area

Let I_y and I_z be the moments of inertia of a cross section with respect to two axes that intersect at point O , and let I_O be the polar moment of inertia of the section with respect to that point. By Considering two barycentric axes (y_o, z_o parallel to the previous ones (see Figure A.5), and apply Steiner's Theorem, leads to the following results:

$$I_O = I_y + I_z = I_{y_o} + d_z^2 A + I_{z_o} + d_y^2 A = I_G + d^2 A \quad (\text{A.21})$$

The polar moment of inertia of a section with respect to any point O can be calculated using the above equations if the polar moment of inertia of the section with respect to the barycentre G and the distance between the two points are known.

A.3.3 Product of inertia

Similarly, if the product of inertia of a section with respect to orthogonal axes (y_o, z_o) barycentric is known and the value of the product of inertia with respect to some coordinate axes (y, z), parallel to the previous ones and located at distances d_y, d_z of the barycentric axes is needed, leads to:

$$I_{yz} = \int_S y z dS = \int_S (y_o + d_y) (z_o + d_z) dS \quad (\text{A.22})$$

$$= \int_S y_o z_o dS + d_y \int_S z_o dS + d_z \int_S y_o dS + d_y d_z \int_S dS \quad (\text{A.23})$$

$$= I_{y_o z_o} + d_y d_z A \quad (\text{A.24})$$

where y, z has been replaced by $y = y_o + d_y, z = z_o + d_z$. The second and third components of Eq. (A.23b) are zero since they are the static moments of the section with respect to two barycentric axes.

Example A.1.4

For the section shown in Figure A.9, calculate (a) the moments of inertia about the y_o, z_o axes passing through the centre of gravity of the section G , (b) the corresponding rotation radius and (c) the product of inertia of the section about these barycentric axes.

Data: $b = 15 \text{ cm}$, $h = 30 \text{ cm}$, $e = 4 \text{ cm}$, $A = 164 \text{ cm}^2$

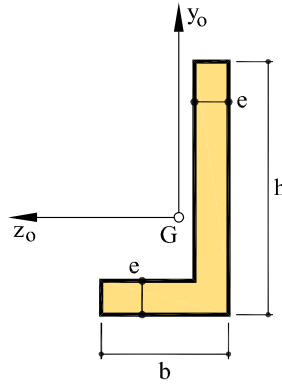


Figure A.9: Cross section of Example A.1.4

(a) The location of the centre of gravity G was determined in Example A.1.1, and its coordinates with respect to the coordinate axes y', z' are:

$$y'_G = 11.51 \text{ cm} \quad z'_G = 4.01 \text{ cm}$$

The moment of inertia with respect to the barycentric axis y_o can be calculated by decomposing the cross-section into the rectangles, $11 \times 4 \text{ cm} \times \text{cm}$ at the base and $4 \times 30 \text{ cm} \times \text{cm}$ at the vertical flange, and applying Steiner's theorem. Taking Eq. (A.14) into account, it follows:

$$\begin{aligned} I_{y_o} &= \frac{4 \cdot 11^3}{12} + 4 \cdot 11 \cdot 5.49^2 + \frac{30 \cdot 4^3}{12} + 30 \cdot 4 \cdot 2.01^2 \\ &= 2414.64 \text{ cm}^4 \end{aligned}$$

Similarly, the moment of inertia of the cross-section with respect to the barycentric axis z_o :

$$\begin{aligned} I_{z_o} &= \frac{11 \cdot 4^3}{12} + 4 \cdot 11 \cdot 9.51^2 + \frac{4 \cdot 30^3}{12} + 30 \cdot 4 \cdot 3.49^2 \\ &= 14499.64 \text{ cm}^4 \end{aligned}$$

(b) The rotation radius corresponding to the barycentric axes y_o, z_o are as follows:

$$\begin{aligned} r_{y_o}^2 &= \frac{I_{y_o}}{A} = \frac{2414.64}{164} = 14.72 \text{ cm}^2 \Rightarrow r_{y_o} = 3.84 \text{ cm} \\ r_{z_o}^2 &= \frac{I_{z_o}}{A} = \frac{14499.64}{164} = 88.41 \text{ cm}^2 \Rightarrow r_{z_o} = 9.40 \text{ cm} \end{aligned}$$

(c) The product of inertia of the section can be determined by applying Eq. (A.12):

$$\begin{aligned} I_{y_o z_o} &= \int_S y_o z_o dS \\ &= (11 \cdot 4) \cdot 5.49 \cdot (-9.51) + (4 \cdot 30) \cdot (-2.01) \cdot 3.49 = -3139.03 \text{ cm}^4 \end{aligned}$$

A.4 Reference axis rotation

Let (y, z) and (y_1, z_1) be two systems of coordinate axes which form an angle to each other as shown in Figure A.10. The coordinates in the system (y_1, z_1) can be expressed as a function of the coordinates in the system (y, z) in the form:

$$y_1 = y \cos \alpha + z \sin \alpha \quad (\text{A.25})$$

$$z_1 = -y \sin \alpha + z \cos \alpha \quad (\text{A.26})$$

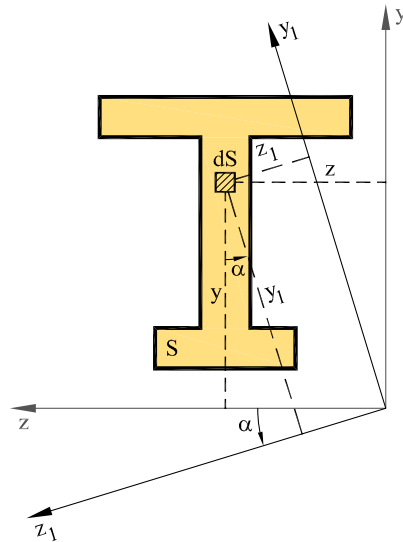


Figure A.10: Reference axis rotation

By using these equations, the moment of inertia of the section about the y_1 -axis is:

$$I_{y_1} = \int_S z_1^2 dS \quad (\text{A.27})$$

$$= \int_S (-y \sin \alpha + z \cos \alpha)^2 dS \quad (\text{A.28})$$

$$= \int_S y^2 \sin^2 \alpha dS - \int_S yz 2 \sin \alpha \cos \alpha dS + \int_S z^2 \cos^2 \alpha dS \quad (\text{A.29})$$

$$= I_z \sin^2 \alpha + I_y \cos^2 \alpha - 2 I_{yz} \sin \alpha \cos \alpha \quad (\text{A.30})$$

Similarly, the expressions can be calculated:

$$I_{z_1} = I_z \cos^2 \alpha + I_y \sin^2 \alpha + 2 I_{yz} \sin \alpha \cos \alpha \quad (\text{A.31})$$

$$I_{y_1 z_1} = (I_y - I_z) \cos \alpha \sin \alpha + I_{yz} (\cos^2 \alpha - \sin^2 \alpha) \quad (\text{A.32})$$

The previous equations can be rewritten as:

$$I_{y_1} = \frac{I_y + I_z}{2} + \frac{I_y - I_z}{2} \cos 2\alpha - I_{yz} \sin 2\alpha \quad (\text{A.33})$$

$$I_{z_1} = \frac{I_y + I_z}{2} - \frac{I_y - I_z}{2} \cos 2\alpha + I_{yz} \sin 2\alpha \quad (\text{A.34})$$

$$I_{y_1 z_1} = \frac{I_y - I_z}{2} \sin 2\alpha + I_{yz} \cos 2\alpha \quad (\text{A.35})$$

where trigonometric equalities have been used:

$$\cos^2 \alpha = \frac{1 + \cos 2\alpha}{2} \quad ; \quad \sin^2 \alpha = \frac{1 - \cos 2\alpha}{2} \quad (\text{A.36})$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha \quad ; \quad \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha \quad (\text{A.37})$$

Example A.1.5

Calculate, for the section depicted in Figure A.11, (a) the moments of inertia about the y_1 and z_1 axes passing through the section's centre of gravity G and forming an angle of $\alpha = 30^\circ$ with the y_o y z_o axes, and (b) the product of inertia about y_1 and z_1 axes. Data: $b = 15 \text{ cm}$, $h = 30 \text{ cm}$, $e = 4 \text{ cm}$, $A = 164 \text{ cm}^2$, $I_{y_o} = 2414.64 \text{ cm}^4$, $I_{z_o} = 14499.64 \text{ cm}^4$, $I_{y_o z_o} = -3139.03 \text{ cm}^4$

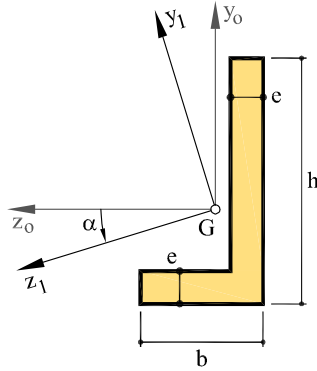


Figure A.11: Cross section of Example A.1.5

(a) The moments of inertia about the y_1 and z_1 axes, which form an angle $= 30^\circ$ $\alpha = 30^\circ$ with the previous ones, can be calculated by applying Eqs. (1.20a) and (A.33b).

$$\begin{aligned}
 I_{y_1} &= \frac{I_{y_o} + I_{z_o}}{2} + \frac{I_{y_o} - I_{z_o}}{2} \cos 2\alpha - I_{y_o z_o} \sin 2\alpha \\
 &= \frac{2414.64 + 14\,499.64}{2} + \frac{2414.64 - 14\,499.64}{2} \cdot 0.5 - (-3139.03) \cdot 0.866 \\
 &= 8\,154.29 \text{ cm}^4 \\
 I_{z_1} &= \frac{I_{y_o} + I_{z_o}}{2} - \frac{I_{y_o} - I_{z_o}}{2} \cos 2\alpha + I_{y_o z_o} \sin 2\alpha \\
 &= \frac{2414.64 + 14\,499.64}{2} - \frac{2414.64 - 14\,499.64}{2} \cdot 0.5 + (-3139.03) \cdot 0.866 \\
 &= 8\,759.60 \text{ cm}^4
 \end{aligned}$$

(b) The product of inertia of the section about y_1 , z_1 axes can be calculated by applying Eq. (A.35):

$$\begin{aligned}
 I_{y_1 z_1} &= \frac{I_{y_o} - I_{z_o}}{2} \sin 2\alpha + I_{y_o z_o} \cos 2\alpha \\
 &= \frac{2414.64 - 14\,499.64}{2} \cdot 0.866 + (-3139.03) \cdot 0.5 \\
 &= -6\,802.33 \text{ cm}^4
 \end{aligned}$$

A.5 Principal axes of inertia

In any plane section, two orthogonal barycentric axes can be identified for which the product of inertia with respect to them is zero. These axes are referred to as principal axes of inertia and are crucial to the study of the

bending of prismatic beam The angle α formed by the principal axes of inertia of a section (y, z) and another system of barycentric axes (v, w) (see Figure A.12) is obtained by setting Eq. (A.35) applied to the axes (v, w) to zero.

$$\tan 2\alpha = -\frac{2I_{vw}}{I_v - I_w} \quad (\text{A.38})$$

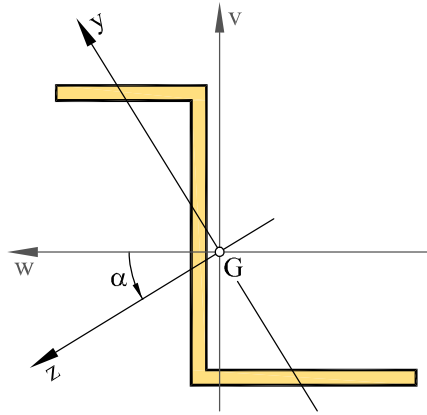


Figure A.12: Cross section principal axis

Observe that if a section is symmetrical about an axis, the system formed by this axis and another barycentric axis perpendicular to it is principal of inertia since the corresponding inertia product is zero. If from Eqs. (A.33) or (A.34) the angle α for which I_{v_1} and I_{w_1} are maximal or minimal is determined, the result is as follows:

$$\frac{\partial I_{v_1}}{\partial \alpha} = -(I_v - I_w) \sin 2\alpha - 2I_{vw} \cos 2\alpha = 0 \quad (\text{A.39})$$

where

$$\tan 2\alpha = -\frac{2I_{vw}}{I_v - I_w} \quad (\text{A.40})$$

This equation coincides with (A.38), so the values of the principal moments of inertia correspond to the maximum and minimum axial moments of inertia of the section with respect to barycentric axes.

When the two principal moments of inertia are equal, $I_y = I_z$, the moment of inertia with respect to any barycentric axis is equal. In addition, the product of inertia with respect to any system of orthogonal barycentric axes is zero, so any barycentric axis is the principal axis of inertia of the section. This is the case, for instance, with circular or square cross sections. By expressing the values $\sin 2\alpha$ and $\cos 2\alpha$, as a function of the value $\tan 2\alpha$ and substituting them into equations (A.33)-(A.34) we obtain:

$$\left. \begin{matrix} I_z \\ I_y \end{matrix} \right\} = \frac{I_v + I_w}{2} \pm \sqrt{\left(\frac{I_v - I_w}{2}\right)^2 + (I_{vw})^2} \quad (\text{A.41})$$

an alternative expression for calculating the principal moments of inertia I_y, I_z given the moments of inertia I_v, I_w, I_{vw} about any axes (v, w) .

The principal rotation radius, r_y and r_z , associated with the principal moments of inertia of a section are as follows:

$$r_y^2 = \frac{I_y}{A} \quad ; \quad r_z^2 = \frac{I_z}{A} \quad (\text{A.42})$$

Ejemplo A.1.6

Calculate, for the section in Figure A.13: (a) the principal directions of inertia and the corresponding moments of inertia, (b) rotation radius associated with the principal directions of inertia of the section.

Data: $b = 15 \text{ cm}$, $h = 30 \text{ cm}$, $e = 4 \text{ cm}$, $A = 164 \text{ cm}^2$

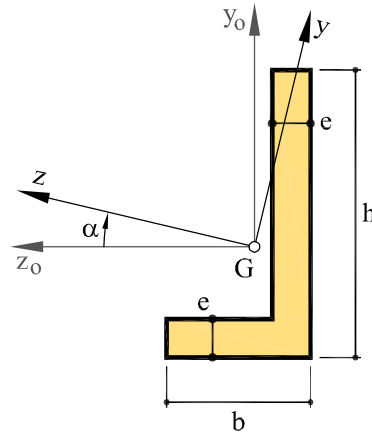


Figure A.13: Cross section for Example A.1.6

(a) The angle α that the principal axes of inertia of a section (y, z) form with another system of barycentric axes y_o, z_o (see Figure A.13) is obtained by setting the Eq. (A.35) applied to the axes to zero (y_o, z_o) . In this way it is given:

$$\tan 2\alpha = -\frac{2I_{y_o z_o}}{I_{y_o} - I_{z_o}}$$

The moments of inertia of the section with respect to the barycentric axes (y_o, z_o) have been calculated in Example A.1.4:

$$I_{y_o} = 2414.64 \text{ cm}^4 \quad I_{z_o} = 14499.64 \text{ cm}^4 \quad I_{y_o z_o} = -3139.03 \text{ cm}^4$$

Therefore, the angle α is equal to:

$$\tan 2\alpha = -\frac{2(-3139.03)}{2414.64 - 14499.64} = -0.5195 \Rightarrow \alpha = -13^\circ 43' 32''$$

The principal axes of inertia of the section (y, z) form an $\alpha = -13^\circ$ with the barycentric axes (y_o, z_o) (Figure A.13). The negative sign indicates that the angle is clockwise. The principal moments of inertia can be calculated by applying Eq. (A.41):

$$\begin{aligned} \left. \begin{matrix} I_z \\ I_y \end{matrix} \right\} &= \frac{I_{y_o} + I_{z_o}}{2} \pm \sqrt{\left(\frac{I_{y_o} - I_{z_o}}{2}\right)^2 + (I_{y_o z_o})^2} \\ &= \frac{2414.64 + 14499.64}{2} \pm \sqrt{\left(\frac{2414.64 - 14499.64}{2}\right)^2 + (-3139.03)^2} \\ &= \begin{cases} 15266.35 \text{ cm}^4 \\ 1647.93 \text{ cm}^4 \end{cases} \end{aligned}$$

(b) The rotation radius associated with the principal directions are as follows:

$$\begin{aligned} r_y^2 &= \frac{I_y}{A} = \frac{1647.93}{164} = 10.05 \text{ cm}^2 \Rightarrow r_y = 3.17 \text{ cm} \\ r_z^2 &= \frac{I_z}{A} = \frac{15266.35}{164} = 93.09 \text{ cm}^2 \Rightarrow r_z = 9.65 \text{ cm} \end{aligned}$$

A.6 Thin cross sections

In practise, Cross-sections with very small thicknesses compared to other dimensions of the cross-section are commonly used. An example of this is metal cross-sections. In such cases, the geometrical characteristics can be calculated in a more simplified way, as can be seen in the following example.

Example A.1.7

Calculate, for the section in Figure A.14a: (a) the position of the centre of gravity and (b) the moments of inertia of the section with respect to the principal axes of the section.

Data: $a = 20 \text{ cm}$, $b = 10 \text{ cm}$, $h = 30 \text{ cm}$, $e = 1.2 \text{ cm}$.

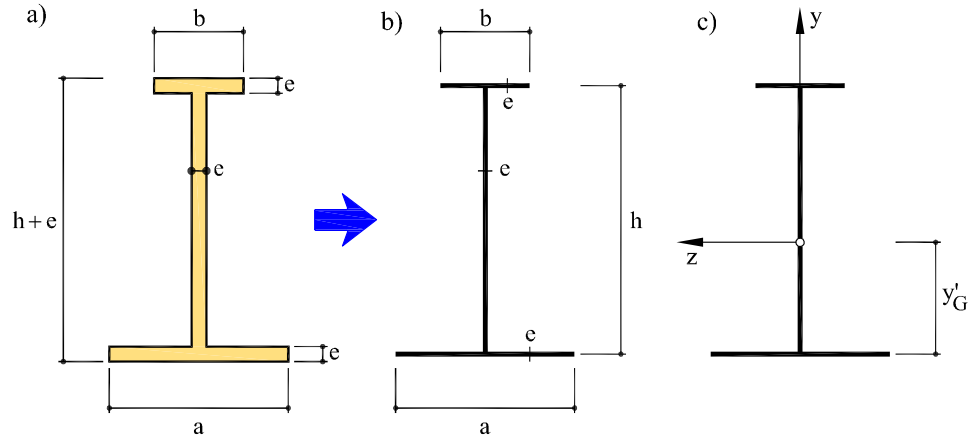


Figure A.14: Thin cross section

(a) It is a thin cross section, and the simplification illustrated in Figure A.14b is used to calculate the geometric characteristics. An axis of symmetry of the section is the vertical axis that coincides with the web axis. Therefore, the centre of gravity will therefore be located on it. The y'_G coordinate with respect to the midline of the lower flange of the section will allow it to be placed on it (Figure A.14c).

Cross section area is

$$A = (20 + 10 + 30) \cdot 1.2 = 72 \text{ cm}^2$$

The first moment of area of the section about the mid-line of the lower flange of the section is:

$$m_{z'}^e = 10 \cdot 1.2 \cdot 30 + 30 \cdot 1.2 \cdot 15 = 750 \text{ cm}^3$$

and y'_G coordinates is:

$$y'_G = \frac{m_{z'}^e}{A} = \frac{900}{72} = 12.5 \text{ cm}$$

The principal axes of the section are the barycentric axes y, z since the y -axis is an axis of symmetry. The principal moments of inertia of the section are:

$$\begin{aligned} I_y &= \frac{1.2 \cdot 20^3}{12} + \frac{30 \cdot 1.2^3}{12} + \frac{1.2 \cdot 10^3}{12} \\ &= 904.32 \text{ cm}^4 \\ I_z &= 20 \cdot 1.2 \cdot 12.5^2 + \frac{1.2 \cdot 30^3}{12} + 30 \cdot 1.2 \cdot 2.5^2 + 10 \cdot 1.2 \cdot 17.5^2 \\ &= 10350 \text{ cm}^4 \end{aligned}$$

Note that the terms corresponding to the barycentric moments of inertia of the lower and upper flanges have been neglected in the calculation of the moment of inertia I_z because they are negligible compared to the remaining terms.

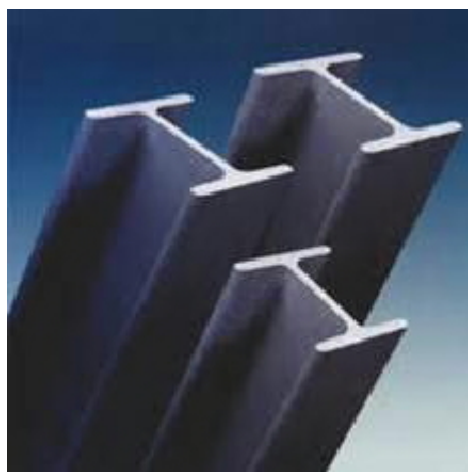
Appendix B

Structural Steel Profiles

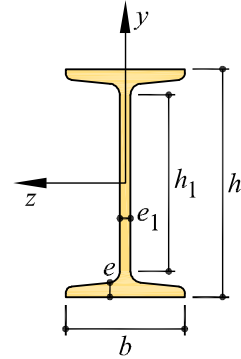
The mechanical properties of some structural steel profiles used in the problems of this Book are presented in a summarized way. A more complete information can be obtained in the handbooks of structural steel profiles.

The y, z axes are the principal axes of the profiles, following the notation used in this Book. For the remaining mechanical properties the following notation is adopted:

- A: cross sectional area (cm^2)
- p: weight per linear meter of the profile (N/m)
- I_y : moment of inertia of the section about the principal y -axis (cm^4)
- I_z : moment of inertia of the section about the principal z -axis (cm^4)
- W_i : section modulus with respect to the i -axis (cm^3)
- I_t : modulus of torsion of the section (cm^4)
- I_w : modulus of warping of the section (cm^6)

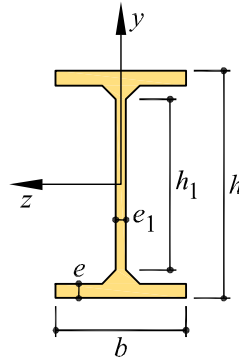


PROFILES IPN



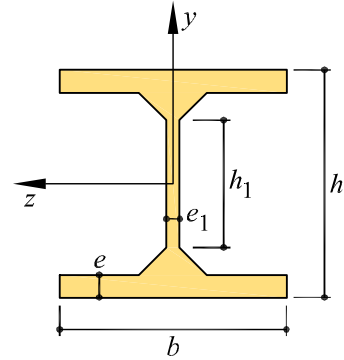
IPN	Dimensions mm					A cm ²	p N/m	I _y cm ⁴	W _y cm ³	I _z cm ⁴	W _z cm ³	I _t cm ⁴	I _w ×10 ³ cm ⁶
	h	b	e	e ₁	h ₁								
80	80	42	5.9	3.9	59	7.58	59.5	6.29	3.00	77.8	19.5	0.93	0.087
100	100	50	6.8	4.5	75	10.6	83.2	12.2	4.88	171	34.2	1.72	0.27
120	120	58	7.7	5.1	92	14.2	111	21.5	7.41	328	54.7	2.92	0.69
140	140	66	8.6	5.7	109	18.3	144	35.2	10.7	573	81.9	4.66	1.54
160	160	74	9.5	6.3	125	22.8	179	54.7	14.8	935	117	7.08	3.14
180	180	82	10.4	6.9	142	27.9	219	81.3	19.8	1450	161	10.3	5.92
200	200	90	11.3	7.5	159	33.5	263	117	26.0	2140	214	14.6	10.5
220	220	98	12.2	8.1	175	39.6	311	162	33.1	3060	278	20.1	17.8
240	240	106	13.1	8.7	192	46.1	362	221	41.7	4250	354	27.0	28.7
260	260	113	14.1	9.4	208	53.4	419	288	51.0	5740	442	36.1	44.1
280	280	119	15.2	10.1	225	61.1	480	364	61.2	7590	542	47.8	64.6
300	300	125	16.2	10.8	241	69.1	542	451	72.2	9800	653	61.2	91.9
320	320	131	17.3	11.5	257	77.8	611	555	84.7	12510	782	78.2	12.8
340	340	137	18.3	12.2	274	86.8	681	674	98.4	15700	923	97.5	176
360	360	143	19.5	13.0	290	97.1	762	818	114	19610	1090	123	240
380	380	149	20.5	13.7	306	107	840	975	131	24010	1260	150	319
400	400	155	21.6	14.4	323	118	926	1160	149	29210	1460	183	420
450	450	170	24.3	16.2	363	147	1150	1730	203	45850	2040	288	791
500	500	185	27.0	18.0	404	180	1410	2480	268	68740	2750	449	1403
550	550	200	30.0	19.0	444	213	1670	3490	349	99180	3610	618	2389
600	600	215	32.4	21.6	485	254	1990	4670	434	139000	4630	875	3821

PROFILES IPE



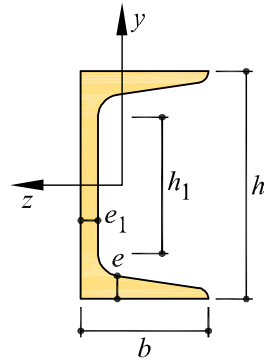
IPE	Dimensions mm					A cm ²	p N/m	I _y cm ⁴	W _y cm ³	I _z cm ⁴	W _z cm ³	I _t cm ⁴	I _w x10 ³ cm ⁶
	h	b	e	e ₁	h ₁								
80	80	46	5.2	3.8	59	7.64	60.0	8.49	3.69	80.1	20.0	0.70	0.12
100	100	55	5.7	4.1	74	10.3	81.0	15.9	5.79	171	34.2	1.20	0.35
120	120	64	6.3	4.4	93	13.2	104	27.7	8.65	318	53.0	1.74	0.89
140	140	73	6.9	4.7	112	16.4	129	44.9	12.3	541	77.3	2.45	1.98
160	160	82	7.4	5.0	127	20.1	158	68.3	16.7	869	109	360	3.96
180	180	91	8.0	5.3	146	23.9	188	101	22.2	1320	146	4.79	7.43
200	200	100	8.5	5.6	159	28.5	224	142	28.5	1940	194	6.98	13.0
220	220	110	9.2	5.9	177	33.4	262	205	37.3	2770	252	9.07	22.7
240	240	120	9.8	6.2	190	39.1	307	284	47.3	3890	324	12.9	37.4
270	270	135	10.2	6.6	219	45.9	361	420	62.2	5790	429	15.9	70.6
300	300	150	10.7	7.1	248	53.8	422	604	80.5	8360	557	20.1	126
330	330	160	11.5	7.5	271	62.6	491	788	98.5	11770	713	28.2	199
360	360	170	12.7	8.0	298	72.7	571	1040	123	16270	904	37.3	314
400	400	180	13.5	8.6	331	84.5	663	1320	146	23130	1160	51.1	490
450	450	190	14.6	9.4	378	98.8	776	1680	176	33740	1500	66.9	791
500	500	200	16.0	10.2	426	116	907	2140	214	48200	1930	89.3	1249
550	550	210	17.2	11.1	467	134	1060	2670	254	67120	2440	123	1884
600	600	220	19.0	12.0	514	156	1220	3390	308	92080	3070	165	2855

PROFILES HEB



HEB	Dimensiones mm					A	p	I_y	W_y	I_z	W_z	I_t	I_w
	h	b	e	e_1	h_1	cm ²	N/m	cm ⁴	cm ³	cm ⁴	cm ³	cm ⁴	$\times 10^3$ cm ⁶
100	100	100	10	6.0	56	26.0	204	167	33.5	450	89.9	9.25	3.38
120	120	120	11	6.5	74	34.0	267	318	52.9	864	144	13.8	9.41
140	140	140	12	7.0	92	43.0	337	550	78.5	1510	216	20.1	22.5
160	160	160	13	8.0	104	54.3	426	889	111	2490	311	31.2	47.9
180	180	180	14	8.5	122	65.3	512	1360	151	3830	426	42.2	93.8
200	200	200	15	9.0	134	78.1	613	2000	200	5700	570	59.3	171
220	220	220	16	9.5	152	91.0	715	2840	258	8090	736	76.6	295
240	240	240	17	10.0	164	106	832	3920	327	11260	938	103	487
260	260	260	17.5	10.0	177	118	930	5130	395	14920	1150	124	754
280	280	280	18	10.5	196	131	1030	6590	471	19270	1380	144	1130
300	300	300	19	11.0	208	149	1170	8560	571	25170	1680	185	1688
320	320	300	20.5	11.5	225	161	1270	9240	616	30820	1930	225	2069
340	340	300	21.5	12.0	243	171	1340	9690	646	36660	2160	257	2454
360	360	300	22.5	12.5	261	181	1420	10140	676	43190	2400	292	2883
400	400	300	24	13.5	298	198	1550	10820	721	57680	2880	356	3817
450	450	300	26	14.0	344	218	1710	11720	781	78890	3550	440	5259
500	500	300	28	14.5	390	239	1870	12620	842	107200	4290	538	7018
550	550	300	29	15.0	438	254	1990	13080	872	136700	4970	600	8856
600	600	300	30	15.5	486	270	2120	13530	902	171000	5700	667	10965

PROFILES UPN



UPN	Dimensiones mm					A	p	I _y	W _y	I _z	W _z	I _t	I _w
	h	b	e	e ₁	h ₁	cm ²	N/m	cm ⁴	cm ³	cm ⁴	cm ³	cm ⁴	×10 ³ cm ⁶
80	80	45	8.0	6.0	46	11.0	86.4	19.4	6.36	106	26.5	2.20	0.18
100	100	50	8.5	6.0	64	13.5	106	29.3	8.49	206	41.2	2.81	0.41
120	120	55	9.0	7.0	82	17.0	134	43.2	11.1	364	60.7	4.15	0.90
140	140	60	10.0	7.0	98	20.4	160	62.7	14.8	605	86.4	5.68	1.80
160	160	65	10.5	7.5	115	24.0	188	85.3	18.3	925	116	7.39	3.26
180	180	70	11.0	8.0	133	28.0	220	114	22.4	1350	150	9.55	5.57
200	200	75	11.5	8.5	151	32.2	253	148	27.0	1910	191	11.9	9.07
220	220	80	12.5	9.0	167	37.4	294	197	33.6	2690	245	16.0	14.6
240	240	85	13.0	9.5	184	42.3	332	248	39.6	3600	300	19.7	22.1
260	260	90	14.0	10.0	200	48.3	379	317	47.7	4820	371	25.5	33.3
280	280	95	15.0	10.0	216	53.3	418	399	57.2	6280	448	31.0	48.5
300	300	100	16.0	10.0	232	58.8	462	495	67.8	8030	535	37.4	69.1
320	320	100	17.5	14.0	246	75.8	595	597	80.6	10870	679	66.7	96.1
350	340	100	16.0	14.0	282	77.3	606	570	75.0	12840	734	61.2	114
380	380	102	16.0	13.5	313	80.4	631	615	78.7	15760	829	59.1	146
400	400	110	18.0	14.0	324	91.5	718	846	102	20350	1020	81.6	221

Appendix C

Tables

C.1 Magnitudes and Units of Mechanics

Fundamental magnitudes or *basic units* of Mechanics are those physical magnitudes from which all the others involved in a mechanical problem, which in turn are called *derived quantities* or *derived units*, are obtained.

The International System of Units (SI) is a system officially adopted in 1960. The meter (m) for length, the second (s) for time, and the kilogram (kg) for mass are chosen as basic units; the remaining magnitudes are derived from them.

The number of basic units varies according to the problem to be analyzed. Thus, for example, in a given problem time or mass may intervene or not.

Until 1983, *meter* was defined as one ten-millionth of the distance from the North Pole to the Equator. There was a platinum-iridium bar with two marks that embodied this definition. In 1983, to avoid the errors that could occur from using a physical bar as a pattern, this definition was changed. At present, the standard of the *meter* is the length of the trajectory that light travels in a vacuum during an interval of $1/299792458$ of a second.

The *second* is defined as the duration of 9192631770 periods of radiation corresponding to the transition between two hyperfine levels of the ground state of the atom of cesium-133.

The mass *kilogram* is the only basic SI unit that is still defined by means of a physical object. The standard in this case is a one kilogram cylinder of iridium platinum called *international prototype kilogram* and is deposited at the International Bureau of Weights and Measures in Sevres, Paris.

The *derived units* used in mechanics are expressed as a function of the basic units: meter, second and kilogram. Thus, for example, the unit of *force* which is the *newton*; taking into account Newton's second law ($F = ma$), it is defined as the necessary force to produce a mass of one kilogram an acceleration of one meter per second squared.

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$$

Magnitude	International System (IS)		
	Unit	Symbol	Fomula
Acceleration (linear)	meter per sec. square		m/s^2
Angle	radian		rad
Area	meter square		m^2
Density	kilog. per meter cubic		kg/m^3
Specific weight	newton per meter cubic		N/m^3
Energy; work	joule	J	$\text{N}\cdot\text{m}$
Force	newton	N	$\text{kg}\cdot\text{m/s}$
Force per unit length	newton per meter		N/m
Length	meter	m	basic unit
Mass	kilogram	kg	basic unit
Moment of a force	newton meter		$\text{N}\cdot\text{m}$
Moment of inertia (section)	meter to the fourth power		m^4
Strength module (section)	metro cubic		m^3
Power	watt	W	$\text{J/s} = \text{N}\cdot\text{m/s}$
Pressure	pascal	Pa	N/m^2
Stress	pascal	Pa	N/m^2
Time	second	s	basic unit
Velocity (lineal)	meter per second		m/s
Volume	meter cubic		m^3

Table C.1: Units used in Mechanics

C.2 Mechanical properties of materials

The following Table presents a summary of the mechanical properties of the most commonly used materials in structural mechanics. The notation adopted is:

E : modulus of elasticity (GPa) ν : poisson's ratio (dimensionless)
 G : shear modulus of stiffness (GPa) α : thermic dilatation coefficient ($^{\circ}\text{C}$)

The modulus of elasticity corresponds to tensile tests, except for the indicated cases of compression (C) and Bending (B). The properties of materials vary remarkably depending on their chemical composition, internal defects, manufacturing processes (treatment thermal, mechanization, etc), of the dimensions and shape of the specimen used in the test, etc. The values in the Table are illustrative; the engineering projects they must be obtained through appropriate tests or follow manufacturers' instructions.

Material	Specific weight	E	G	ν	α
	kN/m ³	GPa	GPa		10 ⁻⁶ /°C
Structural steel	77	210	80	0.27-0.30	10-18
Aluminum alloys	28	70	28	0.33	23
Bronze	86	120	44	0.34	18-21
Rubber	13	0.004	0.001	0.50	130-200
Cooper	87	120	44	0.36	16.9
Cast iron	72	97	39	0.25	10.8
Concrete (C)	23	30		0.1-0.2	7-14
Brass	85	110	41	0.34	19-21
Wood (pine) (F)	5.5	10			5.4
Nickel	87	210	80	0.31	13
Stone: granite. marble (C)	28	70-55	4-3	0.2-0.3	7.2-10.8
Stone: limestone. sandstone (C)	20-28	20-70	2		5-9
Polymers (nylon)	11	2.8		0.4	70-140
Polymers (polyethylene)	14	1.4		0.4	140-190
Titanium alloys	44	120	44	0.33	8-11
Tungsten	190	380	160	0.2	4.3
Glass	28	65	4.1	0.2	5-11

Table C.2: Mechanical properties of materials

C.3 Multiples and Submultiples

Multiples and Submultiples in the IS		
Prefix	Symbol	Multiplicative factor
tera	T	10^{12} = 1 000 000 000 000
giga	G	10^9 = 1 000 000 000
mega	M	10^6 = 1 000 000
kilo	k	10^3 = 1 000
hecto	h	10^2 = 100
deca	da	10^1 = 10
deci	d	10^{-1} = 0.1
centi	c	10^{-2} = 0.01
mili	m	10^{-3} = 0.001
micro	μ	10^{-6} = 0.000 001
nano	n	10^{-9} = 0.000 000 001
pico	p	10^{-12} = 0.000 000 000 001

Table C.3: Multiples and Submultiples in the IS

C.4 Greek alphabet

A	α	Alpha	I	ι	Iota	P	ρ	Rho
B	β	Beta	K	κ	Kappa	Σ	σ	Sigma
Γ	γ	Gamma	Λ	λ	Lambda	T	τ	Tau
Δ	δ	Delta	M	μ	Mu	Υ	υ	Upsilon
E	ϵ	Epsilon	N	ν	Nu	Φ	ϕ	Phi
Z	ζ	Zeta	Ξ	ξ	Xi	X	χ	Chi
H	η	Eta	O	o	Omicron	Ψ	ψ	Psi
Θ	θ	Theta	Π	π	Pi	Ω	ω	Omega

Table C.4: Greek alphabet

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“Strength of Materials” provides engineering and architecture students with a clear overview of the fundamentals of this discipline and the keys for their practical application.

The understanding of the different topics is favored by the inclusion of numerous examples worked step by step.